# Housing Project

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# R Markdown

## Importing libraries

```
# importing Libraries
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.4
## ✓ forcats   1.0.0     ✓ stringr   1.5.1
## ✓ ggplot2   3.5.1     ✓ tibble    3.2.1
## ✓ lubridate 1.9.3     ✓ tidyr     1.3.0
## ✓ purrr     1.0.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✕ dplyr::filter() masks stats::filter()
## ✕ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
e errors
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(ggplot2)
library(corrplot)
```

```
## corrplot 0.92 loaded
```

```
library(broom)
library(lmtest)
```

```
## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(DataExplorer)
library(misty)
```

```
## |--------------------------------------|
## | misty 0.6.3 (2024-05-15)             |
## | Miscellaneous Functions T. Yanagida |
## |--------------------------------------|
```

```
library(dplyr)
library(finalfit)
library(mice)
```

```
##
## Attaching package: 'mice'
##
## The following object is masked from 'package:stats':
##
##     filter
##
## The following objects are masked from 'package:base':
##
##     cbind, rbind
```

```
library(readr)
library(VIM)
```

```
## Loading required package: colorspace
## Loading required package: grid
## VIM is ready to use.
##
## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
##
## Attaching package: 'VIM'
##
## The following object is masked from 'package:datasets':
##
##     sleep
```

```
library(randomForest)
```

```
## randomForest 4.7-1.1
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```
library(FactoMineR)
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

# reading the data

```
housing_data <- read.csv("C:\\Users\\abdelrahman\\Downloads\\House_data.csv",
                         na.strings=c("","NA"))
```

# Lets check some info about the data

the data contain huge number of feature… to make some info about the feature we will cutting it into 3 parts

```
plot_str(housing_data[,0:30])
```

```
plot_str(housing_data[,30:60])
```

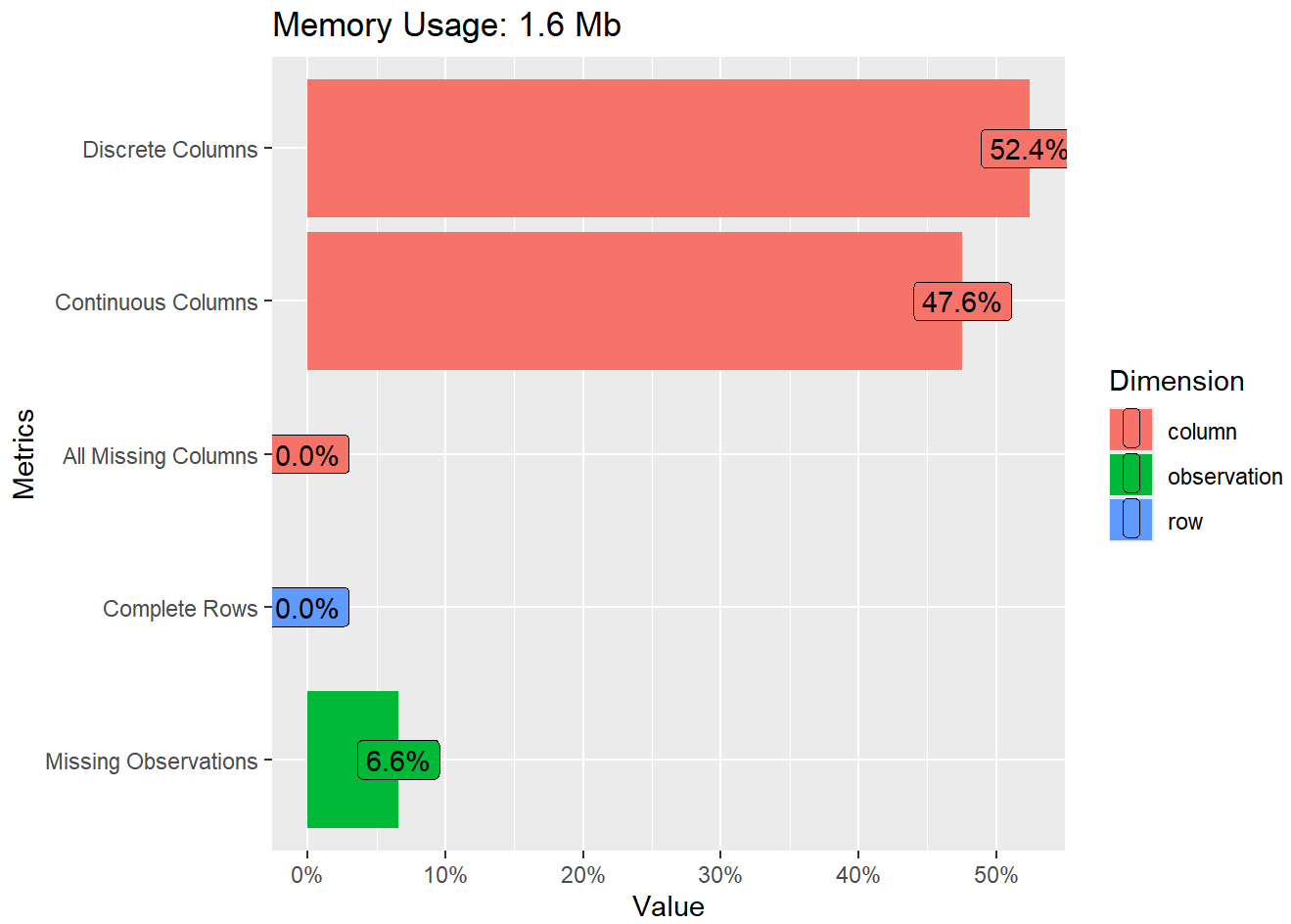
```
plot_str(housing_data[,60:82])
```

## check some info about our data

```
# lets introduce our data
transpose(view(introduce(housing_data)))
```

```
## [[1]]
## [[1]]$rows
## [1] 2919
##
## [[1]]$columns
## [1] 82
##
## [[1]]$discrete_columns
## [1] 43
##
## [[1]]$continuous_columns
## [1] 39
##
## [[1]]$all_missing_columns
## [1] 0
##
## [[1]]$total_missing_values
## [1] 15707
##
## [[1]]$complete_rows
## [1] 0
##
## [[1]]$total_observations
## [1] 239358
##
## [[1]]$memory_usage
## [1] 1630968
```

```
plot_intro(housing_data)
```




```
# we have about 50 of columns discrete and 50 continuous
# 6.6% missing Values Observation
```

# Missing Data

here we see that we have missing obs. lets check for each feature

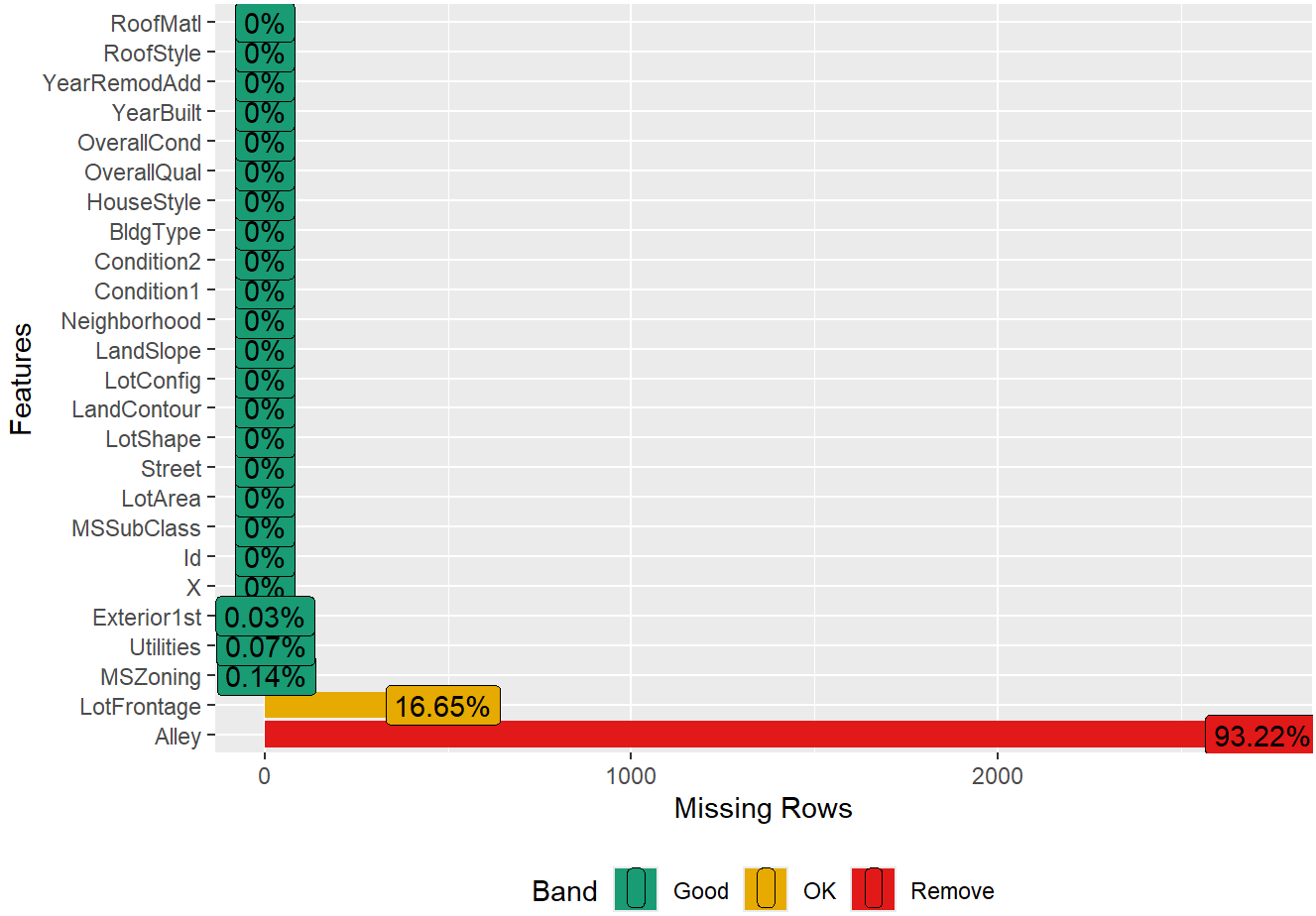
```
missing_percentage <- function(data) {
  colSums(is.na(data)) / nrow(data) * 100
}

missing_perc <- missing_percentage(housing_data)
missing_perc
```

```
##             X            Id    MSSubClass      MSZoning   LotFrontage
##    0.00000000    0.00000000    0.00000000    0.13703323   16.64953751
##       LotArea        Street         Alley      LotShape   LandContour
##    0.00000000    0.00000000   93.21685509    0.00000000    0.00000000
##     Utilities     LotConfig     LandSlope  Neighborhood    Condition1
##    0.06851662    0.00000000    0.00000000    0.00000000    0.00000000
##    Condition2      BldgType    HouseStyle   OverallQual   OverallCond
##    0.00000000    0.00000000    0.00000000    0.00000000    0.00000000
##     YearBuilt  YearRemodAdd     RoofStyle      RoofMatl   Exterior1st
##    0.00000000    0.00000000    0.00000000    0.00000000    0.03425831
##   Exterior2nd    MasVnrType    MasVnrArea     ExterQual     ExterCond
##    0.03425831   60.50017129    0.78794108    0.00000000    0.00000000
##    Foundation      BsmtQual      BsmtCond  BsmtExposure  BsmtFinType1
##    0.00000000    2.77492292    2.80918123    2.80918123    2.70640630
##    BsmtFinSF1  BsmtFinType2    BsmtFinSF2     BsmtUnfSF   TotalBsmtSF
##    0.03425831    2.74066461    0.03425831    0.03425831    0.03425831
##       Heating     HeatingQC    CentralAir    Electrical     X1stFlrSF
##    0.00000000    0.00000000    0.00000000    0.03425831    0.00000000
##     X2ndFlrSF  LowQualFinSF     GrLivArea  BsmtFullBath  BsmtHalfBath
##    0.00000000    0.00000000    0.00000000    0.06851662    0.06851662
##      FullBath      HalfBath  BedroomAbvGr  KitchenAbvGr   KitchenQual
##    0.00000000    0.00000000    0.00000000    0.00000000    0.03425831
##  TotRmsAbvGrd    Functional    Fireplaces   FireplaceQu    GarageType
##    0.00000000    0.06851662    0.00000000   48.64679685    5.37855430
##   GarageYrBlt  GarageFinish    GarageCars    GarageArea    GarageQual
##    5.44707091    5.44707091    0.03425831    0.03425831    5.44707091
##    GarageCond    PavedDrive    WoodDeckSF   OpenPorchSF EnclosedPorch
##    5.44707091    0.00000000    0.00000000    0.00000000    0.00000000
##    X3SsnPorch   ScreenPorch      PoolArea        PoolQC         Fence
##    0.00000000    0.00000000    0.00000000   99.65741692   80.43850634
##   MiscFeature       MiscVal        MoSold        YrSold      SaleType
##   96.40287770    0.00000000    0.00000000    0.00000000    0.03425831
## SaleCondition     SalePrice
##    0.00000000    0.00000000
```

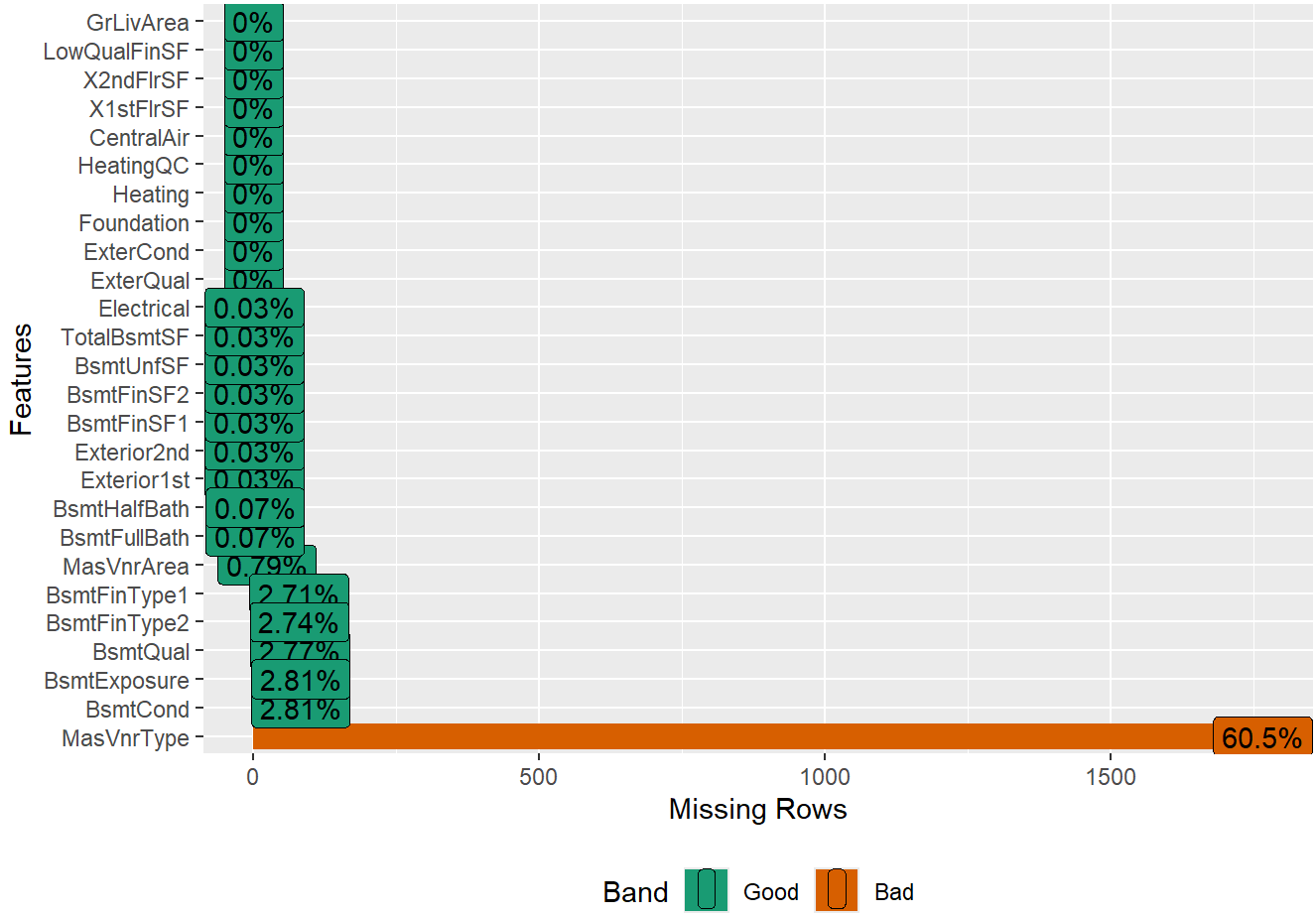
plot the % of Missing in Each Column and Location Of Missing

## % of missing in each column

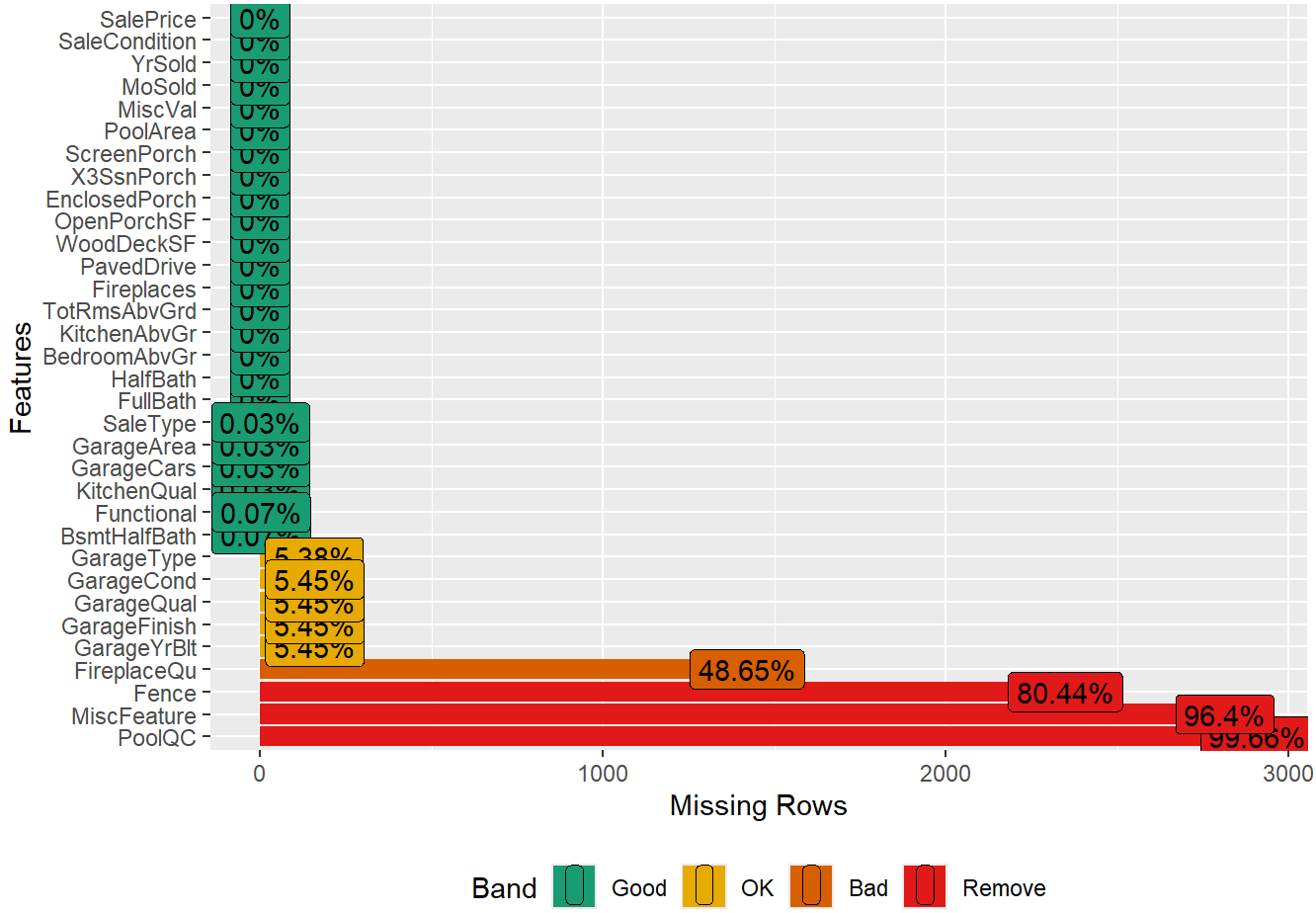
```
plot_missing(housing_data[,0:25])
```




```
plot_missing(housing_data[,25:50])
```

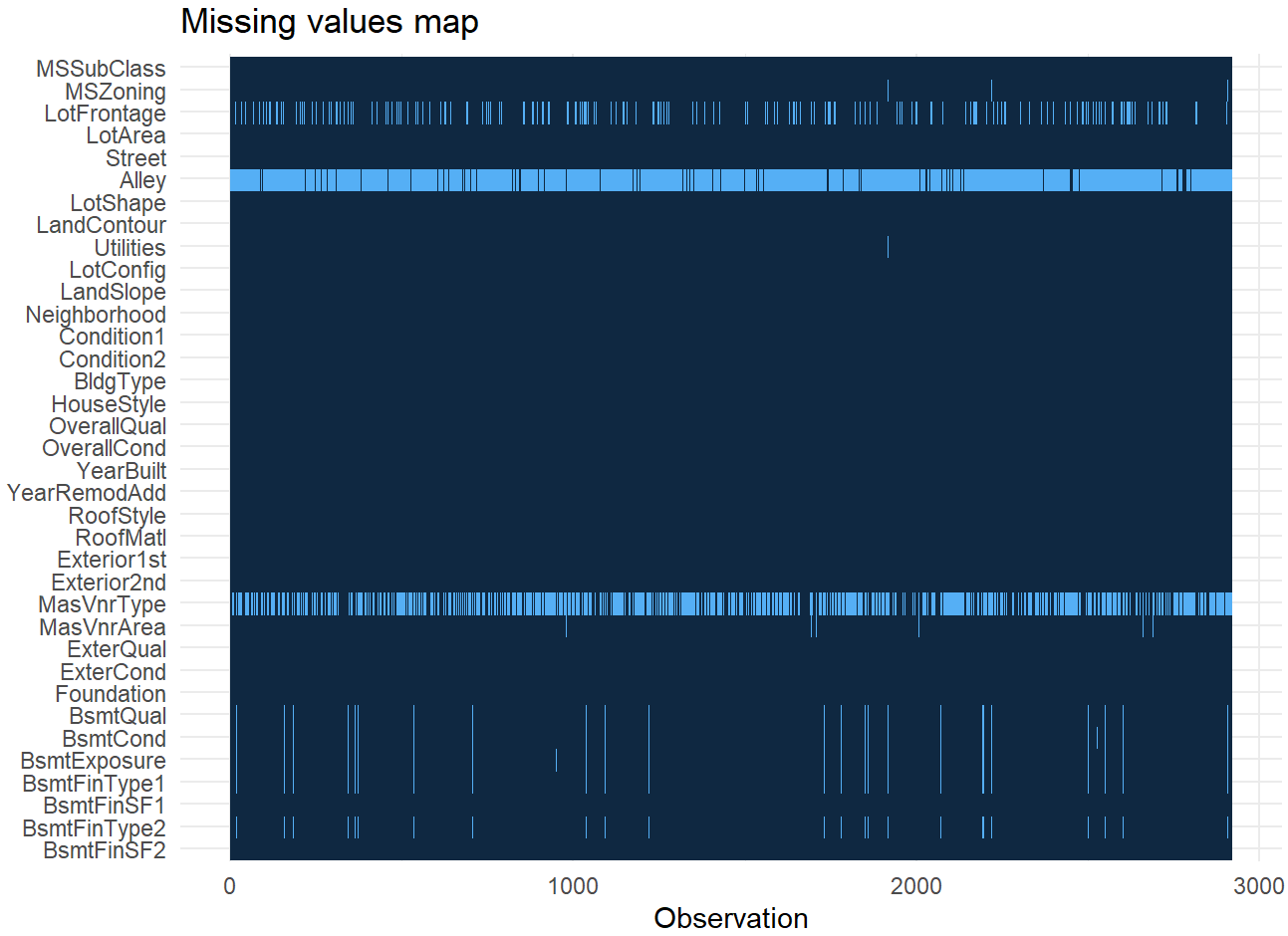



```
plot_missing(housing_data[,50:82])
```

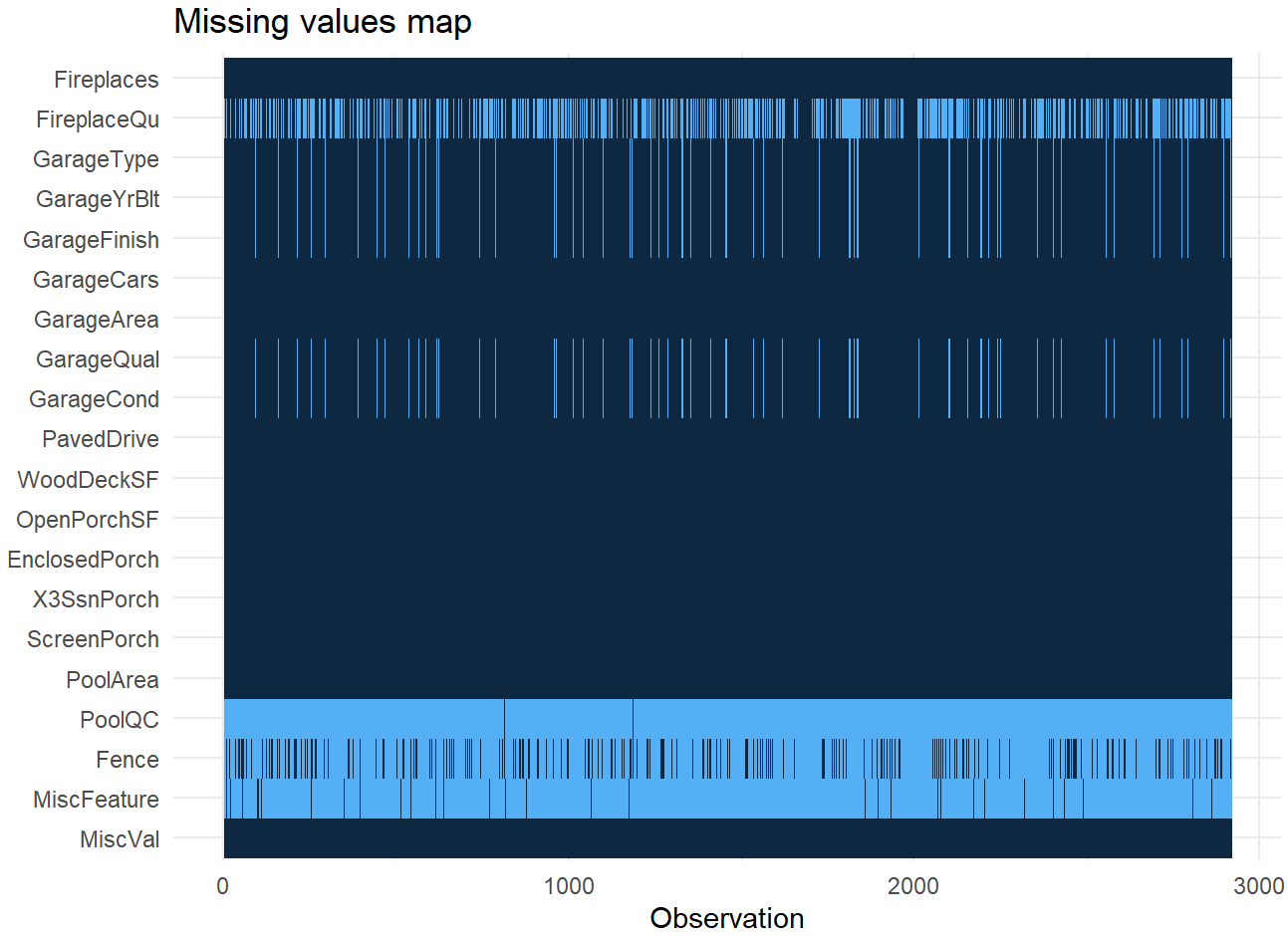



## Location of the Missing

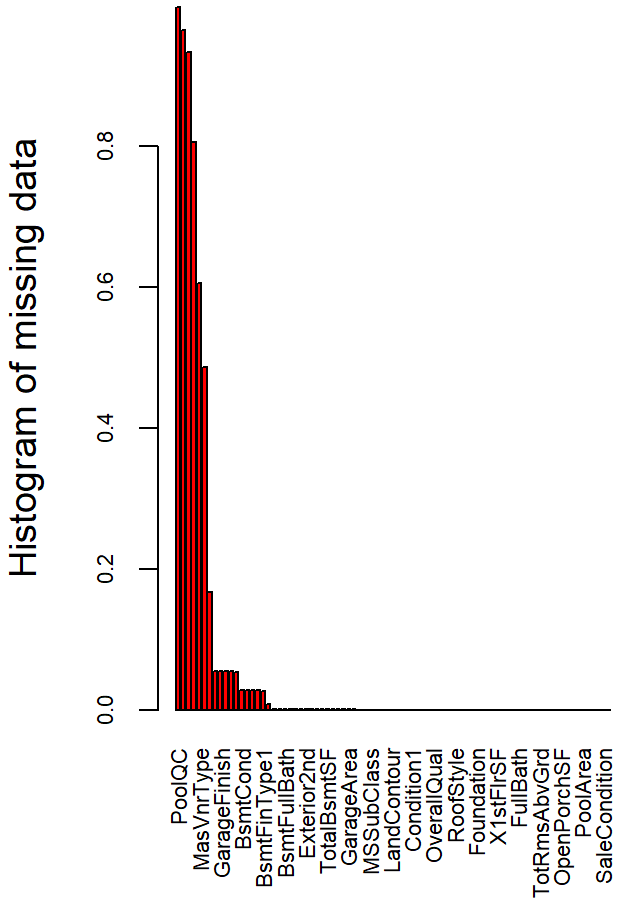
```
housing_data[,3:38]%>%
  missing_plot()
```

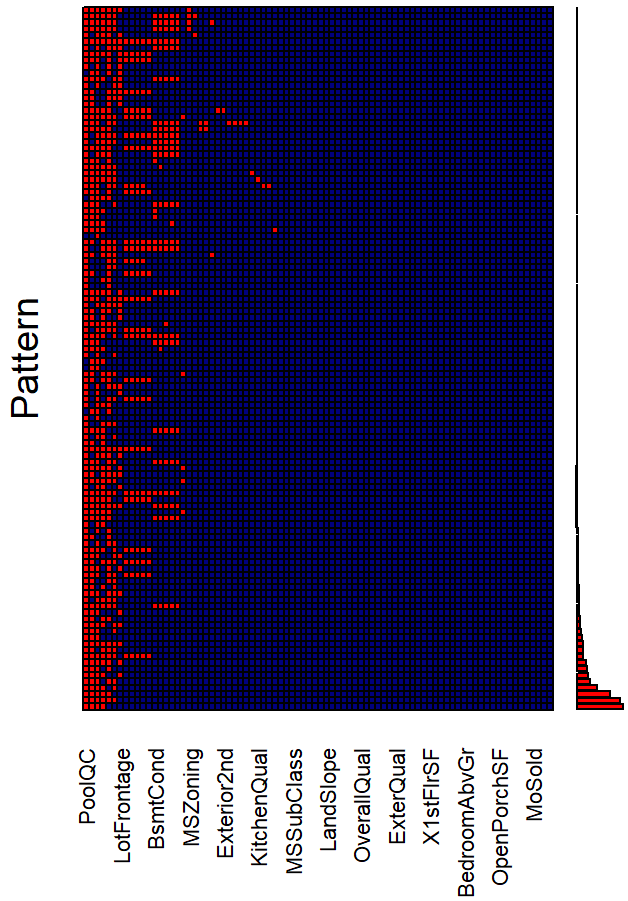



```
housing_data[,58:77]%>%
  missing_plot()
```




```
#pattern of the missing
aggr_plot <- aggr(housing_data, col=c('navyblue','red'),
                  numbers=TRUE,
                  sortVars=TRUE, labels=names(housing_data), cex.axis=.7, gap=3,
                  ylab=c("Histogram of missing data","Pattern"))
```


Histogram of missing data
0.0
0.2
0.4
0.6
0.8
PoolQC
MasVnrType
GarageFinish
BsmtCond
BsmtFinType1
BsmtFullBath
Exterior2nd
TotalBsmtSF
GarageArea
MSSubClass
LandContour
Condition1
OverallQual
RoofStyle
Foundation
X1stFlrSF
FullBath
TotRmsAbvGrd
OpenPorchSF
PoolArea
SaleCondition


Pattern
PoolQC
LotFrontage
BsmtCond
MSZoning
Exterior2nd
KitchenQual
MSSubClass
LandSlope
OverallQual
ExterQual
X1stFlrSF
BedroomAbvGr
OpenPorchSF
MoSold

```
##
##  Variables sorted by number of missings:
##       Variable        Count
##         PoolQC 0.9965741692
##    MiscFeature 0.9640287770
##          Alley 0.9321685509
##          Fence 0.8043850634
##     MasVnrType 0.6050017129
##    FireplaceQu 0.4864679685
##    LotFrontage 0.1664953751
##    GarageYrBlt 0.0544707091
##   GarageFinish 0.0544707091
##     GarageQual 0.0544707091
##     GarageCond 0.0544707091
##     GarageType 0.0537855430
##       BsmtCond 0.0280918123
##   BsmtExposure 0.0280918123
##       BsmtQual 0.0277492292
##   BsmtFinType2 0.0274066461
##   BsmtFinType1 0.0270640630
##     MasVnrArea 0.0078794108
##       MSZoning 0.0013703323
##      Utilities 0.0006851662
##   BsmtFullBath 0.0006851662
##   BsmtHalfBath 0.0006851662
##     Functional 0.0006851662
##    Exterior1st 0.0003425831
##    Exterior2nd 0.0003425831
##     BsmtFinSF1 0.0003425831
##     BsmtFinSF2 0.0003425831
##      BsmtUnfSF 0.0003425831
##    TotalBsmtSF 0.0003425831
##     Electrical 0.0003425831
##    KitchenQual 0.0003425831
##     GarageCars 0.0003425831
##     GarageArea 0.0003425831
##       SaleType 0.0003425831
##              X 0.0000000000
##             Id 0.0000000000
##     MSSubClass 0.0000000000
##        LotArea 0.0000000000
##         Street 0.0000000000
##       LotShape 0.0000000000
##    LandContour 0.0000000000
##      LotConfig 0.0000000000
##      LandSlope 0.0000000000
##   Neighborhood 0.0000000000
##     Condition1 0.0000000000
##     Condition2 0.0000000000
##       BldgType 0.0000000000
##     HouseStyle 0.0000000000
##    OverallQual 0.0000000000
```

```
##    OverallCond 0.0000000000
##      YearBuilt 0.0000000000
##   YearRemodAdd 0.0000000000
##      RoofStyle 0.0000000000
##       RoofMatl 0.0000000000
##      ExterQual 0.0000000000
##      ExterCond 0.0000000000
##     Foundation 0.0000000000
##        Heating 0.0000000000
##      HeatingQC 0.0000000000
##     CentralAir 0.0000000000
##      X1stFlrSF 0.0000000000
##      X2ndFlrSF 0.0000000000
##   LowQualFinSF 0.0000000000
##      GrLivArea 0.0000000000
##       FullBath 0.0000000000
##       HalfBath 0.0000000000
##   BedroomAbvGr 0.0000000000
##   KitchenAbvGr 0.0000000000
##   TotRmsAbvGrd 0.0000000000
##     Fireplaces 0.0000000000
##     PavedDrive 0.0000000000
##     WoodDeckSF 0.0000000000
##    OpenPorchSF 0.0000000000
##  EnclosedPorch 0.0000000000
##     X3SsnPorch 0.0000000000
##    ScreenPorch 0.0000000000
##       PoolArea 0.0000000000
##        MiscVal 0.0000000000
##         MoSold 0.0000000000
##         YrSold 0.0000000000
##  SaleCondition 0.0000000000
##      SalePrice 0.0000000000
```

we will remove the features that have missing more than 40%

```
data_after_removing_high_missing<- subset(housing_data, select = -c(X,
                                                                    Id,
                                                                    Alley,
                                                                    MasVnrType,
                                                                    FireplaceQu,
                                                                    PoolQC,
                                                                    Fence,
                                                                   MiscFeature))
print(dim(data_after_removing_high_missing))
```

```
## [1] 2919   74
```

```
print(dim(housing_data))
```

```
## [1] 2919   82
```

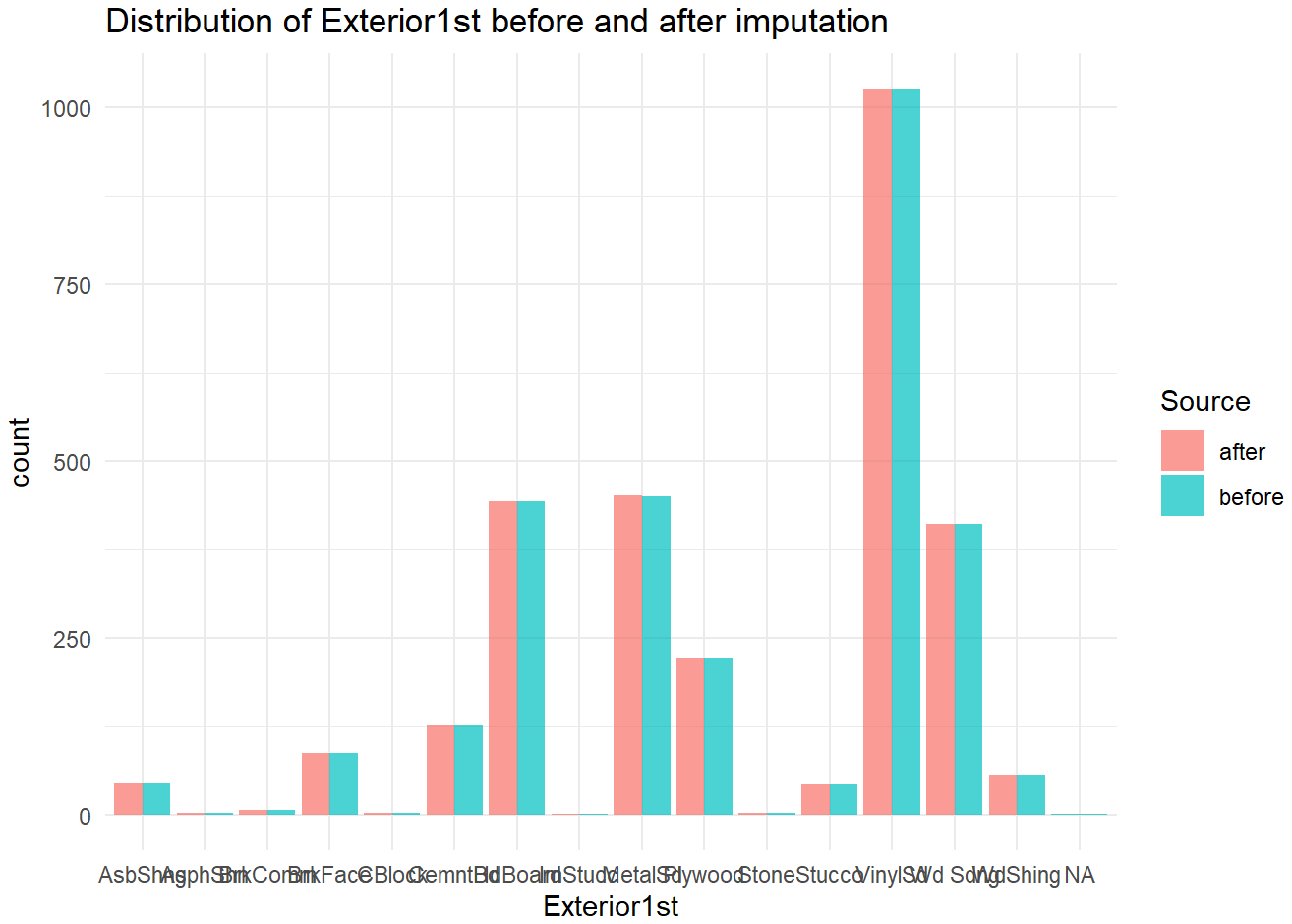
```
# sorting the varibles accourding the number of missing
NAs <- data_after_removing_high_missing %>%
  is.na() %>%
  colSums() %>%
  sort(decreasing = FALSE) %>%
  names()
```

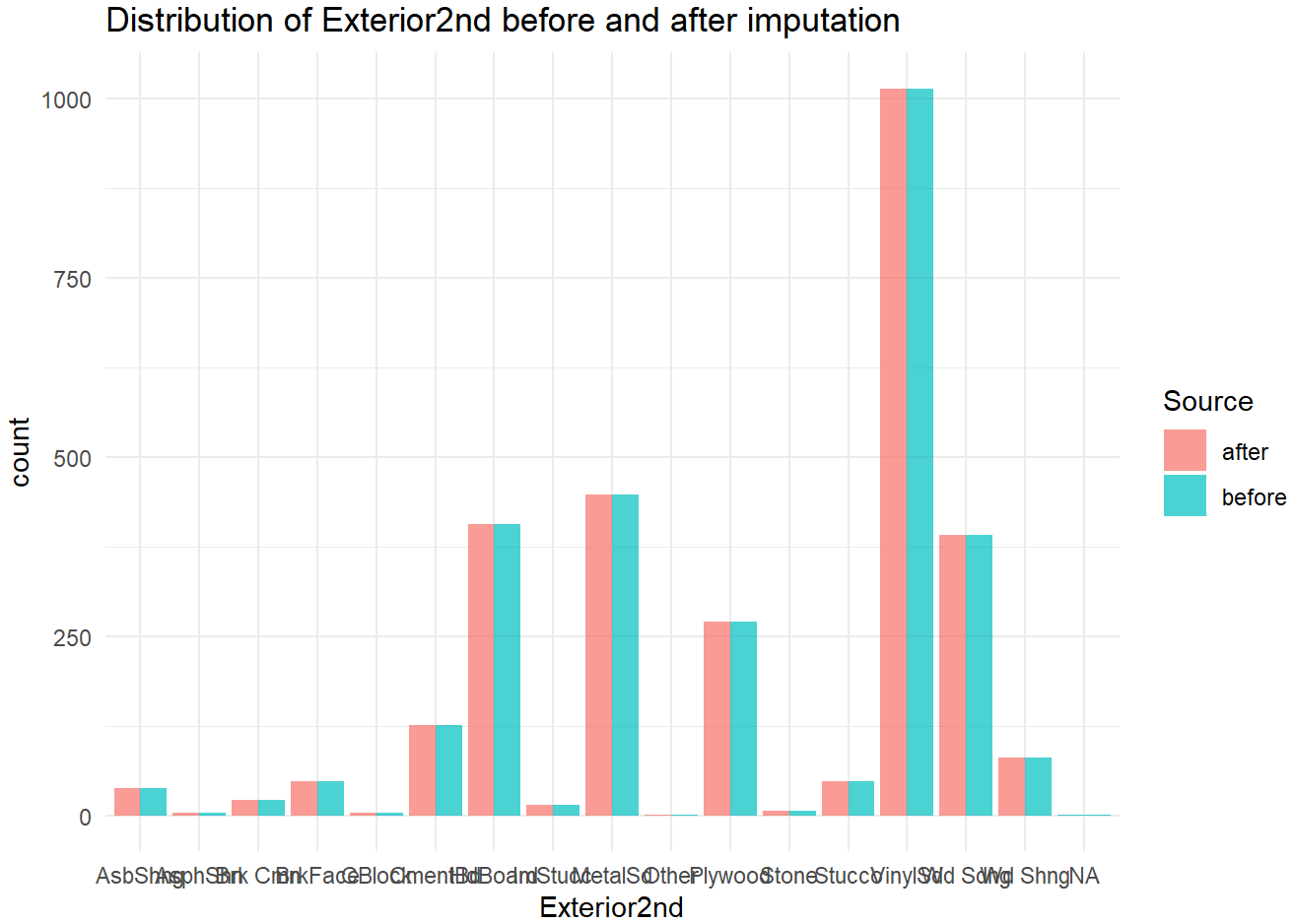
```
# Filter columns with missing values
columns_with_na <- NAs[colSums(is.na(data_after_removing_high_missing[NAs])) > 0]

is_numeric_column <- function(df, col) {
  is.numeric(df[[col]])
}
```

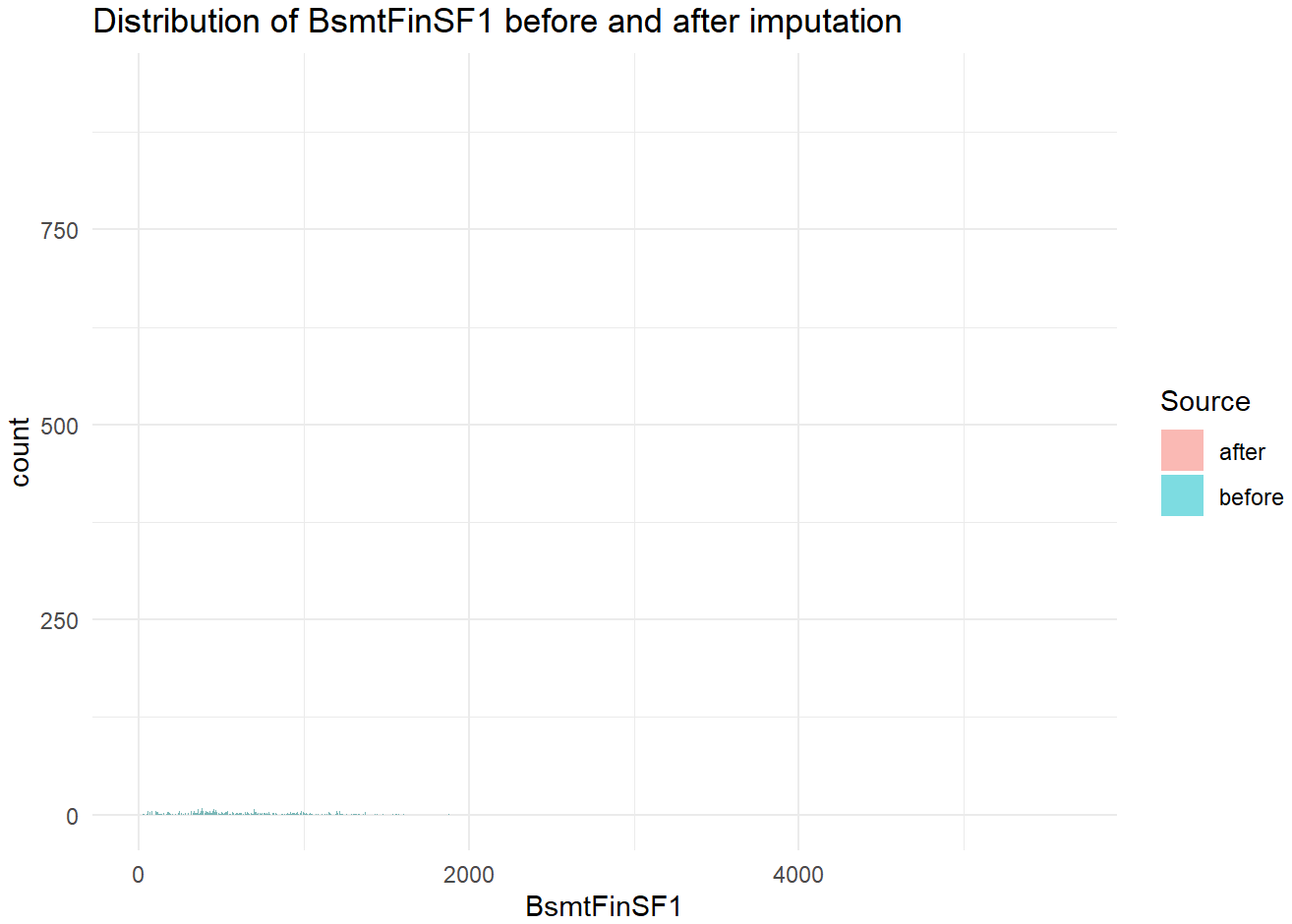
```
# check if the variable is numeric
is_numeric_column <- function(df, col) {
  is.numeric(df[[col]])
}
```

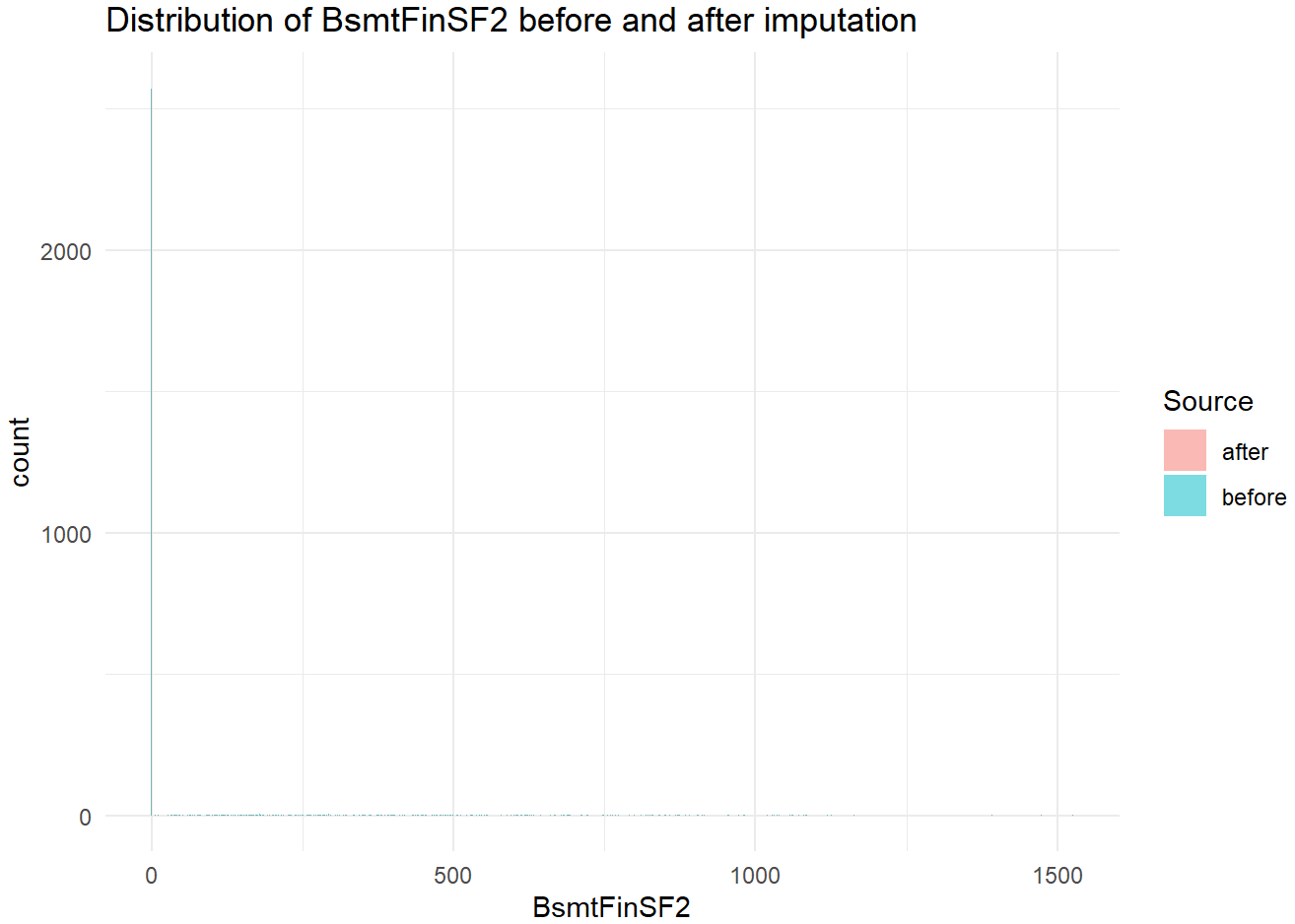
```
# impute the data with KNN imputation
Knnimputaion <- data_after_removing_high_missing %>%
  select(columns_with_na) %>%
  kNN(k = 5)
```

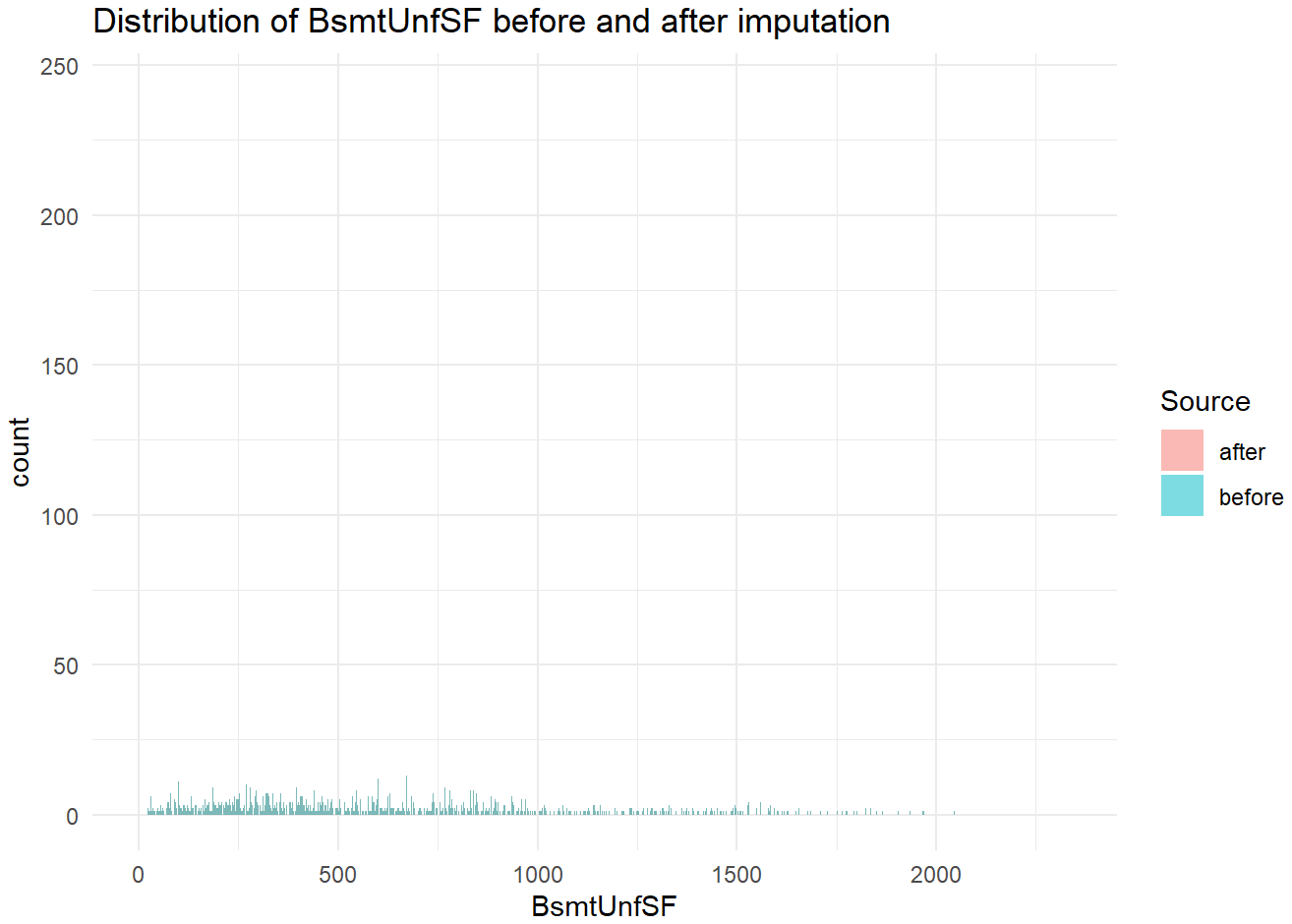
```
# Combine the imputed data with the rest of the dataset
imputed_data <- data_after_removing_high_missing
imputed_data[columns_with_na] <- Knnimputaion[columns_with_na]


combined_data <- data_after_removing_high_missing %>%
  mutate(source = "before") %>%
  bind_rows(imputed_data %>%
              mutate(source = "after"))


plot_combined_distribution <- function(data, cols) {
  for (col in cols) {
    if (is.numeric(data[[col]])) {
      p <- ggplot(data, aes(x = !!sym(col), fill = source)) +
        geom_histogram(position = "identity", alpha = 0.5, binwidth = 1) +
        theme_minimal() +
        labs(title = paste("Distribution of", col, "before and after imputation"),
             fill = "Source")
    } else {
      p <- ggplot(data, aes(x = !!sym(col), fill = source)) +
        geom_bar(position = "dodge", alpha = 0.7) +
        theme_minimal() +
        labs(title = paste("Distribution of", col, "before and after imputation"),
             fill = "Source")
    }
    print(p)
  }
}

# Plot combined distributions of columns with missing values
plot_combined_distribution(combined_data, columns_with_na)
```

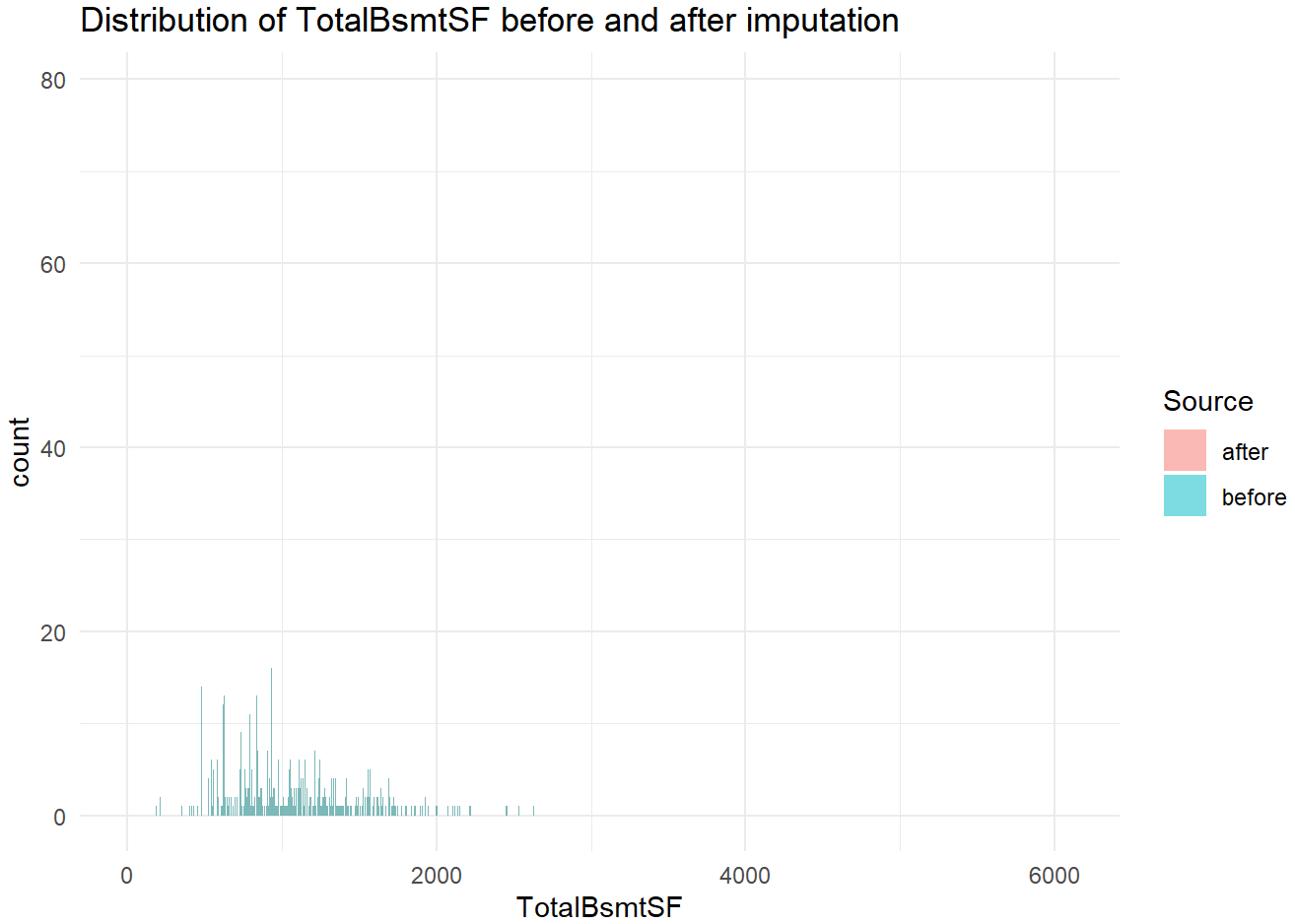

Distribution of Exterior1st before and after imputation
count
1000
750
500
250
0
AsbShng
AsphShn
Brk Cmn
BrkFace
CBlock
CemntBd
HdBoard
ImStucc
MetalSd
Plywood
Stone
Stucco
VinylSd
Wd Sdng
WdShing
NA
Exterior1st
Source
after
before

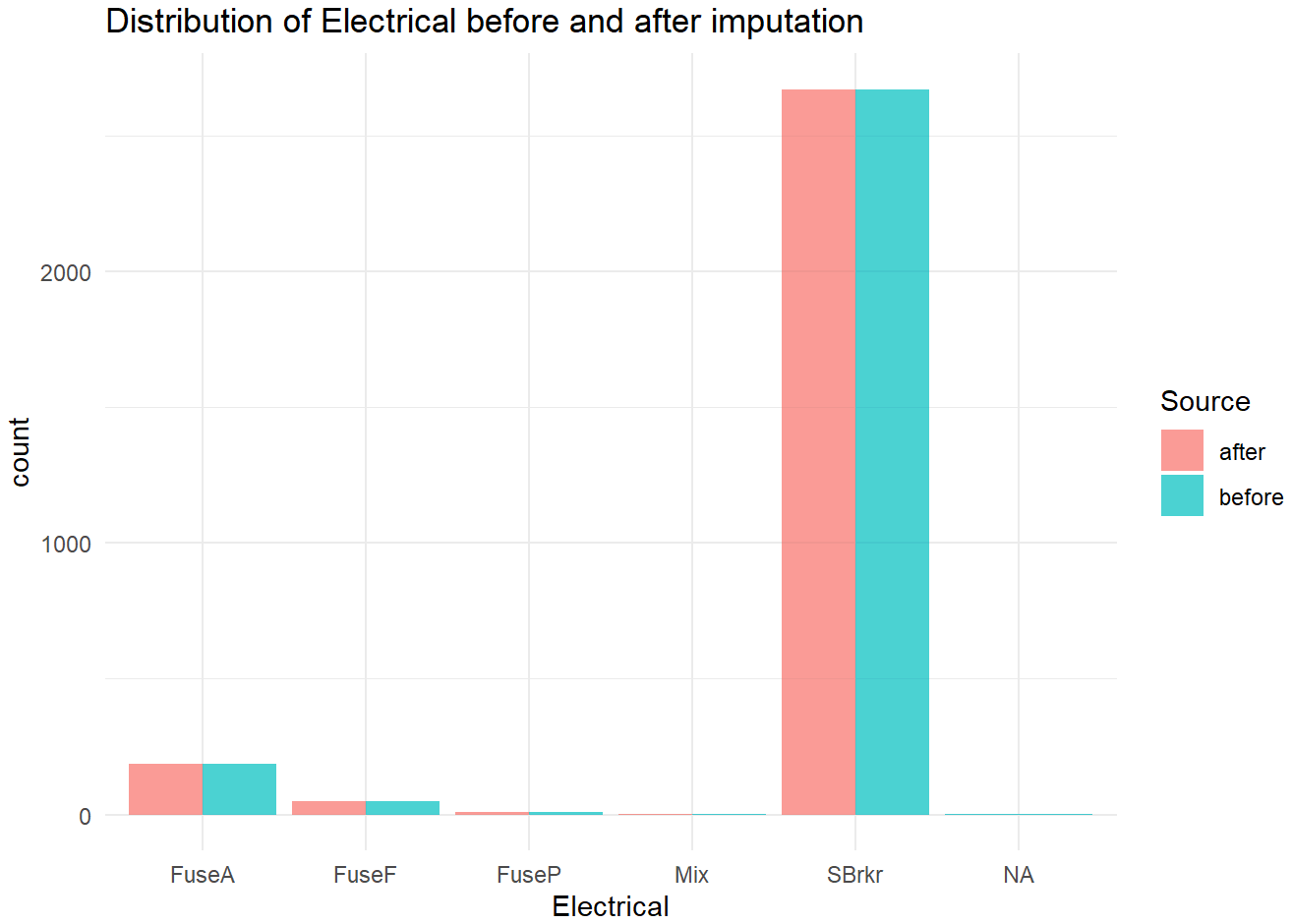

Distribution of Exterior2nd before and after imputation
count
1000
750
500
250
0
AsbShng
AsphShn
Brk Cmn
BrkFace
CBlock
CmentBd
HdBoard
ImStucc
MetalSd
Other
Plywood
Stone
Stucco
VinylSd
Wd Sdng
Wd Shng
NA
Exterior2nd
Source
after
before

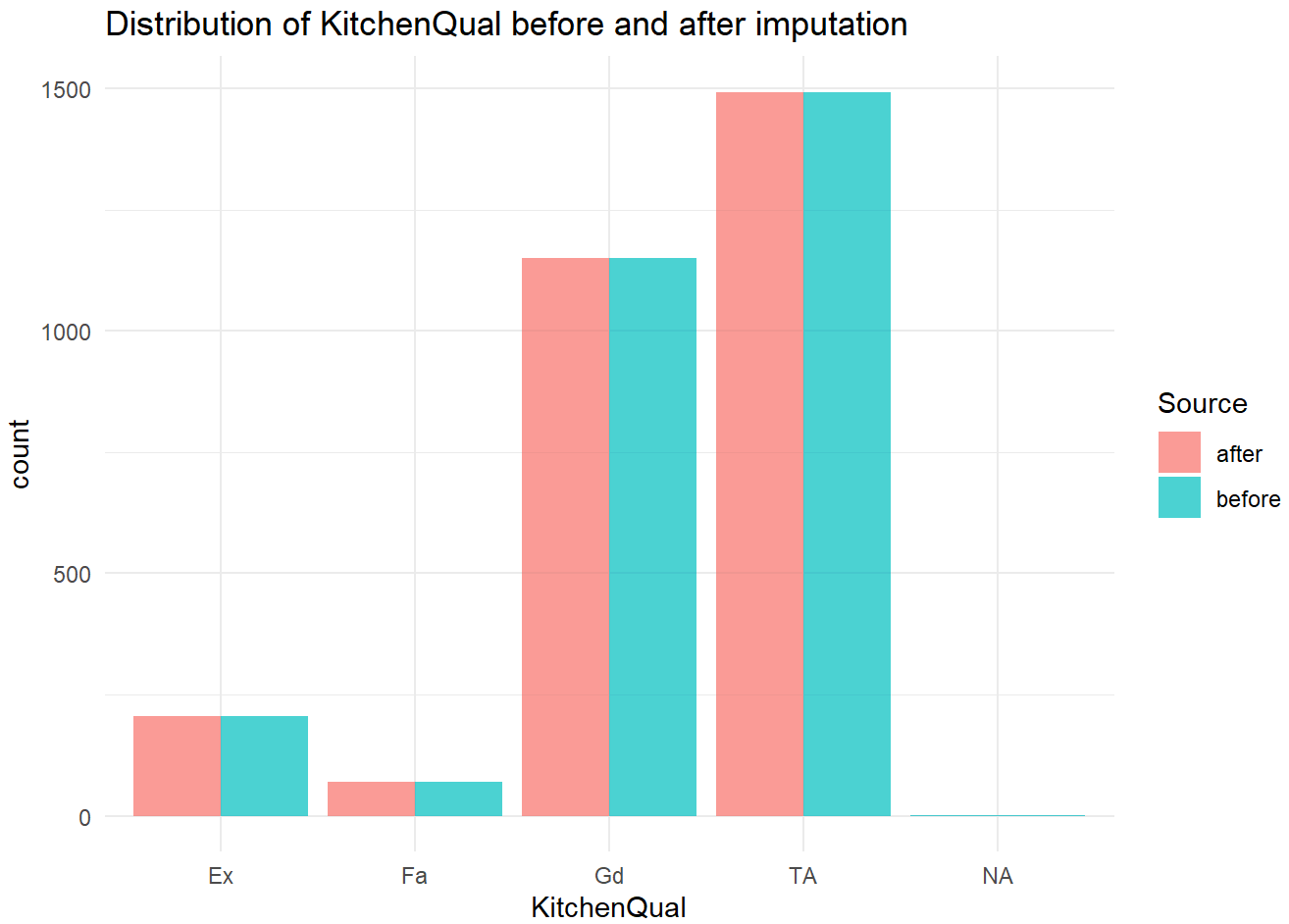

Distribution of BsmtFinSF1 before and after imputation
count
750
500
250
0
0
2000
4000
BsmtFinSF1
Source
after
before

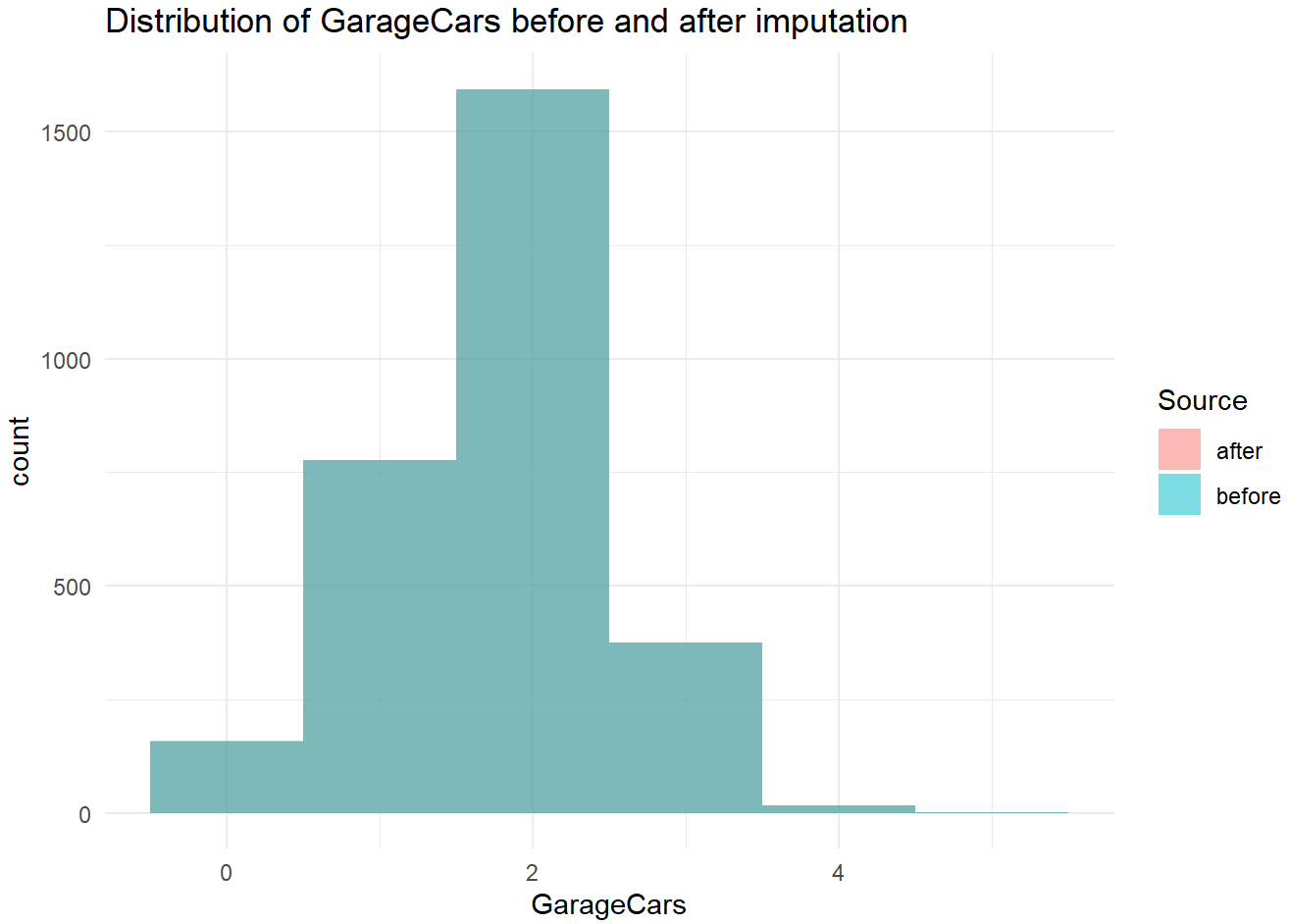

Distribution of BsmtFinSF2 before and after imputation
count
2000
1000
0
0
500
1000
1500
BsmtFinSF2
Source
after
before

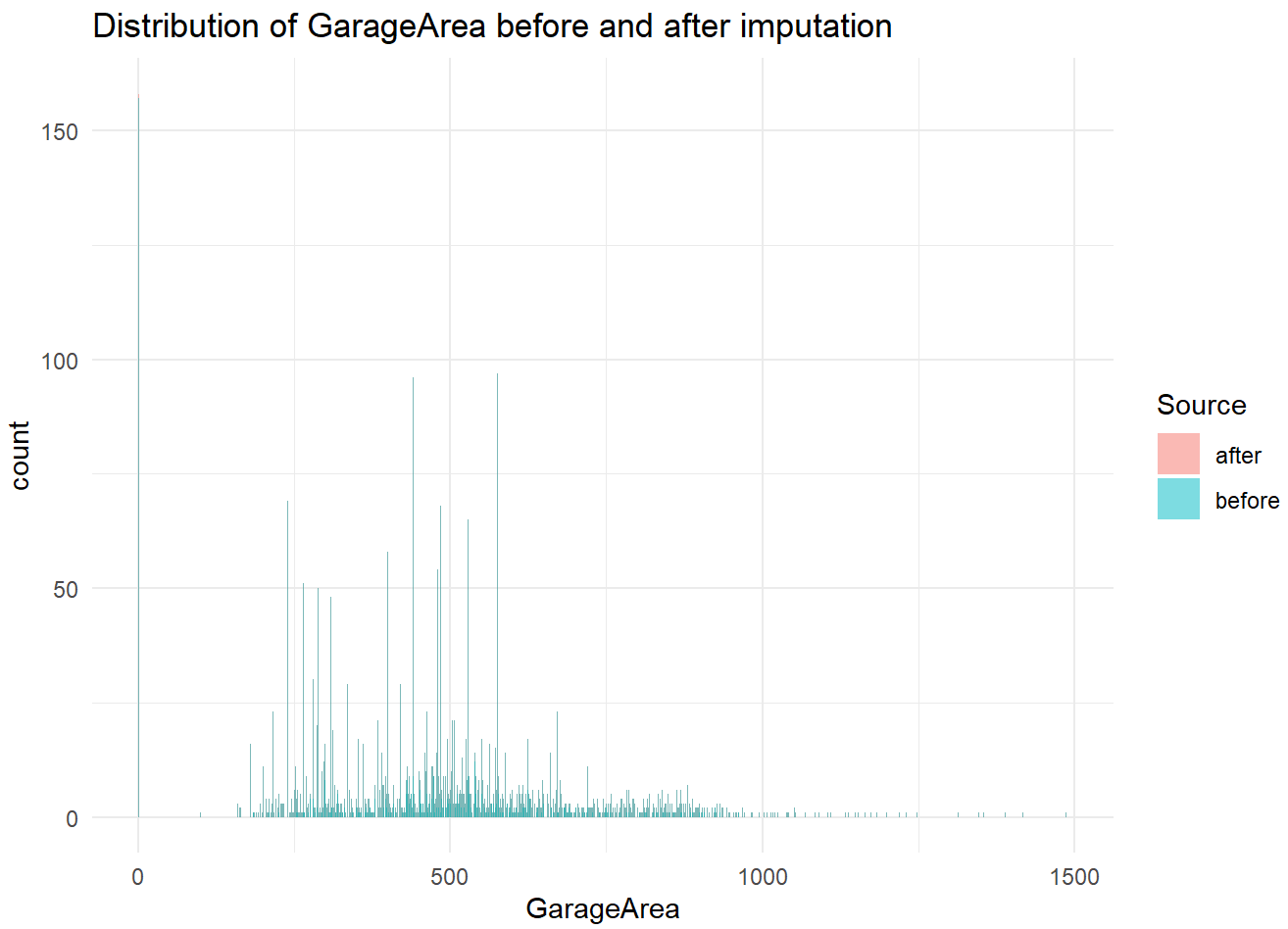

Distribution of BsmtUnfSF before and after imputation
250
200
150
100
50
0
count
0
500
1000
1500
2000
BsmtUnfSF
Source
after
before

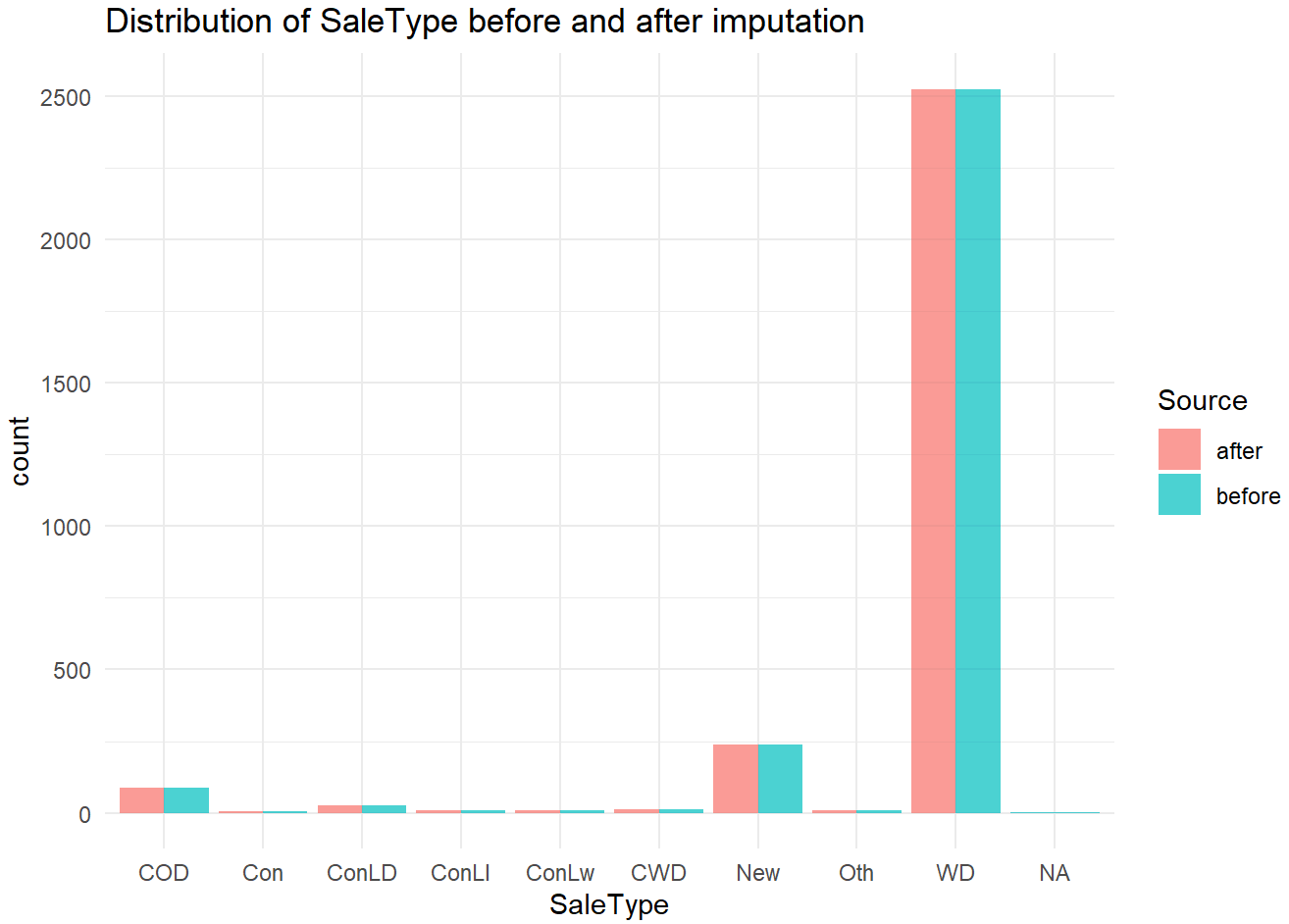

Distribution of TotalBsmtSF before and after imputation
80
60
40
20
0
count
0
2000
4000
6000
TotalBsmtSF
Source
after
before

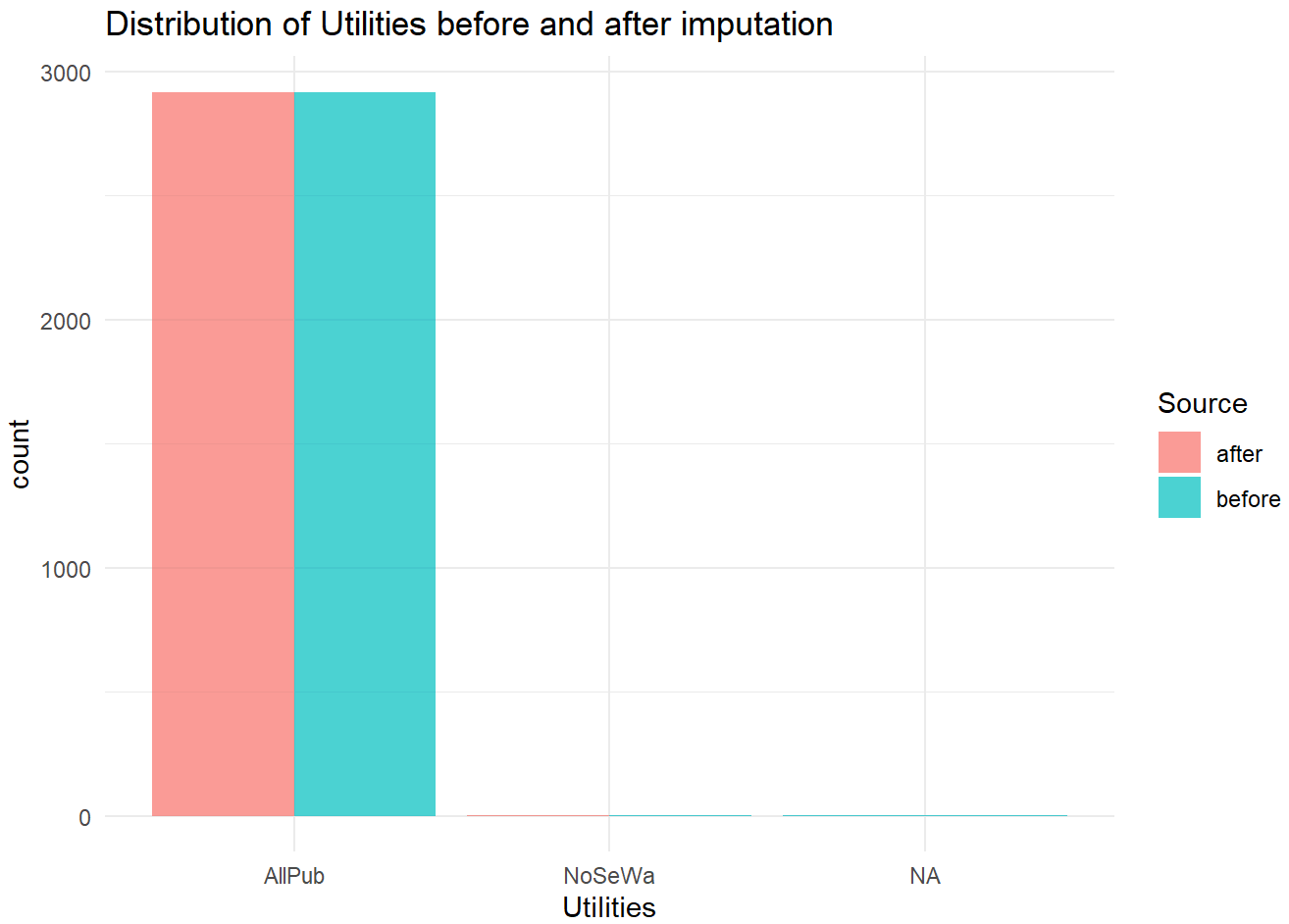

Distribution of Electrical before and after imputation
count
2000
1000
0
FuseA
FuseF
FuseP
Mix
SBrkr
NA
Electrical
Source
after
before

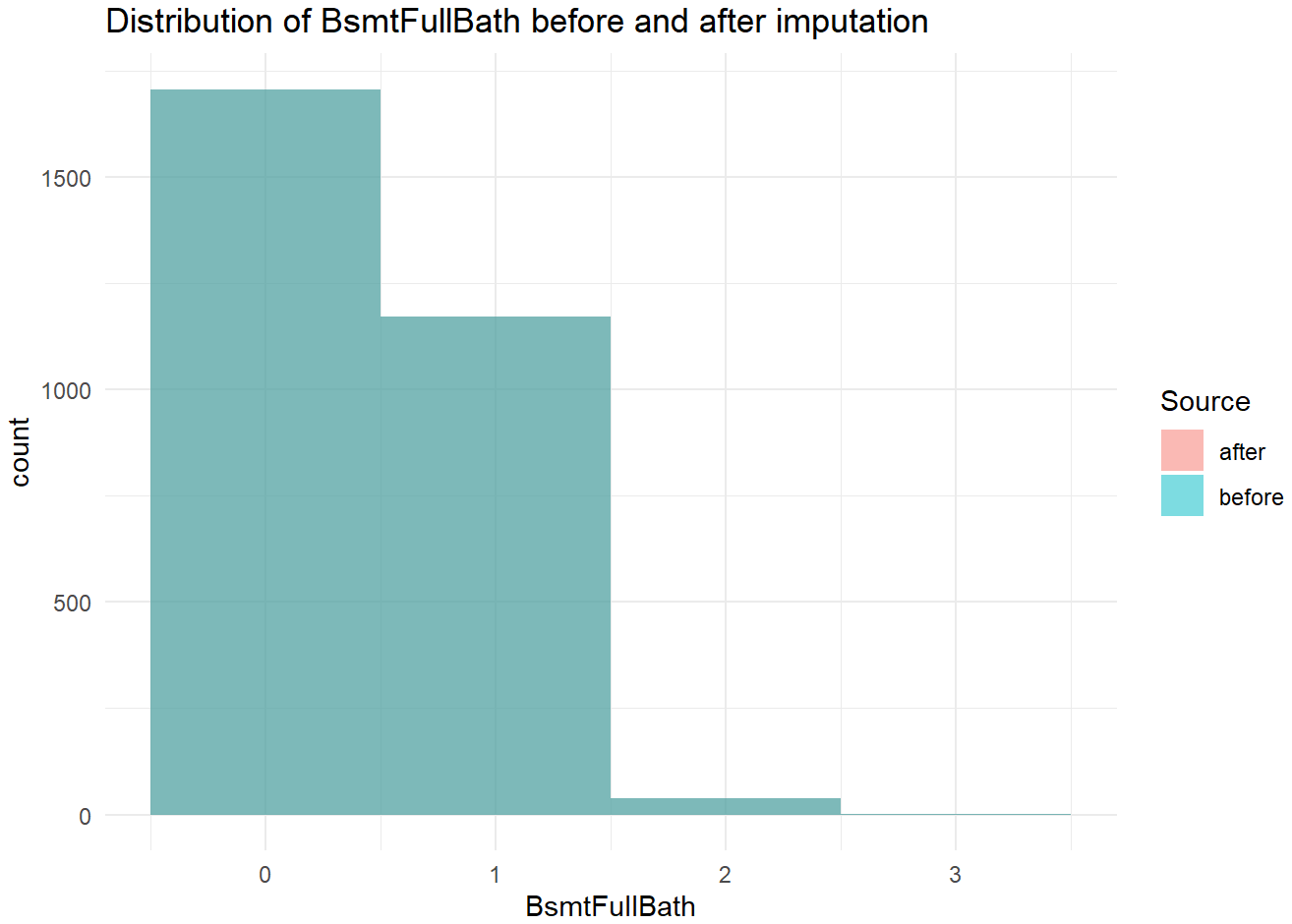

Distribution of KitchenQual before and after imputation
count
1500
1000
500
0
Ex
Fa
Gd
TA
NA
KitchenQual
Source
after
before

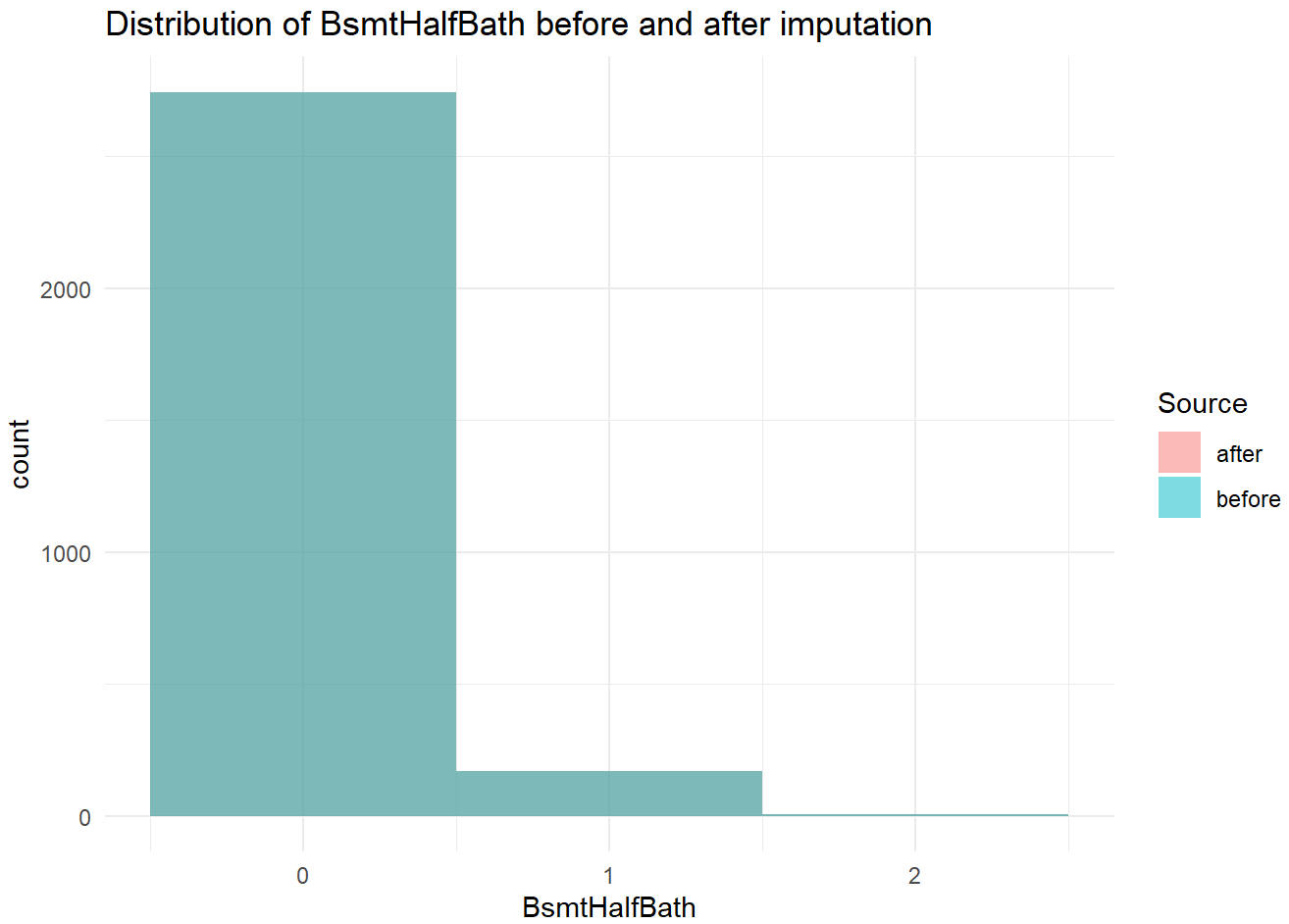

Distribution of GarageCars before and after imputation
1500
1000
500
0
count
0
2
4
GarageCars
Source
after
before

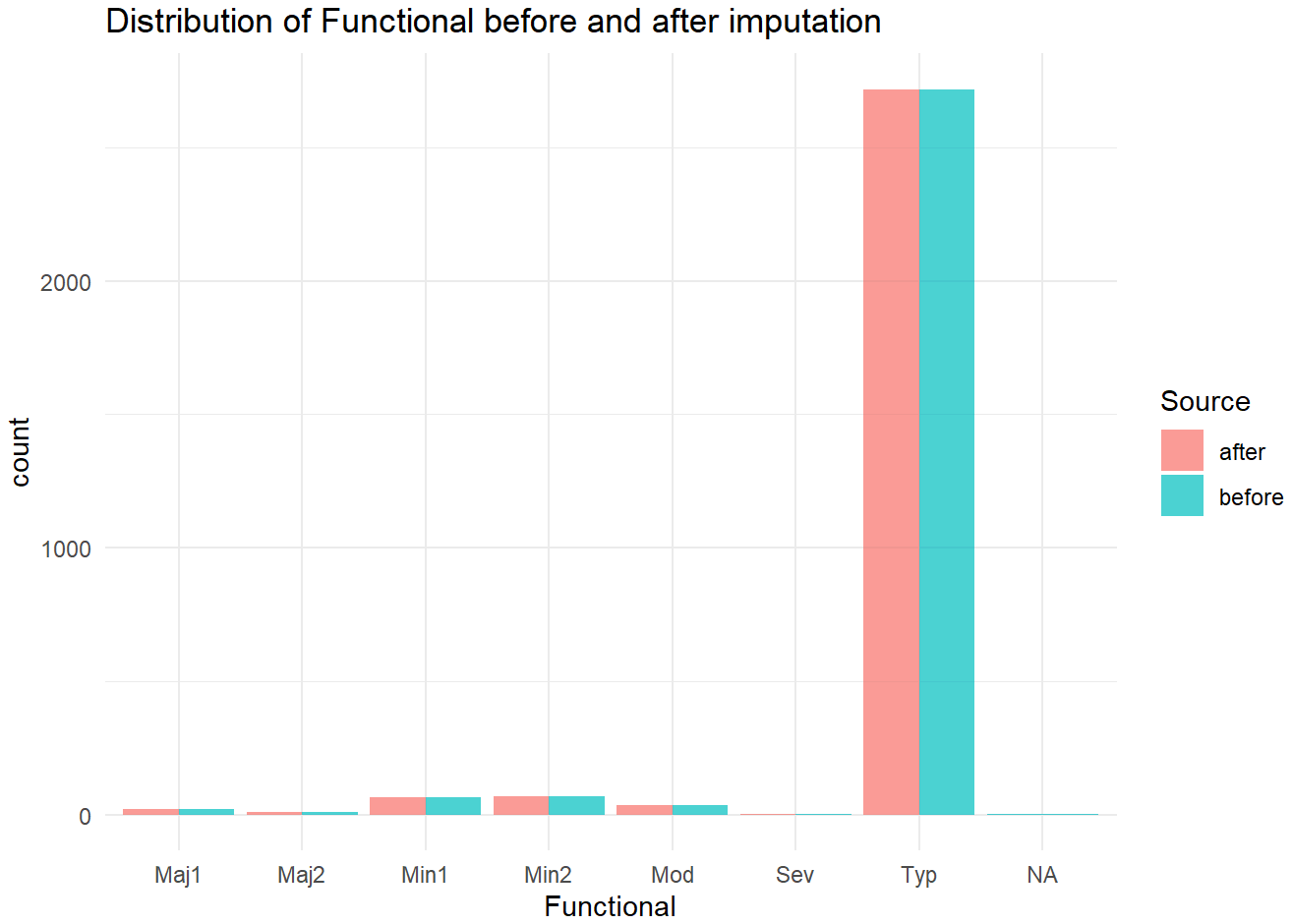

Distribution of GarageArea before and after imputation
150
100
50
0
count
0
500
1000
1500
GarageArea
Source
after
before


Distribution of SaleType before and after imputation
2500
2000
1500
1000
500
0
count
COD
Con
ConLD
ConLI
ConLw
CWD
New
Oth
WD
NA
SaleType
Source
after
before

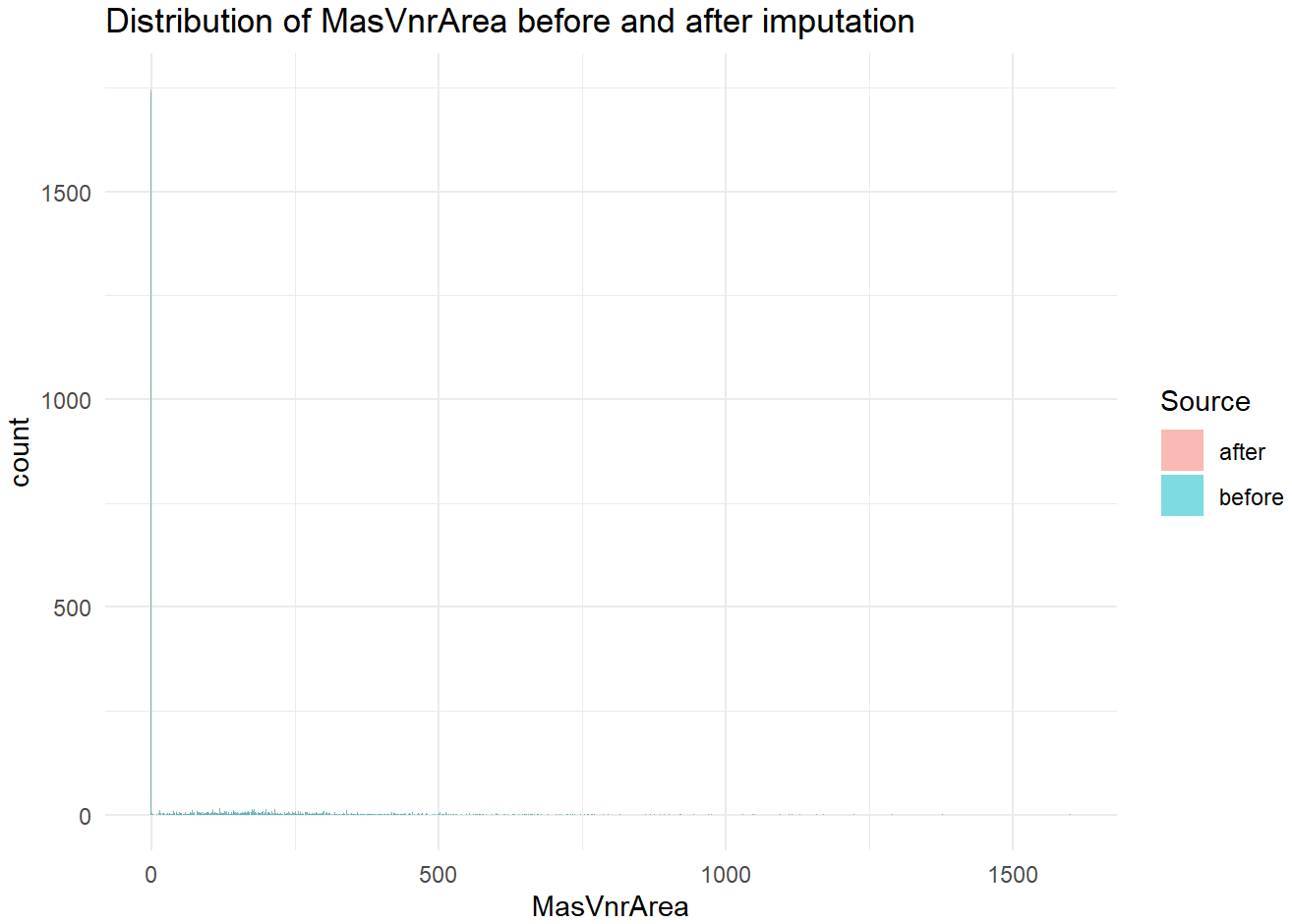

Distribution of Utilities before and after imputation
3000
2000
1000
0
count
AllPub
NoSeWa
NA
Utilities
Source
after
before

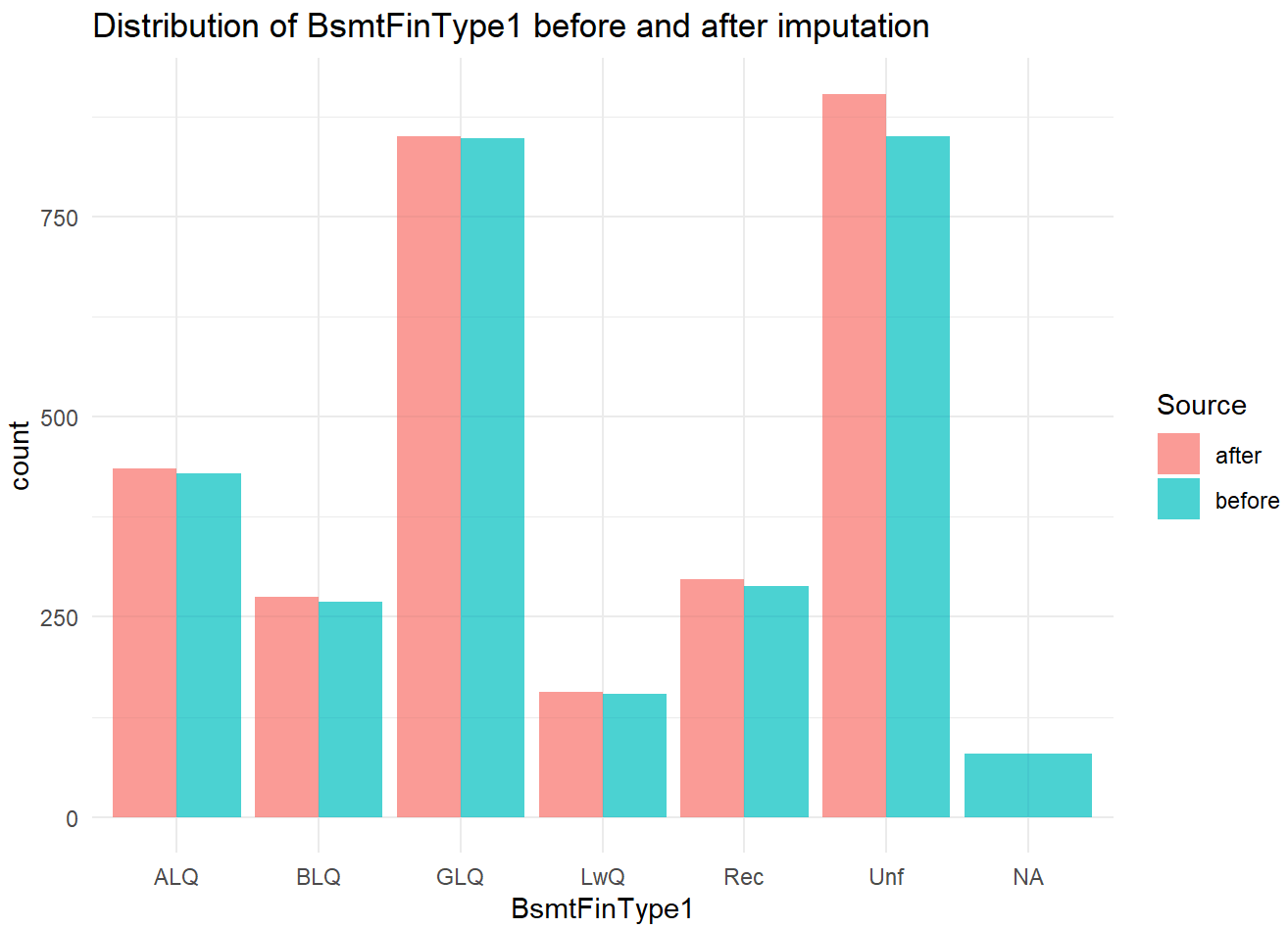

Distribution of BsmtFullBath before and after imputation
count
1500
1000
500
0
0
1
2
3
BsmtFullBath
Source
after
before

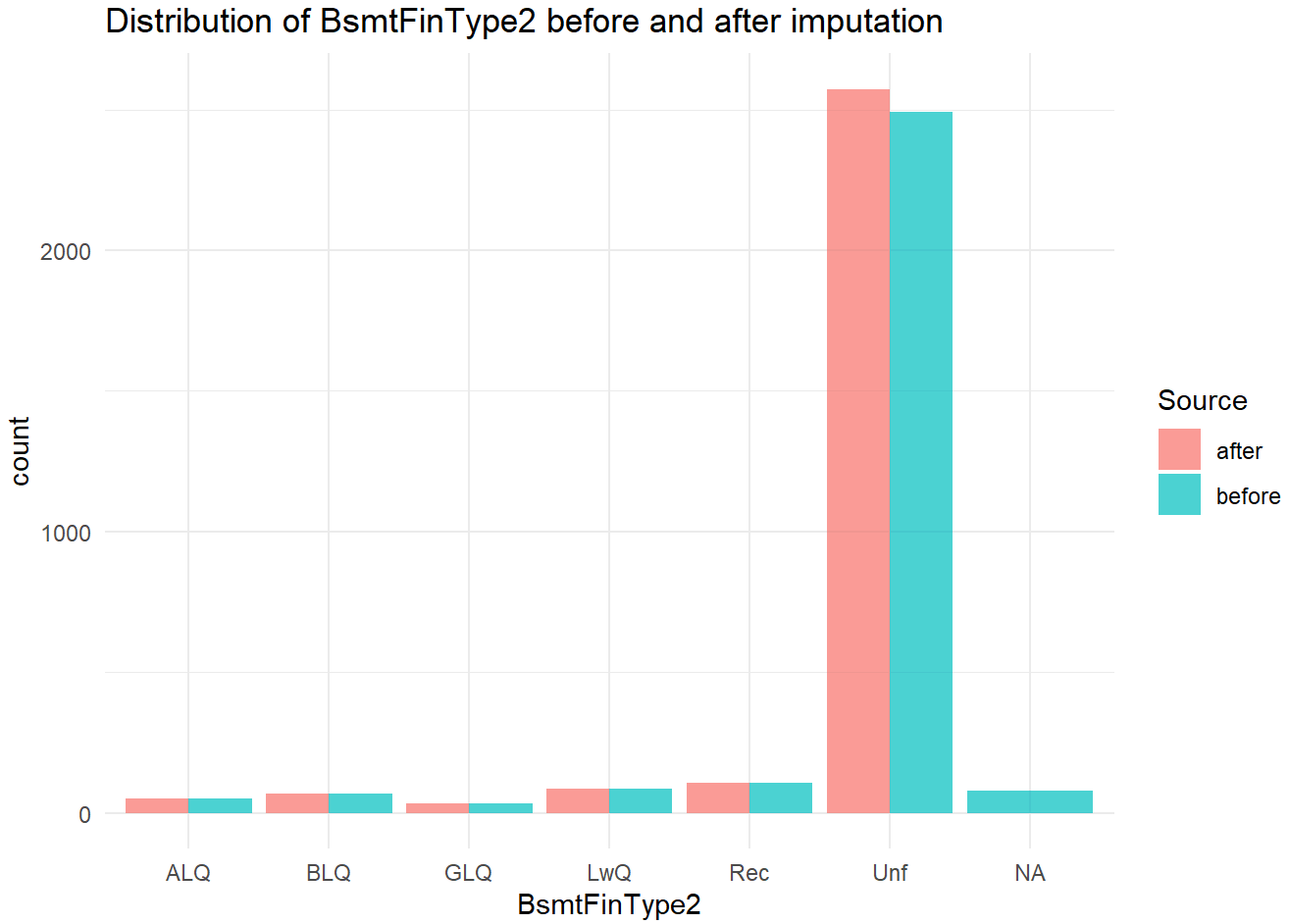

Distribution of BsmtHalfBath before and after imputation
count
2000
1000
0
0
1
2
BsmtHalfBath
Source
after
before

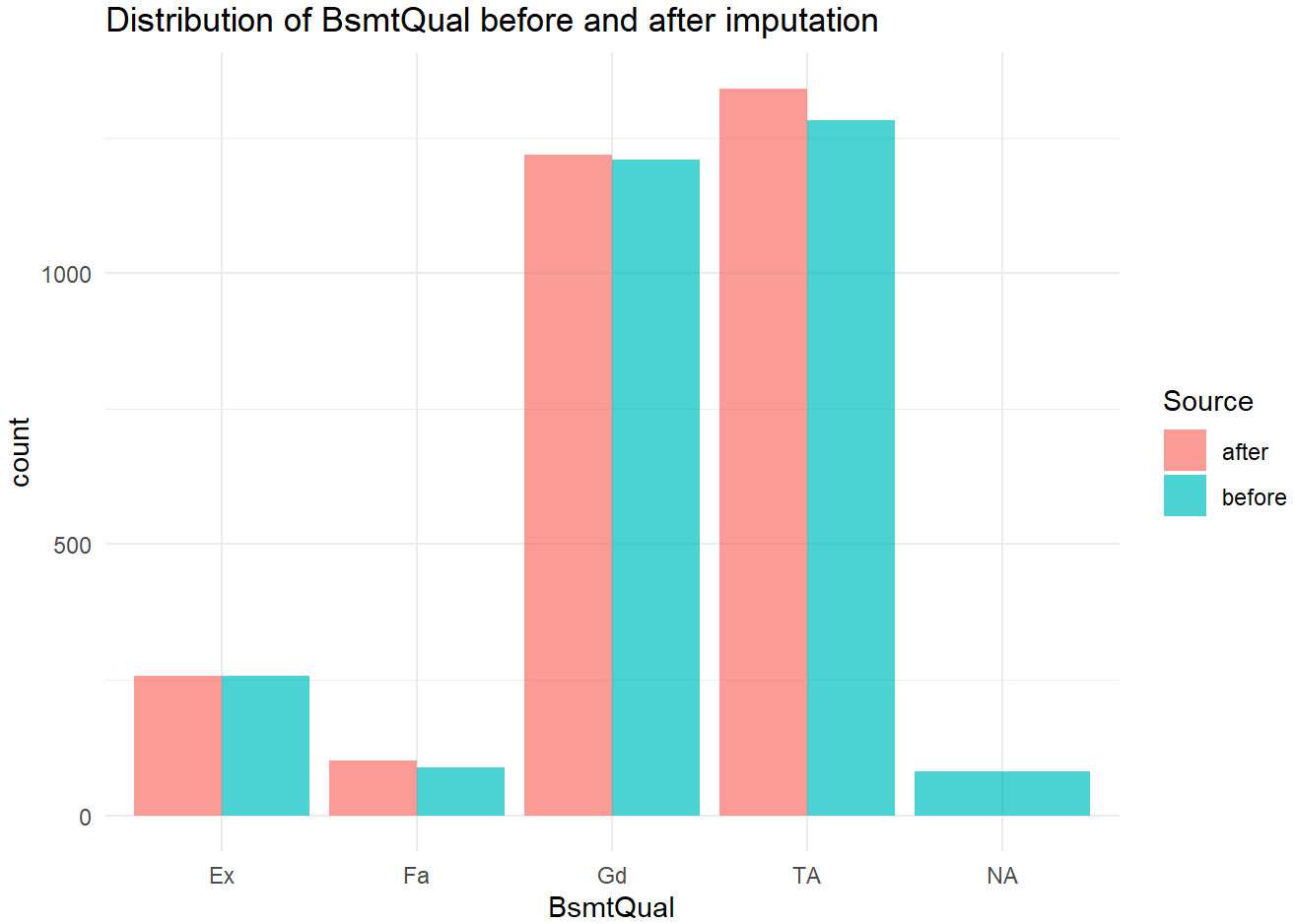

Distribution of Functional before and after imputation
count
2000
1000
0
Maj1
Maj2
Min1
Min2
Mod
Sev
Typ
NA
Functional
Source
after
before

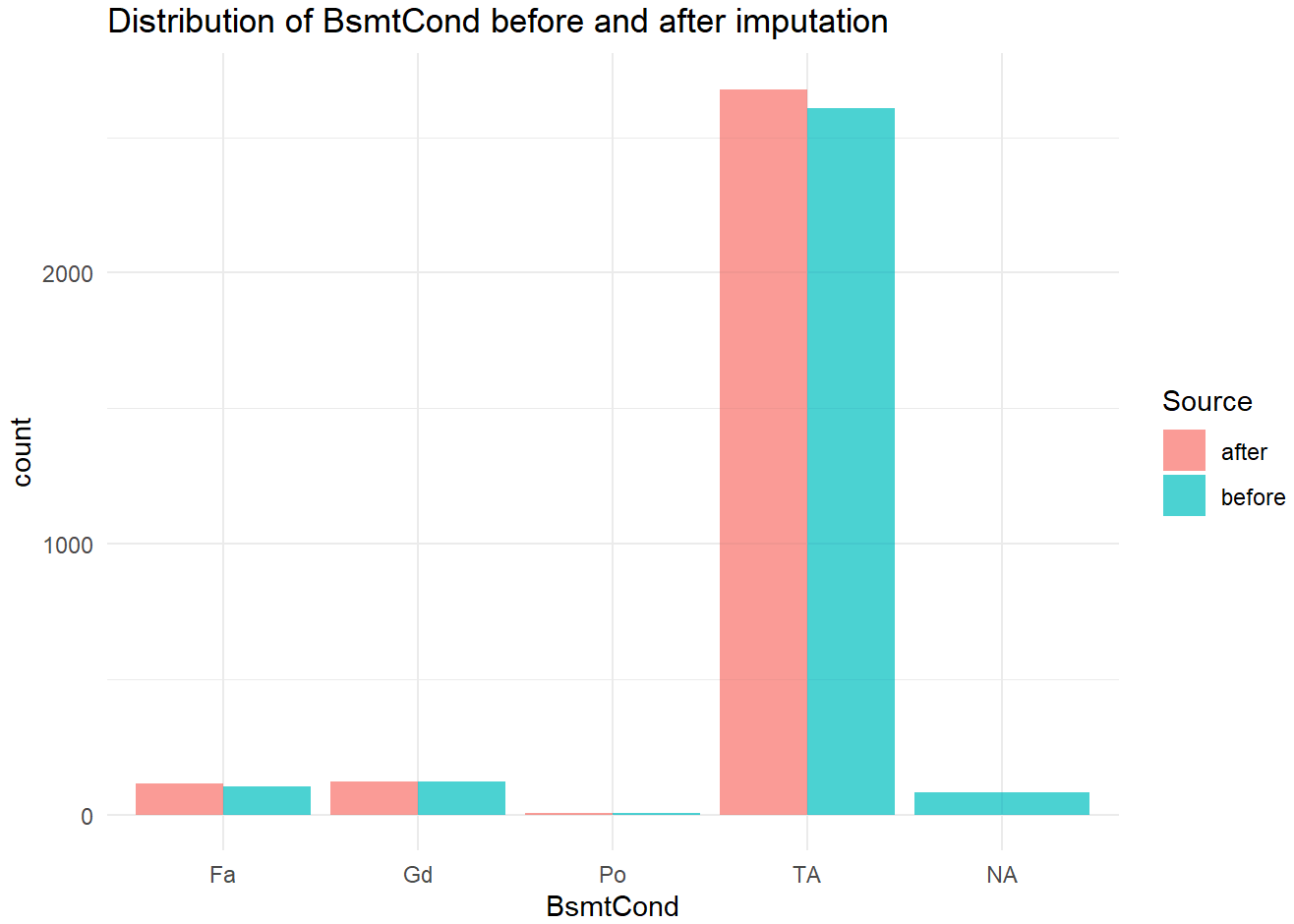
Distribution of MSZoning before and after imputation
count
2000
1500
1000
500
0
C (all)
FV
RH
RL
RM
NA
MSZoning
Source
after
before

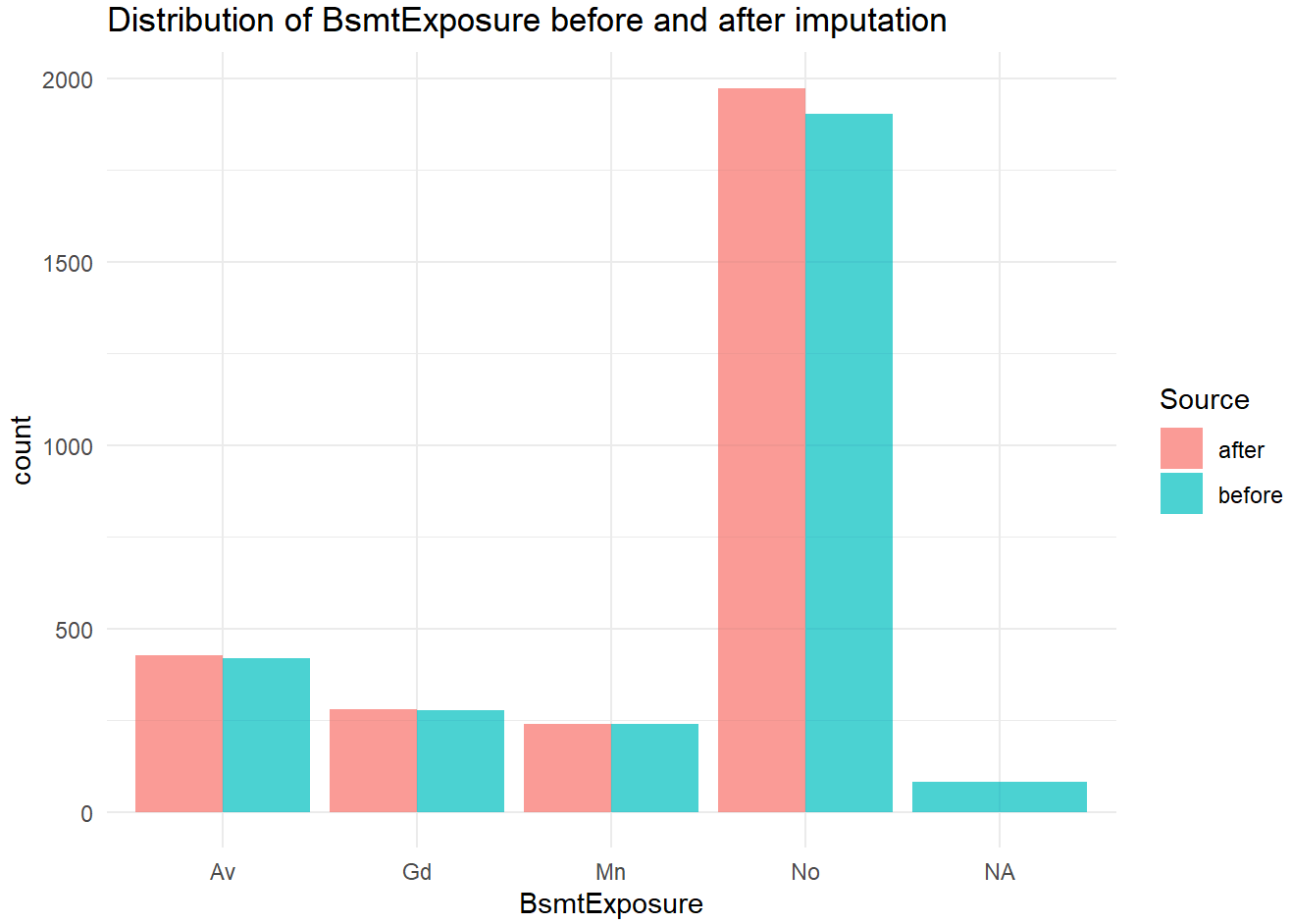

Distribution of MasVnrArea before and after imputation
count
1500
1000
500
0
0
500
1000
1500
MasVnrArea
Source
after
before

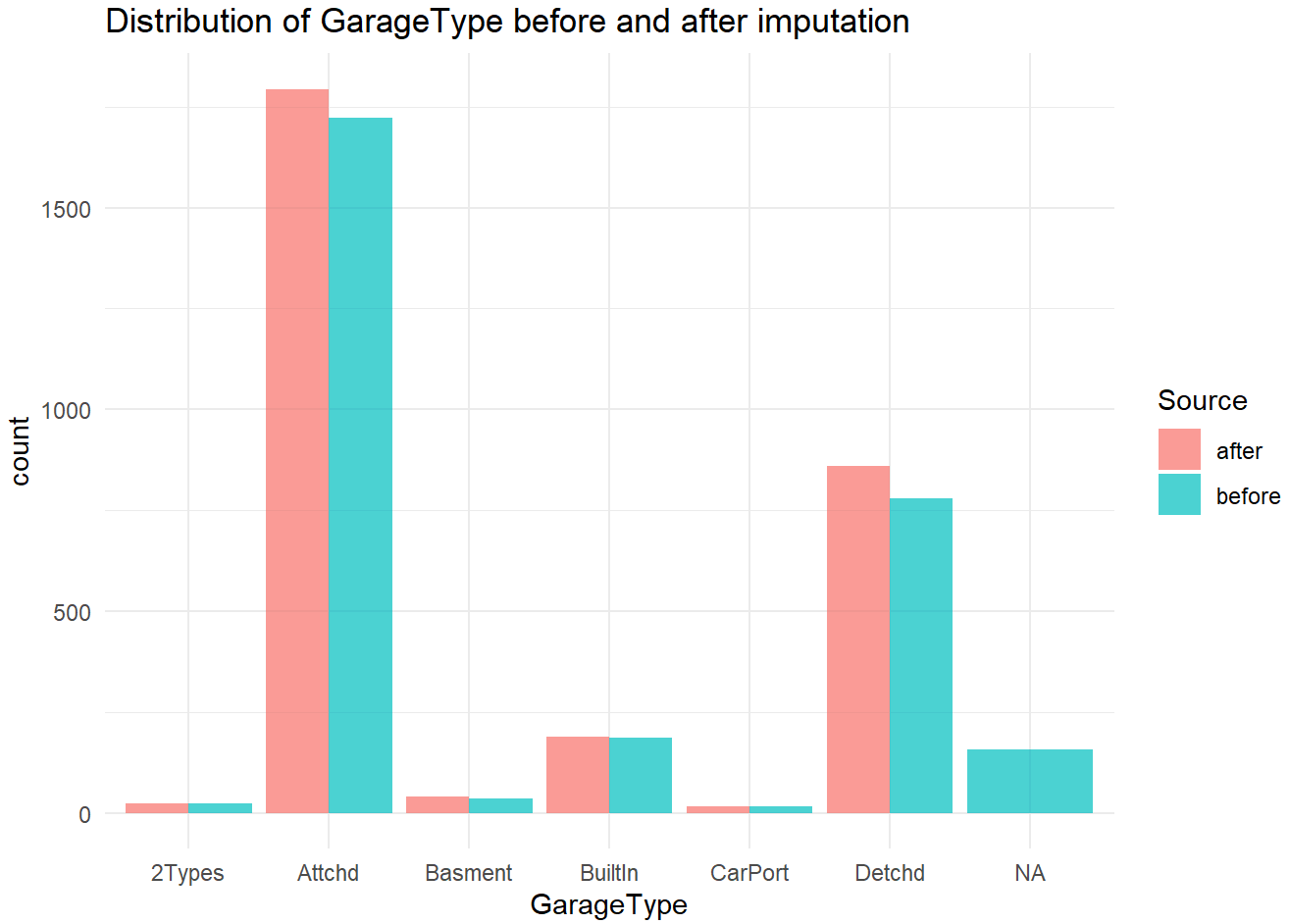

Distribution of BsmtFinType1 before and after imputation
count
750
500
250
0
ALQ
BLQ
GLQ
LwQ
Rec
Unf
NA
BsmtFinType1
Source
after
before

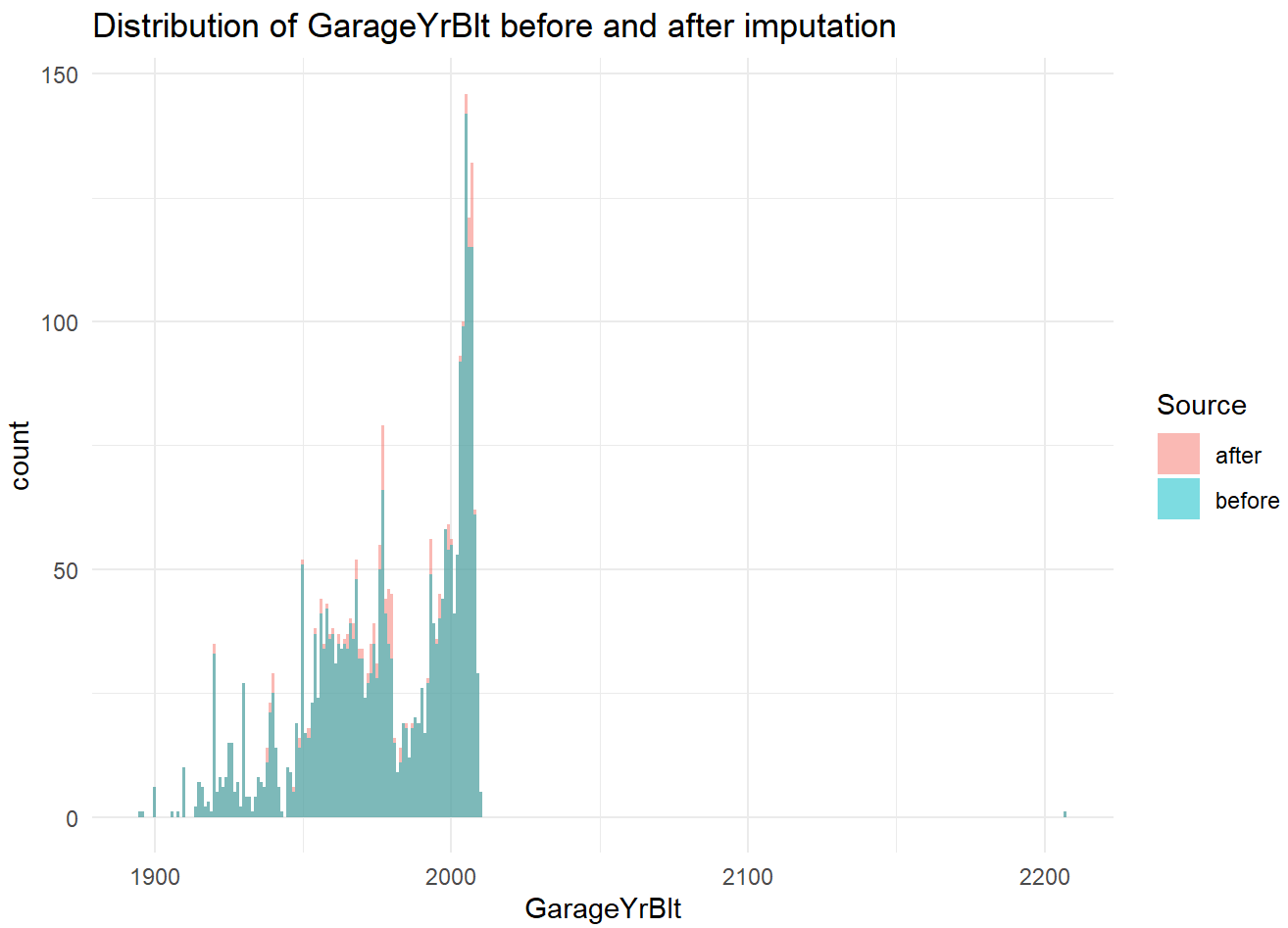

Distribution of BsmtFinType2 before and after imputation
count
2000
1000
0
ALQ
BLQ
GLQ
LwQ
Rec
Unf
NA
BsmtFinType2
Source
after
before

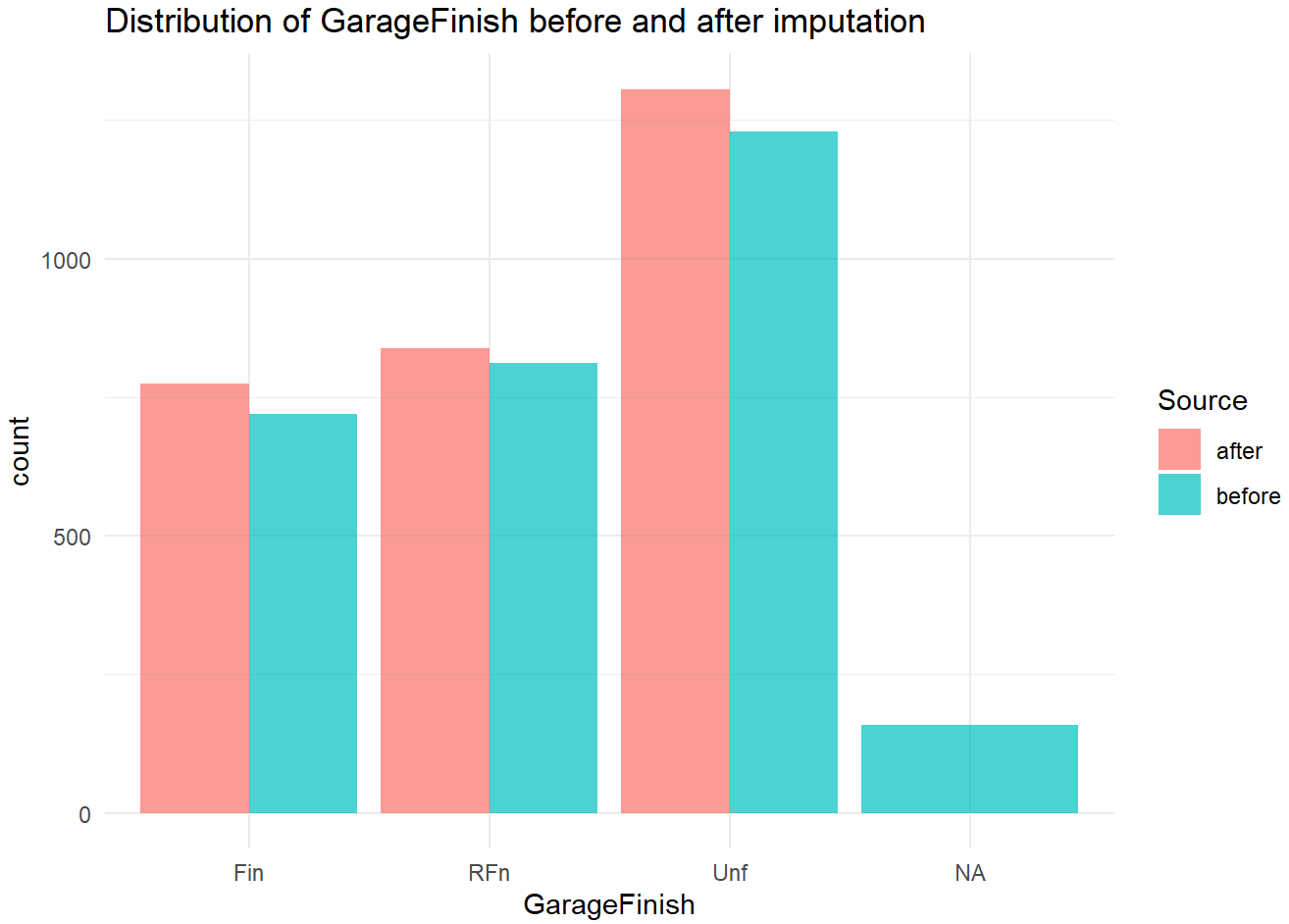

Distribution of BsmtQual before and after imputation
count
1000
500
0
Ex
Fa
Gd
TA
NA
BsmtQual
Source
after
before

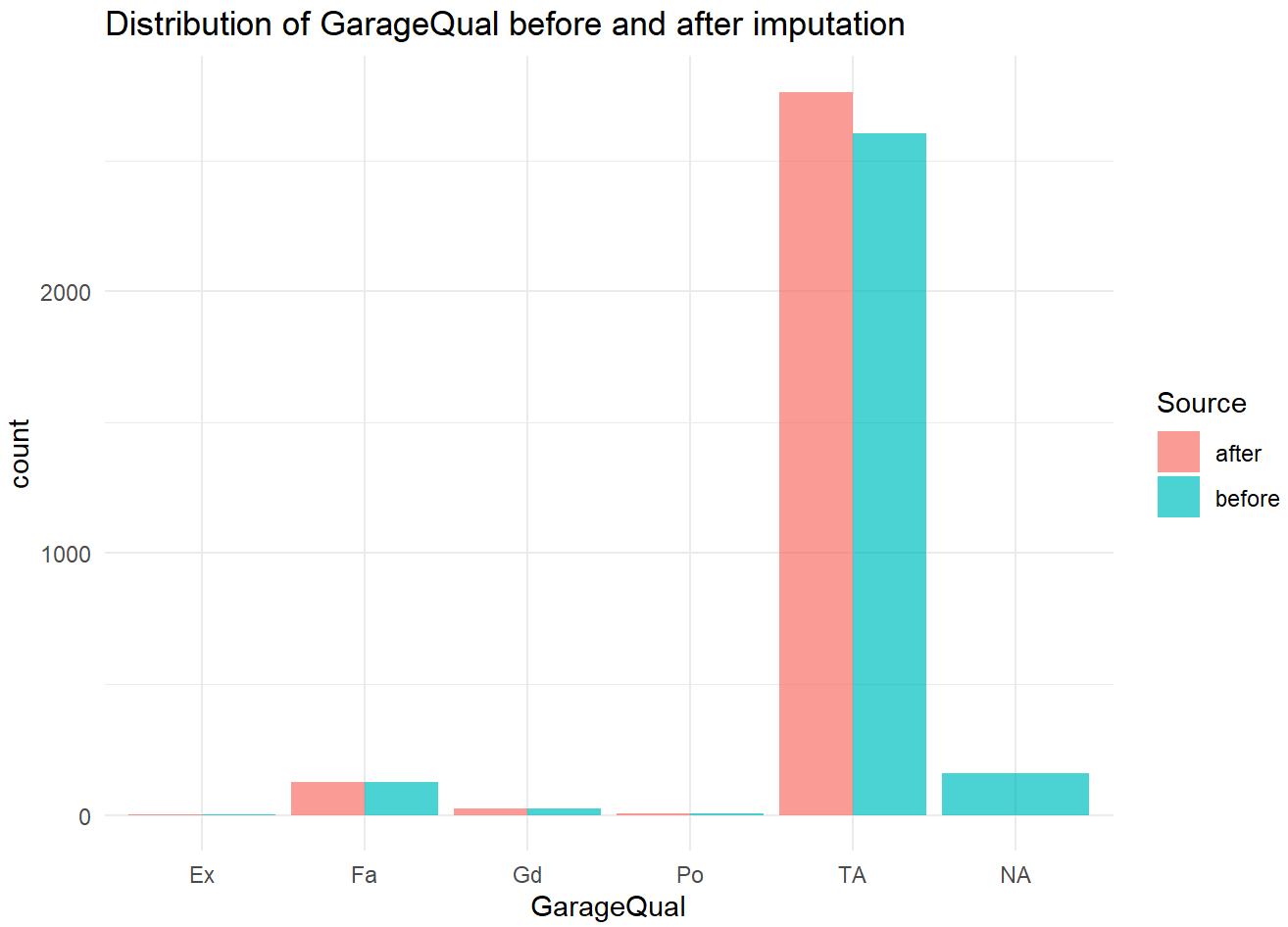

Distribution of BsmtCond before and after imputation
count
2000
1000
0
Fa
Gd
Po
TA
NA
BsmtCond
Source
after
before

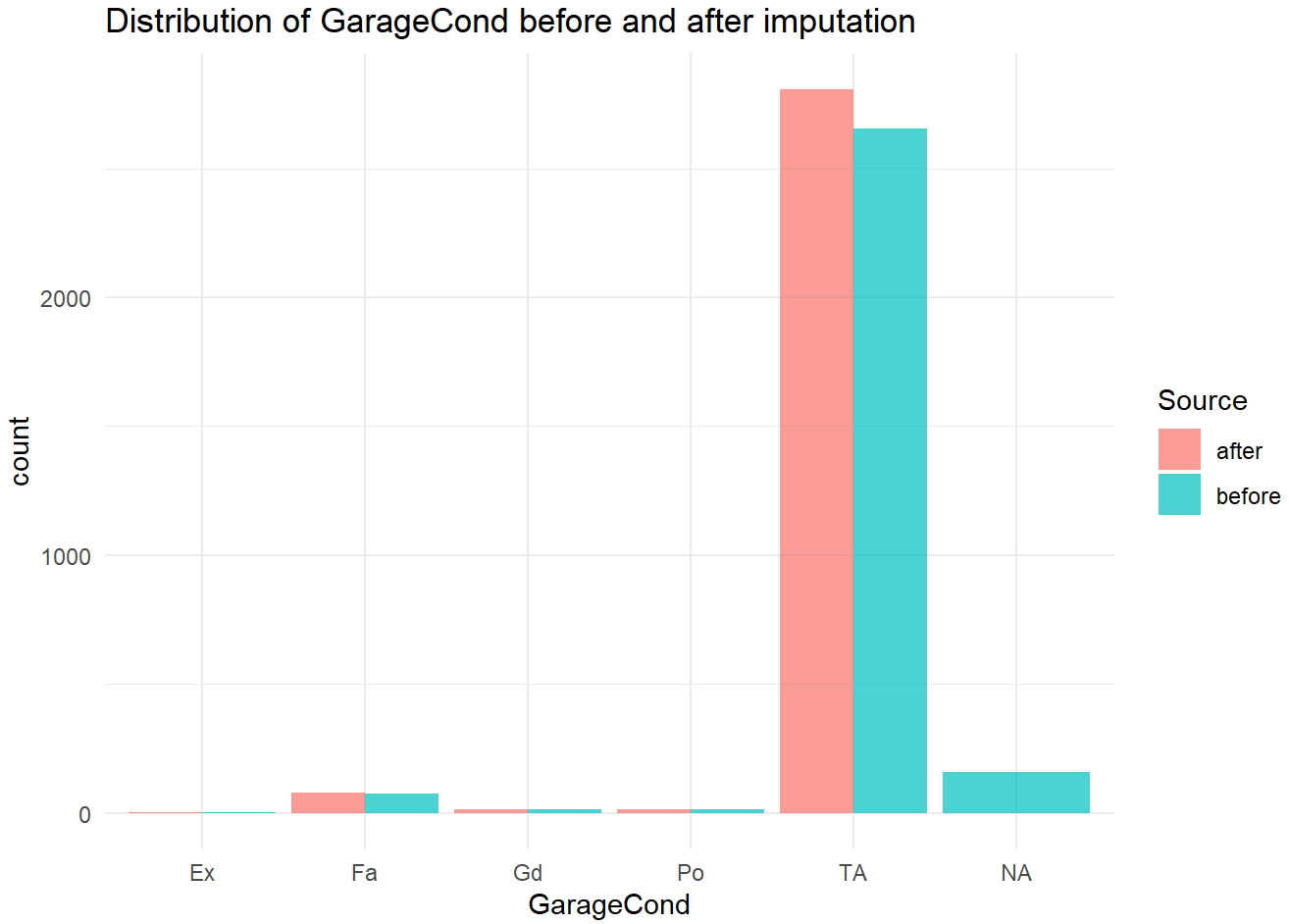

Distribution of BsmtExposure before and after imputation
count
2000
1500
1000
500
0
Av
Gd
Mn
No
NA
BsmtExposure
Source
after
before

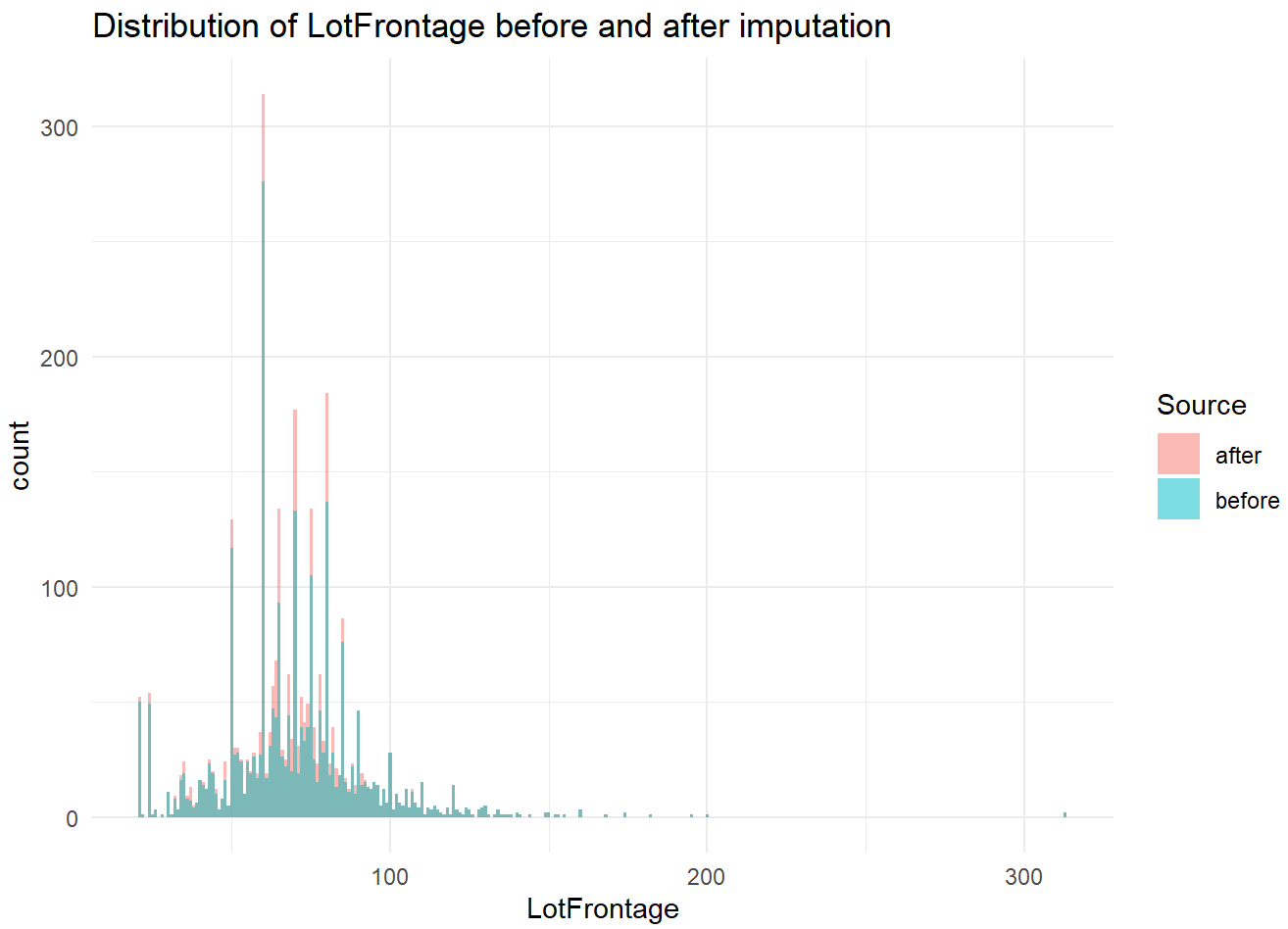

Distribution of GarageType before and after imputation
count
1500
1000
500
0
2Types
Attchd
Basment
BuiltIn
CarPort
Detchd
NA
GarageType
Source
after
before


Distribution of GarageYrBlt before and after imputation
count
150
100
50
0
1900
2000
2100
2200
GarageYrBlt
Source
after
before


Distribution of GarageFinish before and after imputation
count
1000
500
0
Fin
RFn
Unf
NA
GarageFinish
Source
after
before


Distribution of GarageQual before and after imputation
count
2000
1000
0
Ex
Fa
Gd
Po
TA
NA
GarageQual
Source
after
before


Distribution of GarageCond before and after imputation
count
2000
1000
0
Ex
Fa
Gd
Po
TA
NA
GarageCond
Source
after
before


Distribution of LotFrontage before and after imputation
count
300
200
100
0
100
200
300
LotFrontage
Source
after
before

The data has huge number of columns so We need to make dimensionality reduction ## dimensionality reduction ### PCA ##### select the Features with out the target ## select the Features with out the target because This is done by only considering the features (independent variables) and not the target variable (dependent variable). The reason for not including the target variable in PCA is that PCA is an unsupervised dimensionality reduction technique, meaning it focuses solely on the variation within the features of the dataset without taking the target variable into account.

```
X <- select(imputed_data, -c('SalePrice'))
y<- select(imputed_data, SalePrice)
dim(X)
```

```
## [1] 2919   73
```

```
dim(y)
```

```
## [1] 2919    1
```

Select the Numeric data for PCA

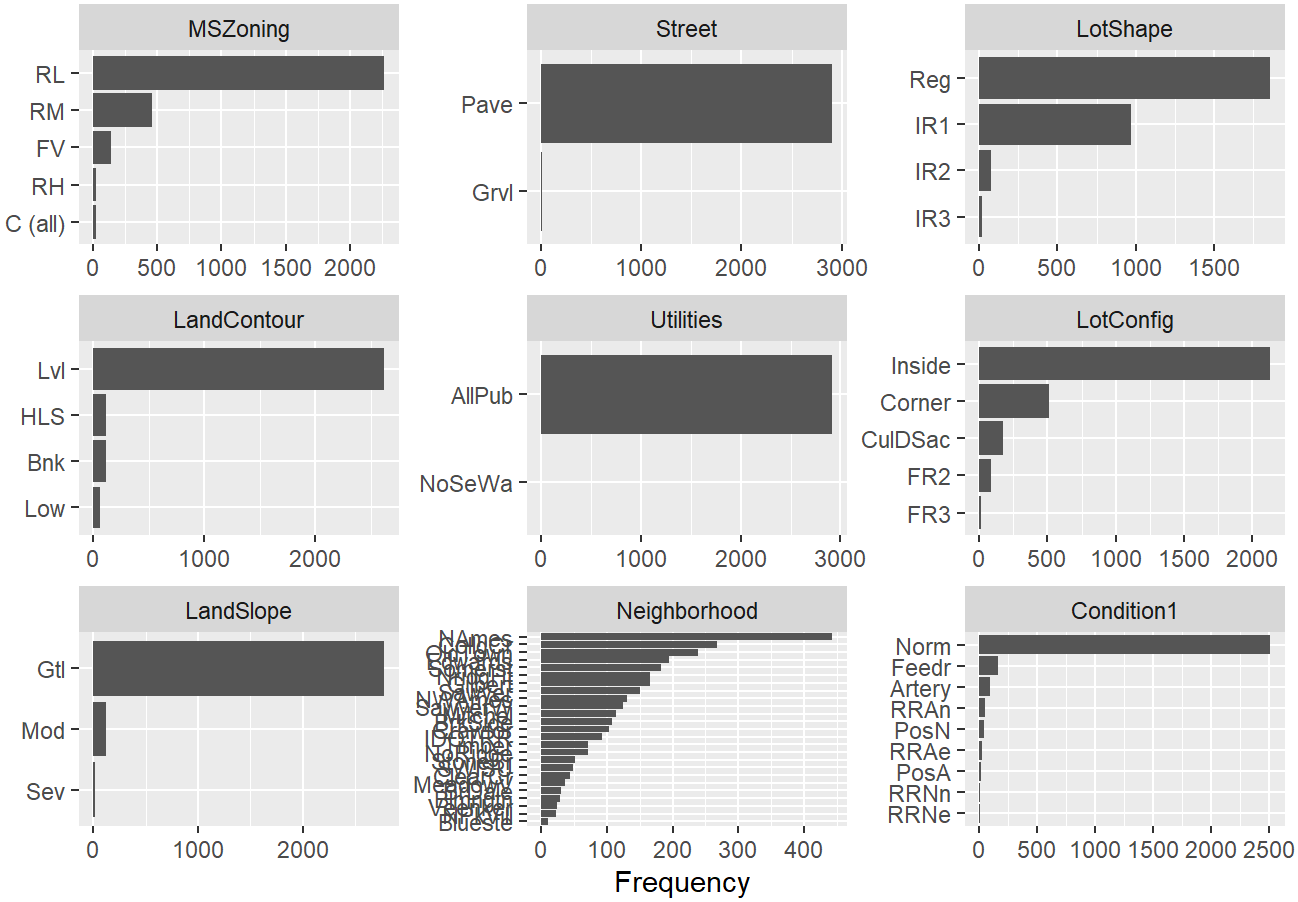
```
numeric_X_for_pca <- select_if(X, is.numeric)
dim(numeric_X_for_pca)
```

```
## [1] 2919   36
```

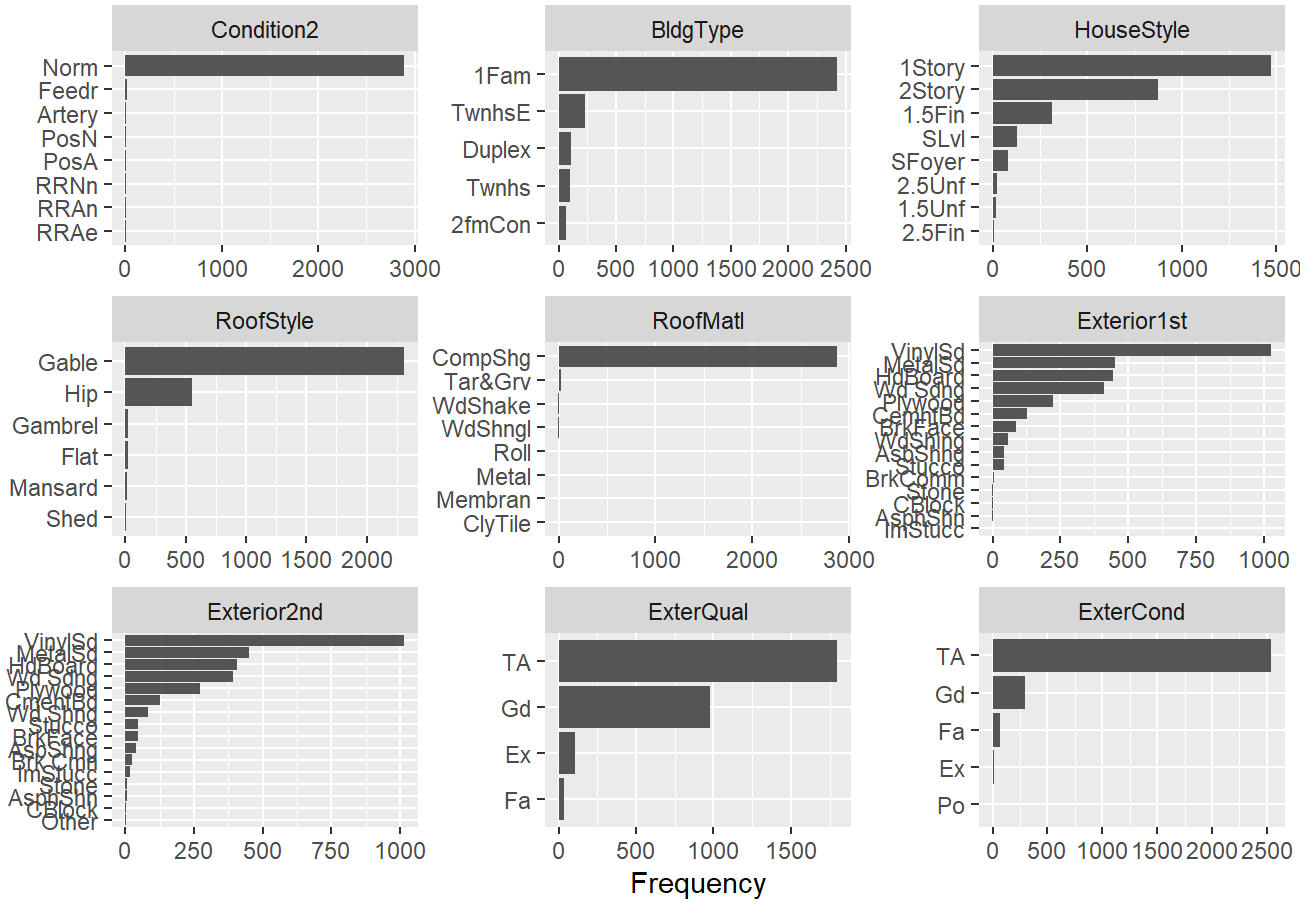
```
categorical_X_pca<- X %>%
  select(-names(numeric_X_for_pca))

dim(categorical_X_pca)
```

```
## [1] 2919   37
```

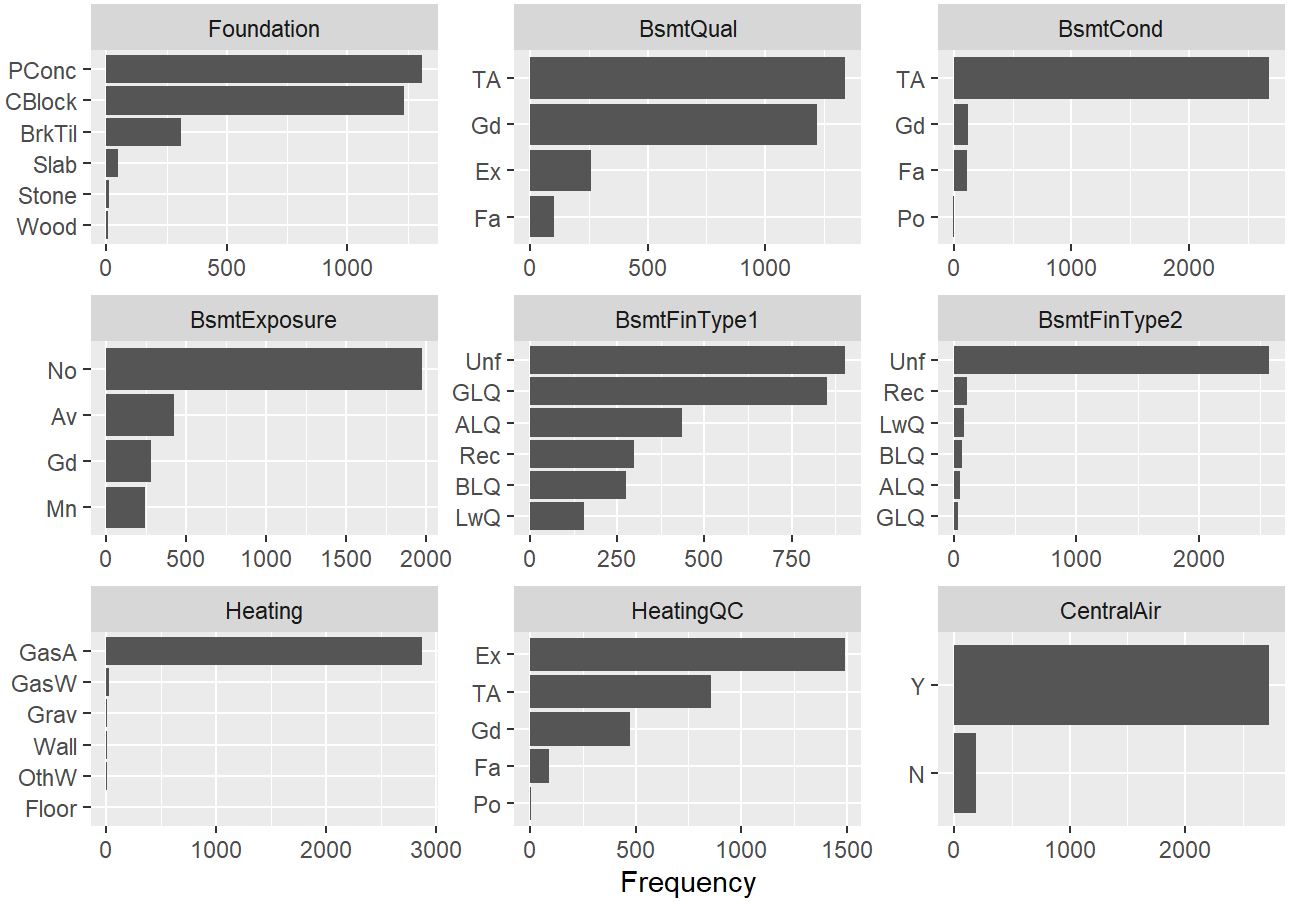
```
plot_bar(categorical_X_pca)
```


MSZoning
RL
RM
FV
RH
C (all)
0 500 1000 1500 2000
Street
Pave
Grvl
0 1000 2000 3000
LotShape
Reg
IR1
IR2
IR3
0 500 1000 1500
LandContour
Lvl
HLS
Bnk
Low
0 1000 2000
Utilities
AllPub
NoSeWa
0 1000 2000 3000
LotConfig
Inside
Corner
CulDSac
FR2
FR3
0 500 1000 1500 2000
LandSlope
Gtl
Mod
Sev
0 1000 2000
Neighborhood
0 100 200 300 400
Condition1
Norm
Feedr
Artery
RRAn
PosN
RRAe
PosA
RRNn
RRNe
0 500 1000 1500 2000 2500
Frequency

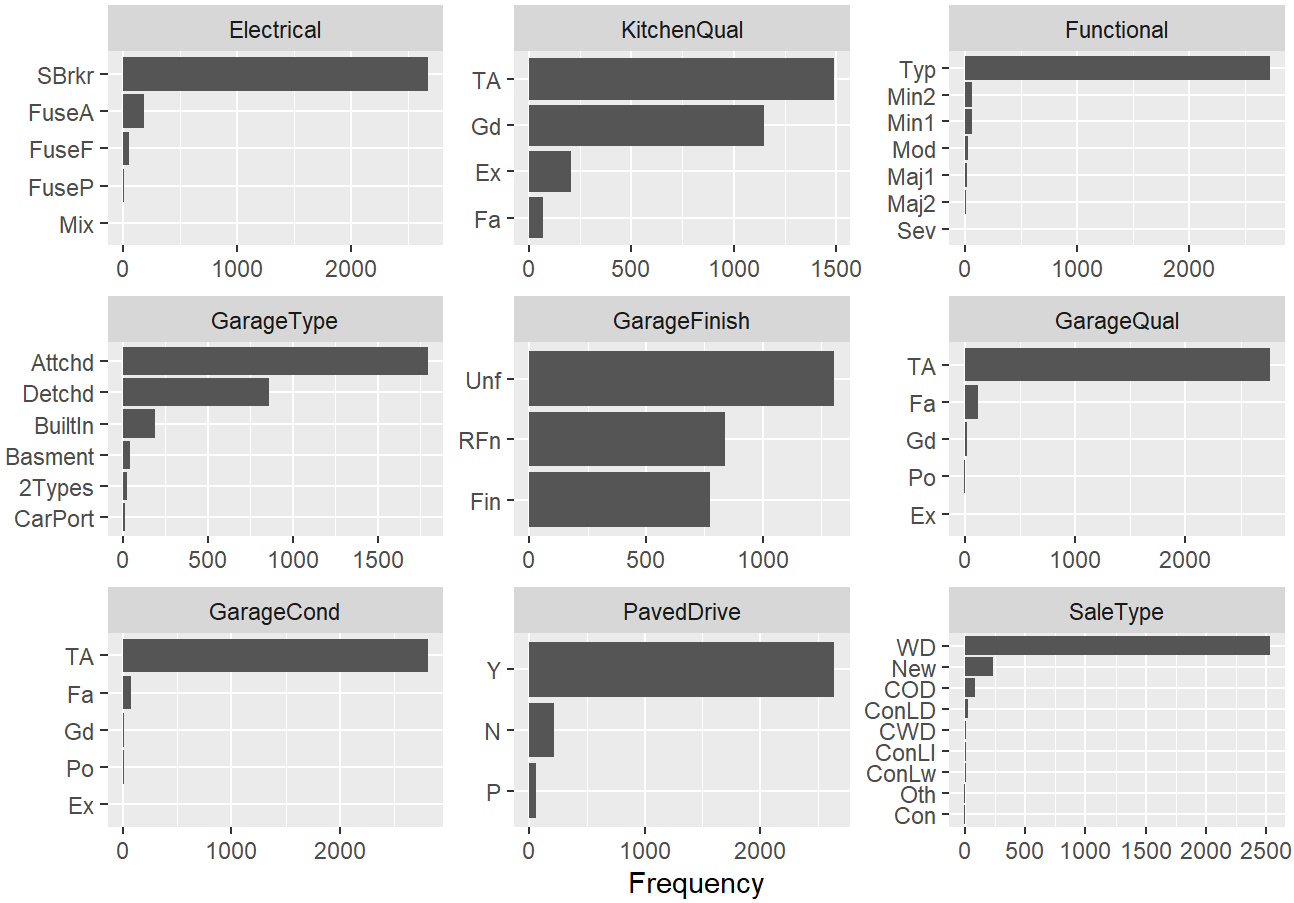
Page 1

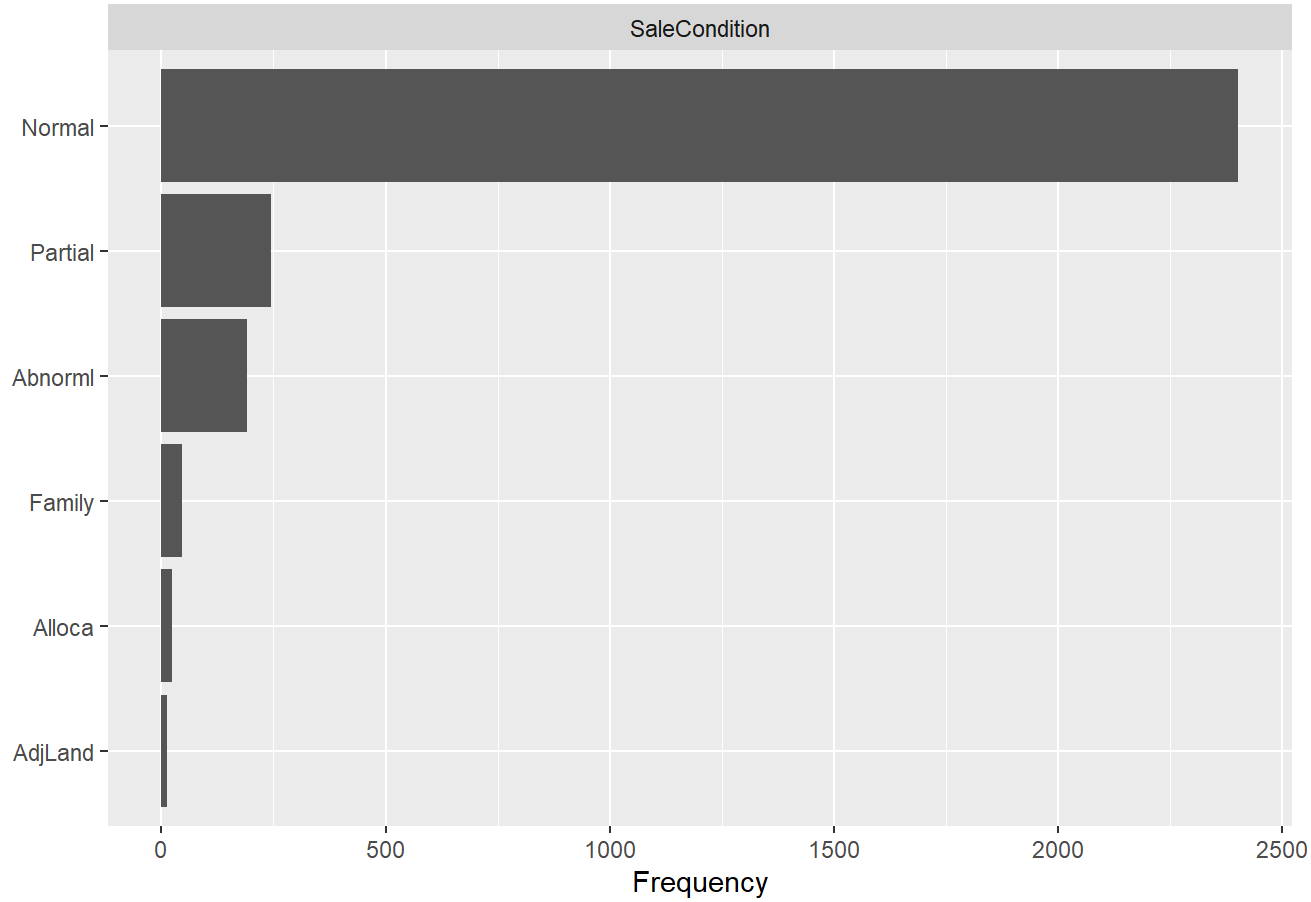

Condition2
Norm
Feedr
Artery
PosN
PosA
RRNn
RRAn
RRAe
0 1000 2000 3000
BldgType
1Fam
TwnhsE
Duplex
Twnhs
2fmCon
0 500 1000 1500 2000 2500
HouseStyle
1Story
2Story
1.5Fin
SLvl
SFoyer
2.5Unf
1.5Unf
2.5Fin
0 500 1000 1500
RoofStyle
Gable
Hip
Gambrel
Flat
Mansard
Shed
0 500 1000 1500 2000
RoofMatl
CompShg
Tar&Grv
WdShake
WdShngl
Roll
Metal
Membran
ClyTile
0 1000 2000 3000
Exterior1st
VinylSd
MetalSd
HdBoard
Wd Sdng
Plywood
CemntBd
BrkFace
WdShing
AsbShng
Stucco
BrkComm
Stone
CBlock
AsphSn
ImStucc
0 250 500 750 1000
Exterior2nd
VinylSd
MetalSd
HdBoard
Wd Sdng
Plywood
CmentBd
Wd Shng
Stucco
BrkFace
AsbShng
BrkComm
ImStucc
Stone
AsphSn
CBlock
Other
0 250 500 750 1000
ExterQual
TA
Gd
Ex
Fa
0 500 1000 1500
ExterCond
TA
Gd
Fa
Ex
Po
0 500 1000 1500 2000 2500
Frequency

Page 2


Foundation
PConc
CBlock
BrkTil
Slab
Stone
Wood
0 500 1000
BsmtQual
TA
Gd
Ex
Fa
0 500 1000
BsmtCond
TA
Gd
Fa
Po
0 1000 2000
BsmtExposure
No
Av
Gd
Mn
0 500 1000 1500 2000
BsmtFinType1
Unf
GLQ
ALQ
Rec
BLQ
LwQ
0 250 500 750
BsmtFinType2
Unf
Rec
LwQ
BLQ
ALQ
GLQ
0 1000 2000
Heating
GasA
GasW
Grav
Wall
OthW
Floor
0 1000 2000 3000
HeatingQC
Ex
TA
Gd
Fa
Po
0 500 1000 1500
CentralAir
Y
N
0 1000 2000
Frequency




Electrical
SBrkr
FuseA
FuseF
FuseP
Mix
0 1000 2000
KitchenQual
TA
Gd
Ex
Fa
0 500 1000 1500
Functional
Typ
Min2
Min1
Mod
Maj1
Maj2
Sev
0 1000 2000
GarageType
Attchd
Detchd
BuiltIn
Basment
2Types
CarPort
0 500 1000 1500
GarageFinish
Unf
RFn
Fin
0 500 1000
GarageQual
TA
Fa
Gd
Po
Ex
0 1000 2000
GarageCond
TA
Fa
Gd
Po
Ex
0 1000 2000
PavedDrive
Y
N
P
0 1000 2000
SaleType
WD
New
COD
ConLD
CWD
ConLI
ConLw
Oth
Con
0 500 1000 1500 2000 2500
Frequency




```
categorical_X_pca<-select(categorical_X_pca,c('BsmtFinType1', 'GarageFinish', 'KitchenQual' , 'N
eighborhood'))
```

### Scaling the data for PCA

##scales the numeric features in the numeric_X_for_pca data frame. #It ensures that each feature has a mean of 0 and a standard deviation of 1. #Without scaling, features with larger scales (e.g., income in thousands vs. age in years) can dominate the principal components.

```
scaled_data_for_pca <- scale(numeric_X_for_pca)
view(scaled_data_for_pca)
```

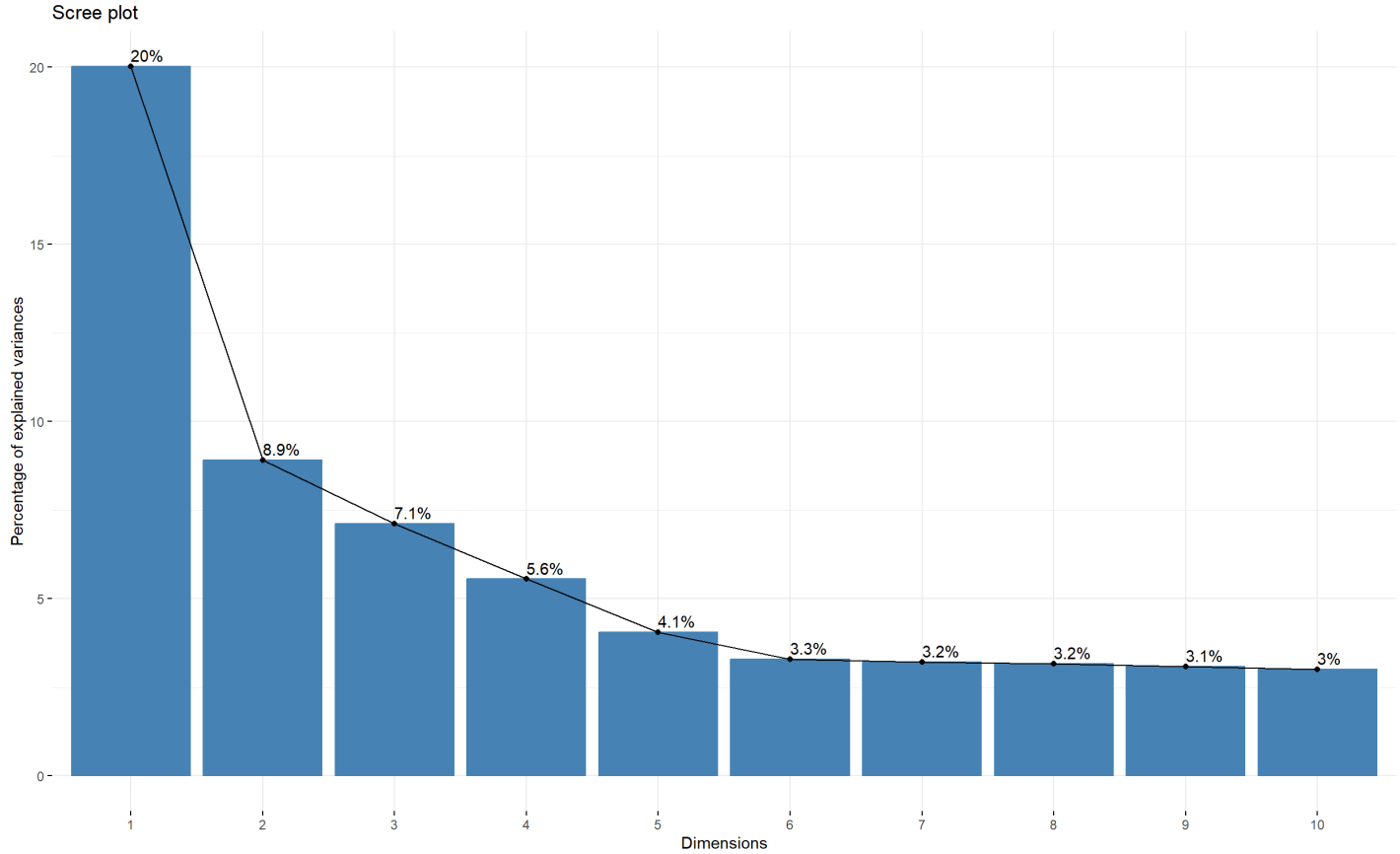
## PCA

```
data_pca <- princomp(scaled_data_for_pca)
#princomp(scaled_data_for_pca): This line computes the principal components of the scaled_data_f
or_pca matrix.
#s assumed to be a numeric matrix where each row represents an observation and each column repre
sents afeature
summary(data_pca)
```

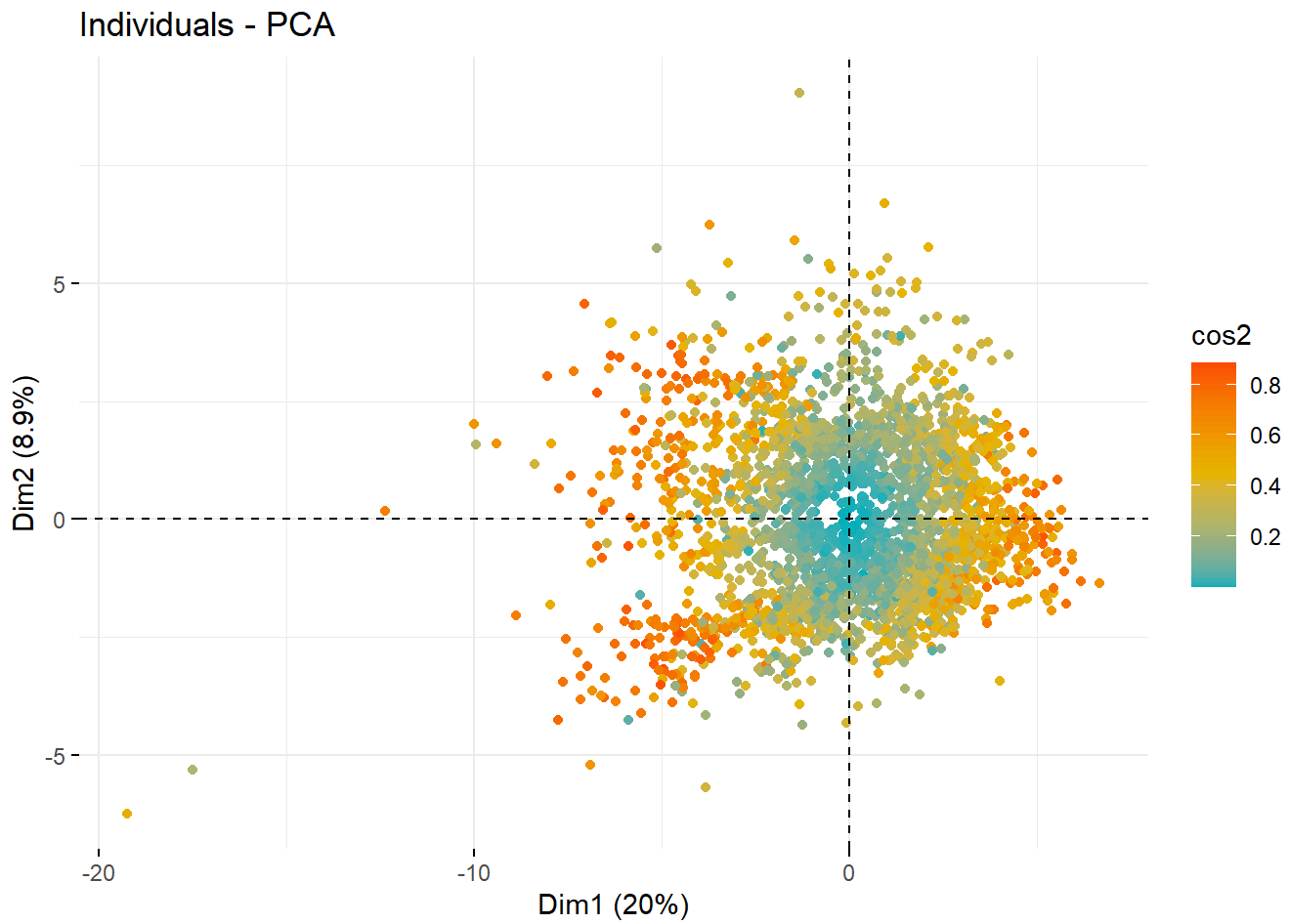
```
## Importance of components:
##                           Comp.1     Comp.2     Comp.3     Comp.4     Comp.5
## Standard deviation     2.6843613 1.79031523 1.60114198 1.41390486 1.20880086
## Proportion of Variance 0.2002296 0.08906464 0.07123706 0.05555034 0.04060279
## Cumulative Proportion  0.2002296 0.28929423 0.36053129 0.41608162 0.45668441
##                            Comp.6     Comp.7     Comp.8     Comp.9    Comp.10
## Standard deviation     1.08808094 1.07504947 1.06686591 1.05431128 1.04033805
## Proportion of Variance 0.03289794 0.03211465 0.03162758 0.03088759 0.03007428
## Cumulative Proportion  0.48958235 0.52169700 0.55332458 0.58421217 0.61428645
##                           Comp.11    Comp.12    Comp.13    Comp.14    Comp.15
## Standard deviation     1.02840617 1.00967915 0.98819971 0.97032027 0.96144596
## Proportion of Variance 0.02938838 0.02832782 0.02713537 0.02616234 0.02568598
## Cumulative Proportion  0.64367483 0.67200265 0.69913802 0.72530035 0.75098633
##                           Comp.16    Comp.17    Comp.18    Comp.19    Comp.20
## Standard deviation     0.94206367 0.92510521 0.90159400 0.87517296 0.84055693
## Proportion of Variance 0.02466078 0.02378091 0.02258751 0.02128306 0.01963272
## Cumulative Proportion  0.77564711 0.79942802 0.82201553 0.84329859 0.86293132
##                           Comp.21    Comp.22    Comp.23    Comp.24    Comp.25
## Standard deviation     0.81192355 0.78319941 0.75320736 0.74019136 0.65101605
## Proportion of Variance 0.01831794 0.01704476 0.01576433 0.01522419 0.01177687
## Cumulative Proportion  0.88124926 0.89829402 0.91405835 0.92928254 0.94105941
##                           Comp.26     Comp.27     Comp.28     Comp.29
## Standard deviation     0.60622792 0.576821287 0.545970676 0.525489728
## Proportion of Variance 0.01021217 0.009245467 0.008282948 0.007673169
## Cumulative Proportion  0.95127158 0.960517047 0.968799996 0.976473165
##                            Comp.30     Comp.31     Comp.32     Comp.33
## Standard deviation     0.499917037 0.462519607 0.398085337 0.365237629
## Proportion of Variance 0.006944519 0.005944381 0.004403507 0.003706784
## Cumulative Proportion  0.983417684 0.989362064 0.993765571 0.997472356
##                            Comp.34      Comp.35 Comp.36
## Standard deviation     0.301602426 1.168500e-07       0
## Proportion of Variance 0.002527644 3.794058e-16       0
## Cumulative Proportion  1.000000000 1.000000e+00       1
```

## Visualization Of PCA

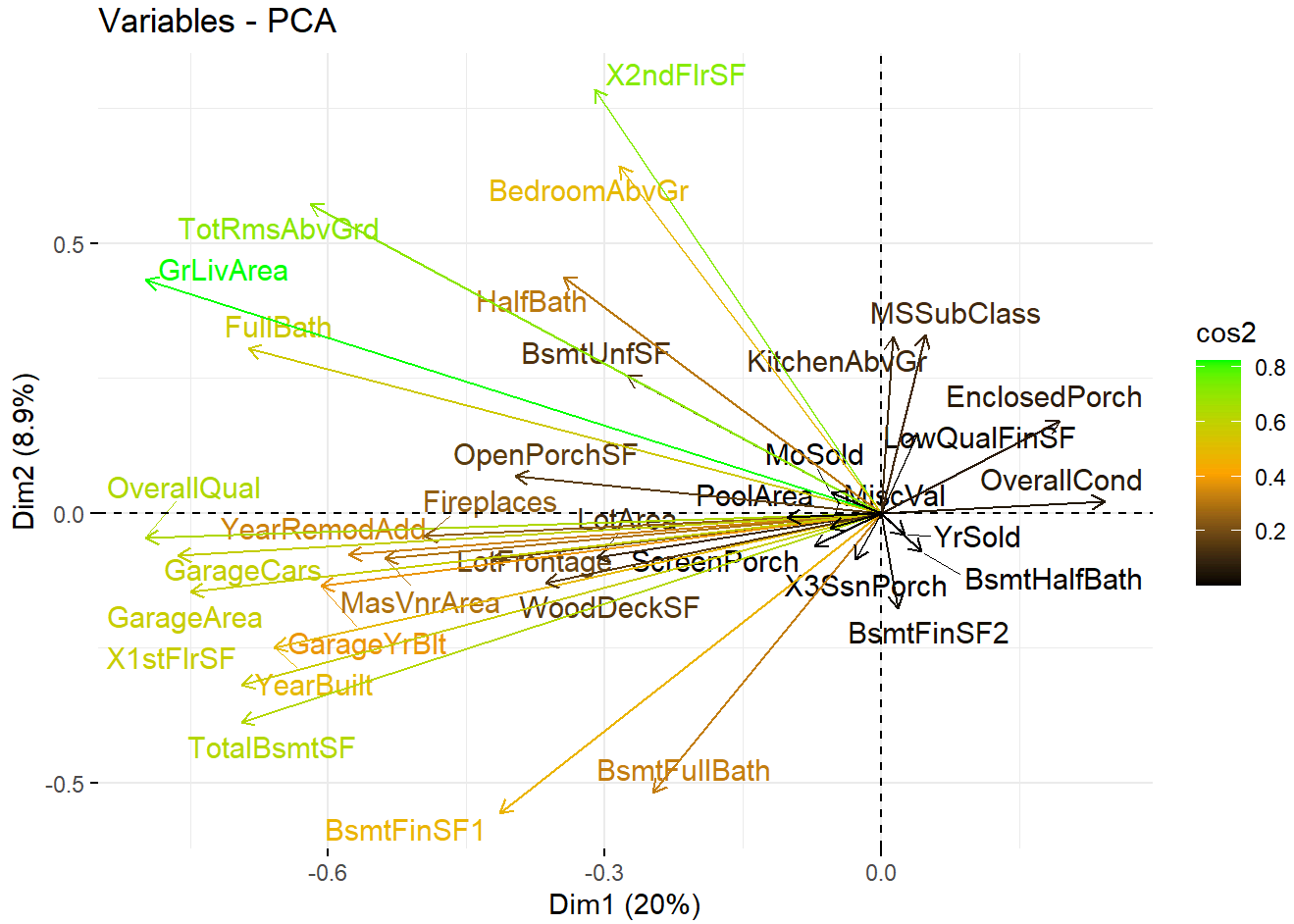
```
fviz_eig(data_pca, addlabels = TRUE)
```




```
# Visualize PCA
pca_plot <- fviz_pca_ind(data_pca,
                         geom.ind = "point",
                         col.ind = "cos2", # Color by the quality of representation
                         gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
                         repel = TRUE)    # Avoid text overlapping
print(pca_plot)
```




```
# Visualize PCA
fviz_pca_var(data_pca, col.var = "cos2",
             gradient.cols = c("black", "orange", "green"),
             repel = TRUE)
```




we will select the first 26 componant which covers 95% of the variation of the data ##### selecting the componants

```
final_pca_data <- view(data.frame(data_pca$scores[,1:26]))
colnames(final_pca_data) <- paste0("PC", 1:26)
view(final_pca_data)
```

compaining the pca data and the categorical data

```
full_pca_X <- cbind(final_pca_data, categorical_X_pca)
dim(full_pca_X)
```

```
## [1] 2919   30
```

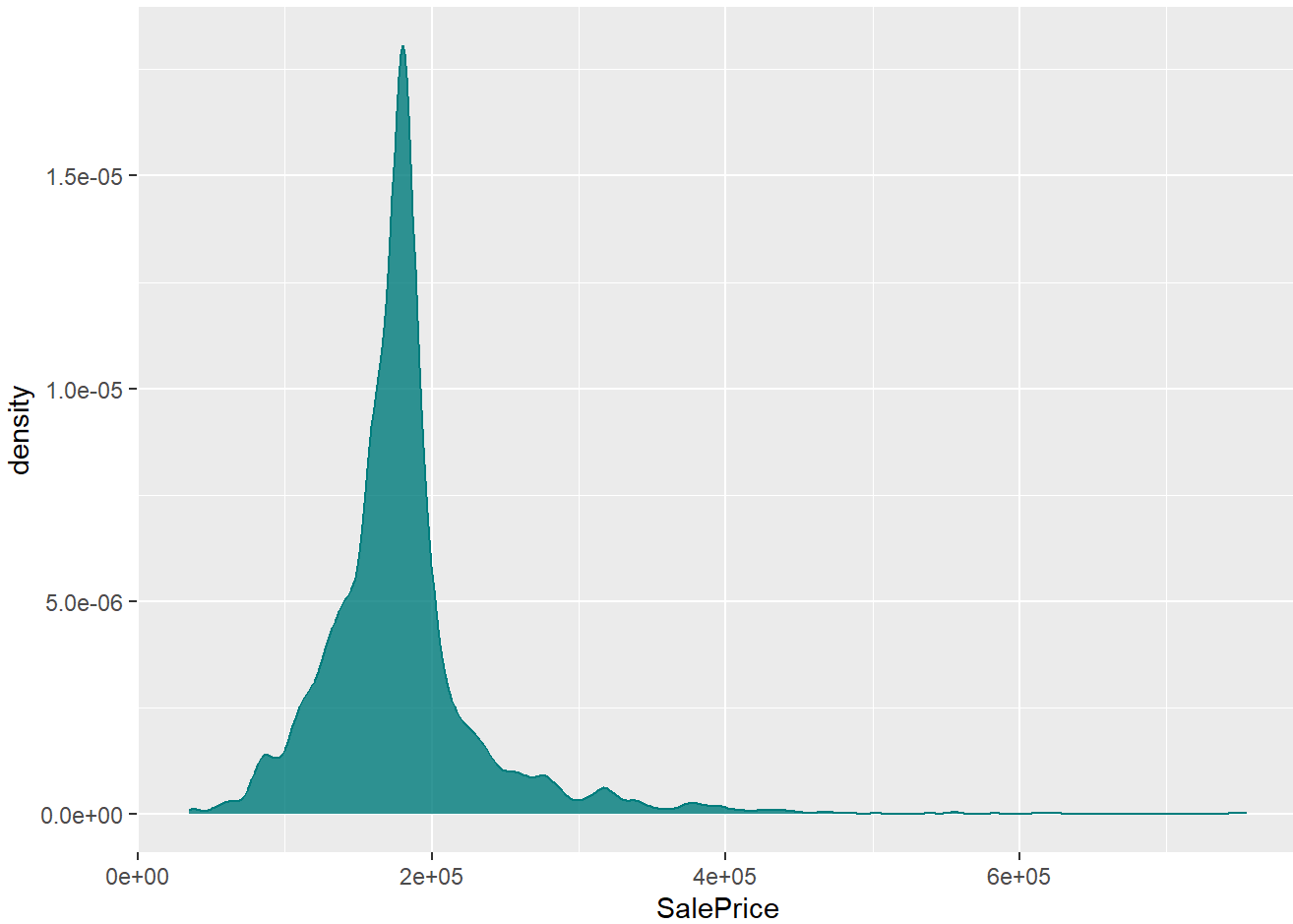
```
full_pca_X_incoded <- full_pca_X %>%
  mutate(across(where(is.factor), as.character)) %>%
  mutate(across(where(is.character), as.factor)) %>%
  model.matrix(~ . - 1, data = .) %>%
  as.data.frame()
view(full_pca_X_incoded)
```

Modeling for PCA

```
New_data_with_pca <- cbind(full_pca_X, y)
```

ploting the dist of our target

```
ggplot(imputed_data, aes(x=SalePrice)) + geom_density(fill="#008080",
                                        color="#008080",
                                        alpha=0.8)
```




our

target is skewed right

```
model <- lm(SalePrice ~ ., data = New_data_with_pca)
summary(model)
```

```
##
## Call:
## lm(formula = SalePrice ~ ., data = New_data_with_pca)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -289073  -22433   -2748   21796  376855
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         203253.79    9332.88  21.778  < 2e-16 ***
## PC1                  -9929.05     613.88 -16.174  < 2e-16 ***
## PC2                   2941.68     574.48   5.121 3.25e-07 ***
## PC3                  -5879.01     791.39  -7.429 1.44e-13 ***
## PC4                    347.60     708.28   0.491 0.623630
## PC5                  -4491.40     695.37  -6.459 1.23e-10 ***
## PC6                    468.83     717.95   0.653 0.513804
## PC7                    297.87     755.91   0.394 0.693573
## PC8                    372.51     809.26   0.460 0.645334
## PC9                  -1264.12     772.75  -1.636 0.101978
## PC10                 -2077.62     765.17  -2.715 0.006662 **
## PC11                  1937.08     773.70   2.504 0.012347 *
## PC12                    85.35     776.84   0.110 0.912523
## PC13                   206.50     809.07   0.255 0.798564
## PC14                   367.90     810.08   0.454 0.649756
## PC15                 -1950.60     825.69  -2.362 0.018225 *
## PC16                  1587.32     843.60   1.882 0.059991 .
## PC17                  -572.54     869.55  -0.658 0.510312
## PC18                   405.31     866.98   0.467 0.640178
## PC19                  1998.42     935.04   2.137 0.032662 *
## PC20                 -1838.67    1003.52  -1.832 0.067021 .
## PC21                  2083.87     972.64   2.142 0.032239 *
## PC22                  2046.50    1015.53   2.015 0.043977 *
## PC23                  -598.42    1032.65  -0.579 0.562298
## PC24                  5003.83    1080.00   4.633 3.76e-06 ***
## PC25                  -787.48    1203.80  -0.654 0.513058
## PC26                 -3679.91    1321.82  -2.784 0.005405 **
## BsmtFinType1BLQ      -1357.02    3287.42  -0.413 0.679789
## BsmtFinType1GLQ       5328.63    2876.89   1.852 0.064097 .
## BsmtFinType1LwQ      -1804.76    4141.72  -0.436 0.663050
## BsmtFinType1Rec        438.67    3296.35   0.133 0.894142
## BsmtFinType1Unf      -2190.44    3183.13  -0.688 0.491420
## GarageFinishRFn      -4363.61    2250.48  -1.939 0.052603 .
## GarageFinishUnf      -4975.05    2457.13  -2.025 0.042987 *
## KitchenQualFa       -26966.81    6734.59  -4.004 6.38e-05 ***
## KitchenQualGd       -19152.37    3737.69  -5.124 3.19e-07 ***
## KitchenQualTA       -22108.42    4295.29  -5.147 2.82e-07 ***
## NeighborhoodBlueste  -1862.17   15438.48  -0.121 0.904001
## NeighborhoodBrDale   -8801.01   11819.13  -0.745 0.456550
## NeighborhoodBrkSide  -9811.02    9808.16  -1.000 0.317254
## NeighborhoodClearCr  -1636.75   10681.30  -0.153 0.878224
## NeighborhoodCollgCr   1951.64    8677.24   0.225 0.822062
```

```
## NeighborhoodCrawfor   5721.32    9582.81   0.597 0.550528
## NeighborhoodEdwards -10375.08    9181.65  -1.130 0.258579
## NeighborhoodGilbert  -2152.36    8896.43  -0.242 0.808848
## NeighborhoodIDOTRR   -5081.35   10012.70  -0.507 0.611850
## NeighborhoodMeadowV  -1515.24   10999.83  -0.138 0.890446
## NeighborhoodMitchel  -1891.64    9310.09  -0.203 0.839008
## NeighborhoodNAmes    -5193.37    8996.46  -0.577 0.563804
## NeighborhoodNoRidge  37218.18    9916.51   3.753 0.000178 ***
## NeighborhoodNPkVill  -2924.43   12167.39  -0.240 0.810076
## NeighborhoodNridgHt  15455.09    8931.58   1.730 0.083669 .
## NeighborhoodNWAmes   -8936.68    9284.14  -0.963 0.335843
## NeighborhoodOldTown -10392.38    9415.87  -1.104 0.269813
## NeighborhoodSawyer   -2530.95    9358.65  -0.270 0.786842
## NeighborhoodSawyerW    431.44    9152.92   0.047 0.962407
## NeighborhoodSomerst   4987.92    8710.71   0.573 0.566948
## NeighborhoodStoneBr  24491.51    9967.24   2.457 0.014061 *
## NeighborhoodSWISU    -7798.45   10880.81  -0.717 0.473608
## NeighborhoodTimber    1662.64    9595.69   0.173 0.862452
## NeighborhoodVeenker   -131.34   11990.68  -0.011 0.991261
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41130 on 2858 degrees of freedom
## Multiple R-squared:  0.4969, Adjusted R-squared:  0.4863
## F-statistic: 47.04 on 60 and 2858 DF,  p-value: < 2.2e-16
```

```
coef(model)
```

```
##         (Intercept)                 PC1                 PC2                 PC3
##        203253.79308         -9929.04904          2941.68112         -5879.00944
##                 PC4                 PC5                 PC6                 PC7
##           347.60047         -4491.40045           468.82803           297.86675
##                 PC8                 PC9                PC10                PC11
##           372.50529         -1264.11613         -2077.61944          1937.08401
##                PC12                PC13                PC14                PC15
##            85.34955           206.50000           367.89554         -1950.60137
##                PC16                PC17                PC18                PC19
##          1587.32357          -572.54309           405.31435          1998.41963
##                PC20                PC21                PC22                PC23
##         -1838.67384          2083.86610          2046.50030          -598.41829
##                PC24                PC25                PC26     BsmtFinType1BLQ
##          5003.83109          -787.48301         -3679.90562         -1357.02367
##     BsmtFinType1GLQ     BsmtFinType1LwQ     BsmtFinType1Rec     BsmtFinType1Unf
##          5328.63473         -1804.75732           438.66954         -2190.44051
##     GarageFinishRFn     GarageFinishUnf       KitchenQualFa       KitchenQualGd
##         -4363.61170         -4975.04520        -26966.81102        -19152.36625
##       KitchenQualTA NeighborhoodBlueste  NeighborhoodBrDale NeighborhoodBrkSide
##        -22108.42176         -1862.17252         -8801.00815         -9811.01503
## NeighborhoodClearCr NeighborhoodCollgCr NeighborhoodCrawfor NeighborhoodEdwards
##         -1636.74568          1951.63801          5721.31794        -10375.07715
## NeighborhoodGilbert  NeighborhoodIDOTRR NeighborhoodMeadowV NeighborhoodMitchel
##         -2152.36044         -5081.34561         -1515.24432         -1891.63674
##   NeighborhoodNAmes NeighborhoodNoRidge NeighborhoodNPkVill NeighborhoodNridgHt
##         -5193.36976         37218.17955         -2924.43331         15455.09150
##  NeighborhoodNWAmes NeighborhoodOldTown  NeighborhoodSawyer NeighborhoodSawyerW
##         -8936.68077        -10392.37537         -2530.94804           431.44033
## NeighborhoodSomerst NeighborhoodStoneBr   NeighborhoodSWISU  NeighborhoodTimber
##          4987.91513         24491.50827         -7798.45345          1662.63930
## NeighborhoodVeenker
##          -131.34004
```

```
summary(model)
```

```
##
## Call:
## lm(formula = SalePrice ~ ., data = New_data_with_pca)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -289073  -22433   -2748   21796  376855
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         203253.79    9332.88  21.778  < 2e-16 ***
## PC1                  -9929.05     613.88 -16.174  < 2e-16 ***
## PC2                   2941.68     574.48   5.121 3.25e-07 ***
## PC3                  -5879.01     791.39  -7.429 1.44e-13 ***
## PC4                    347.60     708.28   0.491 0.623630
## PC5                  -4491.40     695.37  -6.459 1.23e-10 ***
## PC6                    468.83     717.95   0.653 0.513804
## PC7                    297.87     755.91   0.394 0.693573
## PC8                    372.51     809.26   0.460 0.645334
## PC9                  -1264.12     772.75  -1.636 0.101978
## PC10                 -2077.62     765.17  -2.715 0.006662 **
## PC11                  1937.08     773.70   2.504 0.012347 *
## PC12                    85.35     776.84   0.110 0.912523
## PC13                   206.50     809.07   0.255 0.798564
## PC14                   367.90     810.08   0.454 0.649756
## PC15                 -1950.60     825.69  -2.362 0.018225 *
## PC16                  1587.32     843.60   1.882 0.059991 .
## PC17                  -572.54     869.55  -0.658 0.510312
## PC18                   405.31     866.98   0.467 0.640178
## PC19                  1998.42     935.04   2.137 0.032662 *
## PC20                 -1838.67    1003.52  -1.832 0.067021 .
## PC21                  2083.87     972.64   2.142 0.032239 *
## PC22                  2046.50    1015.53   2.015 0.043977 *
## PC23                  -598.42    1032.65  -0.579 0.562298
## PC24                  5003.83    1080.00   4.633 3.76e-06 ***
## PC25                  -787.48    1203.80  -0.654 0.513058
## PC26                 -3679.91    1321.82  -2.784 0.005405 **
## BsmtFinType1BLQ      -1357.02    3287.42  -0.413 0.679789
## BsmtFinType1GLQ       5328.63    2876.89   1.852 0.064097 .
## BsmtFinType1LwQ      -1804.76    4141.72  -0.436 0.663050
## BsmtFinType1Rec        438.67    3296.35   0.133 0.894142
## BsmtFinType1Unf      -2190.44    3183.13  -0.688 0.491420
## GarageFinishRFn      -4363.61    2250.48  -1.939 0.052603 .
## GarageFinishUnf      -4975.05    2457.13  -2.025 0.042987 *
## KitchenQualFa       -26966.81    6734.59  -4.004 6.38e-05 ***
## KitchenQualGd       -19152.37    3737.69  -5.124 3.19e-07 ***
## KitchenQualTA       -22108.42    4295.29  -5.147 2.82e-07 ***
## NeighborhoodBlueste  -1862.17   15438.48  -0.121 0.904001
## NeighborhoodBrDale   -8801.01   11819.13  -0.745 0.456550
## NeighborhoodBrkSide  -9811.02    9808.16  -1.000 0.317254
## NeighborhoodClearCr  -1636.75   10681.30  -0.153 0.878224
## NeighborhoodCollgCr   1951.64    8677.24   0.225 0.822062
```

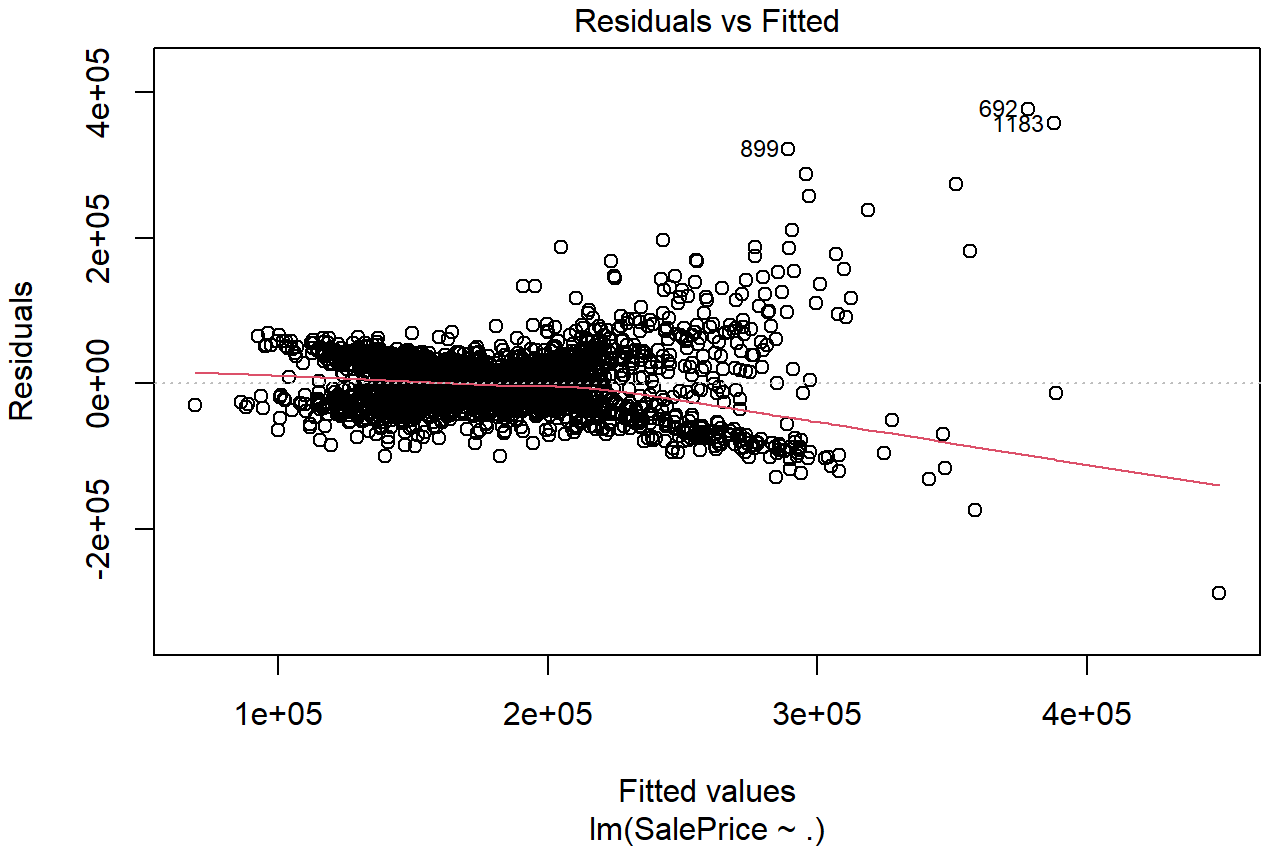
```
## NeighborhoodCrawfor   5721.32    9582.81   0.597 0.550528
## NeighborhoodEdwards -10375.08    9181.65  -1.130 0.258579
## NeighborhoodGilbert  -2152.36    8896.43  -0.242 0.808848
## NeighborhoodIDOTRR   -5081.35   10012.70  -0.507 0.611850
## NeighborhoodMeadowV  -1515.24   10999.83  -0.138 0.890446
## NeighborhoodMitchel  -1891.64    9310.09  -0.203 0.839008
## NeighborhoodNAmes    -5193.37    8996.46  -0.577 0.563804
## NeighborhoodNoRidge  37218.18    9916.51   3.753 0.000178 ***
## NeighborhoodNPkVill  -2924.43   12167.39  -0.240 0.810076
## NeighborhoodNridgHt  15455.09    8931.58   1.730 0.083669 .
## NeighborhoodNWAmes   -8936.68    9284.14  -0.963 0.335843
## NeighborhoodOldTown -10392.38    9415.87  -1.104 0.269813
## NeighborhoodSawyer   -2530.95    9358.65  -0.270 0.786842
## NeighborhoodSawyerW    431.44    9152.92   0.047 0.962407
## NeighborhoodSomerst   4987.92    8710.71   0.573 0.566948
## NeighborhoodStoneBr  24491.51    9967.24   2.457 0.014061 *
## NeighborhoodSWISU    -7798.45   10880.81  -0.717 0.473608
## NeighborhoodTimber    1662.64    9595.69   0.173 0.862452
## NeighborhoodVeenker   -131.34   11990.68  -0.011 0.991261
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41130 on 2858 degrees of freedom
## Multiple R-squared:  0.4969, Adjusted R-squared:  0.4863
## F-statistic: 47.04 on 60 and 2858 DF,  p-value: < 2.2e-16
```

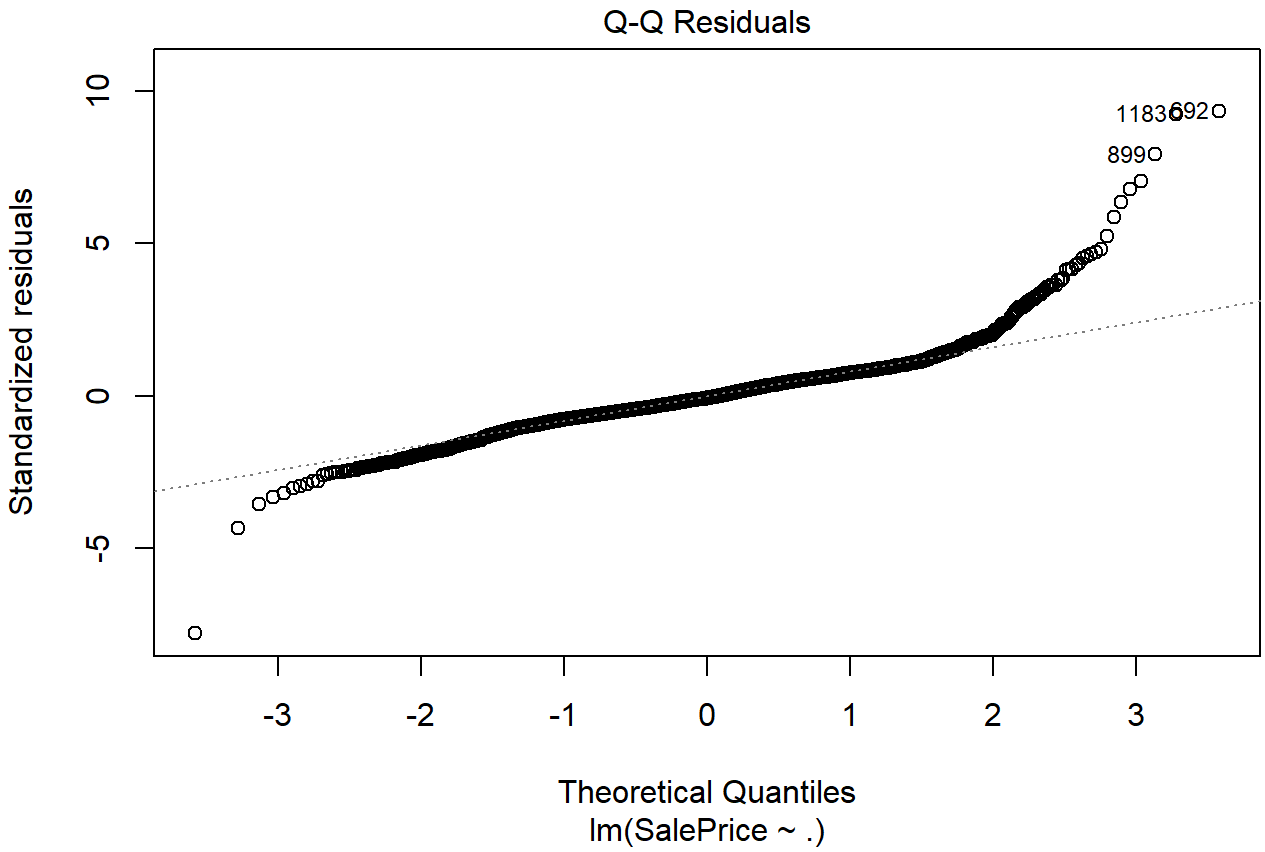
## Check Auto Correlection

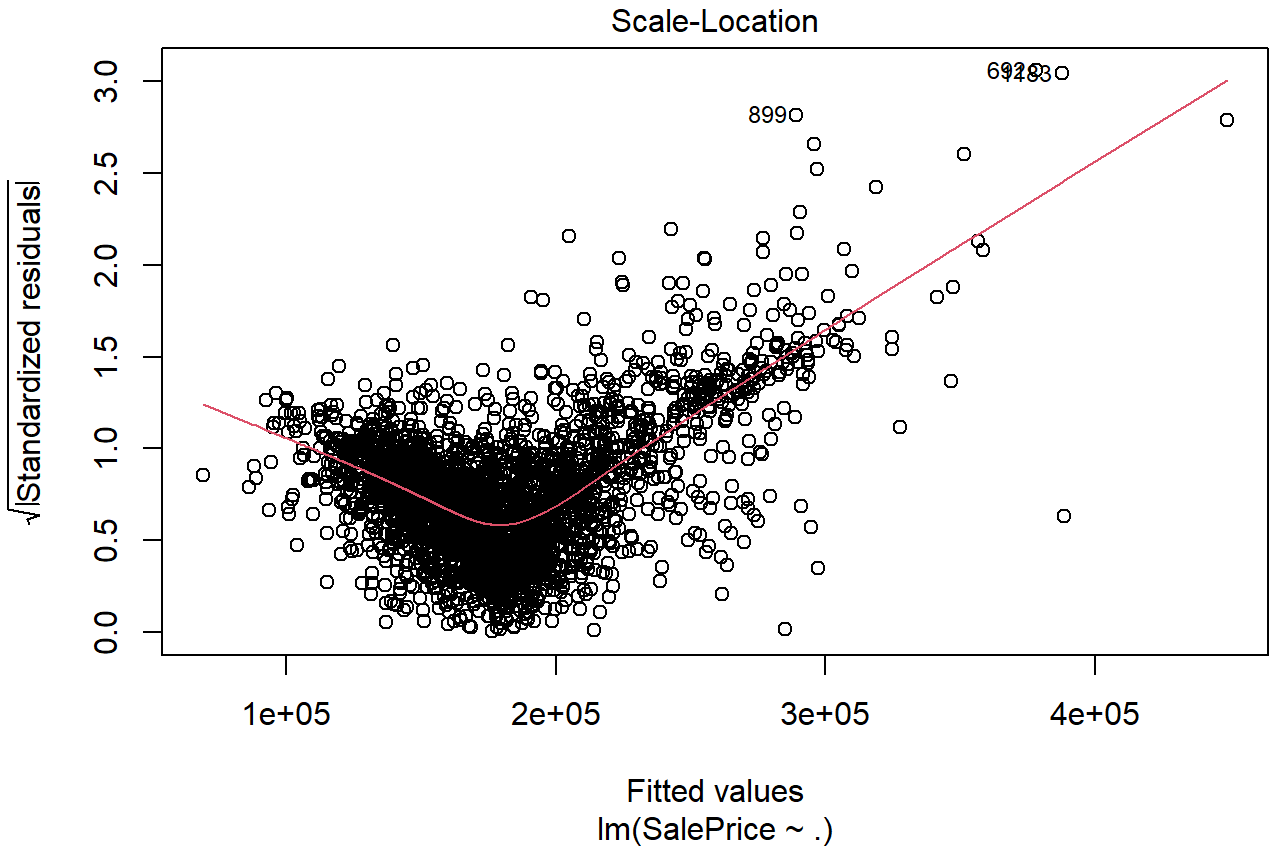
```
dw_result <- dwtest(model)
print(dw_result)
```

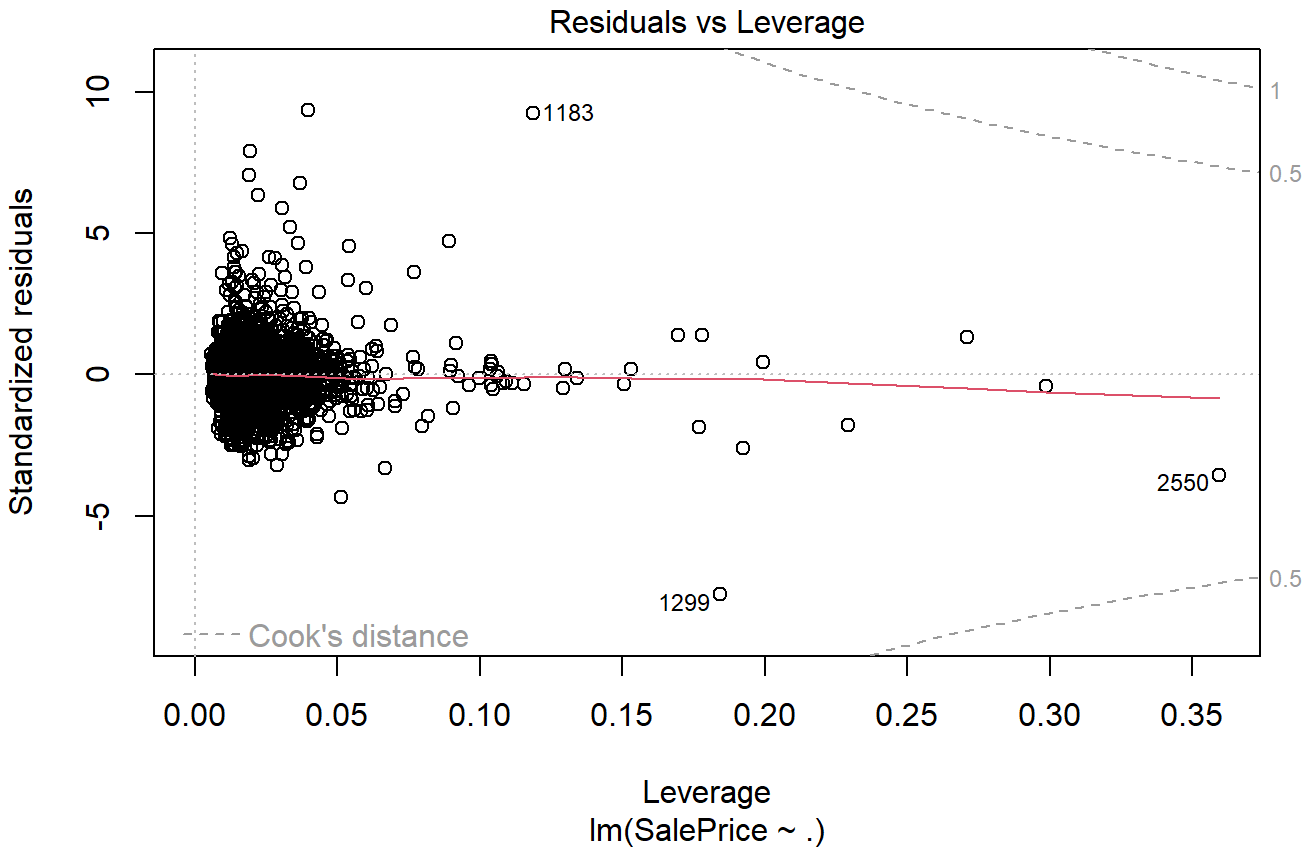
```
##
##  Durbin-Watson test
##
## data:  model
## DW = 1.5373, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

```
plot(lm(SalePrice ~ ., data = New_data_with_pca))
```


Residuals vs Fitted
Residuals
4e+05
2e+05
0e+00
-2e+05
692
1183
899
1e+05
2e+05
3e+05
4e+05
Fitted values
lm(SalePrice ~ .)

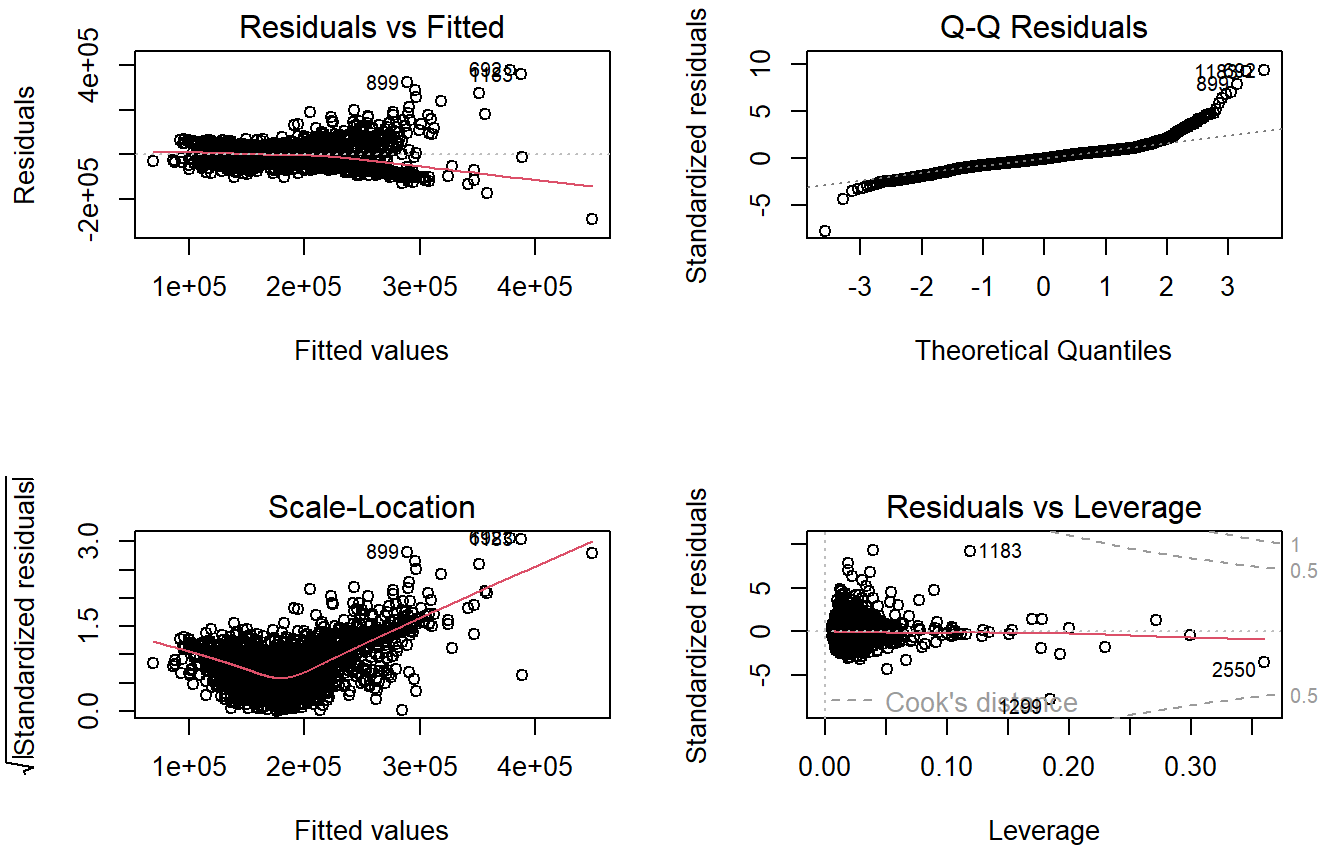

Q-Q Residuals
Standardized residuals
10
5
0
-5
1183
692
899
-3
-2
-1
0
1
2
3
Theoretical Quantiles
lm(SalePrice ~ .)




##### may be we need to take log y

```
par(mfrow = c(2, 2))
plot(model)
```




check multicollinearity

```
library(car)
```

```
## Loading required package: carData
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##     recode
```

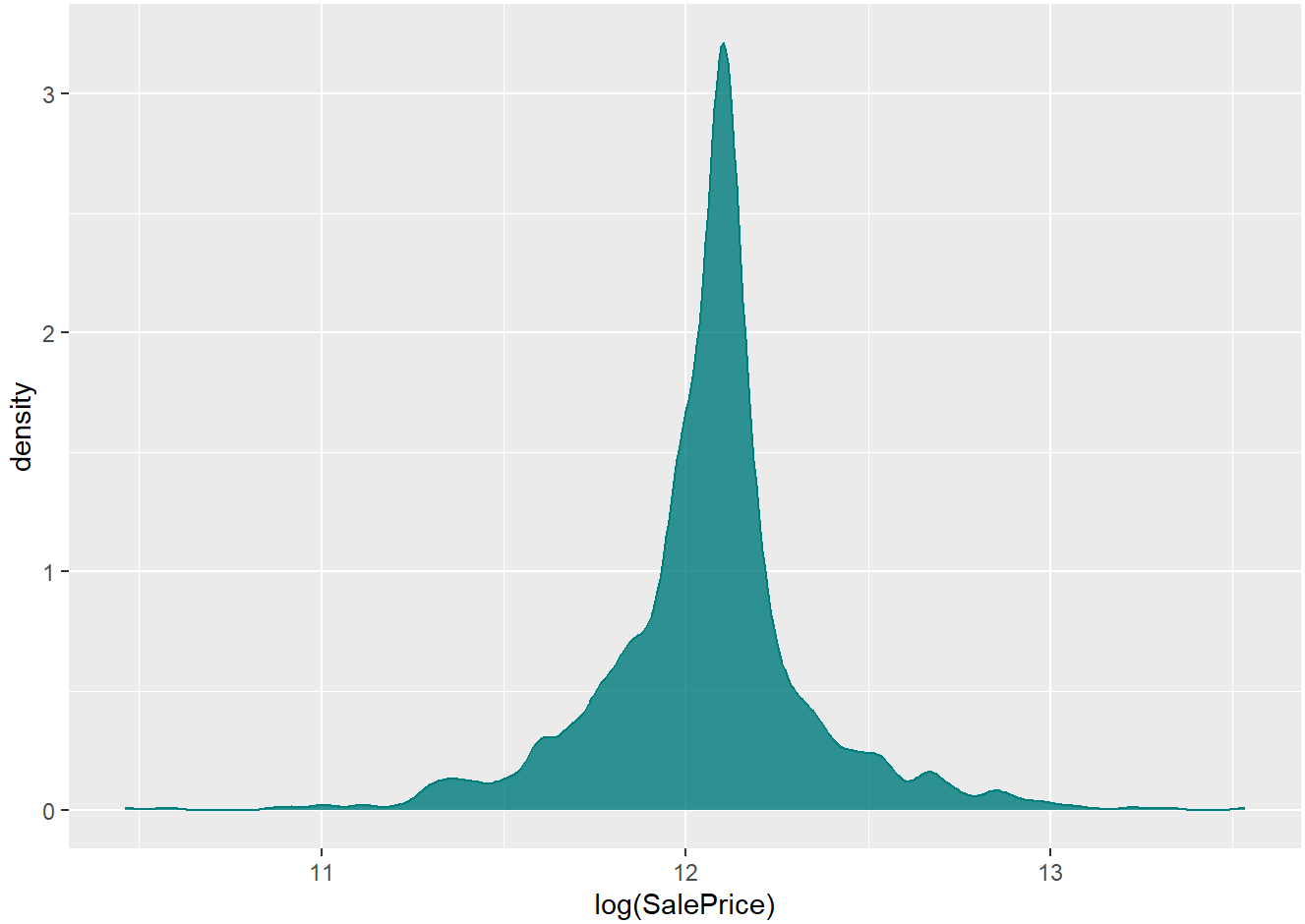
```
## The following object is masked from 'package:purrr':
##
##     some
```

```
VIF <- vif(model)
print(VIF)
```

```
##                   GVIF Df GVIF^(1/(2*Df))
## PC1           4.686506  1        2.164834
## PC2           1.825650  1        1.351166
## PC3           2.771010  1        1.664635
## PC4           1.730837  1        1.315613
## PC5           1.219390  1        1.104260
## PC6           1.053213  1        1.026261
## PC7           1.139724  1        1.067579
## PC8           1.286472  1        1.134227
## PC9           1.145565  1        1.070311
## PC10          1.093610  1        1.045758
## PC11          1.092648  1        1.045298
## PC12          1.061786  1        1.030430
## PC13          1.103235  1        1.050350
## PC14          1.066315  1        1.032625
## PC15          1.087649  1        1.042904
## PC16          1.090026  1        1.044043
## PC17          1.116798  1        1.056787
## PC18          1.054499  1        1.026888
## PC19          1.155710  1        1.075039
## PC20          1.227961  1        1.108134
## PC21          1.076310  1        1.037454
## PC22          1.091771  1        1.044878
## PC23          1.044083  1        1.021804
## PC24          1.102913  1        1.050197
## PC25          1.059971  1        1.029549
## PC26          1.108197  1        1.052709
## BsmtFinType1  5.453150  5        1.184854
## GarageFinish  2.152654  2        1.211277
## KitchenQual   3.524722  3        1.233637
## Neighborhood 55.124687 24        1.087121
```

## Remove Neighborhood and take log y

```
ggplot(imputed_data, aes(x=log(SalePrice))) + geom_density(fill="#008080",
                                          color="#008080",
                                          alpha=0.8)
```




its

looks better after taking log y

```
New_data_with_pca<- subset(New_data_with_pca, select = -c(Neighborhood))
```

```
model2 <- lm(log(SalePrice) ~ ., data = New_data_with_pca)
summary(model)
```

```
##
## Call:
## lm(formula = SalePrice ~ ., data = New_data_with_pca)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -289073  -22433   -2748   21796  376855
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         203253.79    9332.88  21.778  < 2e-16 ***
## PC1                  -9929.05     613.88 -16.174  < 2e-16 ***
## PC2                   2941.68     574.48   5.121 3.25e-07 ***
## PC3                  -5879.01     791.39  -7.429 1.44e-13 ***
## PC4                    347.60     708.28   0.491 0.623630
## PC5                  -4491.40     695.37  -6.459 1.23e-10 ***
## PC6                    468.83     717.95   0.653 0.513804
## PC7                    297.87     755.91   0.394 0.693573
## PC8                    372.51     809.26   0.460 0.645334
## PC9                  -1264.12     772.75  -1.636 0.101978
## PC10                 -2077.62     765.17  -2.715 0.006662 **
## PC11                  1937.08     773.70   2.504 0.012347 *
## PC12                    85.35     776.84   0.110 0.912523
## PC13                   206.50     809.07   0.255 0.798564
## PC14                   367.90     810.08   0.454 0.649756
## PC15                 -1950.60     825.69  -2.362 0.018225 *
## PC16                  1587.32     843.60   1.882 0.059991 .
## PC17                  -572.54     869.55  -0.658 0.510312
## PC18                   405.31     866.98   0.467 0.640178
## PC19                  1998.42     935.04   2.137 0.032662 *
## PC20                 -1838.67    1003.52  -1.832 0.067021 .
## PC21                  2083.87     972.64   2.142 0.032239 *
## PC22                  2046.50    1015.53   2.015 0.043977 *
## PC23                  -598.42    1032.65  -0.579 0.562298
## PC24                  5003.83    1080.00   4.633 3.76e-06 ***
## PC25                  -787.48    1203.80  -0.654 0.513058
## PC26                 -3679.91    1321.82  -2.784 0.005405 **
## BsmtFinType1BLQ      -1357.02    3287.42  -0.413 0.679789
## BsmtFinType1GLQ       5328.63    2876.89   1.852 0.064097 .
## BsmtFinType1LwQ      -1804.76    4141.72  -0.436 0.663050
## BsmtFinType1Rec        438.67    3296.35   0.133 0.894142
## BsmtFinType1Unf      -2190.44    3183.13  -0.688 0.491420
## GarageFinishRFn      -4363.61    2250.48  -1.939 0.052603 .
## GarageFinishUnf      -4975.05    2457.13  -2.025 0.042987 *
## KitchenQualFa       -26966.81    6734.59  -4.004 6.38e-05 ***
## KitchenQualGd       -19152.37    3737.69  -5.124 3.19e-07 ***
## KitchenQualTA       -22108.42    4295.29  -5.147 2.82e-07 ***
## NeighborhoodBlueste  -1862.17   15438.48  -0.121 0.904001
## NeighborhoodBrDale   -8801.01   11819.13  -0.745 0.456550
## NeighborhoodBrkSide  -9811.02    9808.16  -1.000 0.317254
## NeighborhoodClearCr  -1636.75   10681.30  -0.153 0.878224
## NeighborhoodCollgCr   1951.64    8677.24   0.225 0.822062
```

```
## NeighborhoodCrawfor   5721.32    9582.81   0.597 0.550528
## NeighborhoodEdwards -10375.08    9181.65  -1.130 0.258579
## NeighborhoodGilbert  -2152.36    8896.43  -0.242 0.808848
## NeighborhoodIDOTRR   -5081.35   10012.70  -0.507 0.611850
## NeighborhoodMeadowV  -1515.24   10999.83  -0.138 0.890446
## NeighborhoodMitchel  -1891.64    9310.09  -0.203 0.839008
## NeighborhoodNAmes    -5193.37    8996.46  -0.577 0.563804
## NeighborhoodNoRidge  37218.18    9916.51   3.753 0.000178 ***
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## NeighborhoodOldTown -10392.38    9415.87  -1.104 0.269813
## NeighborhoodSawyer   -2530.95    9358.65  -0.270 0.786842
## NeighborhoodSawyerW    431.44    9152.92   0.047 0.962407
## NeighborhoodSomerst   4987.92    8710.71   0.573 0.566948
## NeighborhoodStoneBr  24491.51    9967.24   2.457 0.014061 *
## NeighborhoodSWISU    -7798.45   10880.81  -0.717 0.473608
## NeighborhoodTimber    1662.64    9595.69   0.173 0.862452
## NeighborhoodVeenker   -131.34   11990.68  -0.011 0.991261
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 41130 on 2858 degrees of freedom
## Multiple R-squared:  0.4969, Adjusted R-squared:  0.4863
## F-statistic: 47.04 on 60 and 2858 DF,  p-value: < 2.2e-16
```

```
coef(model2)
```

```
##     (Intercept)             PC1             PC2             PC3             PC4
##   12.1185631352   -0.0622967409    0.0161031690   -0.0212337213   -0.0005943853
##             PC5             PC6             PC7             PC8             PC9
##   -0.0285725473    0.0045734735    0.0012253064    0.0001755180   -0.0096041724
##            PC10            PC11            PC12            PC13            PC14
##   -0.0117896131    0.0160464688   -0.0052774147    0.0057655388    0.0049913350
##            PC15            PC16            PC17            PC18            PC19
##   -0.0081638337    0.0069737610   -0.0023027625   -0.0001858448    0.0105733913
##            PC20            PC21            PC22            PC23            PC24
##   -0.0184345893    0.0088882771    0.0032568200    0.0033572974    0.0249436178
##            PC25            PC26 BsmtFinType1BLQ BsmtFinType1GLQ BsmtFinType1LwQ
##    0.0029816825    0.0005229627   -0.0088609564    0.0276034302   -0.0093832288
## BsmtFinType1Rec BsmtFinType1Unf GarageFinishRFn GarageFinishUnf   KitchenQualFa
##    0.0070700371   -0.0335819792   -0.0047487961   -0.0249269865   -0.1352543490
##   KitchenQualGd   KitchenQualTA
##   -0.0291472267   -0.0592297636
```

```
summary(model2)
```

```
##
## Call:
## lm(formula = log(SalePrice) ~ ., data = New_data_with_pca)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.36980 -0.12654  0.00076  0.13858  0.83029
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)     12.1185631  0.0215829 561.488  < 2e-16 ***
## PC1             -0.0622967  0.0023580 -26.419  < 2e-16 ***
## PC2              0.0161032  0.0027278   5.903 3.98e-09 ***
## PC3             -0.0212337  0.0029085  -7.301 3.69e-13 ***
## PC4             -0.0005944  0.0034035  -0.175 0.861375
## PC5             -0.0285725  0.0034124  -8.373  < 2e-16 ***
## PC6              0.0045735  0.0035582   1.285 0.198775
## PC7              0.0012253  0.0036601   0.335 0.737823
## PC8              0.0001755  0.0037064   0.047 0.962233
## PC9             -0.0096042  0.0037190  -2.582 0.009858 **
## PC10            -0.0117896  0.0037549  -3.140 0.001708 **
## PC11             0.0160465  0.0037863   4.238 2.33e-05 ***
## PC12            -0.0052774  0.0038685  -1.364 0.172612
## PC13             0.0057655  0.0039254   1.469 0.142005
## PC14             0.0049913  0.0040031   1.247 0.212546
## PC15            -0.0081638  0.0041123  -1.985 0.047212 *
## PC16             0.0069738  0.0041378   1.685 0.092022 .
## PC17            -0.0023028  0.0043219  -0.533 0.594206
## PC18            -0.0001858  0.0043023  -0.043 0.965548
## PC19             0.0105734  0.0044639   2.369 0.017919 *
## PC20            -0.0184346  0.0047686  -3.866 0.000113 ***
## PC21             0.0088883  0.0047821   1.859 0.063179 .
## PC22             0.0032568  0.0049461   0.658 0.510292
## PC23             0.0033573  0.0051272   0.655 0.512644
## PC24             0.0249436  0.0053263   4.683 2.96e-06 ***
## PC25             0.0029817  0.0059534   0.501 0.616522
## PC26             0.0005230  0.0064582   0.081 0.935467
## BsmtFinType1BLQ -0.0088610  0.0163663  -0.541 0.588263
## BsmtFinType1GLQ  0.0276034  0.0139943   1.972 0.048651 *
## BsmtFinType1LwQ -0.0093832  0.0205032  -0.458 0.647241
## BsmtFinType1Rec  0.0070700  0.0163003   0.434 0.664513
## BsmtFinType1Unf -0.0335820  0.0154899  -2.168 0.030241 *
## GarageFinishRFn -0.0047488  0.0107481  -0.442 0.658649
## GarageFinishUnf -0.0249270  0.0119683  -2.083 0.037362 *
## KitchenQualFa   -0.1352543  0.0333134  -4.060 5.04e-05 ***
## KitchenQualGd   -0.0291472  0.0176139  -1.655 0.098076 .
## KitchenQualTA   -0.0592298  0.0203887  -2.905 0.003700 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2079 on 2882 degrees of freedom
```

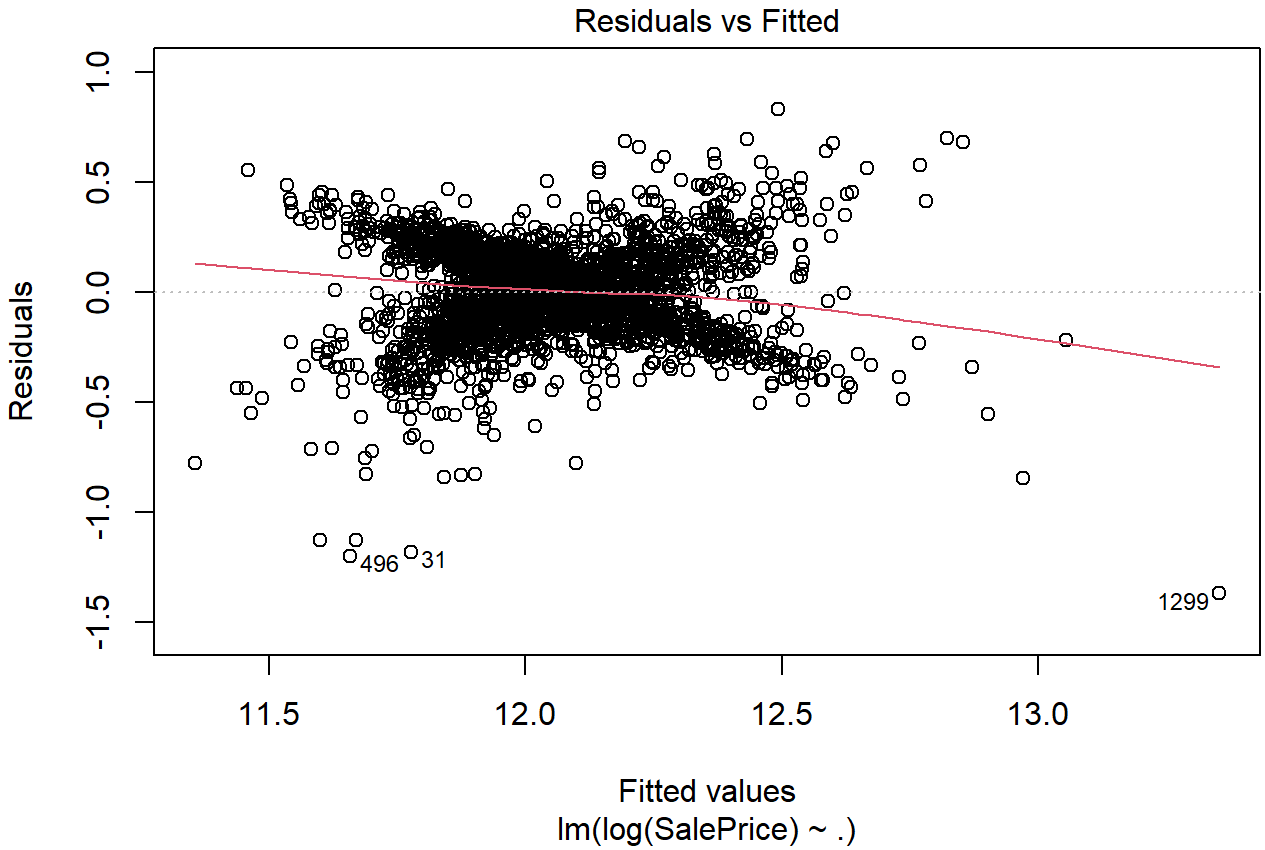
```
## Multiple R-squared:  0.4975, Adjusted R-squared:  0.4912
## F-statistic: 79.26 on 36 and 2882 DF,  p-value: < 2.2e-16
```

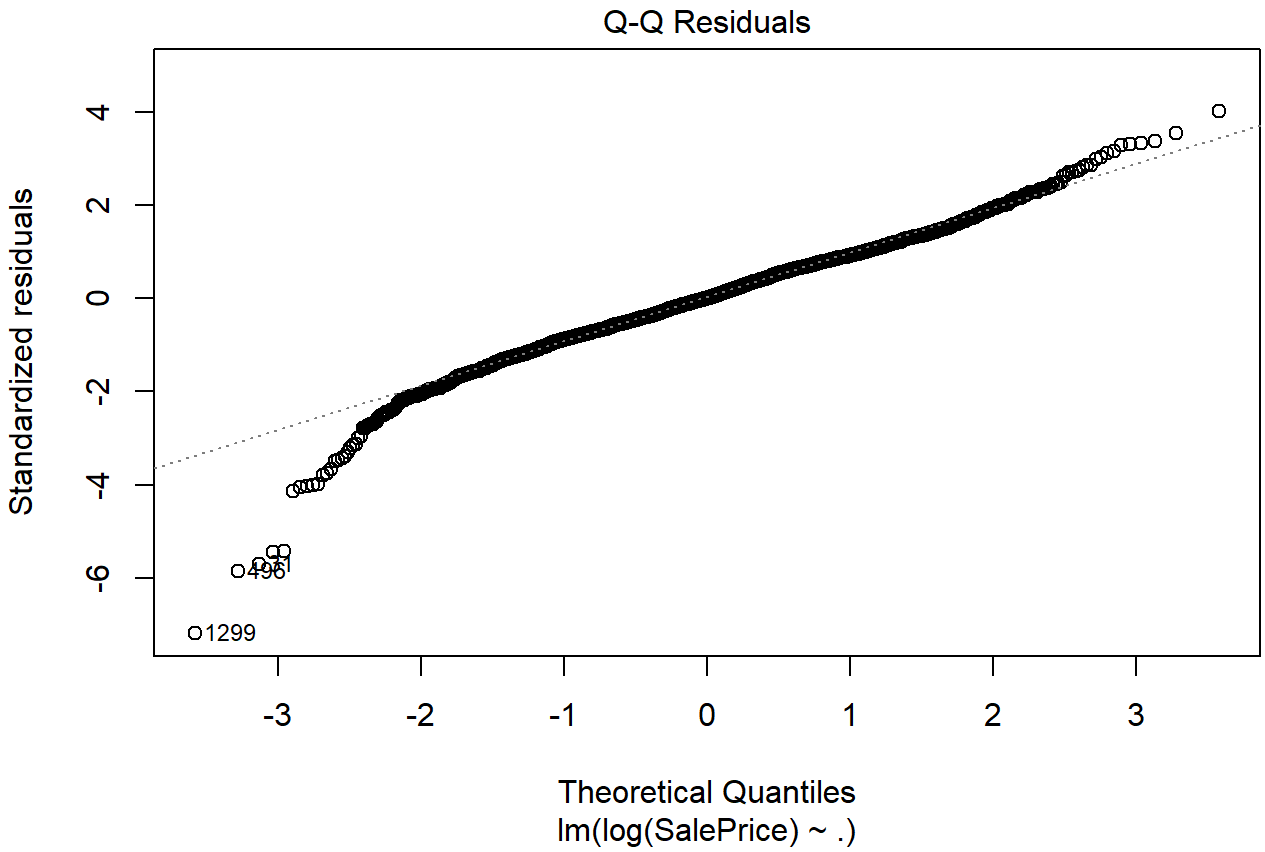
## Check Auto Correlection

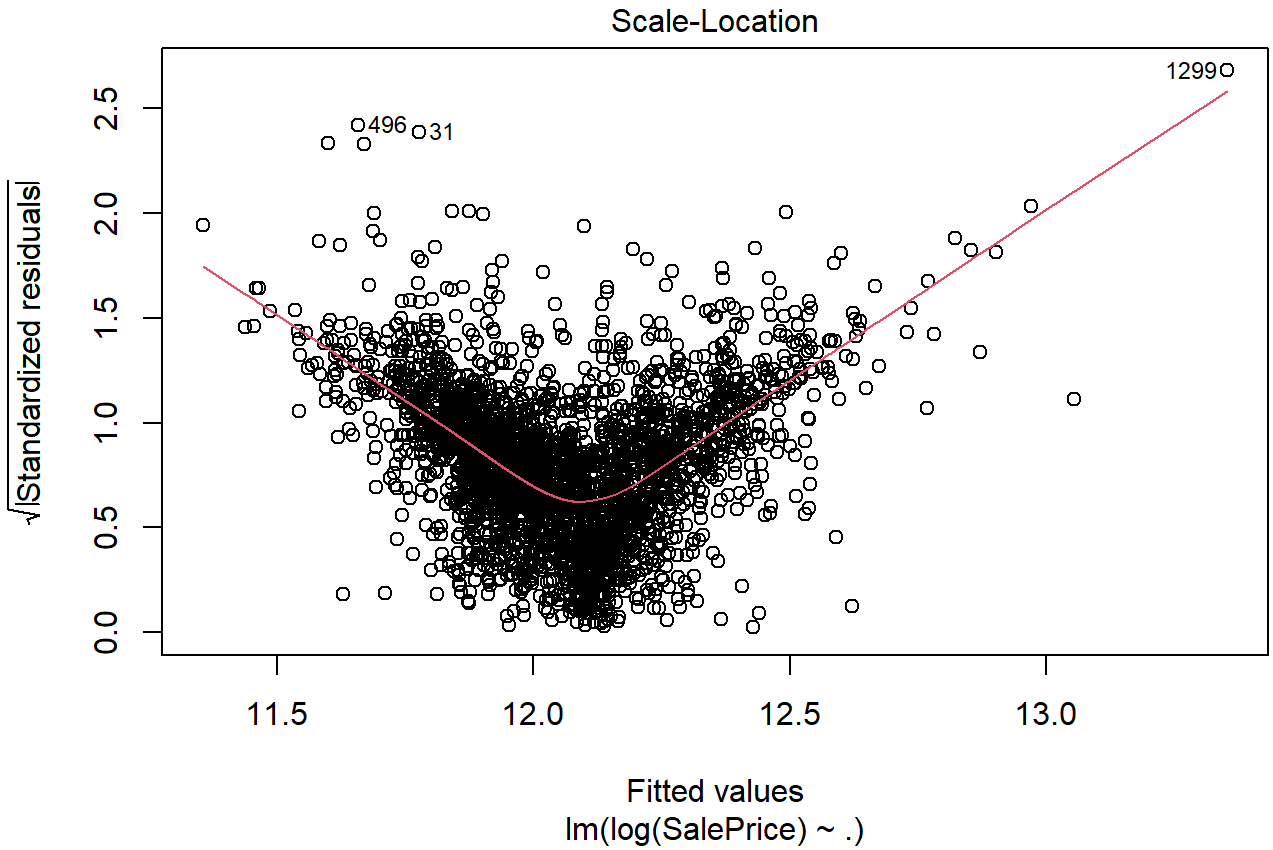
```
dw_result <- dwtest(model2)
print(dw_result)
```

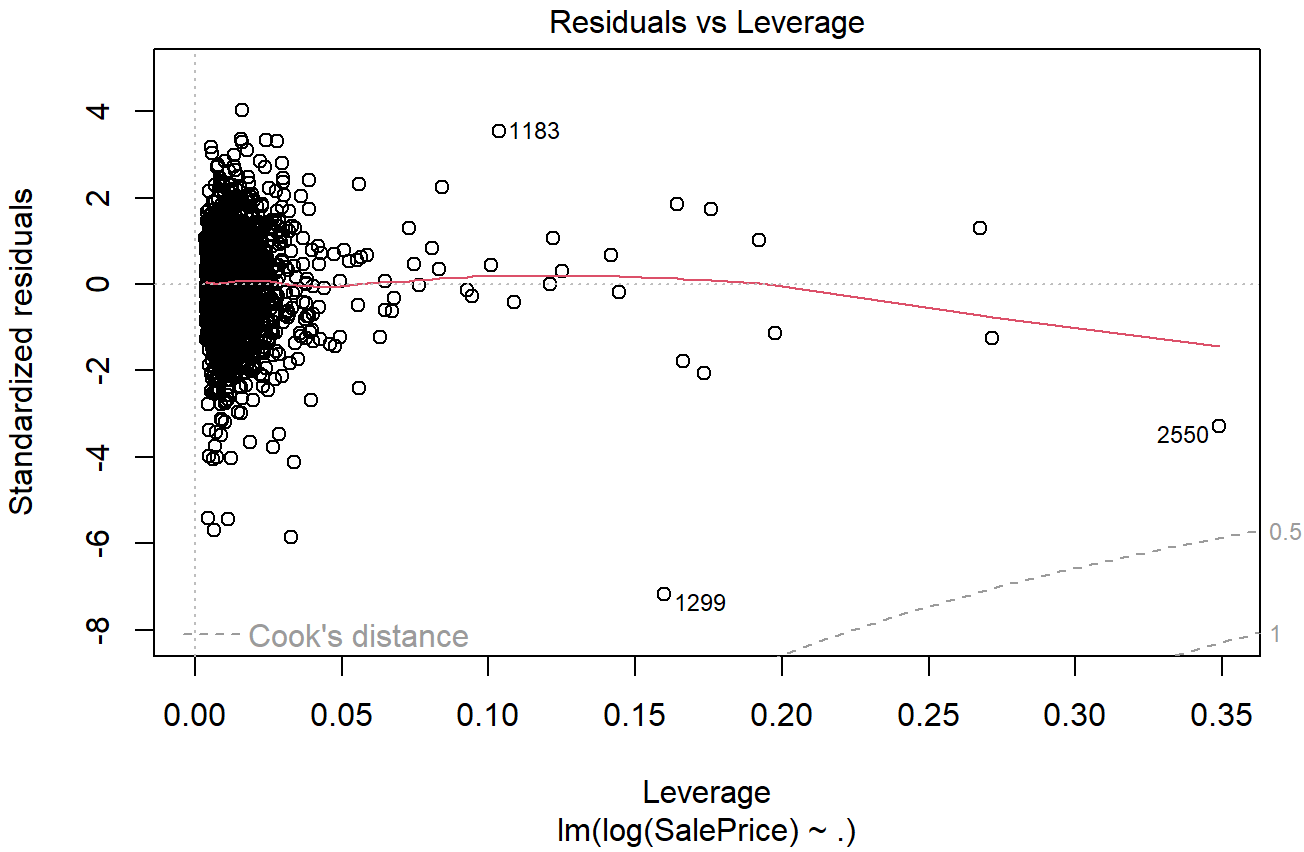
```
##
##  Durbin-Watson test
##
## data:  model2
## DW = 1.502, p-value < 2.2e-16
## alternative hypothesis: true autocorrelation is greater than 0
```

```
plot(lm(log(SalePrice) ~ ., data = New_data_with_pca))
```


Residuals vs Fitted
Residuals
1.0
0.5
0.0
-0.5
-1.0
-1.5
496
31
1299
11.5
12.0
12.5
13.0
Fitted values
lm(log(SalePrice) ~ .)

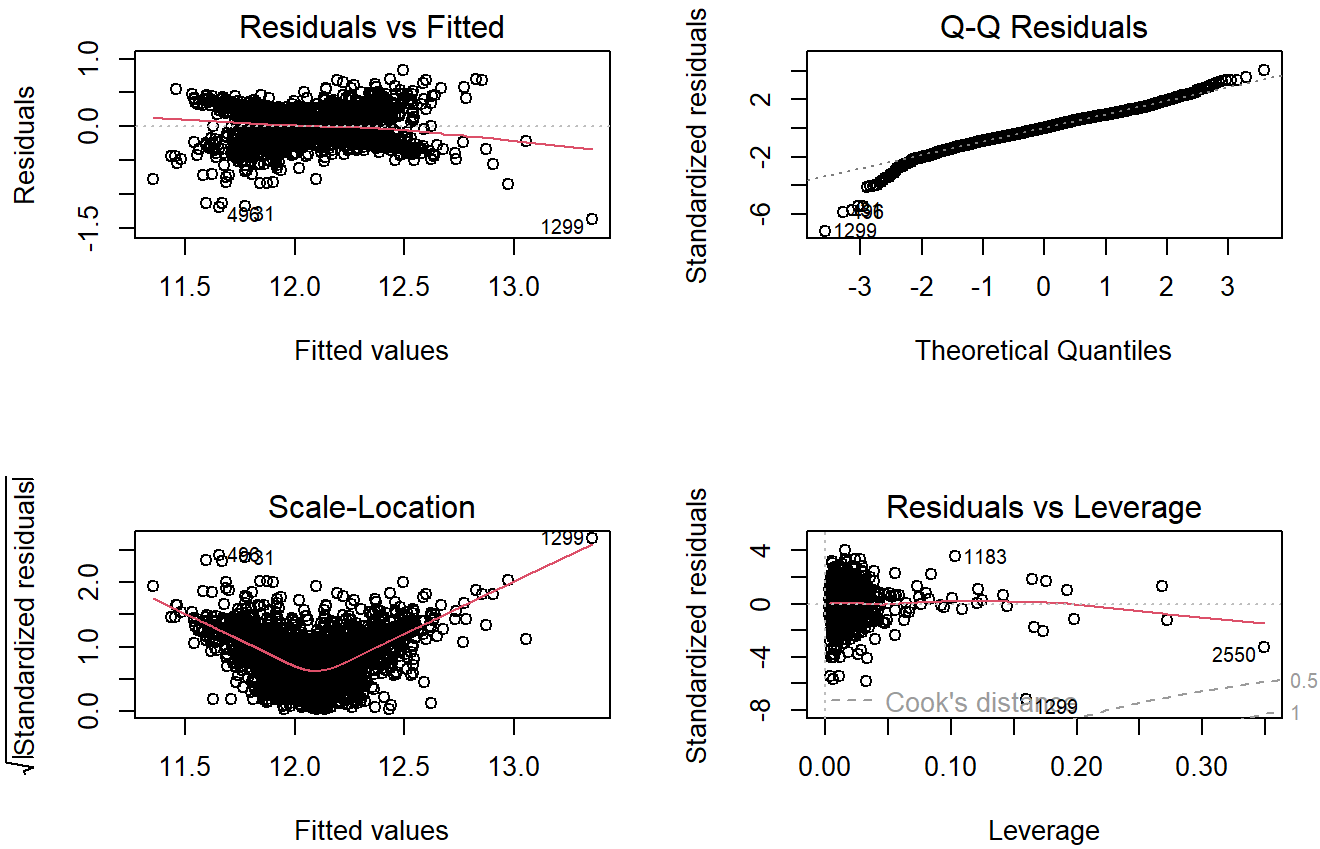

Q-Q Residuals
Standardized residuals
4
2
0
-2
-4
-6
496
1299
-3
-2
-1
0
1
2
3
Theoretical Quantiles
lm(log(SalePrice) ~ .)




##### may be we need to take log y

```
par(mfrow = c(2, 2))
plot(model2)
```




check multicollinearity

```
library(car)
VIF <- vif(model2)
print(VIF)
```

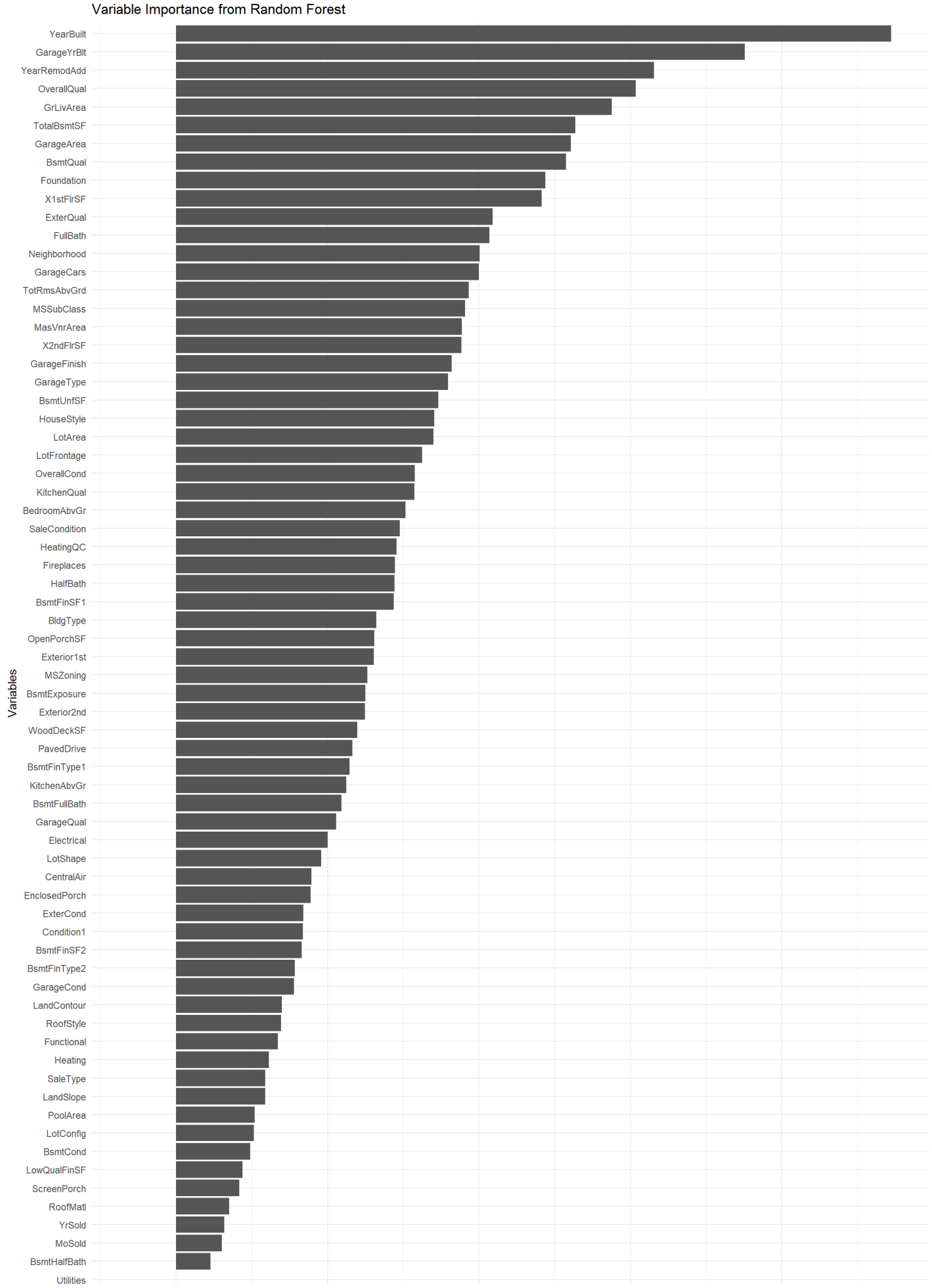
```
##                  GVIF Df GVIF^(1/(2*Df))
## PC1          2.704653  1        1.644583
## PC2          1.609965  1        1.268844
## PC3          1.463947  1        1.209937
## PC4          1.563248  1        1.250299
## PC5          1.148606  1        1.071730
## PC6          1.011846  1        1.005906
## PC7          1.045184  1        1.022342
## PC8          1.055524  1        1.027387
## PC9          1.037831  1        1.018740
## PC10         1.030101  1        1.014939
## PC11         1.023550  1        1.011706
## PC12         1.029897  1        1.014839
## PC13         1.015787  1        1.007863
## PC14         1.018496  1        1.009206
## PC15         1.055236  1        1.027247
## PC16         1.025730  1        1.012783
## PC17         1.079134  1        1.038814
## PC18         1.015696  1        1.007817
## PC19         1.030289  1        1.015032
## PC20         1.084559  1        1.041422
## PC21         1.017681  1        1.008802
## PC22         1.012991  1        1.006474
## PC23         1.006748  1        1.003368
## PC24         1.049268  1        1.024338
## PC25         1.014025  1        1.006988
## PC26         1.034763  1        1.017233
## BsmtFinType1 4.532721  5        1.163150
## GarageFinish 1.833961  2        1.163717
## KitchenQual  2.787072  3        1.186291
```

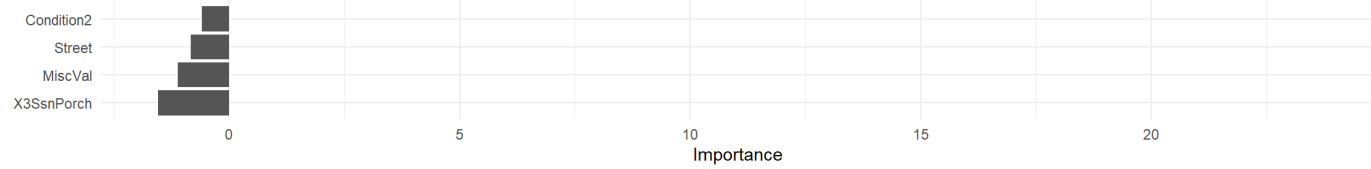
the model not Satsified lets make feature selection using Random Forest

# Feature Selection

## Random Forest

```
# Create Random Forest model
set.seed(123)  # For reproducibility
rf_model <- randomForest(x = X, ntree = 100, importance = TRUE)

# Get variable importance
importance <- importance(rf_model, type = 1)
var_importance <- data.frame(Variables = rownames(importance), Importance = importance[, 1])

# Visualize variable importance
var_importance_plot <- ggplot(var_importance, aes(x = reorder(Variables, Importance), y = Import
ance)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  theme_minimal() +
  labs(title = "Variable Importance from Random Forest",
       x = "Variables",
       y = "Importance")
print(var_importance_plot)
```


Variable Importance from Random Forest
Variables
YearBuilt
GarageYrBlt
YearRemodAdd
OverallQual
GrLivArea
TotalBsmtSF
GarageArea
BsmtQual
Foundation
X1stFlrSF
ExterQual
FullBath
Neighborhood
GarageCars
TotRmsAbvGrd
MSSubClass
MasVnrArea
X2ndFlrSF
GarageFinish
GarageType
BsmtUnfSF
HouseStyle
LotArea
LotFrontage
OverallCond
KitchenQual
BedroomAbvGr
SaleCondition
HeatingQC
Fireplaces
HalfBath
BsmtFinSF1
BldgType
OpenPorchSF
Exterior1st
MSZoning
BsmtExposure
Exterior2nd
WoodDeckSF
PavedDrive
BsmtFinType1
KitchenAbvGr
BsmtFullBath
GarageQual
Electrical
LotShape
CentralAir
EnclosedPorch
ExterCond
Condition1
BsmtFinSF2
BsmtFinType2
GarageCond
LandContour
RoofStyle
Functional
Heating
SaleType
LandSlope
PoolArea
LotConfig
BsmtCond
LowQualFinSF
ScreenPorch
RoofMatl
YrSold
MoSold
BsmtHalfBath
Utilities




# Modelling

```
# Select 7 features
selected_features <- c('LotArea', 'YearBuilt', 'BedroomAbvGr', 'GarageArea', 'OverallQual', 'Yea
rRemodAdd')
X <- X[, selected_features]
y <- log(imputed_data$SalePrice)

# Split the data into training and test sets
set.seed(42)
train_index <- createDataPartition(y, p = 0.8, list = FALSE)
X_train <- X[train_index, ]
y_train <- y[train_index]
X_test <- X[-train_index, ]
y_test <- y[-train_index]

# Combine X_train and y_train into a single data frame
train_data <- data.frame(X_train, SalePrice = y_train)

# Fit the linear regression model
model <- lm(SalePrice ~ ., data = train_data)

# Combine X_test and y_test into a single data frame for prediction
test_data <- data.frame(X_test)

# Evaluate the model
y_pred <- predict(model, newdata = test_data)
r2 <- R2(y_test, y_pred)
mse <- mean((y_test - y_pred)^2)
cat(sprintf('R-squared: %.2f\n', r2))
```

```
## R-squared: 0.44
```

```
cat(sprintf('Mean Squared Error: %.2f\n', mse))
```

```
## Mean Squared Error: 0.05
```

```
# Check for normality of residuals
shapiro.test(resid(model))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid(model)
## W = 0.98044, p-value < 2.2e-16
```

```
# Check for independence of residuals
durbinWatsonTest(model)
```

```
##  lag Autocorrelation D-W Statistic p-value
##    1         0.22224      1.555452       0
##  Alternative hypothesis: rho != 0
```

```
# Check for homoscedasticity

bptest(model)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  model
## BP = 83.037, df = 6, p-value = 8.416e-16
```

```
# Check for multicollinearity
vif(model)
```

```
##      LotArea    YearBuilt BedroomAbvGr   GarageArea  OverallQual YearRemodAdd
##     1.058237     1.941192     1.039225     1.586456     2.029417     1.766240
```

```
libraries(MASS)
```

```
##  Load and Attach Packages
##
##   MASS 7.3-60 loaded
```

```
# Fit the full linear regression model
full_model <- lm(SalePrice ~ ., data = imputed_data)

# Perform stepwise regression
step_model <- stepAIC(full_model, direction = "both")
```

```
## Start:  AIC=62057.88
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + Exterior1st + Exterior2nd + MasVnrArea + ExterQual +
##     ExterCond + Foundation + BsmtQual + BsmtCond + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     TotalBsmtSF + Heating + HeatingQC + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + GrLivArea + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageType +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleType + SaleCondition
##
##
## Step:  AIC=62057.88
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + Exterior1st + Exterior2nd + MasVnrArea + ExterQual +
##     ExterCond + Foundation + BsmtQual + BsmtCond + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     TotalBsmtSF + Heating + HeatingQC + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Functional + Fireplaces + GarageType + GarageYrBlt +
##     GarageFinish + GarageCars + GarageArea + GarageQual + GarageCond +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleType +
##     SaleCondition
##
##
## Step:  AIC=62057.88
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + Exterior1st + Exterior2nd + MasVnrArea + ExterQual +
##     ExterCond + Foundation + BsmtQual + BsmtCond + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + HeatingQC + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Functional + Fireplaces + GarageType + GarageYrBlt +
##     GarageFinish + GarageCars + GarageArea + GarageQual + GarageCond +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleType +
##     SaleCondition
```

```
##
##                 Df  Sum of Sq        RSS   AIC
## - Exterior2nd   15 1.9736e+10 4.2903e+12 62041
## - Exterior1st   14 2.6478e+10 4.2970e+12 62048
## - Foundation     5 1.7340e+09 4.2723e+12 62049
## - RoofStyle      5 2.0841e+09 4.2726e+12 62049
## - SaleType       8 1.1912e+10 4.2824e+12 62050
## - Functional     6 6.1522e+09 4.2767e+12 62050
## - GarageType     5 3.5527e+09 4.2741e+12 62050
## - ExterCond      4 6.6808e+08 4.2712e+12 62050
## - HeatingQC      4 8.3207e+08 4.2714e+12 62050
## - SaleCondition  5 3.8472e+09 4.2744e+12 62051
## - HouseStyle     7 1.0658e+10 4.2812e+12 62051
## - Condition1     8 1.4528e+10 4.2851e+12 62052
## - BsmtCond       3 5.7688e+08 4.2711e+12 62052
## - GarageQual     4 3.6033e+09 4.2741e+12 62052
## - MSZoning       4 4.0568e+09 4.2746e+12 62053
## - BldgType       4 4.1321e+09 4.2747e+12 62053
## - LotConfig      4 4.3082e+09 4.2748e+12 62053
## - BsmtFinType1   5 7.9507e+09 4.2785e+12 62053
## - Electrical     4 5.8522e+09 4.2764e+12 62054
## - GarageCond     4 6.3289e+09 4.2769e+12 62054
## - GarageFinish   2 1.7197e+09 4.2722e+12 62055
## - BsmtExposure   3 4.8205e+09 4.2753e+12 62055
## - YrSold         1 1.4781e+06 4.2705e+12 62056
## - MSSubClass     1 1.6802e+06 4.2705e+12 62056
## - EnclosedPorch  1 5.8823e+06 4.2705e+12 62056
## - BsmtHalfBath   1 1.8119e+07 4.2705e+12 62056
## - GarageCars     1 2.1395e+07 4.2705e+12 62056
## - HalfBath       1 3.4037e+07 4.2706e+12 62056
## - BedroomAbvGr   1 1.1390e+08 4.2706e+12 62056
## - Fireplaces     1 1.7779e+08 4.2707e+12 62056
## - OpenPorchSF    1 1.9016e+08 4.2707e+12 62056
## - WoodDeckSF     1 2.0613e+08 4.2707e+12 62056
## - YearRemodAdd   1 3.0867e+08 4.2708e+12 62056
## - LowQualFinSF   1 3.5280e+08 4.2709e+12 62056
## - MasVnrArea     1 4.9395e+08 4.2710e+12 62056
## - X3SsnPorch     1 5.8424e+08 4.2711e+12 62056
## - BsmtFinType2   5 1.2368e+10 4.2829e+12 62056
## - BsmtFullBath   1 7.1315e+08 4.2712e+12 62056
## - Heating        5 1.2592e+10 4.2831e+12 62056
## - LotFrontage    1 1.1825e+09 4.2717e+12 62057
## - Utilities      1 1.7646e+09 4.2723e+12 62057
## - CentralAir     1 1.9468e+09 4.2725e+12 62057
## - BsmtFinSF2     1 2.3784e+09 4.2729e+12 62058
## - Street         1 2.4300e+09 4.2730e+12 62058
## <none>                        4.2705e+12 62058
## - ScreenPorch    1 3.3381e+09 4.2739e+12 62058
## - LotShape       3 9.2928e+09 4.2798e+12 62058
## - GarageArea     1 3.4430e+09 4.2740e+12 62058
## - BsmtQual       3 9.4437e+09 4.2800e+12 62058
## - FullBath       1 4.1790e+09 4.2747e+12 62059
```

```
## - GarageYrBlt    1 6.2988e+09 4.2768e+12 62060
## - OverallQual    1 6.6305e+09 4.2772e+12 62060
## - Condition2     7 2.6484e+10 4.2970e+12 62062
## - ExterQual      3 1.5555e+10 4.2861e+12 62062
## - KitchenQual    3 1.5905e+10 4.2864e+12 62063
## - PavedDrive     2 1.2981e+10 4.2835e+12 62063
## - MoSold         1 1.1709e+10 4.2822e+12 62064
## - MiscVal        1 1.3417e+10 4.2839e+12 62065
## - LandSlope      2 1.6457e+10 4.2870e+12 62065
## - PoolArea       1 1.3585e+10 4.2841e+12 62065
## - LandContour    3 1.9678e+10 4.2902e+12 62065
## - YearBuilt      1 1.5716e+10 4.2862e+12 62067
## - X1stFlrSF      1 1.7272e+10 4.2878e+12 62068
## - BsmtUnfSF      1 1.8137e+10 4.2887e+12 62068
## - BsmtFinSF1     1 1.8540e+10 4.2891e+12 62069
## - Neighborhood  24 8.7188e+10 4.3577e+12 62069
## - OverallCond    1 2.2098e+10 4.2926e+12 62071
## - KitchenAbvGr   1 2.2130e+10 4.2927e+12 62071
## - TotRmsAbvGrd   1 2.9999e+10 4.3005e+12 62076
## - X2ndFlrSF      1 6.5263e+10 4.3358e+12 62100
## - LotArea        1 8.5325e+10 4.3558e+12 62114
## - RoofMatl       7 1.6695e+11 4.4375e+12 62156
##
## Step:  AIC=62041.34
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + Exterior1st + MasVnrArea + ExterQual + ExterCond +
##     Foundation + BsmtQual + BsmtCond + BsmtExposure + BsmtFinType1 +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     HeatingQC + CentralAir + Electrical + X1stFlrSF + X2ndFlrSF +
##     LowQualFinSF + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Functional + Fireplaces + GarageType + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleType + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Exterior1st   14 2.1219e+10 4.3115e+12 62028
## - Foundation     5 2.0762e+09 4.2923e+12 62033
## - RoofStyle      5 2.2307e+09 4.2925e+12 62033
## - SaleType       8 1.1218e+10 4.3015e+12 62033
## - Functional     6 5.4535e+09 4.2957e+12 62033
## - GarageType     5 3.6911e+09 4.2940e+12 62034
## - SaleCondition  5 3.6932e+09 4.2940e+12 62034
## - ExterCond      4 7.5809e+08 4.2910e+12 62034
## - HeatingQC      4 9.5297e+08 4.2912e+12 62034
## - HouseStyle     7 1.0743e+10 4.3010e+12 62035
## - Condition1     8 1.3710e+10 4.3040e+12 62035
## - GarageQual     4 3.4182e+09 4.2937e+12 62036
```

```
## - BsmtCond       3 4.9085e+08 4.2908e+12 62036
## - LotConfig      4 3.7565e+09 4.2940e+12 62036
## - BldgType       4 4.5634e+09 4.2948e+12 62036
## - BsmtFinType1   5 7.5143e+09 4.2978e+12 62036
## - MSZoning       4 5.2956e+09 4.2956e+12 62037
## - Electrical     4 5.7077e+09 4.2960e+12 62037
## - GarageCond     4 5.8705e+09 4.2961e+12 62037
## - GarageFinish   2 1.9730e+09 4.2922e+12 62039
## - BsmtExposure   3 5.7047e+09 4.2960e+12 62039
## - EnclosedPorch  1 3.5246e+04 4.2903e+12 62039
## - YrSold         1 3.4488e+05 4.2903e+12 62039
## - BsmtHalfBath   1 1.9156e+06 4.2903e+12 62039
## - MSSubClass     1 3.3254e+06 4.2903e+12 62039
## - GarageCars     1 1.9711e+07 4.2903e+12 62039
## - OpenPorchSF    1 6.3406e+07 4.2903e+12 62039
## - BedroomAbvGr   1 7.1688e+07 4.2903e+12 62039
## - WoodDeckSF     1 1.2150e+08 4.2904e+12 62039
## - YearRemodAdd   1 1.3524e+08 4.2904e+12 62039
## - Fireplaces     1 1.7929e+08 4.2904e+12 62039
## - HalfBath       1 1.9309e+08 4.2905e+12 62039
## - LowQualFinSF   1 3.4353e+08 4.2906e+12 62040
## - MasVnrArea     1 5.5544e+08 4.2908e+12 62040
## - X3SsnPorch     1 9.3461e+08 4.2912e+12 62040
## - BsmtFullBath   1 1.0042e+09 4.2913e+12 62040
## - BsmtFinType2   5 1.2783e+10 4.3030e+12 62040
## - LotFrontage    1 1.2401e+09 4.2915e+12 62040
## - CentralAir     1 1.6714e+09 4.2919e+12 62040
## - Utilities      1 1.7232e+09 4.2920e+12 62041
## - Heating        5 1.3573e+10 4.3038e+12 62041
## - Street         1 2.2598e+09 4.2925e+12 62041
## - BsmtFinSF2     1 2.2761e+09 4.2925e+12 62041
## <none>                        4.2903e+12 62041
## - ScreenPorch    1 3.0716e+09 4.2933e+12 62041
## - BsmtQual       3 8.9886e+09 4.2992e+12 62041
## - GarageArea     1 3.3738e+09 4.2936e+12 62042
## - FullBath       1 3.6460e+09 4.2939e+12 62042
## - LotShape       3 9.7812e+09 4.3000e+12 62042
## - GarageYrBlt    1 4.7567e+09 4.2950e+12 62043
## - OverallQual    1 7.5549e+09 4.2978e+12 62044
## - KitchenQual    3 1.4466e+10 4.3047e+12 62045
## - PavedDrive     2 1.2174e+10 4.3024e+12 62046
## - Condition2     7 2.7918e+10 4.3182e+12 62046
## - ExterQual      3 1.6110e+10 4.3064e+12 62046
## - MoSold         1 1.0451e+10 4.3007e+12 62046
## - MiscVal        1 1.3482e+10 4.3037e+12 62048
## - LandSlope      2 1.6819e+10 4.3071e+12 62049
## - PoolArea       1 1.3894e+10 4.3042e+12 62049
## - YearBuilt      1 1.4198e+10 4.3045e+12 62049
## - LandContour    3 2.1499e+10 4.3118e+12 62050
## - BsmtUnfSF      1 1.6988e+10 4.3072e+12 62051
## - X1stFlrSF      1 1.7737e+10 4.3080e+12 62051
## - BsmtFinSF1     1 1.9411e+10 4.3097e+12 62053
```

```
## - OverallCond    1 2.2628e+10 4.3129e+12 62055
## - KitchenAbvGr   1 2.3040e+10 4.3133e+12 62055
## - Neighborhood  24 9.5411e+10 4.3857e+12 62058
## + Exterior2nd   15 1.9736e+10 4.2705e+12 62058
## - TotRmsAbvGrd   1 2.9303e+10 4.3196e+12 62059
## - X2ndFlrSF      1 6.6814e+10 4.3571e+12 62084
## - LotArea        1 8.9186e+10 4.3794e+12 62099
## - RoofMatl       7 1.6794e+11 4.4582e+12 62139
##
## Step:  AIC=62027.74
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + MasVnrArea + ExterQual + ExterCond + Foundation +
##     BsmtQual + BsmtCond + BsmtExposure + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
##     BsmtFullBath + BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Functional +
##     Fireplaces + GarageType + GarageYrBlt + GarageFinish + GarageCars +
##     GarageArea + GarageQual + GarageCond + PavedDrive + WoodDeckSF +
##     OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleType + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Foundation     5 1.9015e+09 4.3134e+12 62019
## - RoofStyle      5 2.0842e+09 4.3136e+12 62019
## - SaleType       8 1.1256e+10 4.3227e+12 62019
## - Functional     6 5.8759e+09 4.3174e+12 62020
## - GarageType     5 3.4262e+09 4.3149e+12 62020
## - HeatingQC      4 6.1651e+08 4.3121e+12 62020
## - ExterCond      4 1.0466e+09 4.3125e+12 62020
## - SaleCondition  5 4.3931e+09 4.3159e+12 62021
## - Condition1     8 1.3913e+10 4.3254e+12 62021
## - HouseStyle     7 1.1853e+10 4.3233e+12 62022
## - GarageQual     4 3.3598e+09 4.3148e+12 62022
## - BsmtCond       3 4.2921e+08 4.3119e+12 62022
## - LotConfig      4 3.9224e+09 4.3154e+12 62022
## - BsmtFinType1   5 7.2431e+09 4.3187e+12 62023
## - BldgType       4 4.4878e+09 4.3160e+12 62023
## - GarageCond     4 5.3665e+09 4.3168e+12 62023
## - MSZoning       4 5.8946e+09 4.3174e+12 62024
## - Electrical     4 5.9611e+09 4.3174e+12 62024
## - GarageFinish   2 2.2180e+09 4.3137e+12 62025
## - BsmtExposure   3 5.3491e+09 4.3168e+12 62025
## - YrSold         1 8.1515e+06 4.3115e+12 62026
## - WoodDeckSF     1 1.0636e+07 4.3115e+12 62026
## - MSSubClass     1 1.8478e+07 4.3115e+12 62026
## - EnclosedPorch  1 2.1521e+07 4.3115e+12 62026
## - BsmtHalfBath   1 3.8224e+07 4.3115e+12 62026
## - BedroomAbvGr   1 6.6827e+07 4.3115e+12 62026
```

```
## - GarageCars     1 6.7766e+07 4.3115e+12 62026
## - MasVnrArea     1 7.6675e+07 4.3116e+12 62026
## - OpenPorchSF    1 1.0305e+08 4.3116e+12 62026
## - HalfBath       1 1.1628e+08 4.3116e+12 62026
## - LowQualFinSF   1 3.2471e+08 4.3118e+12 62026
## - YearRemodAdd   1 5.6448e+08 4.3120e+12 62026
## - Fireplaces     1 5.7970e+08 4.3121e+12 62026
## - X3SsnPorch     1 1.0243e+09 4.3125e+12 62026
## - BsmtFinType2   5 1.2972e+10 4.3245e+12 62027
## - LotFrontage    1 1.2137e+09 4.3127e+12 62027
## - BsmtFullBath   1 1.5407e+09 4.3130e+12 62027
## - CentralAir     1 1.7278e+09 4.3132e+12 62027
## - BsmtQual       3 7.6432e+09 4.3191e+12 62027
## - Utilities      1 1.8605e+09 4.3133e+12 62027
## - Heating        5 1.3930e+10 4.3254e+12 62027
## - BsmtFinSF2     1 2.6759e+09 4.3142e+12 62028
## - Street         1 2.7195e+09 4.3142e+12 62028
## <none>                        4.3115e+12 62028
## - ScreenPorch    1 3.0570e+09 4.3145e+12 62028
## - GarageArea     1 3.3465e+09 4.3148e+12 62028
## - FullBath       1 3.6517e+09 4.3151e+12 62028
## - LotShape       3 9.9970e+09 4.3215e+12 62028
## - GarageYrBlt    1 5.6595e+09 4.3171e+12 62030
## - OverallQual    1 8.2583e+09 4.3197e+12 62031
## - PavedDrive     2 1.1575e+10 4.3231e+12 62032
## - ExterQual      3 1.5443e+10 4.3269e+12 62032
## - KitchenQual    3 1.6369e+10 4.3278e+12 62033
## - Condition2     7 2.8275e+10 4.3398e+12 62033
## - MoSold         1 1.0474e+10 4.3220e+12 62033
## - PoolArea       1 1.2477e+10 4.3240e+12 62034
## - YearBuilt      1 1.2525e+10 4.3240e+12 62034
## - LandSlope      2 1.6910e+10 4.3284e+12 62035
## - MiscVal        1 1.4355e+10 4.3258e+12 62035
## - LandContour    3 2.2539e+10 4.3340e+12 62037
## - BsmtUnfSF      1 1.6737e+10 4.3282e+12 62037
## - BsmtFinSF1     1 1.9415e+10 4.3309e+12 62039
## - OverallCond    1 2.1980e+10 4.3335e+12 62041
## - X1stFlrSF      1 2.2391e+10 4.3339e+12 62041
## + Exterior1st   14 2.1219e+10 4.2903e+12 62041
## - KitchenAbvGr   1 2.3440e+10 4.3349e+12 62042
## - Neighborhood  24 9.5029e+10 4.4065e+12 62043
## - TotRmsAbvGrd   1 2.8173e+10 4.3397e+12 62045
## + Exterior2nd   15 1.4477e+10 4.2970e+12 62048
## - X2ndFlrSF      1 6.8172e+10 4.3797e+12 62072
## - LotArea        1 9.3611e+10 4.4051e+12 62088
## - RoofMatl       7 1.7209e+11 4.4836e+12 62128
##
## Step:  AIC=62019.03
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
```

```
##     RoofMatl + MasVnrArea + ExterQual + ExterCond + BsmtQual +
##     BsmtCond + BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageType +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleType + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - RoofStyle      5 2.1093e+09 4.3155e+12 62010
## - SaleType       8 1.1293e+10 4.3247e+12 62011
## - Functional     6 5.8749e+09 4.3193e+12 62011
## - GarageType     5 3.2270e+09 4.3166e+12 62011
## - HeatingQC      4 6.6300e+08 4.3140e+12 62011
## - ExterCond      4 1.1594e+09 4.3145e+12 62012
## - SaleCondition  5 4.6738e+09 4.3181e+12 62012
## - Condition1     8 1.4073e+10 4.3275e+12 62013
## - HouseStyle     7 1.1802e+10 4.3252e+12 62013
## - BsmtCond       3 3.7313e+08 4.3138e+12 62013
## - GarageQual     4 3.4258e+09 4.3168e+12 62013
## - LotConfig      4 3.9035e+09 4.3173e+12 62014
## - BsmtFinType1   5 7.2277e+09 4.3206e+12 62014
## - BldgType       4 4.4610e+09 4.3178e+12 62014
## - GarageCond     4 5.4664e+09 4.3188e+12 62015
## - MSZoning       4 5.7499e+09 4.3191e+12 62015
## - Electrical     4 6.2085e+09 4.3196e+12 62015
## - GarageFinish   2 2.3138e+09 4.3157e+12 62017
## - BsmtExposure   3 5.3067e+09 4.3187e+12 62017
## - YrSold         1 5.9064e+06 4.3134e+12 62017
## - WoodDeckSF     1 8.1344e+06 4.3134e+12 62017
## - MSSubClass     1 2.2213e+07 4.3134e+12 62017
## - BsmtHalfBath   1 3.7302e+07 4.3134e+12 62017
## - EnclosedPorch  1 4.0507e+07 4.3134e+12 62017
## - GarageCars     1 4.9277e+07 4.3134e+12 62017
## - BedroomAbvGr   1 5.7220e+07 4.3134e+12 62017
## - MasVnrArea     1 6.4890e+07 4.3134e+12 62017
## - OpenPorchSF    1 7.2006e+07 4.3135e+12 62017
## - HalfBath       1 1.7835e+08 4.3136e+12 62017
## - LowQualFinSF   1 3.5055e+08 4.3137e+12 62017
## - YearRemodAdd   1 5.9817e+08 4.3140e+12 62017
## - Fireplaces     1 6.1032e+08 4.3140e+12 62017
## - X3SsnPorch     1 8.9314e+08 4.3143e+12 62018
## - LotFrontage    1 1.0093e+09 4.3144e+12 62018
## - BsmtFinType2   5 1.3074e+10 4.3265e+12 62018
## - BsmtFullBath   1 1.4836e+09 4.3149e+12 62018
## - CentralAir     1 1.8164e+09 4.3152e+12 62018
## - Utilities      1 1.8390e+09 4.3152e+12 62018
## - BsmtQual       3 7.9638e+09 4.3213e+12 62018
## - Heating        5 1.4263e+10 4.3276e+12 62019
```

```
## - BsmtFinSF2     1 2.5103e+09 4.3159e+12 62019
## - Street         1 2.6570e+09 4.3160e+12 62019
## <none>                        4.3134e+12 62019
## - ScreenPorch    1 3.1293e+09 4.3165e+12 62019
## - GarageArea     1 3.3364e+09 4.3167e+12 62019
## - FullBath       1 3.6763e+09 4.3171e+12 62020
## - LotShape       3 1.0470e+10 4.3239e+12 62020
## - GarageYrBlt    1 5.3315e+09 4.3187e+12 62021
## - OverallQual    1 8.4392e+09 4.3218e+12 62023
## - PavedDrive     2 1.1868e+10 4.3252e+12 62023
## - ExterQual      3 1.5468e+10 4.3288e+12 62023
## - MoSold         1 1.0187e+10 4.3236e+12 62024
## - KitchenQual    3 1.6240e+10 4.3296e+12 62024
## - Condition2     7 2.8274e+10 4.3417e+12 62024
## - PoolArea       1 1.2192e+10 4.3256e+12 62025
## - LandSlope      2 1.6583e+10 4.3300e+12 62026
## - MiscVal        1 1.4090e+10 4.3275e+12 62027
## - YearBuilt      1 1.4458e+10 4.3278e+12 62027
## + Foundation     5 1.9015e+09 4.3115e+12 62028
## - LandContour    3 2.2355e+10 4.3357e+12 62028
## - BsmtUnfSF      1 1.9959e+10 4.3333e+12 62031
## - BsmtFinSF1     1 2.1147e+10 4.3345e+12 62031
## - OverallCond    1 2.2107e+10 4.3355e+12 62032
## - KitchenAbvGr   1 2.2980e+10 4.3364e+12 62033
## + Exterior1st   14 2.1044e+10 4.2923e+12 62033
## - X1stFlrSF      1 2.5660e+10 4.3390e+12 62034
## - Neighborhood  24 9.6602e+10 4.4100e+12 62036
## - TotRmsAbvGrd   1 2.8500e+10 4.3419e+12 62036
## + Exterior2nd   15 1.4092e+10 4.2993e+12 62039
## - X2ndFlrSF      1 6.8459e+10 4.3818e+12 62063
## - LotArea        1 9.3404e+10 4.4068e+12 62080
## - RoofMatl       7 1.7120e+11 4.4846e+12 62119
##
## Step:  AIC=62010.45
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + ExterCond + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageType +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleType + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - SaleType       8 1.1553e+10 4.3270e+12 62002
## - Functional     6 5.9219e+09 4.3214e+12 62002
```

```
## - GarageType     5 3.1087e+09 4.3186e+12 62003
## - HeatingQC      4 6.9196e+08 4.3162e+12 62003
## - ExterCond      4 1.1670e+09 4.3167e+12 62003
## - SaleCondition  5 4.8140e+09 4.3203e+12 62004
## - Condition1     8 1.4013e+10 4.3295e+12 62004
## - HouseStyle     7 1.1618e+10 4.3271e+12 62004
## - BsmtCond       3 2.8207e+08 4.3158e+12 62005
## - GarageQual     4 3.2651e+09 4.3188e+12 62005
## - LotConfig      4 4.0356e+09 4.3195e+12 62005
## - BsmtFinType1   5 7.2567e+09 4.3227e+12 62005
## - BldgType       4 4.6834e+09 4.3202e+12 62006
## - GarageCond     4 5.2607e+09 4.3207e+12 62006
## - MSZoning       4 5.8449e+09 4.3213e+12 62006
## - Electrical     4 6.0853e+09 4.3216e+12 62007
## - GarageFinish   2 2.2854e+09 4.3178e+12 62008
## - BsmtExposure   3 5.3793e+09 4.3209e+12 62008
## - YrSold         1 8.8531e+06 4.3155e+12 62008
## - WoodDeckSF     1 1.6498e+07 4.3155e+12 62008
## - MSSubClass     1 2.9110e+07 4.3155e+12 62008
## - MasVnrArea     1 2.9641e+07 4.3155e+12 62008
## - EnclosedPorch  1 3.6642e+07 4.3155e+12 62008
## - BsmtHalfBath   1 5.6845e+07 4.3155e+12 62008
## - GarageCars     1 6.0867e+07 4.3156e+12 62008
## - OpenPorchSF    1 6.2730e+07 4.3156e+12 62008
## - BedroomAbvGr   1 7.7621e+07 4.3156e+12 62009
## - HalfBath       1 2.0836e+08 4.3157e+12 62009
## - LowQualFinSF   1 3.4408e+08 4.3158e+12 62009
## - Fireplaces     1 6.0257e+08 4.3161e+12 62009
## - YearRemodAdd   1 6.1609e+08 4.3161e+12 62009
## - X3SsnPorch     1 8.9765e+08 4.3164e+12 62009
## - BsmtFinType2   5 1.3013e+10 4.3285e+12 62009
## - LotFrontage    1 1.2792e+09 4.3168e+12 62009
## - BsmtFullBath   1 1.4614e+09 4.3170e+12 62009
## - CentralAir     1 1.7544e+09 4.3172e+12 62010
## - Utilities      1 1.7643e+09 4.3173e+12 62010
## - BsmtQual       3 7.6859e+09 4.3232e+12 62010
## - BsmtFinSF2     1 2.5599e+09 4.3180e+12 62010
## - Street         1 2.6062e+09 4.3181e+12 62010
## - Heating        5 1.4631e+10 4.3301e+12 62010
## <none>                        4.3155e+12 62010
## - ScreenPorch    1 2.9776e+09 4.3185e+12 62010
## - GarageArea     1 3.3872e+09 4.3189e+12 62011
## - FullBath       1 3.6163e+09 4.3191e+12 62011
## - LotShape       3 1.0558e+10 4.3260e+12 62012
## - GarageYrBlt    1 5.1154e+09 4.3206e+12 62012
## - OverallQual    1 8.2852e+09 4.3238e+12 62014
## - PavedDrive     2 1.1638e+10 4.3271e+12 62014
## - ExterQual      3 1.5320e+10 4.3308e+12 62015
## - Condition2     7 2.7722e+10 4.3432e+12 62015
## - MoSold         1 1.0048e+10 4.3255e+12 62015
## - KitchenQual    3 1.6075e+10 4.3316e+12 62015
## - LandSlope      2 1.5009e+10 4.3305e+12 62017
```

```
## - PoolArea       1 1.2603e+10 4.3281e+12 62017
## - MiscVal        1 1.4628e+10 4.3301e+12 62018
## - YearBuilt      1 1.5143e+10 4.3306e+12 62019
## + RoofStyle      5 2.1093e+09 4.3134e+12 62019
## + Foundation     5 1.9265e+09 4.3136e+12 62019
## - LandContour    3 2.2533e+10 4.3380e+12 62020
## - BsmtUnfSF      1 1.9461e+10 4.3349e+12 62022
## - BsmtFinSF1     1 2.0696e+10 4.3362e+12 62022
## - OverallCond    1 2.2389e+10 4.3379e+12 62024
## - KitchenAbvGr   1 2.3407e+10 4.3389e+12 62024
## + Exterior1st   14 2.0878e+10 4.2946e+12 62024
## - X1stFlrSF      1 2.6290e+10 4.3418e+12 62026
## - Neighborhood  24 9.6832e+10 4.4123e+12 62027
## - TotRmsAbvGrd   1 2.7998e+10 4.3435e+12 62027
## + Exterior2nd   15 1.3858e+10 4.3016e+12 62031
## - X2ndFlrSF      1 6.9363e+10 4.3849e+12 62055
## - LotArea        1 9.1609e+10 4.4071e+12 62070
## - RoofMatl       7 1.7603e+11 4.4915e+12 62113
##
## Step:  AIC=62002.26
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + ExterCond + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageType +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Functional     6 6.0339e+09 4.3331e+12 61994
## - GarageType     5 3.1360e+09 4.3302e+12 61994
## - HeatingQC      4 6.0551e+08 4.3276e+12 61995
## - ExterCond      4 1.2513e+09 4.3283e+12 61995
## - Condition1     8 1.4174e+10 4.3412e+12 61996
## - HouseStyle     7 1.1983e+10 4.3390e+12 61996
## - BsmtCond       3 2.7241e+08 4.3273e+12 61996
## - GarageQual     4 3.2689e+09 4.3303e+12 61996
## - BsmtFinType1   5 6.9680e+09 4.3340e+12 61997
## - LotConfig      4 4.3880e+09 4.3314e+12 61997
## - BldgType       4 4.7456e+09 4.3318e+12 61997
## - GarageCond     4 5.0628e+09 4.3321e+12 61998
## - MSZoning       4 6.0533e+09 4.3331e+12 61998
## - Electrical     4 6.2249e+09 4.3333e+12 61998
## - GarageFinish   2 2.2639e+09 4.3293e+12 62000
## - BsmtExposure   3 5.3870e+09 4.3324e+12 62000
```

```
## - YrSold         1 2.5838e+05 4.3270e+12 62000
## - WoodDeckSF     1 7.0606e+06 4.3270e+12 62000
## - MasVnrArea     1 1.1435e+07 4.3271e+12 62000
## - EnclosedPorch  1 2.8707e+07 4.3271e+12 62000
## - BsmtHalfBath   1 5.5565e+07 4.3271e+12 62000
## - MSSubClass     1 6.4576e+07 4.3271e+12 62000
## - GarageCars     1 7.0739e+07 4.3271e+12 62000
## - OpenPorchSF    1 8.4424e+07 4.3271e+12 62000
## - BedroomAbvGr   1 1.0411e+08 4.3271e+12 62000
## - HalfBath       1 2.2008e+08 4.3273e+12 62000
## - LowQualFinSF   1 3.9029e+08 4.3274e+12 62001
## - YearRemodAdd   1 6.1570e+08 4.3277e+12 62001
## - Fireplaces     1 6.8424e+08 4.3277e+12 62001
## - X3SsnPorch     1 8.2721e+08 4.3279e+12 62001
## - LotFrontage    1 1.2588e+09 4.3283e+12 62001
## - Utilities      1 1.6448e+09 4.3287e+12 62001
## - BsmtFullBath   1 1.7470e+09 4.3288e+12 62001
## - CentralAir     1 1.8140e+09 4.3289e+12 62001
## - BsmtQual       3 7.9535e+09 4.3350e+12 62002
## - BsmtFinType2   5 1.4004e+10 4.3410e+12 62002
## - Street         1 2.4371e+09 4.3295e+12 62002
## - Heating        5 1.4404e+10 4.3414e+12 62002
## - ScreenPorch    1 2.6979e+09 4.3297e+12 62002
## - BsmtFinSF2     1 2.7586e+09 4.3298e+12 62002
## <none>                        4.3270e+12 62002
## - FullBath       1 3.5456e+09 4.3306e+12 62003
## - GarageArea     1 3.6944e+09 4.3307e+12 62003
## - LotShape       3 1.0863e+10 4.3379e+12 62004
## - GarageYrBlt    1 5.5461e+09 4.3326e+12 62004
## - OverallQual    1 8.5122e+09 4.3356e+12 62006
## - PavedDrive     2 1.2066e+10 4.3391e+12 62006
## - KitchenQual    3 1.5406e+10 4.3424e+12 62007
## - Condition2     7 2.7738e+10 4.3548e+12 62007
## - MoSold         1 1.0357e+10 4.3374e+12 62007
## - ExterQual      3 1.6483e+10 4.3435e+12 62007
## - SaleCondition  5 2.2540e+10 4.3496e+12 62007
## - PoolArea       1 1.2529e+10 4.3396e+12 62009
## - LandSlope      2 1.6830e+10 4.3439e+12 62010
## - MiscVal        1 1.4790e+10 4.3418e+12 62010
## + SaleType       8 1.1553e+10 4.3155e+12 62010
## + RoofStyle      5 2.3693e+09 4.3247e+12 62011
## - YearBuilt      1 1.5763e+10 4.3428e+12 62011
## + Foundation     5 1.9595e+09 4.3251e+12 62011
## - LandContour    3 2.2788e+10 4.3498e+12 62012
## - BsmtUnfSF      1 1.8765e+10 4.3458e+12 62013
## - BsmtFinSF1     1 2.0999e+10 4.3480e+12 62014
## - OverallCond    1 2.1764e+10 4.3488e+12 62015
## - KitchenAbvGr   1 2.2960e+10 4.3500e+12 62016
## + Exterior1st   14 2.0942e+10 4.3061e+12 62016
## - X1stFlrSF      1 2.5610e+10 4.3527e+12 62017
## - Neighborhood  24 9.5339e+10 4.4224e+12 62018
## - TotRmsAbvGrd   1 2.9134e+10 4.3562e+12 62020
```

```
## + Exterior2nd   15 1.4236e+10 4.3128e+12 62023
## - X2ndFlrSF      1 6.9509e+10 4.3966e+12 62047
## - LotArea        1 9.3618e+10 4.4207e+12 62063
## - RoofMatl       7 1.7677e+11 4.5038e+12 62105
##
## Step:  AIC=61994.32
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + ExterCond + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageType + GarageYrBlt +
##     GarageFinish + GarageCars + GarageArea + GarageQual + GarageCond +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - GarageType     5 3.0247e+09 4.3361e+12 61986
## - HeatingQC      4 6.2268e+08 4.3337e+12 61987
## - ExterCond      4 1.2136e+09 4.3343e+12 61987
## - Condition1     8 1.3357e+10 4.3464e+12 61987
## - BsmtCond       3 2.1245e+08 4.3333e+12 61988
## - GarageQual     4 3.4057e+09 4.3365e+12 61989
## - HouseStyle     7 1.2519e+10 4.3456e+12 61989
## - BsmtFinType1   5 7.1713e+09 4.3402e+12 61989
## - LotConfig      4 4.4395e+09 4.3375e+12 61989
## - GarageCond     4 4.9171e+09 4.3380e+12 61990
## - BldgType       4 5.4872e+09 4.3386e+12 61990
## - MSZoning       4 5.8783e+09 4.3390e+12 61990
## - Electrical     4 6.6335e+09 4.3397e+12 61991
## - GarageFinish   2 2.2981e+09 4.3354e+12 61992
## - BsmtExposure   3 5.5745e+09 4.3387e+12 61992
## - WoodDeckSF     1 9.6780e+03 4.3331e+12 61992
## - EnclosedPorch  1 9.7052e+05 4.3331e+12 61992
## - YrSold         1 4.7601e+06 4.3331e+12 61992
## - BedroomAbvGr   1 4.0238e+07 4.3331e+12 61992
## - BsmtHalfBath   1 4.8104e+07 4.3331e+12 61992
## - MasVnrArea     1 5.8336e+07 4.3331e+12 61992
## - OpenPorchSF    1 6.1768e+07 4.3331e+12 61992
## - GarageCars     1 9.5572e+07 4.3332e+12 61992
## - MSSubClass     1 1.1906e+08 4.3332e+12 61992
## - HalfBath       1 1.7078e+08 4.3332e+12 61992
## - LowQualFinSF   1 2.5693e+08 4.3333e+12 61992
## - Fireplaces     1 6.0933e+08 4.3337e+12 61993
## - X3SsnPorch     1 7.2072e+08 4.3338e+12 61993
## - YearRemodAdd   1 8.8830e+08 4.3340e+12 61993
## - LotFrontage    1 1.3419e+09 4.3344e+12 61993
## - Utilities      1 1.6330e+09 4.3347e+12 61993
```

```
## - CentralAir     1 1.6852e+09 4.3348e+12 61993
## - BsmtQual       3 7.6566e+09 4.3407e+12 61993
## - BsmtFullBath   1 1.8810e+09 4.3350e+12 61994
## - BsmtFinType2   5 1.4376e+10 4.3475e+12 61994
## - Street         1 2.5691e+09 4.3356e+12 61994
## - ScreenPorch    1 2.7905e+09 4.3359e+12 61994
## - Heating        5 1.4696e+10 4.3478e+12 61994
## <none>                        4.3331e+12 61994
## - FullBath       1 3.4075e+09 4.3365e+12 61995
## - BsmtFinSF2     1 3.4153e+09 4.3365e+12 61995
## - GarageArea     1 3.8175e+09 4.3369e+12 61995
## - GarageYrBlt    1 5.0990e+09 4.3382e+12 61996
## - LotShape       3 1.1104e+10 4.3442e+12 61996
## - PavedDrive     2 1.2066e+10 4.3451e+12 61998
## - KitchenQual    3 1.5609e+10 4.3487e+12 61999
## - ExterQual      3 1.5633e+10 4.3487e+12 61999
## - Condition2     7 2.7757e+10 4.3608e+12 61999
## - OverallQual    1 1.0135e+10 4.3432e+12 61999
## - MoSold         1 1.0529e+10 4.3436e+12 61999
## - SaleCondition  5 2.3017e+10 4.3561e+12 62000
## - PoolArea       1 1.2861e+10 4.3459e+12 62001
## - MiscVal        1 1.4509e+10 4.3476e+12 62002
## - LandSlope      2 1.7745e+10 4.3508e+12 62002
## + Functional     6 6.0339e+09 4.3270e+12 62002
## + SaleType       8 1.1665e+10 4.3214e+12 62002
## + RoofStyle      5 2.4028e+09 4.3307e+12 62003
## + Foundation     5 1.9600e+09 4.3311e+12 62003
## - YearBuilt      1 1.6005e+10 4.3491e+12 62003
## - LandContour    3 2.2788e+10 4.3559e+12 62004
## - BsmtUnfSF      1 2.1538e+10 4.3546e+12 62007
## - X1stFlrSF      1 2.2591e+10 4.3557e+12 62008
## - KitchenAbvGr   1 2.2965e+10 4.3560e+12 62008
## + Exterior1st   14 2.1356e+10 4.3117e+12 62008
## - BsmtFinSF1     1 2.3798e+10 4.3569e+12 62008
## - OverallCond    1 2.4796e+10 4.3579e+12 62009
## - TotRmsAbvGrd   1 2.8549e+10 4.3616e+12 62011
## - Neighborhood  24 9.7942e+10 4.4310e+12 62012
## + Exterior2nd   15 1.3792e+10 4.3193e+12 62015
## - X2ndFlrSF      1 6.8506e+10 4.4016e+12 62038
## - LotArea        1 9.5138e+10 4.4282e+12 62056
## - RoofMatl       7 1.7807e+11 4.5111e+12 62098
##
## Step:  AIC=61986.36
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + ExterCond + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
```

```
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - HeatingQC      4 5.6739e+08 4.3367e+12 61979
## - ExterCond      4 1.1694e+09 4.3373e+12 61979
## - Condition1     8 1.3345e+10 4.3494e+12 61979
## - HouseStyle     7 1.1655e+10 4.3478e+12 61980
## - BsmtCond       3 1.9916e+08 4.3363e+12 61980
## - GarageQual     4 3.3251e+09 4.3394e+12 61981
## - LotConfig      4 4.5215e+09 4.3406e+12 61981
## - BsmtFinType1   5 7.5815e+09 4.3437e+12 61981
## - GarageCond     4 4.6577e+09 4.3408e+12 61981
## - BldgType       4 5.0639e+09 4.3412e+12 61982
## - MSZoning       4 5.9286e+09 4.3420e+12 61982
## - Electrical     4 6.6623e+09 4.3428e+12 61983
## - GarageFinish   2 1.9184e+09 4.3380e+12 61984
## - BsmtExposure   3 5.4259e+09 4.3415e+12 61984
## - EnclosedPorch  1 2.3520e+05 4.3361e+12 61984
## - WoodDeckSF     1 1.9677e+06 4.3361e+12 61984
## - BedroomAbvGr   1 4.5515e+06 4.3361e+12 61984
## - YrSold         1 9.3301e+06 4.3361e+12 61984
## - BsmtHalfBath   1 7.2283e+07 4.3362e+12 61984
## - OpenPorchSF    1 8.1983e+07 4.3362e+12 61984
## - MasVnrArea     1 8.4272e+07 4.3362e+12 61984
## - MSSubClass     1 1.1563e+08 4.3362e+12 61984
## - GarageCars     1 1.4865e+08 4.3362e+12 61984
## - HalfBath       1 2.0411e+08 4.3363e+12 61984
## - LowQualFinSF   1 2.2012e+08 4.3363e+12 61985
## - Fireplaces     1 5.4379e+08 4.3366e+12 61985
## - X3SsnPorch     1 7.2357e+08 4.3368e+12 61985
## - YearRemodAdd   1 1.0451e+09 4.3371e+12 61985
## - LotFrontage    1 1.1733e+09 4.3373e+12 61985
## - Utilities      1 1.5151e+09 4.3376e+12 61985
## - CentralAir     1 1.6866e+09 4.3378e+12 61985
## - BsmtFullBath   1 1.9877e+09 4.3381e+12 61986
## - BsmtQual       3 7.9437e+09 4.3440e+12 61986
## - BsmtFinType2   5 1.4333e+10 4.3504e+12 61986
## - Heating        5 1.4508e+10 4.3506e+12 61986
## - Street         1 2.6041e+09 4.3387e+12 61986
## - ScreenPorch    1 2.7941e+09 4.3389e+12 61986
## <none>                        4.3361e+12 61986
## - FullBath       1 3.3224e+09 4.3394e+12 61987
## - BsmtFinSF2     1 3.4775e+09 4.3396e+12 61987
## - GarageArea     1 4.4167e+09 4.3405e+12 61987
## - GarageYrBlt    1 4.8590e+09 4.3410e+12 61988
## - LotShape       3 1.1406e+10 4.3475e+12 61988
## - KitchenQual    3 1.5412e+10 4.3515e+12 61991
## - Condition2     7 2.7412e+10 4.3635e+12 61991
## - ExterQual      3 1.5566e+10 4.3517e+12 61991
```

```
## - PavedDrive     2 1.2700e+10 4.3488e+12 61991
## - OverallQual    1 1.0068e+10 4.3462e+12 61991
## - MoSold         1 1.0606e+10 4.3467e+12 61991
## - SaleCondition  5 2.3260e+10 4.3594e+12 61992
## - PoolArea       1 1.3661e+10 4.3498e+12 61994
## - LandSlope      2 1.7459e+10 4.3536e+12 61994
## - MiscVal        1 1.4816e+10 4.3509e+12 61994
## + GarageType     5 3.0247e+09 4.3331e+12 61994
## + Functional     6 5.9226e+09 4.3302e+12 61994
## + SaleType       8 1.1703e+10 4.3244e+12 61994
## + RoofStyle      5 2.2464e+09 4.3339e+12 61995
## - YearBuilt      1 1.5618e+10 4.3517e+12 61995
## + Foundation     5 1.7815e+09 4.3343e+12 61995
## - LandContour    3 2.2183e+10 4.3583e+12 61995
## - BsmtUnfSF      1 2.2499e+10 4.3586e+12 61999
## - X1stFlrSF      1 2.2536e+10 4.3586e+12 61999
## - KitchenAbvGr   1 2.3118e+10 4.3592e+12 62000
## + Exterior1st   14 2.1137e+10 4.3150e+12 62000
## - BsmtFinSF1     1 2.4685e+10 4.3608e+12 62001
## - OverallCond    1 2.5526e+10 4.3616e+12 62001
## - TotRmsAbvGrd   1 2.8448e+10 4.3645e+12 62003
## - Neighborhood  24 9.8698e+10 4.4348e+12 62004
## + Exterior2nd   15 1.3755e+10 4.3223e+12 62007
## - X2ndFlrSF      1 7.1160e+10 4.4073e+12 62032
## - LotArea        1 9.4764e+10 4.4309e+12 62047
## - RoofMatl       7 1.7866e+11 4.5148e+12 62090
##
## Step:  AIC=61978.74
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + ExterCond + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - ExterCond      4 1.1575e+09 4.3378e+12 61972
## - Condition1     8 1.3376e+10 4.3500e+12 61972
## - HouseStyle     7 1.1413e+10 4.3481e+12 61972
## - BsmtCond       3 1.7504e+08 4.3368e+12 61973
## - GarageQual     4 3.4206e+09 4.3401e+12 61973
## - LotConfig      4 4.5317e+09 4.3412e+12 61974
## - BsmtFinType1   5 7.5737e+09 4.3442e+12 61974
## - GarageCond     4 4.6807e+09 4.3413e+12 61974
## - BldgType       4 5.0461e+09 4.3417e+12 61974
```

```
## - MSZoning       4 6.0559e+09 4.3427e+12 61975
## - Electrical     4 6.7045e+09 4.3434e+12 61975
## - GarageFinish   2 1.9427e+09 4.3386e+12 61976
## - BsmtExposure   3 5.5027e+09 4.3422e+12 61976
## - EnclosedPorch  1 8.0908e+05 4.3367e+12 61977
## - WoodDeckSF     1 2.8465e+06 4.3367e+12 61977
## - BedroomAbvGr   1 4.7235e+06 4.3367e+12 61977
## - YrSold         1 1.4059e+07 4.3367e+12 61977
## - OpenPorchSF    1 6.8428e+07 4.3367e+12 61977
## - BsmtHalfBath   1 7.3220e+07 4.3367e+12 61977
## - MasVnrArea     1 9.0662e+07 4.3368e+12 61977
## - MSSubClass     1 1.1383e+08 4.3368e+12 61977
## - GarageCars     1 1.3304e+08 4.3368e+12 61977
## - LowQualFinSF   1 2.1218e+08 4.3369e+12 61977
## - HalfBath       1 2.2029e+08 4.3369e+12 61977
## - Fireplaces     1 5.9578e+08 4.3373e+12 61977
## - X3SsnPorch     1 7.1536e+08 4.3374e+12 61977
## - YearRemodAdd   1 1.0297e+09 4.3377e+12 61977
## - LotFrontage    1 1.2213e+09 4.3379e+12 61978
## - Utilities      1 1.5648e+09 4.3382e+12 61978
## - CentralAir     1 1.7875e+09 4.3385e+12 61978
## - BsmtFullBath   1 1.9633e+09 4.3386e+12 61978
## - BsmtQual       3 7.9953e+09 4.3447e+12 61978
## - BsmtFinType2   5 1.4434e+10 4.3511e+12 61978
## - Street         1 2.6251e+09 4.3393e+12 61979
## - ScreenPorch    1 2.8330e+09 4.3395e+12 61979
## <none>                        4.3367e+12 61979
## - Heating        5 1.5076e+10 4.3517e+12 61979
## - FullBath       1 3.3615e+09 4.3400e+12 61979
## - BsmtFinSF2     1 3.4866e+09 4.3402e+12 61979
## - GarageArea     1 4.3735e+09 4.3410e+12 61980
## - GarageYrBlt    1 4.9719e+09 4.3416e+12 61980
## - LotShape       3 1.1383e+10 4.3481e+12 61980
## - Condition2     7 2.7303e+10 4.3640e+12 61983
## - KitchenQual    3 1.5398e+10 4.3521e+12 61983
## - ExterQual      3 1.5622e+10 4.3523e+12 61983
## - PavedDrive     2 1.2664e+10 4.3493e+12 61983
## - OverallQual    1 1.0143e+10 4.3468e+12 61984
## - MoSold         1 1.0602e+10 4.3473e+12 61984
## - SaleCondition  5 2.3372e+10 4.3600e+12 61984
## - PoolArea       1 1.3633e+10 4.3503e+12 61986
## + HeatingQC      4 5.6739e+08 4.3361e+12 61986
## - LandSlope      2 1.7490e+10 4.3542e+12 61986
## - MiscVal        1 1.4765e+10 4.3514e+12 61987
## + Functional     6 5.9393e+09 4.3307e+12 61987
## + GarageType     5 2.9694e+09 4.3337e+12 61987
## + SaleType       8 1.1622e+10 4.3250e+12 61987
## + RoofStyle      5 2.2601e+09 4.3344e+12 61987
## + Foundation     5 1.8261e+09 4.3348e+12 61988
## - YearBuilt      1 1.6088e+10 4.3528e+12 61988
## - LandContour    3 2.2314e+10 4.3590e+12 61988
## - X1stFlrSF      1 2.2389e+10 4.3591e+12 61992
```

```
## - BsmtUnfSF      1 2.2616e+10 4.3593e+12 61992
## - KitchenAbvGr   1 2.3339e+10 4.3600e+12 61992
## + Exterior1st   14 2.0789e+10 4.3159e+12 61993
## - BsmtFinSF1     1 2.4636e+10 4.3613e+12 61993
## - OverallCond    1 2.5750e+10 4.3624e+12 61994
## - TotRmsAbvGrd   1 2.8663e+10 4.3653e+12 61996
## - Neighborhood  24 9.9576e+10 4.4362e+12 61997
## + Exterior2nd   15 1.3208e+10 4.3235e+12 62000
## - X2ndFlrSF      1 7.1030e+10 4.4077e+12 62024
## - LotArea        1 9.4793e+10 4.4315e+12 62040
## - RoofMatl       7 1.7863e+11 4.5153e+12 62083
##
## Step:  AIC=61971.52
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition1 + Condition2 + BldgType + HouseStyle +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtCond + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + CentralAir + Electrical + X1stFlrSF + X2ndFlrSF +
##     LowQualFinSF + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageFinish + GarageCars + GarageArea +
##     GarageQual + GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Condition1     8 1.3259e+10 4.3511e+12 61964
## - HouseStyle     7 1.1266e+10 4.3491e+12 61965
## - BsmtCond       3 1.8931e+08 4.3380e+12 61966
## - GarageQual     4 3.5014e+09 4.3413e+12 61966
## - BsmtFinType1   5 7.4782e+09 4.3453e+12 61967
## - LotConfig      4 4.5287e+09 4.3424e+12 61967
## - GarageCond     4 4.5894e+09 4.3424e+12 61967
## - BldgType       4 5.0735e+09 4.3429e+12 61967
## - MSZoning       4 6.0287e+09 4.3439e+12 61968
## - Electrical     4 6.7863e+09 4.3446e+12 61968
## - GarageFinish   2 1.9388e+09 4.3398e+12 61969
## - BsmtExposure   3 5.5117e+09 4.3433e+12 61969
## - WoodDeckSF     1 2.7985e+06 4.3378e+12 61970
## - EnclosedPorch  1 3.4438e+06 4.3378e+12 61970
## - BedroomAbvGr   1 6.1702e+06 4.3378e+12 61970
## - YrSold         1 1.7063e+07 4.3378e+12 61970
## - BsmtHalfBath   1 6.0386e+07 4.3379e+12 61970
## - MasVnrArea     1 7.8622e+07 4.3379e+12 61970
## - OpenPorchSF    1 9.7563e+07 4.3379e+12 61970
## - MSSubClass     1 1.0224e+08 4.3379e+12 61970
## - GarageCars     1 1.1530e+08 4.3379e+12 61970
## - LowQualFinSF   1 2.1885e+08 4.3380e+12 61970
## - HalfBath       1 2.4397e+08 4.3381e+12 61970
## - Fireplaces     1 5.7662e+08 4.3384e+12 61970
```

```
## - X3SsnPorch     1 6.6742e+08 4.3385e+12 61970
## - YearRemodAdd   1 1.0361e+09 4.3389e+12 61970
## - LotFrontage    1 1.2291e+09 4.3391e+12 61970
## - Utilities      1 1.5661e+09 4.3394e+12 61971
## - CentralAir     1 1.7744e+09 4.3396e+12 61971
## - BsmtFullBath   1 1.8594e+09 4.3397e+12 61971
## - BsmtQual       3 8.2084e+09 4.3460e+12 61971
## - Street         1 2.6172e+09 4.3404e+12 61971
## - BsmtFinType2   5 1.4578e+10 4.3524e+12 61971
## - ScreenPorch    1 2.8815e+09 4.3407e+12 61971
## <none>                        4.3378e+12 61972
## - FullBath       1 3.1335e+09 4.3410e+12 61972
## - Heating        5 1.5078e+10 4.3529e+12 61972
## - BsmtFinSF2     1 3.5314e+09 4.3414e+12 61972
## - GarageArea     1 4.2690e+09 4.3421e+12 61972
## - GarageYrBlt    1 4.8923e+09 4.3427e+12 61973
## - LotShape       3 1.1384e+10 4.3492e+12 61973
## - KitchenQual    3 1.5063e+10 4.3529e+12 61976
## - ExterQual      3 1.5725e+10 4.3536e+12 61976
## - PavedDrive     2 1.2752e+10 4.3506e+12 61976
## - Condition2     7 2.8117e+10 4.3659e+12 61976
## - OverallQual    1 1.0281e+10 4.3481e+12 61976
## - MoSold         1 1.0752e+10 4.3486e+12 61977
## - SaleCondition  5 2.3347e+10 4.3612e+12 61977
## - PoolArea       1 1.3702e+10 4.3515e+12 61979
## + ExterCond      4 1.1575e+09 4.3367e+12 61979
## + HeatingQC      4 5.5549e+08 4.3373e+12 61979
## - LandSlope      2 1.7398e+10 4.3552e+12 61979
## - MiscVal        1 1.4680e+10 4.3525e+12 61979
## + Functional     6 5.9069e+09 4.3319e+12 61980
## + GarageType     5 2.9245e+09 4.3349e+12 61980
## + SaleType       8 1.1696e+10 4.3261e+12 61980
## + RoofStyle      5 2.2606e+09 4.3356e+12 61980
## + Foundation     5 1.9476e+09 4.3359e+12 61980
## - YearBuilt      1 1.6126e+10 4.3540e+12 61980
## - LandContour    3 2.2102e+10 4.3599e+12 61980
## - BsmtUnfSF      1 2.2648e+10 4.3605e+12 61985
## - X1stFlrSF      1 2.2999e+10 4.3608e+12 61985
## - KitchenAbvGr   1 2.3330e+10 4.3612e+12 61985
## + Exterior1st   14 2.1120e+10 4.3167e+12 61985
## - BsmtFinSF1     1 2.4473e+10 4.3623e+12 61986
## - OverallCond    1 2.6085e+10 4.3639e+12 61987
## - TotRmsAbvGrd   1 2.8603e+10 4.3664e+12 61989
## - Neighborhood  24 1.0048e+11 4.4383e+12 61990
## + Exterior2nd   15 1.3448e+10 4.3244e+12 61992
## - X2ndFlrSF      1 7.1213e+10 4.4090e+12 62017
## - LotArea        1 9.4485e+10 4.4323e+12 62032
## - RoofMatl       7 1.7855e+11 4.5164e+12 62075
##
## Step:  AIC=61964.43
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
```

```
##     Neighborhood + Condition2 + BldgType + HouseStyle + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea +
##     ExterQual + BsmtQual + BsmtCond + BsmtExposure + BsmtFinType1 +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
##     BsmtFullBath + BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BsmtCond       3 7.7376e+07 4.3512e+12 61958
## - HouseStyle     7 1.2026e+10 4.3631e+12 61958
## - GarageQual     4 3.5895e+09 4.3547e+12 61959
## - GarageCond     4 3.8907e+09 4.3550e+12 61959
## - LotConfig      4 4.3210e+09 4.3554e+12 61959
## - BsmtFinType1   5 7.7326e+09 4.3588e+12 61960
## - BldgType       4 5.3251e+09 4.3564e+12 61960
## - MSZoning       4 6.0986e+09 4.3572e+12 61961
## - Electrical     4 6.6464e+09 4.3577e+12 61961
## - GarageFinish   2 1.9800e+09 4.3531e+12 61962
## - BsmtExposure   3 5.4563e+09 4.3565e+12 61962
## - YrSold         1 9.8657e+04 4.3511e+12 61962
## - WoodDeckSF     1 1.8793e+07 4.3511e+12 61962
## - EnclosedPorch  1 2.4087e+07 4.3511e+12 61962
## - MasVnrArea     1 6.6702e+07 4.3512e+12 61962
## - BedroomAbvGr   1 8.1283e+07 4.3512e+12 61962
## - GarageCars     1 8.1371e+07 4.3512e+12 61962
## - BsmtHalfBath   1 9.3377e+07 4.3512e+12 61962
## - MSSubClass     1 9.9715e+07 4.3512e+12 61962
## - HalfBath       1 2.0902e+08 4.3513e+12 61963
## - OpenPorchSF    1 2.4310e+08 4.3513e+12 61963
## - LowQualFinSF   1 2.5145e+08 4.3513e+12 61963
## - Fireplaces     1 4.3512e+08 4.3515e+12 61963
## - X3SsnPorch     1 6.3357e+08 4.3517e+12 61963
## - YearRemodAdd   1 9.3096e+08 4.3520e+12 61963
## - LotFrontage    1 1.0993e+09 4.3522e+12 61963
## - CentralAir     1 1.4984e+09 4.3526e+12 61963
## - Utilities      1 1.5568e+09 4.3526e+12 61963
## - Street         1 1.8755e+09 4.3530e+12 61964
## - BsmtQual       3 8.0934e+09 4.3592e+12 61964
## - BsmtFinType2   5 1.4155e+10 4.3652e+12 61964
## - BsmtFullBath   1 2.3401e+09 4.3534e+12 61964
## - Heating        5 1.4717e+10 4.3658e+12 61964
## - ScreenPorch    1 2.8929e+09 4.3540e+12 61964
## <none>                        4.3511e+12 61964
## - FullBath       1 3.1172e+09 4.3542e+12 61965
## - GarageArea     1 3.9527e+09 4.3550e+12 61965
## - LotShape       3 1.0038e+10 4.3611e+12 61965
## - BsmtFinSF2     1 4.4305e+09 4.3555e+12 61965
```

```
## - GarageYrBlt    1 4.5891e+09 4.3557e+12 61966
## - KitchenQual    3 1.4920e+10 4.3660e+12 61968
## - PavedDrive     2 1.2009e+10 4.3631e+12 61968
## - ExterQual      3 1.5402e+10 4.3665e+12 61969
## - OverallQual    1 1.0091e+10 4.3612e+12 61969
## - MoSold         1 1.0238e+10 4.3613e+12 61969
## - SaleCondition  5 2.2245e+10 4.3733e+12 61969
## - PoolArea       1 1.3355e+10 4.3644e+12 61971
## - LandSlope      2 1.6484e+10 4.3676e+12 61971
## + Condition1     8 1.3259e+10 4.3378e+12 61972
## + ExterCond      4 1.0405e+09 4.3500e+12 61972
## - MiscVal        1 1.4274e+10 4.3654e+12 61972
## + HeatingQC      4 5.8790e+08 4.3505e+12 61972
## + SaleType       8 1.1859e+10 4.3392e+12 61972
## + GarageType     5 2.9175e+09 4.3482e+12 61972
## + RoofStyle      5 2.2494e+09 4.3488e+12 61973
## + Functional     6 5.0871e+09 4.3460e+12 61973
## + Foundation     5 2.0577e+09 4.3490e+12 61973
## - YearBuilt      1 1.6212e+10 4.3673e+12 61973
## - LandContour    3 2.2970e+10 4.3741e+12 61974
## - Condition2     7 4.0338e+10 4.3914e+12 61977
## - X1stFlrSF      1 2.2372e+10 4.3735e+12 61977
## - KitchenAbvGr   1 2.2911e+10 4.3740e+12 61978
## + Exterior1st   14 2.1265e+10 4.3298e+12 61978
## - BsmtUnfSF      1 2.3532e+10 4.3746e+12 61978
## - BsmtFinSF1     1 2.5499e+10 4.3766e+12 61979
## - OverallCond    1 2.6439e+10 4.3775e+12 61980
## - TotRmsAbvGrd   1 2.9772e+10 4.3809e+12 61982
## + Exterior2nd   15 1.3721e+10 4.3374e+12 61985
## - Neighborhood  24 1.0642e+11 4.4575e+12 61987
## - X2ndFlrSF      1 7.1581e+10 4.4227e+12 62010
## - LotArea        1 9.4484e+10 4.4456e+12 62025
## - RoofMatl       7 1.7526e+11 4.5263e+12 62066
##
## Step:  AIC=61958.48
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition2 + BldgType + HouseStyle + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea +
##     ExterQual + BsmtQual + BsmtExposure + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - HouseStyle     7 1.2078e+10 4.3632e+12 61953
## - GarageQual     4 3.5778e+09 4.3547e+12 61953
## - GarageCond     4 3.9661e+09 4.3551e+12 61953
```

```
## - LotConfig      4 4.3192e+09 4.3555e+12 61953
## - BsmtFinType1   5 7.7447e+09 4.3589e+12 61954
## - BldgType       4 5.2844e+09 4.3564e+12 61954
## - MSZoning       4 6.1512e+09 4.3573e+12 61955
## - Electrical     4 7.8404e+09 4.3590e+12 61956
## - GarageFinish   2 1.9994e+09 4.3532e+12 61956
## - BsmtExposure   3 5.4793e+09 4.3566e+12 61956
## - YrSold         1 1.7452e+04 4.3512e+12 61956
## - WoodDeckSF     1 2.0800e+07 4.3512e+12 61956
## - EnclosedPorch  1 2.5185e+07 4.3512e+12 61956
## - MasVnrArea     1 7.1717e+07 4.3512e+12 61957
## - BedroomAbvGr   1 7.8583e+07 4.3512e+12 61957
## - GarageCars     1 7.8680e+07 4.3512e+12 61957
## - BsmtHalfBath   1 9.6231e+07 4.3513e+12 61957
## - MSSubClass     1 1.0449e+08 4.3513e+12 61957
## - HalfBath       1 2.0428e+08 4.3514e+12 61957
## - LowQualFinSF   1 2.4794e+08 4.3514e+12 61957
## - OpenPorchSF    1 2.4804e+08 4.3514e+12 61957
## - Fireplaces     1 4.2961e+08 4.3516e+12 61957
## - X3SsnPorch     1 6.3974e+08 4.3518e+12 61957
## - YearRemodAdd   1 9.7020e+08 4.3521e+12 61957
## - LotFrontage    1 1.1096e+09 4.3523e+12 61957
## - CentralAir     1 1.4857e+09 4.3526e+12 61957
## - Utilities      1 1.5473e+09 4.3527e+12 61958
## - Street         1 1.8591e+09 4.3530e+12 61958
## - BsmtQual       3 8.0726e+09 4.3592e+12 61958
## - BsmtFinType2   5 1.4116e+10 4.3653e+12 61958
## - BsmtFullBath   1 2.3461e+09 4.3535e+12 61958
## - Heating        5 1.4731e+10 4.3659e+12 61958
## - ScreenPorch    1 2.8796e+09 4.3540e+12 61958
## <none>                        4.3512e+12 61958
## - FullBath       1 3.0974e+09 4.3543e+12 61959
## - GarageArea     1 3.9406e+09 4.3551e+12 61959
## - LotShape       3 1.0062e+10 4.3612e+12 61959
## - BsmtFinSF2     1 4.4757e+09 4.3556e+12 61959
## - GarageYrBlt    1 4.5899e+09 4.3558e+12 61960
## - PavedDrive     2 1.1970e+10 4.3631e+12 61963
## - KitchenQual    3 1.5028e+10 4.3662e+12 61963
## - ExterQual      3 1.5370e+10 4.3665e+12 61963
## - OverallQual    1 1.0068e+10 4.3612e+12 61963
## - MoSold         1 1.0230e+10 4.3614e+12 61963
## - SaleCondition  5 2.2213e+10 4.3734e+12 61963
## + BsmtCond       3 7.7376e+07 4.3511e+12 61964
## - PoolArea       1 1.3466e+10 4.3646e+12 61966
## + Condition1     8 1.3147e+10 4.3380e+12 61966
## + ExterCond      4 1.0585e+09 4.3501e+12 61966
## - LandSlope      2 1.6885e+10 4.3680e+12 61966
## - MiscVal        1 1.4257e+10 4.3654e+12 61966
## + HeatingQC      4 5.6959e+08 4.3506e+12 61966
## + SaleType       8 1.1873e+10 4.3393e+12 61967
## + GarageType     5 2.9163e+09 4.3482e+12 61967
## + RoofStyle      5 2.2087e+09 4.3490e+12 61967
```

```
## + Functional     6 5.0720e+09 4.3461e+12 61967
## + Foundation     5 2.0234e+09 4.3491e+12 61967
## - YearBuilt      1 1.6613e+10 4.3678e+12 61968
## - LandContour    3 2.3000e+10 4.3742e+12 61968
## - Condition2     7 4.0289e+10 4.3915e+12 61971
## - X1stFlrSF      1 2.2350e+10 4.3735e+12 61971
## - KitchenAbvGr   1 2.2854e+10 4.3740e+12 61972
## + Exterior1st   14 2.1237e+10 4.3299e+12 61972
## - BsmtUnfSF      1 2.3641e+10 4.3748e+12 61972
## - BsmtFinSF1     1 2.5613e+10 4.3768e+12 61974
## - OverallCond    1 2.7404e+10 4.3786e+12 61975
## - TotRmsAbvGrd   1 2.9752e+10 4.3809e+12 61976
## + Exterior2nd   15 1.3699e+10 4.3375e+12 61979
## - Neighborhood  24 1.0649e+11 4.4577e+12 61981
## - X2ndFlrSF      1 7.1575e+10 4.4227e+12 62004
## - LotArea        1 9.5010e+10 4.4462e+12 62020
## - RoofMatl       7 1.7538e+11 4.5265e+12 62060
##
## Step:  AIC=61952.57
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea + ExterQual +
##     BsmtQual + BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - GarageCond     4 3.9160e+09 4.3672e+12 61947
## - GarageQual     4 3.9239e+09 4.3672e+12 61947
## - LotConfig      4 4.2686e+09 4.3675e+12 61947
## - BsmtFinType1   5 7.9511e+09 4.3712e+12 61948
## - BldgType       4 4.9958e+09 4.3682e+12 61948
## - MSZoning       4 6.8408e+09 4.3701e+12 61949
## - Electrical     4 7.2949e+09 4.3705e+12 61949
## - GarageFinish   2 2.0614e+09 4.3653e+12 61950
## - YrSold         1 2.0878e+06 4.3632e+12 61951
## - WoodDeckSF     1 1.7667e+07 4.3633e+12 61951
## - EnclosedPorch  1 2.4860e+07 4.3633e+12 61951
## - LowQualFinSF   1 5.2071e+07 4.3633e+12 61951
## - BedroomAbvGr   1 8.4203e+07 4.3633e+12 61951
## - BsmtHalfBath   1 8.9360e+07 4.3633e+12 61951
## - GarageCars     1 1.1671e+08 4.3634e+12 61951
## - MasVnrArea     1 1.4490e+08 4.3634e+12 61951
## - BsmtExposure   3 6.1713e+09 4.3694e+12 61951
## - OpenPorchSF    1 2.3669e+08 4.3635e+12 61951
## - Fireplaces     1 4.3195e+08 4.3637e+12 61951
```

```
## - X3SsnPorch     1 6.1564e+08 4.3639e+12 61951
## - HalfBath       1 9.3423e+08 4.3642e+12 61951
## - YearRemodAdd   1 9.6300e+08 4.3642e+12 61951
## - LotFrontage    1 1.3346e+09 4.3646e+12 61951
## - CentralAir     1 1.4744e+09 4.3647e+12 61952
## - BsmtFinType2   5 1.3540e+10 4.3768e+12 61952
## - Street         1 1.6085e+09 4.3648e+12 61952
## - BsmtFullBath   1 1.9288e+09 4.3652e+12 61952
## - Utilities      1 2.1539e+09 4.3654e+12 61952
## - Heating        5 1.4338e+10 4.3776e+12 61952
## - BsmtQual       3 8.4449e+09 4.3717e+12 61952
## - ScreenPorch    1 2.9761e+09 4.3662e+12 61953
## <none>                        4.3632e+12 61953
## - MSSubClass     1 3.0470e+09 4.3663e+12 61953
## - FullBath       1 3.1492e+09 4.3664e+12 61953
## - LotShape       3 9.4484e+09 4.3727e+12 61953
## - BsmtFinSF2     1 4.0156e+09 4.3673e+12 61953
## - GarageArea     1 4.1472e+09 4.3674e+12 61953
## - GarageYrBlt    1 4.5600e+09 4.3678e+12 61954
## - OverallQual    1 8.4653e+09 4.3717e+12 61956
## - ExterQual      3 1.5099e+10 4.3783e+12 61957
## - KitchenQual    3 1.5234e+10 4.3785e+12 61957
## - SaleCondition  5 2.1523e+10 4.3848e+12 61957
## - PavedDrive     2 1.2612e+10 4.3759e+12 61957
## - MoSold         1 1.0549e+10 4.3738e+12 61958
## + HouseStyle     7 1.2078e+10 4.3512e+12 61958
## + BsmtCond       3 1.2974e+08 4.3631e+12 61958
## + Condition1     8 1.3889e+10 4.3494e+12 61959
## - LandSlope      2 1.6879e+10 4.3801e+12 61960
## + ExterCond      4 8.6762e+08 4.3624e+12 61960
## - MiscVal        1 1.4484e+10 4.3777e+12 61960
## + SaleType       8 1.2302e+10 4.3509e+12 61960
## + HeatingQC      4 3.3666e+08 4.3629e+12 61960
## - PoolArea       1 1.4948e+10 4.3782e+12 61961
## + Functional     6 5.6939e+09 4.3575e+12 61961
## + GarageType     5 2.1467e+09 4.3611e+12 61961
## + RoofStyle      5 1.9850e+09 4.3613e+12 61961
## + Foundation     5 1.8586e+09 4.3614e+12 61961
## - LandContour    3 2.4340e+10 4.3876e+12 61963
## - YearBuilt      1 1.9494e+10 4.3827e+12 61964
## - Condition2     7 3.9669e+10 4.4029e+12 61965
## - KitchenAbvGr   1 2.2225e+10 4.3855e+12 61965
## + Exterior1st   14 2.2450e+10 4.3408e+12 61966
## - BsmtUnfSF      1 2.2896e+10 4.3861e+12 61966
## - BsmtFinSF1     1 2.4162e+10 4.3874e+12 61967
## - OverallCond    1 2.7306e+10 4.3905e+12 61969
## - TotRmsAbvGrd   1 2.7662e+10 4.3909e+12 61969
## - X1stFlrSF      1 2.9269e+10 4.3925e+12 61970
## + Exterior2nd   15 1.4458e+10 4.3488e+12 61973
## - Neighborhood  24 1.1185e+11 4.4751e+12 61978
## - X2ndFlrSF      1 6.7944e+10 4.4312e+12 61996
## - LotArea        1 9.1806e+10 4.4550e+12 62011
```

```
## - RoofMatl       7 1.8099e+11 4.5442e+12 62057
##
## Step:  AIC=61947.19
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea + ExterQual +
##     BsmtQual + BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + PavedDrive + WoodDeckSF +
##     OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - GarageQual     4 2.4174e+09 4.3696e+12 61941
## - LotConfig      4 3.9732e+09 4.3711e+12 61942
## - BldgType       4 5.0387e+09 4.3722e+12 61943
## - BsmtFinType1   5 8.0910e+09 4.3752e+12 61943
## - MSZoning       4 6.7824e+09 4.3739e+12 61944
## - Electrical     4 7.1019e+09 4.3743e+12 61944
## - GarageFinish   2 1.8539e+09 4.3690e+12 61944
## - YrSold         1 2.7493e+06 4.3672e+12 61945
## - EnclosedPorch  1 1.1817e+07 4.3672e+12 61945
## - LowQualFinSF   1 2.1101e+07 4.3672e+12 61945
## - WoodDeckSF     1 3.0817e+07 4.3672e+12 61945
## - BsmtHalfBath   1 7.3607e+07 4.3672e+12 61945
## - BedroomAbvGr   1 1.0769e+08 4.3673e+12 61945
## - GarageCars     1 1.3751e+08 4.3673e+12 61945
## - MasVnrArea     1 1.3761e+08 4.3673e+12 61945
## - OpenPorchSF    1 2.2678e+08 4.3674e+12 61945
## - BsmtExposure   3 6.4128e+09 4.3736e+12 61945
## - Fireplaces     1 5.3076e+08 4.3677e+12 61946
## - X3SsnPorch     1 6.5844e+08 4.3678e+12 61946
## - HalfBath       1 9.1551e+08 4.3681e+12 61946
## - YearRemodAdd   1 1.0479e+09 4.3682e+12 61946
## - CentralAir     1 1.3729e+09 4.3685e+12 61946
## - LotFrontage    1 1.5060e+09 4.3687e+12 61946
## - Street         1 1.6029e+09 4.3688e+12 61946
## - BsmtFinType2   5 1.3785e+10 4.3809e+12 61946
## - BsmtFullBath   1 1.8514e+09 4.3690e+12 61946
## - Utilities      1 2.1572e+09 4.3693e+12 61947
## - BsmtQual       3 8.5624e+09 4.3757e+12 61947
## - Heating        5 1.4733e+10 4.3819e+12 61947
## <none>                        4.3672e+12 61947
## - MSSubClass     1 3.0670e+09 4.3702e+12 61947
## - FullBath       1 3.2499e+09 4.3704e+12 61947
## - ScreenPorch    1 3.3875e+09 4.3705e+12 61947
## - GarageYrBlt    1 3.7317e+09 4.3709e+12 61948
## - LotShape       3 9.9248e+09 4.3771e+12 61948
```

```
## - GarageArea     1 4.0219e+09 4.3712e+12 61948
## - BsmtFinSF2     1 4.0534e+09 4.3712e+12 61948
## - KitchenQual    3 1.4682e+10 4.3818e+12 61951
## - PavedDrive     2 1.2023e+10 4.3792e+12 61951
## - OverallQual    1 9.1661e+09 4.3763e+12 61951
## - ExterQual      3 1.5433e+10 4.3826e+12 61951
## - SaleCondition  5 2.1621e+10 4.3888e+12 61952
## - MoSold         1 1.0112e+10 4.3773e+12 61952
## + GarageCond     4 3.9160e+09 4.3632e+12 61953
## + BsmtCond       3 2.2290e+08 4.3669e+12 61953
## + HouseStyle     7 1.2028e+10 4.3551e+12 61953
## + Condition1     8 1.3176e+10 4.3540e+12 61954
## - LandSlope      2 1.6781e+10 4.3839e+12 61954
## + ExterCond      4 7.9356e+08 4.3664e+12 61955
## - PoolArea       1 1.4298e+10 4.3815e+12 61955
## - MiscVal        1 1.4528e+10 4.3817e+12 61955
## + HeatingQC      4 3.6522e+08 4.3668e+12 61955
## + SaleType       8 1.2209e+10 4.3549e+12 61955
## + Functional     6 5.6136e+09 4.3615e+12 61955
## + GarageType     5 2.0663e+09 4.3651e+12 61956
## + Foundation     5 1.8805e+09 4.3653e+12 61956
## + RoofStyle      5 1.7994e+09 4.3654e+12 61956
## - LandContour    3 2.4302e+10 4.3915e+12 61957
## - YearBuilt      1 1.9270e+10 4.3864e+12 61958
## - Condition2     7 3.9383e+10 4.4065e+12 61959
## - KitchenAbvGr   1 2.2560e+10 4.3897e+12 61960
## - BsmtUnfSF      1 2.2743e+10 4.3899e+12 61960
## + Exterior1st   14 2.2044e+10 4.3451e+12 61960
## - BsmtFinSF1     1 2.3979e+10 4.3911e+12 61961
## - TotRmsAbvGrd   1 2.7895e+10 4.3951e+12 61964
## - OverallCond    1 2.9076e+10 4.3962e+12 61965
## - X1stFlrSF      1 2.9795e+10 4.3970e+12 61965
## + Exterior2nd   15 1.4061e+10 4.3531e+12 61968
## - Neighborhood  24 1.1129e+11 4.4785e+12 61973
## - X2ndFlrSF      1 6.8264e+10 4.4354e+12 61990
## - LotArea        1 9.2166e+10 4.4593e+12 62006
## - RoofMatl       7 1.8877e+11 4.5559e+12 62057
##
## Step:  AIC=61940.81
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LotConfig + LandSlope +
##     Neighborhood + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea + ExterQual +
##     BsmtQual + BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
```

```
##                 Df  Sum of Sq        RSS   AIC
## - LotConfig      4 3.9296e+09 4.3735e+12 61935
## - BldgType       4 4.7228e+09 4.3743e+12 61936
## - BsmtFinType1   5 8.1295e+09 4.3777e+12 61936
## - MSZoning       4 6.9031e+09 4.3765e+12 61937
## - Electrical     4 7.8820e+09 4.3775e+12 61938
## - GarageFinish   2 1.8905e+09 4.3715e+12 61938
## - YrSold         1 8.9600e+03 4.3696e+12 61939
## - LowQualFinSF   1 1.1705e+07 4.3696e+12 61939
## - EnclosedPorch  1 1.3821e+07 4.3696e+12 61939
## - WoodDeckSF     1 2.0869e+07 4.3696e+12 61939
## - BsmtHalfBath   1 7.9579e+07 4.3697e+12 61939
## - GarageCars     1 9.9918e+07 4.3697e+12 61939
## - BedroomAbvGr   1 1.1361e+08 4.3697e+12 61939
## - MasVnrArea     1 1.1940e+08 4.3697e+12 61939
## - OpenPorchSF    1 2.3790e+08 4.3698e+12 61939
## - Fireplaces     1 4.6532e+08 4.3700e+12 61939
## - BsmtExposure   3 6.5962e+09 4.3762e+12 61939
## - X3SsnPorch     1 6.8881e+08 4.3703e+12 61939
## - HalfBath       1 8.6639e+08 4.3704e+12 61939
## - YearRemodAdd   1 1.0492e+09 4.3706e+12 61940
## - LotFrontage    1 1.4629e+09 4.3710e+12 61940
## - BsmtFinType2   5 1.3574e+10 4.3831e+12 61940
## - CentralAir     1 1.6169e+09 4.3712e+12 61940
## - Street         1 1.7118e+09 4.3713e+12 61940
## - BsmtFullBath   1 1.9235e+09 4.3715e+12 61940
## - Utilities      1 2.1983e+09 4.3718e+12 61940
## - BsmtQual       3 8.5537e+09 4.3781e+12 61941
## - Heating        5 1.4799e+10 4.3844e+12 61941
## <none>                        4.3696e+12 61941
## - MSSubClass     1 3.1393e+09 4.3727e+12 61941
## - FullBath       1 3.2911e+09 4.3729e+12 61941
## - ScreenPorch    1 3.5379e+09 4.3731e+12 61941
## - BsmtFinSF2     1 4.0243e+09 4.3736e+12 61941
## - GarageArea     1 4.1343e+09 4.3737e+12 61942
## - LotShape       3 1.0247e+10 4.3798e+12 61942
## - GarageYrBlt    1 5.6565e+09 4.3752e+12 61943
## - OverallQual    1 8.9319e+09 4.3785e+12 61945
## - KitchenQual    3 1.5196e+10 4.3848e+12 61945
## - ExterQual      3 1.5344e+10 4.3849e+12 61945
## - PavedDrive     2 1.2572e+10 4.3821e+12 61945
## - SaleCondition  5 2.1753e+10 4.3913e+12 61945
## - MoSold         1 9.9627e+09 4.3795e+12 61945
## + BsmtCond       3 1.8855e+08 4.3694e+12 61947
## + HouseStyle     7 1.2097e+10 4.3575e+12 61947
## + GarageQual     4 2.4174e+09 4.3672e+12 61947
## + GarageCond     4 2.4096e+09 4.3672e+12 61947
## + Condition1     8 1.3581e+10 4.3560e+12 61948
## - LandSlope      2 1.7020e+10 4.3866e+12 61948
## - PoolArea       1 1.4077e+10 4.3837e+12 61948
## + ExterCond      4 9.1374e+08 4.3687e+12 61948
## - MiscVal        1 1.4382e+10 4.3840e+12 61948
```

```
## + HeatingQC      4 4.5522e+08 4.3691e+12 61949
## + SaleType       8 1.2306e+10 4.3573e+12 61949
## + Functional     6 5.7670e+09 4.3638e+12 61949
## + GarageType     5 2.0938e+09 4.3675e+12 61949
## + Foundation     5 2.0899e+09 4.3675e+12 61949
## + RoofStyle      5 1.7437e+09 4.3678e+12 61950
## - LandContour    3 2.4614e+10 4.3942e+12 61951
## - YearBuilt      1 1.9251e+10 4.3888e+12 61952
## - Condition2     7 3.9418e+10 4.4090e+12 61953
## - KitchenAbvGr   1 2.2548e+10 4.3921e+12 61954
## + Exterior1st   14 2.2056e+10 4.3475e+12 61954
## - BsmtUnfSF      1 2.3524e+10 4.3931e+12 61954
## - BsmtFinSF1     1 2.4280e+10 4.3939e+12 61955
## - TotRmsAbvGrd   1 2.8418e+10 4.3980e+12 61958
## - OverallCond    1 2.8806e+10 4.3984e+12 61958
## - X1stFlrSF      1 2.8874e+10 4.3984e+12 61958
## + Exterior2nd   15 1.3884e+10 4.3557e+12 61962
## - Neighborhood  24 1.1190e+11 4.4815e+12 61967
## - X2ndFlrSF      1 6.7718e+10 4.4373e+12 61984
## - LotArea        1 9.2243e+10 4.4618e+12 62000
## - RoofMatl       7 1.9480e+11 4.5644e+12 62054
##
## Step:  AIC=61935.43
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LandSlope + Neighborhood +
##     Condition2 + BldgType + OverallQual + OverallCond + YearBuilt +
##     YearRemodAdd + RoofMatl + MasVnrArea + ExterQual + BsmtQual +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BldgType       4 4.7942e+09 4.3783e+12 61931
## - BsmtFinType1   5 7.8659e+09 4.3814e+12 61931
## - MSZoning       4 7.0768e+09 4.3806e+12 61932
## - Electrical     4 7.8228e+09 4.3813e+12 61933
## - GarageFinish   2 1.9266e+09 4.3754e+12 61933
## - YrSold         1 7.2163e+05 4.3735e+12 61933
## - EnclosedPorch  1 3.8947e+06 4.3735e+12 61933
## - LowQualFinSF   1 1.0016e+07 4.3735e+12 61933
## - WoodDeckSF     1 2.9981e+07 4.3735e+12 61933
## - GarageCars     1 5.9871e+07 4.3736e+12 61933
## - BsmtHalfBath   1 8.5808e+07 4.3736e+12 61933
## - BedroomAbvGr   1 9.9166e+07 4.3736e+12 61933
## - MasVnrArea     1 1.3424e+08 4.3736e+12 61934
## - OpenPorchSF    1 2.7509e+08 4.3738e+12 61934
## - Fireplaces     1 4.8533e+08 4.3740e+12 61934
```

```
## - X3SsnPorch     1 7.4524e+08 4.3742e+12 61934
## - LotFrontage    1 7.8323e+08 4.3743e+12 61934
## - HalfBath       1 8.0442e+08 4.3743e+12 61934
## - YearRemodAdd   1 1.0692e+09 4.3746e+12 61934
## - BsmtExposure   3 7.1873e+09 4.3807e+12 61934
## - BsmtFinType2   5 1.3385e+10 4.3869e+12 61934
## - CentralAir     1 1.6874e+09 4.3752e+12 61935
## - Utilities      1 1.7825e+09 4.3753e+12 61935
## - Street         1 1.8041e+09 4.3753e+12 61935
## - BsmtFullBath   1 2.0275e+09 4.3755e+12 61935
## - Heating        5 1.4843e+10 4.3883e+12 61935
## <none>                        4.3735e+12 61935
## - BsmtQual       3 9.0353e+09 4.3825e+12 61935
## - FullBath       1 3.1046e+09 4.3766e+12 61936
## - MSSubClass     1 3.1898e+09 4.3767e+12 61936
## - ScreenPorch    1 3.4372e+09 4.3769e+12 61936
## - LotShape       3 9.6379e+09 4.3831e+12 61936
## - BsmtFinSF2     1 3.8686e+09 4.3774e+12 61936
## - GarageArea     1 4.0710e+09 4.3776e+12 61936
## - GarageYrBlt    1 5.7581e+09 4.3793e+12 61937
## - KitchenQual    3 1.4557e+10 4.3881e+12 61939
## - OverallQual    1 8.9644e+09 4.3825e+12 61939
## - ExterQual      3 1.5462e+10 4.3890e+12 61940
## - SaleCondition  5 2.1721e+10 4.3952e+12 61940
## - PavedDrive     2 1.2710e+10 4.3862e+12 61940
## - MoSold         1 9.7540e+09 4.3833e+12 61940
## + LotConfig      4 3.9296e+09 4.3696e+12 61941
## + BsmtCond       3 1.8421e+08 4.3733e+12 61941
## + HouseStyle     7 1.2056e+10 4.3614e+12 61941
## + GarageQual     4 2.3739e+09 4.3711e+12 61942
## + GarageCond     4 2.0516e+09 4.3715e+12 61942
## + Condition1     8 1.3424e+10 4.3601e+12 61942
## - LandSlope      2 1.7004e+10 4.3905e+12 61943
## + ExterCond      4 9.0659e+08 4.3726e+12 61943
## - MiscVal        1 1.4194e+10 4.3877e+12 61943
## - PoolArea       1 1.4266e+10 4.3878e+12 61943
## + SaleType       8 1.2638e+10 4.3609e+12 61943
## + HeatingQC      4 5.0364e+08 4.3730e+12 61943
## + Functional     6 5.8691e+09 4.3676e+12 61944
## + GarageType     5 2.2501e+09 4.3713e+12 61944
## + Foundation     5 1.9833e+09 4.3715e+12 61944
## + RoofStyle      5 1.8407e+09 4.3717e+12 61944
## - LandContour    3 2.4679e+10 4.3982e+12 61946
## - YearBuilt      1 1.9752e+10 4.3933e+12 61947
## - Condition2     7 4.0118e+10 4.4136e+12 61948
## - KitchenAbvGr   1 2.2421e+10 4.3959e+12 61948
## + Exterior1st   14 2.2307e+10 4.3512e+12 61949
## - BsmtUnfSF      1 2.2688e+10 4.3962e+12 61949
## - BsmtFinSF1     1 2.3961e+10 4.3975e+12 61949
## - TotRmsAbvGrd   1 2.8213e+10 4.4017e+12 61952
## - OverallCond    1 2.9683e+10 4.4032e+12 61953
## - X1stFlrSF      1 3.0159e+10 4.4037e+12 61953
```

```
## + Exterior2nd   15 1.3801e+10 4.3597e+12 61956
## - Neighborhood  24 1.1307e+11 4.4866e+12 61962
## - X2ndFlrSF      1 6.8434e+10 4.4419e+12 61979
## - LotArea        1 9.7330e+10 4.4708e+12 61998
## - RoofMatl       7 1.9373e+11 4.5672e+12 62048
##
## Step:  AIC=61930.63
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LandSlope + Neighborhood +
##     Condition2 + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofMatl + MasVnrArea + ExterQual + BsmtQual + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + CentralAir + Electrical + X1stFlrSF + X2ndFlrSF +
##     LowQualFinSF + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageFinish + GarageCars + GarageArea +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BsmtFinType1   5 7.4179e+09 4.3857e+12 61926
## - GarageFinish   2 1.8195e+09 4.3801e+12 61928
## - Electrical     4 7.8520e+09 4.3861e+12 61928
## - MSZoning       4 8.6426e+09 4.3869e+12 61928
## - YrSold         1 7.2390e+03 4.3783e+12 61929
## - BedroomAbvGr   1 7.9132e+05 4.3783e+12 61929
## - EnclosedPorch  1 1.2323e+07 4.3783e+12 61929
## - LowQualFinSF   1 3.5818e+07 4.3783e+12 61929
## - WoodDeckSF     1 4.8743e+07 4.3783e+12 61929
## - BsmtHalfBath   1 6.4073e+07 4.3784e+12 61929
## - GarageCars     1 1.1159e+08 4.3784e+12 61929
## - MasVnrArea     1 1.2073e+08 4.3784e+12 61929
## - Fireplaces     1 3.0290e+08 4.3786e+12 61929
## - OpenPorchSF    1 3.9257e+08 4.3787e+12 61929
## - HalfBath       1 6.3668e+08 4.3789e+12 61929
## - X3SsnPorch     1 7.5257e+08 4.3791e+12 61929
## - YearRemodAdd   1 1.1988e+09 4.3795e+12 61929
## - BsmtFinType2   5 1.3230e+10 4.3915e+12 61929
## - LotFrontage    1 1.3995e+09 4.3797e+12 61930
## - Street         1 1.4880e+09 4.3798e+12 61930
## - BsmtFullBath   1 1.6361e+09 4.3799e+12 61930
## - Utilities      1 1.6370e+09 4.3799e+12 61930
## - BsmtExposure   3 8.2149e+09 4.3865e+12 61930
## - CentralAir     1 2.2142e+09 4.3805e+12 61930
## - Heating        5 1.4722e+10 4.3930e+12 61930
## <none>                        4.3783e+12 61931
## - BsmtQual       3 9.2028e+09 4.3875e+12 61931
## - ScreenPorch    1 3.2799e+09 4.3816e+12 61931
## - FullBath       1 3.4306e+09 4.3817e+12 61931
## - LotShape       3 9.7476e+09 4.3880e+12 61931
## - BsmtFinSF2     1 3.8531e+09 4.3822e+12 61931
## - GarageArea     1 4.3487e+09 4.3826e+12 61932
```

```
## - GarageYrBlt    1 5.4191e+09 4.3837e+12 61932
## - MSSubClass     1 6.8514e+09 4.3851e+12 61933
## - KitchenQual    3 1.4446e+10 4.3927e+12 61934
## - OverallQual    1 8.7117e+09 4.3870e+12 61934
## - PavedDrive     2 1.2463e+10 4.3908e+12 61935
## - ExterQual      3 1.5506e+10 4.3938e+12 61935
## - MoSold         1 9.6370e+09 4.3879e+12 61935
## + BldgType       4 4.7942e+09 4.3735e+12 61935
## + LotConfig      4 4.0010e+09 4.3743e+12 61936
## - SaleCondition  5 2.3331e+10 4.4016e+12 61936
## + BsmtCond       3 9.7630e+07 4.3782e+12 61937
## + HouseStyle     7 1.1767e+10 4.3665e+12 61937
## + GarageQual     4 2.0392e+09 4.3763e+12 61937
## + GarageCond     4 1.9748e+09 4.3763e+12 61937
## + Condition1     8 1.3582e+10 4.3647e+12 61938
## - PoolArea       1 1.3948e+10 4.3922e+12 61938
## + ExterCond      4 9.3477e+08 4.3774e+12 61938
## - LandSlope      2 1.7124e+10 4.3954e+12 61938
## + SaleType       8 1.2773e+10 4.3655e+12 61938
## + HeatingQC      4 5.6155e+08 4.3777e+12 61938
## - MiscVal        1 1.4482e+10 4.3928e+12 61938
## + Functional     6 6.3428e+09 4.3720e+12 61938
## + RoofStyle      5 2.0793e+09 4.3762e+12 61939
## + Foundation     5 1.8919e+09 4.3764e+12 61939
## + GarageType     5 1.8701e+09 4.3764e+12 61939
## - LandContour    3 2.4286e+10 4.4026e+12 61941
## - YearBuilt      1 2.1218e+10 4.3995e+12 61943
## - Condition2     7 4.0221e+10 4.4185e+12 61943
## + Exterior1st   14 2.2190e+10 4.3561e+12 61944
## - KitchenAbvGr   1 2.3490e+10 4.4018e+12 61944
## - BsmtUnfSF      1 2.3823e+10 4.4021e+12 61944
## - BsmtFinSF1     1 2.5925e+10 4.4042e+12 61946
## - TotRmsAbvGrd   1 2.9292e+10 4.4076e+12 61948
## - X1stFlrSF      1 2.9803e+10 4.4081e+12 61948
## - OverallCond    1 3.1227e+10 4.4095e+12 61949
## + Exterior2nd   15 1.3825e+10 4.3645e+12 61951
## - Neighborhood  24 1.1238e+11 4.4907e+12 61957
## - X2ndFlrSF      1 6.9096e+10 4.4474e+12 61974
## - LotArea        1 1.0186e+11 4.4802e+12 61996
## - RoofMatl       7 1.9792e+11 4.5762e+12 62046
##
## Step:  AIC=61925.57
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LandSlope + Neighborhood +
##     Condition2 + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofMatl + MasVnrArea + ExterQual + BsmtQual + BsmtExposure +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
##     BsmtFullBath + BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
```

```
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Electrical     4 7.5398e+09 4.3933e+12 61923
## - GarageFinish   2 1.8296e+09 4.3875e+12 61923
## - MSZoning       4 8.6505e+09 4.3944e+12 61923
## - YrSold         1 8.0699e+06 4.3857e+12 61924
## - BsmtHalfBath   1 1.1080e+07 4.3857e+12 61924
## - BedroomAbvGr   1 1.1122e+07 4.3857e+12 61924
## - EnclosedPorch  1 1.2801e+07 4.3857e+12 61924
## - LowQualFinSF   1 3.2328e+07 4.3857e+12 61924
## - WoodDeckSF     1 1.2250e+08 4.3858e+12 61924
## - GarageCars     1 1.4683e+08 4.3859e+12 61924
## - Fireplaces     1 1.5982e+08 4.3859e+12 61924
## - MasVnrArea     1 1.8593e+08 4.3859e+12 61924
## - OpenPorchSF    1 3.6382e+08 4.3861e+12 61924
## - BsmtFullBath   1 7.0262e+08 4.3864e+12 61924
## - X3SsnPorch     1 7.8712e+08 4.3865e+12 61924
## - HalfBath       1 8.0923e+08 4.3865e+12 61924
## - LotFrontage    1 1.1500e+09 4.3869e+12 61924
## - YearRemodAdd   1 1.2142e+09 4.3869e+12 61924
## - Utilities      1 1.3143e+09 4.3870e+12 61924
## - Street         1 1.6190e+09 4.3873e+12 61925
## - CentralAir     1 2.0781e+09 4.3878e+12 61925
## - BsmtQual       3 8.4941e+09 4.3942e+12 61925
## - BsmtFinType2   5 1.4810e+10 4.4005e+12 61925
## - Heating        5 1.4978e+10 4.4007e+12 61926
## <none>                        4.3857e+12 61926
## - BsmtExposure   3 9.0835e+09 4.3948e+12 61926
## - ScreenPorch    1 3.1604e+09 4.3889e+12 61926
## - FullBath       1 3.5946e+09 4.3893e+12 61926
## - BsmtFinSF2     1 3.6879e+09 4.3894e+12 61926
## - LotShape       3 9.7850e+09 4.3955e+12 61926
## - GarageArea     1 4.4391e+09 4.3902e+12 61927
## - GarageYrBlt    1 5.0911e+09 4.3908e+12 61927
## - MSSubClass     1 6.5245e+09 4.3922e+12 61928
## - KitchenQual    3 1.4490e+10 4.4002e+12 61929
## - PavedDrive     2 1.1936e+10 4.3977e+12 61930
## - ExterQual      3 1.5639e+10 4.4014e+12 61930
## - MoSold         1 9.6507e+09 4.3954e+12 61930
## - SaleCondition  5 2.1827e+10 4.4075e+12 61930
## - OverallQual    1 1.0108e+10 4.3958e+12 61930
## + BsmtFinType1   5 7.4179e+09 4.3783e+12 61931
## + BldgType       4 4.3462e+09 4.3814e+12 61931
## + LotConfig      4 3.7320e+09 4.3820e+12 61931
## + BsmtCond       3 1.3152e+08 4.3856e+12 61931
## + HouseStyle     7 1.2059e+10 4.3737e+12 61932
## + GarageQual     4 2.1062e+09 4.3836e+12 61932
## + GarageCond     4 2.0462e+09 4.3837e+12 61932
## + Condition1     8 1.3830e+10 4.3719e+12 61932
## - PoolArea       1 1.3486e+10 4.3992e+12 61933
## - LandSlope      2 1.6899e+10 4.4026e+12 61933
```

```
## + ExterCond      4 8.5321e+08 4.3849e+12 61933
## + HeatingQC      4 5.2407e+08 4.3852e+12 61933
## + Functional     6 6.3785e+09 4.3793e+12 61933
## + SaleType       8 1.2373e+10 4.3733e+12 61933
## - MiscVal        1 1.5340e+10 4.4011e+12 61934
## + RoofStyle      5 2.1134e+09 4.3836e+12 61934
## + GarageType     5 1.9306e+09 4.3838e+12 61934
## + Foundation     5 1.8041e+09 4.3839e+12 61934
## - LandContour    3 2.3317e+10 4.4090e+12 61935
## - BsmtUnfSF      1 2.1118e+10 4.4068e+12 61938
## - YearBuilt      1 2.2346e+10 4.4081e+12 61938
## + Exterior1st   14 2.1928e+10 4.3638e+12 61939
## - Condition2     7 4.1350e+10 4.4271e+12 61939
## - KitchenAbvGr   1 2.4493e+10 4.4102e+12 61940
## - TotRmsAbvGrd   1 2.8836e+10 4.4146e+12 61943
## - X1stFlrSF      1 2.9736e+10 4.4155e+12 61943
## - OverallCond    1 3.1429e+10 4.4171e+12 61944
## + Exterior2nd   15 1.3225e+10 4.3725e+12 61947
## - BsmtFinSF1     1 3.7866e+10 4.4236e+12 61949
## - Neighborhood  24 1.1652e+11 4.5022e+12 61954
## - X2ndFlrSF      1 6.8709e+10 4.4544e+12 61969
## - LotArea        1 1.0135e+11 4.4871e+12 61990
## - RoofMatl       7 2.0383e+11 4.5895e+12 62044
##
## Step:  AIC=61922.59
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LandSlope + Neighborhood +
##     Condition2 + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofMatl + MasVnrArea + ExterQual + BsmtQual + BsmtExposure +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     CentralAir + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - GarageFinish   2 1.8410e+09 4.3951e+12 61920
## - MSZoning       4 8.5355e+09 4.4018e+12 61920
## - EnclosedPorch  1 2.7680e+06 4.3933e+12 61921
## - YrSold         1 8.2940e+06 4.3933e+12 61921
## - LowQualFinSF   1 1.7636e+07 4.3933e+12 61921
## - BsmtHalfBath   1 1.9435e+07 4.3933e+12 61921
## - BedroomAbvGr   1 3.9785e+07 4.3933e+12 61921
## - GarageCars     1 1.1282e+08 4.3934e+12 61921
## - WoodDeckSF     1 1.2097e+08 4.3934e+12 61921
## - Fireplaces     1 1.9799e+08 4.3935e+12 61921
## - MasVnrArea     1 2.0505e+08 4.3935e+12 61921
## - OpenPorchSF    1 3.5207e+08 4.3936e+12 61921
## - BsmtFullBath   1 7.0886e+08 4.3940e+12 61921
## - X3SsnPorch     1 7.7547e+08 4.3940e+12 61921
```

```
## - HalfBath       1 8.7287e+08 4.3941e+12 61921
## - YearRemodAdd   1 1.1384e+09 4.3944e+12 61921
## - LotFrontage    1 1.2141e+09 4.3945e+12 61921
## - Utilities      1 1.4065e+09 4.3947e+12 61922
## - Street         1 1.6172e+09 4.3949e+12 61922
## - BsmtQual       3 8.0072e+09 4.4013e+12 61922
## - CentralAir     1 2.3018e+09 4.3956e+12 61922
## - BsmtFinType2   5 1.4598e+10 4.4079e+12 61922
## <none>                        4.3933e+12 61923
## - BsmtExposure   3 9.1240e+09 4.4024e+12 61923
## - ScreenPorch    1 3.2247e+09 4.3965e+12 61923
## - BsmtFinSF2     1 3.6062e+09 4.3969e+12 61923
## - Heating        5 1.5737e+10 4.4090e+12 61923
## - FullBath       1 3.6746e+09 4.3969e+12 61923
## - LotShape       3 9.7899e+09 4.4030e+12 61923
## - GarageArea     1 4.3031e+09 4.3976e+12 61923
## - GarageYrBlt    1 4.3696e+09 4.3976e+12 61923
## - MSSubClass     1 6.1932e+09 4.3994e+12 61925
## + Electrical     4 7.5398e+09 4.3857e+12 61926
## - KitchenQual    3 1.4257e+10 4.4075e+12 61926
## - PavedDrive     2 1.1443e+10 4.4047e+12 61926
## - MoSold         1 9.1277e+09 4.4024e+12 61927
## - OverallQual    1 9.6947e+09 4.4029e+12 61927
## - ExterQual      3 1.5887e+10 4.4091e+12 61927
## - SaleCondition  5 2.2259e+10 4.4155e+12 61927
## + BsmtCond       3 1.4265e+09 4.3918e+12 61928
## + BldgType       4 4.3743e+09 4.3889e+12 61928
## + BsmtFinType1   5 7.1057e+09 4.3861e+12 61928
## + LotConfig      4 3.7320e+09 4.3895e+12 61928
## + GarageQual     4 2.9739e+09 4.3903e+12 61929
## + HouseStyle     7 1.1718e+10 4.3815e+12 61929
## + Condition1     8 1.4079e+10 4.3792e+12 61929
## + GarageCond     4 1.9340e+09 4.3913e+12 61929
## - PoolArea       1 1.3577e+10 4.4068e+12 61930
## - LandSlope      2 1.6777e+10 4.4100e+12 61930
## + Functional     6 7.1415e+09 4.3861e+12 61930
## + ExterCond      4 9.2593e+08 4.3923e+12 61930
## + SaleType       8 1.2896e+10 4.3804e+12 61930
## + HeatingQC      4 6.9314e+08 4.3926e+12 61930
## - MiscVal        1 1.5289e+10 4.4085e+12 61931
## + RoofStyle      5 2.0607e+09 4.3912e+12 61931
## + GarageType     5 1.8995e+09 4.3914e+12 61931
## + Foundation     5 1.8659e+09 4.3914e+12 61931
## - LandContour    3 2.3528e+10 4.4168e+12 61932
## - BsmtUnfSF      1 2.0885e+10 4.4141e+12 61934
## + Exterior1st   14 2.2407e+10 4.3708e+12 61936
## - YearBuilt      1 2.3006e+10 4.4163e+12 61936
## - Condition2     7 4.1320e+10 4.4346e+12 61936
## - KitchenAbvGr   1 2.4508e+10 4.4178e+12 61937
## - TotRmsAbvGrd   1 2.8255e+10 4.4215e+12 61939
## - X1stFlrSF      1 2.9923e+10 4.4232e+12 61940
## + Exterior2nd   15 1.3581e+10 4.3797e+12 61944
```

```
## - OverallCond    1 3.4789e+10 4.4280e+12 61944
## - BsmtFinSF1     1 3.7964e+10 4.4312e+12 61946
## - Neighborhood  24 1.1597e+11 4.5092e+12 61951
## - X2ndFlrSF      1 6.8986e+10 4.4622e+12 61966
## - LotArea        1 1.0137e+11 4.4946e+12 61987
## - RoofMatl       7 2.0469e+11 4.5979e+12 62042
##
## Step:  AIC=61919.81
## SalePrice ~ MSSubClass + MSZoning + LotFrontage + LotArea + Street +
##     LotShape + LandContour + Utilities + LandSlope + Neighborhood +
##     Condition2 + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofMatl + MasVnrArea + ExterQual + BsmtQual + BsmtExposure +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     CentralAir + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageCars +
##     GarageArea + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - MSZoning       4 8.0714e+09 4.4032e+12 61917
## - EnclosedPorch  1 6.2516e+06 4.3951e+12 61918
## - YrSold         1 1.1776e+07 4.3951e+12 61918
## - BsmtHalfBath   1 1.2788e+07 4.3951e+12 61918
## - LowQualFinSF   1 2.3987e+07 4.3951e+12 61918
## - BedroomAbvGr   1 3.7329e+07 4.3951e+12 61918
## - GarageCars     1 8.4456e+07 4.3952e+12 61918
## - WoodDeckSF     1 1.0302e+08 4.3952e+12 61918
## - MasVnrArea     1 2.3028e+08 4.3953e+12 61918
## - Fireplaces     1 2.4762e+08 4.3953e+12 61918
## - OpenPorchSF    1 3.4560e+08 4.3954e+12 61918
## - X3SsnPorch     1 6.8840e+08 4.3958e+12 61918
## - BsmtFullBath   1 7.0680e+08 4.3958e+12 61918
## - HalfBath       1 8.0372e+08 4.3959e+12 61918
## - YearRemodAdd   1 1.1179e+09 4.3962e+12 61919
## - LotFrontage    1 1.2486e+09 4.3963e+12 61919
## - Utilities      1 1.4586e+09 4.3966e+12 61919
## - Street         1 1.6679e+09 4.3968e+12 61919
## - CentralAir     1 2.3254e+09 4.3974e+12 61919
## - BsmtFinType2   5 1.4601e+10 4.4097e+12 61919
## - BsmtQual       3 8.6116e+09 4.4037e+12 61920
## <none>                        4.3951e+12 61920
## - ScreenPorch    1 3.2563e+09 4.3984e+12 61920
## - BsmtExposure   3 9.2973e+09 4.4044e+12 61920
## - FullBath       1 3.6627e+09 4.3988e+12 61920
## - BsmtFinSF2     1 3.7093e+09 4.3988e+12 61920
## - GarageYrBlt    1 3.7104e+09 4.3988e+12 61920
## - GarageArea     1 3.8758e+09 4.3990e+12 61920
## - LotShape       3 9.9342e+09 4.4050e+12 61920
## - Heating        5 1.6047e+10 4.4111e+12 61920
## - MSSubClass     1 6.4204e+09 4.4015e+12 61922
```

```
## + GarageFinish   2 1.8410e+09 4.3933e+12 61923
## + Electrical     4 7.5512e+09 4.3875e+12 61923
## - KitchenQual    3 1.4803e+10 4.4099e+12 61924
## - PavedDrive     2 1.1793e+10 4.4069e+12 61924
## - MoSold         1 8.9052e+09 4.4040e+12 61924
## - ExterQual      3 1.6053e+10 4.4111e+12 61924
## - OverallQual    1 1.0063e+10 4.4052e+12 61924
## + BsmtCond       3 1.4551e+09 4.3936e+12 61925
## - SaleCondition  5 2.2704e+10 4.4178e+12 61925
## + BldgType       4 4.2820e+09 4.3908e+12 61925
## + BsmtFinType1   5 7.1122e+09 4.3880e+12 61925
## + LotConfig      4 3.7081e+09 4.3914e+12 61925
## + GarageQual     4 2.9963e+09 4.3921e+12 61926
## + HouseStyle     7 1.1810e+10 4.3833e+12 61926
## + Condition1     8 1.4044e+10 4.3811e+12 61926
## + GarageCond     4 1.8955e+09 4.3932e+12 61927
## - PoolArea       1 1.3625e+10 4.4087e+12 61927
## - LandSlope      2 1.6824e+10 4.4119e+12 61927
## + Functional     6 7.1746e+09 4.3879e+12 61927
## + ExterCond      4 9.4079e+08 4.3942e+12 61927
## + SaleType       8 1.2876e+10 4.3822e+12 61927
## + HeatingQC      4 6.6126e+08 4.3944e+12 61927
## - MiscVal        1 1.5524e+10 4.4106e+12 61928
## + RoofStyle      5 2.0618e+09 4.3930e+12 61928
## + GarageType     5 1.9566e+09 4.3931e+12 61929
## + Foundation     5 1.9165e+09 4.3932e+12 61929
## - LandContour    3 2.3922e+10 4.4190e+12 61930
## - BsmtUnfSF      1 2.0355e+10 4.4155e+12 61931
## + Exterior1st   14 2.2603e+10 4.3725e+12 61933
## - Condition2     7 4.1507e+10 4.4366e+12 61933
## - YearBuilt      1 2.3874e+10 4.4190e+12 61934
## - KitchenAbvGr   1 2.4569e+10 4.4197e+12 61934
## - TotRmsAbvGrd   1 2.8426e+10 4.4235e+12 61937
## - X1stFlrSF      1 3.0576e+10 4.4257e+12 61938
## - OverallCond    1 3.4605e+10 4.4297e+12 61941
## + Exterior2nd   15 1.3605e+10 4.3815e+12 61941
## - BsmtFinSF1     1 3.7946e+10 4.4330e+12 61943
## - Neighborhood  24 1.1624e+11 4.5113e+12 61948
## - X2ndFlrSF      1 6.9897e+10 4.4650e+12 61964
## - LotArea        1 1.0145e+11 4.4965e+12 61984
## - RoofMatl       7 2.0490e+11 4.6000e+12 62039
##
## Step:  AIC=61917.16
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageCars + GarageArea +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
```

```
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - YrSold         1 1.6526e+06 4.4032e+12 61915
## - BsmtHalfBath   1 3.5521e+06 4.4032e+12 61915
## - LowQualFinSF   1 2.0985e+07 4.4032e+12 61915
## - EnclosedPorch  1 2.2790e+07 4.4032e+12 61915
## - WoodDeckSF     1 6.6375e+07 4.4032e+12 61915
## - BedroomAbvGr   1 7.6594e+07 4.4032e+12 61915
## - GarageCars     1 1.5487e+08 4.4033e+12 61915
## - MasVnrArea     1 1.5854e+08 4.4033e+12 61915
## - Fireplaces     1 3.8083e+08 4.4035e+12 61915
## - HalfBath       1 3.9100e+08 4.4036e+12 61915
## - OpenPorchSF    1 4.0151e+08 4.4036e+12 61915
## - BsmtFullBath   1 5.9940e+08 4.4038e+12 61916
## - X3SsnPorch     1 6.7601e+08 4.4038e+12 61916
## - YearRemodAdd   1 1.2919e+09 4.4045e+12 61916
## - Utilities      1 1.4877e+09 4.4047e+12 61916
## - LotFrontage    1 1.8365e+09 4.4050e+12 61916
## - CentralAir     1 1.9783e+09 4.4051e+12 61916
## - BsmtQual       3 8.0562e+09 4.4112e+12 61917
## - Street         1 2.0524e+09 4.4052e+12 61917
## - BsmtFinType2   5 1.4781e+10 4.4179e+12 61917
## <none>                        4.4032e+12 61917
## - LotShape       3 9.1268e+09 4.4123e+12 61917
## - Heating        5 1.5423e+10 4.4186e+12 61917
## - ScreenPorch    1 3.3896e+09 4.4066e+12 61917
## - GarageYrBlt    1 3.5252e+09 4.4067e+12 61918
## - BsmtExposure   3 9.6453e+09 4.4128e+12 61918
## - BsmtFinSF2     1 3.7540e+09 4.4069e+12 61918
## - GarageArea     1 4.3019e+09 4.4075e+12 61918
## - FullBath       1 4.3373e+09 4.4075e+12 61918
## + MSZoning       4 8.0714e+09 4.3951e+12 61920
## + Electrical     4 7.4209e+09 4.3957e+12 61920
## + GarageFinish   2 1.3769e+09 4.4018e+12 61920
## - MoSold         1 8.3326e+09 4.4115e+12 61921
## - MSSubClass     1 8.8819e+09 4.4120e+12 61921
## - PavedDrive     2 1.1941e+10 4.4151e+12 61921
## - KitchenQual    3 1.5217e+10 4.4184e+12 61921
## + BldgType       4 5.6717e+09 4.3975e+12 61921
## - OverallQual    1 9.7803e+09 4.4129e+12 61922
## - ExterQual      3 1.6194e+10 4.4194e+12 61922
## + BsmtCond       3 1.5089e+09 4.4017e+12 61922
## + BsmtFinType1   5 7.1529e+09 4.3960e+12 61922
## - SaleCondition  5 2.3079e+10 4.4262e+12 61922
## + LotConfig      4 4.0926e+09 4.3991e+12 61922
## + GarageQual     4 3.0452e+09 4.4001e+12 61923
## + HouseStyle     7 1.1654e+10 4.3915e+12 61923
## + Condition1     8 1.4136e+10 4.3890e+12 61924
## + GarageCond     4 1.7964e+09 4.4014e+12 61924
## - PoolArea       1 1.3634e+10 4.4168e+12 61924
## + SaleType       8 1.3234e+10 4.3899e+12 61924
```

```
## - LandSlope      2 1.7046e+10 4.4202e+12 61924
## + Functional     6 7.0176e+09 4.3961e+12 61925
## + ExterCond      4 9.7471e+08 4.4022e+12 61925
## + HeatingQC      4 7.8492e+08 4.4024e+12 61925
## - MiscVal        1 1.5300e+10 4.4185e+12 61925
## + RoofStyle      5 2.2577e+09 4.4009e+12 61926
## + GarageType     5 2.0070e+09 4.4012e+12 61926
## + Foundation     5 1.6012e+09 4.4016e+12 61926
## - LandContour    3 2.4701e+10 4.4279e+12 61927
## - BsmtUnfSF      1 2.0949e+10 4.4241e+12 61929
## + Exterior1st   14 2.3387e+10 4.3798e+12 61930
## - YearBuilt      1 2.2363e+10 4.4255e+12 61930
## - Condition2     7 4.1273e+10 4.4444e+12 61930
## - KitchenAbvGr   1 2.6042e+10 4.4292e+12 61932
## - TotRmsAbvGrd   1 2.8840e+10 4.4320e+12 61934
## - X1stFlrSF      1 2.9535e+10 4.4327e+12 61935
## + Exterior2nd   15 1.4640e+10 4.3885e+12 61937
## - OverallCond    1 3.4797e+10 4.4380e+12 61938
## - BsmtFinSF1     1 3.7701e+10 4.4409e+12 61940
## - Neighborhood  24 1.1807e+11 4.5212e+12 61946
## - X2ndFlrSF      1 6.7700e+10 4.4709e+12 61960
## - LotArea        1 1.0463e+11 4.5078e+12 61984
## - RoofMatl       7 2.0414e+11 4.6073e+12 62035
##
## Step:  AIC=61915.17
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageCars + GarageArea +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BsmtHalfBath   1 3.5265e+06 4.4032e+12 61913
## - LowQualFinSF   1 2.0984e+07 4.4032e+12 61913
## - EnclosedPorch  1 2.2769e+07 4.4032e+12 61913
## - WoodDeckSF     1 6.6433e+07 4.4032e+12 61913
## - BedroomAbvGr   1 7.6769e+07 4.4032e+12 61913
## - GarageCars     1 1.5566e+08 4.4033e+12 61913
## - MasVnrArea     1 1.5795e+08 4.4033e+12 61913
## - Fireplaces     1 3.8154e+08 4.4036e+12 61913
## - HalfBath       1 3.8991e+08 4.4036e+12 61913
## - OpenPorchSF    1 4.0413e+08 4.4036e+12 61913
## - BsmtFullBath   1 5.9787e+08 4.4038e+12 61914
## - X3SsnPorch     1 6.7816e+08 4.4038e+12 61914
## - YearRemodAdd   1 1.2912e+09 4.4045e+12 61914
## - Utilities      1 1.4865e+09 4.4047e+12 61914
## - LotFrontage    1 1.8373e+09 4.4050e+12 61914
```

```
## - CentralAir     1 1.9786e+09 4.4051e+12 61914
## - BsmtQual       3 8.0554e+09 4.4112e+12 61915
## - Street         1 2.0511e+09 4.4052e+12 61915
## - BsmtFinType2   5 1.4781e+10 4.4180e+12 61915
## <none>                        4.4032e+12 61915
## - LotShape       3 9.1259e+09 4.4123e+12 61915
## - Heating        5 1.5425e+10 4.4186e+12 61915
## - ScreenPorch    1 3.3898e+09 4.4066e+12 61915
## - GarageYrBlt    1 3.5250e+09 4.4067e+12 61916
## - BsmtExposure   3 9.6478e+09 4.4128e+12 61916
## - BsmtFinSF2     1 3.7524e+09 4.4069e+12 61916
## - GarageArea     1 4.3074e+09 4.4075e+12 61916
## - FullBath       1 4.3422e+09 4.4075e+12 61916
## + YrSold         1 1.6526e+06 4.4032e+12 61917
## + MSZoning       4 8.0613e+09 4.3951e+12 61918
## + Electrical     4 7.4217e+09 4.3957e+12 61918
## + GarageFinish   2 1.3780e+09 4.4018e+12 61918
## - MoSold         1 8.4710e+09 4.4116e+12 61919
## - PavedDrive     2 1.1939e+10 4.4151e+12 61919
## - MSSubClass     1 8.9255e+09 4.4121e+12 61919
## - KitchenQual    3 1.5219e+10 4.4184e+12 61919
## + BldgType       4 5.6667e+09 4.3975e+12 61919
## - OverallQual    1 9.7787e+09 4.4129e+12 61920
## - ExterQual      3 1.6285e+10 4.4195e+12 61920
## + BsmtCond       3 1.5083e+09 4.4017e+12 61920
## + BsmtFinType1   5 7.1522e+09 4.3960e+12 61920
## + LotConfig      4 4.0935e+09 4.3991e+12 61920
## - SaleCondition  5 2.3523e+10 4.4267e+12 61921
## + GarageQual     4 3.0413e+09 4.4001e+12 61921
## + HouseStyle     7 1.1655e+10 4.3915e+12 61921
## + Condition1     8 1.4100e+10 4.3891e+12 61922
## + GarageCond     4 1.7980e+09 4.4014e+12 61922
## - PoolArea       1 1.3661e+10 4.4168e+12 61922
## + SaleType       8 1.3229e+10 4.3899e+12 61922
## - LandSlope      2 1.7048e+10 4.4202e+12 61922
## + Functional     6 7.0189e+09 4.3962e+12 61923
## + ExterCond      4 9.7590e+08 4.4022e+12 61923
## + HeatingQC      4 7.8652e+08 4.4024e+12 61923
## - MiscVal        1 1.5300e+10 4.4185e+12 61923
## + RoofStyle      5 2.2547e+09 4.4009e+12 61924
## + GarageType     5 2.0087e+09 4.4012e+12 61924
## + Foundation     5 1.6014e+09 4.4016e+12 61924
## - LandContour    3 2.4706e+10 4.4279e+12 61925
## - BsmtUnfSF      1 2.0947e+10 4.4241e+12 61927
## + Exterior1st   14 2.3368e+10 4.3798e+12 61928
## - YearBuilt      1 2.2366e+10 4.4255e+12 61928
## - Condition2     7 4.1272e+10 4.4444e+12 61928
## - KitchenAbvGr   1 2.6099e+10 4.4293e+12 61930
## - TotRmsAbvGrd   1 2.8845e+10 4.4320e+12 61932
## - X1stFlrSF      1 2.9535e+10 4.4327e+12 61933
## + Exterior2nd   15 1.4628e+10 4.3885e+12 61935
## - OverallCond    1 3.4795e+10 4.4380e+12 61936
```

```
## - BsmtFinSF1     1 3.7700e+10 4.4409e+12 61938
## - Neighborhood  24 1.1807e+11 4.5212e+12 61944
## - X2ndFlrSF      1 6.7700e+10 4.4709e+12 61958
## - LotArea        1 1.0465e+11 4.5078e+12 61982
## - RoofMatl       7 2.0438e+11 4.6076e+12 62034
##
## Step:  AIC=61913.17
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + FullBath +
##     HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageCars + GarageArea + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - LowQualFinSF   1 2.0884e+07 4.4032e+12 61911
## - EnclosedPorch  1 2.2601e+07 4.4032e+12 61911
## - WoodDeckSF     1 6.7616e+07 4.4032e+12 61911
## - BedroomAbvGr   1 7.4983e+07 4.4032e+12 61911
## - GarageCars     1 1.5548e+08 4.4033e+12 61911
## - MasVnrArea     1 1.5706e+08 4.4033e+12 61911
## - Fireplaces     1 3.8012e+08 4.4036e+12 61911
## - HalfBath       1 3.8747e+08 4.4036e+12 61911
## - OpenPorchSF    1 4.0405e+08 4.4036e+12 61911
## - BsmtFullBath   1 6.3123e+08 4.4038e+12 61912
## - X3SsnPorch     1 6.7642e+08 4.4038e+12 61912
## - YearRemodAdd   1 1.2954e+09 4.4045e+12 61912
## - Utilities      1 1.5052e+09 4.4047e+12 61912
## - LotFrontage    1 1.8455e+09 4.4050e+12 61912
## - CentralAir     1 1.9833e+09 4.4052e+12 61912
## - BsmtQual       3 8.0524e+09 4.4112e+12 61913
## - Street         1 2.0508e+09 4.4052e+12 61913
## - BsmtFinType2   5 1.4787e+10 4.4180e+12 61913
## <none>                        4.4032e+12 61913
## - LotShape       3 9.1232e+09 4.4123e+12 61913
## - Heating        5 1.5423e+10 4.4186e+12 61913
## - ScreenPorch    1 3.3903e+09 4.4066e+12 61913
## - GarageYrBlt    1 3.5244e+09 4.4067e+12 61914
## - BsmtExposure   3 9.6573e+09 4.4128e+12 61914
## - BsmtFinSF2     1 3.7500e+09 4.4069e+12 61914
## - GarageArea     1 4.3046e+09 4.4075e+12 61914
## - FullBath       1 4.3521e+09 4.4075e+12 61914
## + BsmtHalfBath   1 3.5265e+06 4.4032e+12 61915
## + YrSold         1 1.6270e+06 4.4032e+12 61915
## + MSZoning       4 8.0522e+09 4.3951e+12 61916
## + Electrical     4 7.4241e+09 4.3957e+12 61916
## + GarageFinish   2 1.3752e+09 4.4018e+12 61916
## - MoSold         1 8.4681e+09 4.4116e+12 61917
```

```
## - PavedDrive     2 1.1941e+10 4.4151e+12 61917
## - MSSubClass     1 8.9929e+09 4.4122e+12 61917
## - KitchenQual    3 1.5231e+10 4.4184e+12 61917
## + BldgType       4 5.6604e+09 4.3975e+12 61917
## - OverallQual    1 9.7807e+09 4.4130e+12 61918
## - ExterQual      3 1.6284e+10 4.4195e+12 61918
## + BsmtCond       3 1.5077e+09 4.4017e+12 61918
## + BsmtFinType1   5 7.1089e+09 4.3961e+12 61918
## + LotConfig      4 4.0940e+09 4.3991e+12 61918
## - SaleCondition  5 2.3534e+10 4.4267e+12 61919
## + GarageQual     4 3.0429e+09 4.4001e+12 61919
## + HouseStyle     7 1.1651e+10 4.3915e+12 61919
## + Condition1     8 1.4103e+10 4.3891e+12 61920
## + GarageCond     4 1.7917e+09 4.4014e+12 61920
## - PoolArea       1 1.3694e+10 4.4169e+12 61920
## + SaleType       8 1.3232e+10 4.3899e+12 61920
## - LandSlope      2 1.7080e+10 4.4203e+12 61920
## + Functional     6 7.0157e+09 4.3962e+12 61921
## + ExterCond      4 9.7305e+08 4.4022e+12 61921
## + HeatingQC      4 7.8834e+08 4.4024e+12 61921
## - MiscVal        1 1.5306e+10 4.4185e+12 61921
## + RoofStyle      5 2.2578e+09 4.4009e+12 61922
## + GarageType     5 2.0067e+09 4.4012e+12 61922
## + Foundation     5 1.6020e+09 4.4016e+12 61922
## - LandContour    3 2.4703e+10 4.4279e+12 61923
## - BsmtUnfSF      1 2.0945e+10 4.4241e+12 61925
## + Exterior1st   14 2.3368e+10 4.3798e+12 61926
## - YearBuilt      1 2.2367e+10 4.4255e+12 61926
## - Condition2     7 4.1272e+10 4.4444e+12 61926
## - KitchenAbvGr   1 2.6111e+10 4.4293e+12 61928
## - TotRmsAbvGrd   1 2.8852e+10 4.4320e+12 61930
## - X1stFlrSF      1 2.9635e+10 4.4328e+12 61931
## + Exterior2nd   15 1.4624e+10 4.3885e+12 61933
## - OverallCond    1 3.4813e+10 4.4380e+12 61934
## - BsmtFinSF1     1 3.8528e+10 4.4417e+12 61937
## - Neighborhood  24 1.1821e+11 4.5214e+12 61943
## - X2ndFlrSF      1 6.7811e+10 4.4710e+12 61956
## - LotArea        1 1.0468e+11 4.5079e+12 61980
## - RoofMatl       7 2.0469e+11 4.6079e+12 62032
##
## Step:  AIC=61911.18
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + BsmtFullBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageCars + GarageArea + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + SaleCondition
##
```

```
##                 Df  Sum of Sq        RSS   AIC
## - EnclosedPorch  1 2.3581e+07 4.4032e+12 61909
## - WoodDeckSF     1 6.8023e+07 4.4033e+12 61909
## - BedroomAbvGr   1 7.5594e+07 4.4033e+12 61909
## - GarageCars     1 1.5363e+08 4.4033e+12 61909
## - MasVnrArea     1 1.5901e+08 4.4034e+12 61909
## - Fireplaces     1 3.7708e+08 4.4036e+12 61909
## - HalfBath       1 3.9156e+08 4.4036e+12 61909
## - OpenPorchSF    1 4.0326e+08 4.4036e+12 61909
## - BsmtFullBath   1 6.3333e+08 4.4038e+12 61910
## - X3SsnPorch     1 6.7372e+08 4.4039e+12 61910
## - YearRemodAdd   1 1.2978e+09 4.4045e+12 61910
## - Utilities      1 1.5041e+09 4.4047e+12 61910
## - LotFrontage    1 1.8421e+09 4.4050e+12 61910
## - CentralAir     1 2.0044e+09 4.4052e+12 61911
## - BsmtQual       3 8.0669e+09 4.4113e+12 61911
## - Street         1 2.0380e+09 4.4052e+12 61911
## - BsmtFinType2   5 1.4772e+10 4.4180e+12 61911
## <none>                        4.4032e+12 61911
## - LotShape       3 9.1135e+09 4.4123e+12 61911
## - Heating        5 1.5430e+10 4.4186e+12 61911
## - ScreenPorch    1 3.3839e+09 4.4066e+12 61911
## - GarageYrBlt    1 3.5806e+09 4.4068e+12 61912
## - BsmtExposure   3 9.6902e+09 4.4129e+12 61912
## - BsmtFinSF2     1 3.7466e+09 4.4069e+12 61912
## - GarageArea     1 4.2976e+09 4.4075e+12 61912
## - FullBath       1 4.3320e+09 4.4075e+12 61912
## + LowQualFinSF   1 2.0884e+07 4.4032e+12 61913
## + GrLivArea      1 2.0884e+07 4.4032e+12 61913
## + BsmtHalfBath   1 3.4266e+06 4.4032e+12 61913
## + YrSold         1 1.6267e+06 4.4032e+12 61913
## + MSZoning       4 8.0493e+09 4.3951e+12 61914
## + Electrical     4 7.4095e+09 4.3958e+12 61914
## + GarageFinish   2 1.3805e+09 4.4018e+12 61914
## - MoSold         1 8.4573e+09 4.4117e+12 61915
## - PavedDrive     2 1.2033e+10 4.4152e+12 61915
## - MSSubClass     1 9.0981e+09 4.4123e+12 61915
## - KitchenQual    3 1.5213e+10 4.4184e+12 61915
## + BldgType       4 5.6784e+09 4.3975e+12 61915
## - OverallQual    1 9.7911e+09 4.4130e+12 61916
## - ExterQual      3 1.6268e+10 4.4195e+12 61916
## + BsmtCond       3 1.5046e+09 4.4017e+12 61916
## + BsmtFinType1   5 7.1073e+09 4.3961e+12 61916
## + LotConfig      4 4.0912e+09 4.3991e+12 61916
## - SaleCondition  5 2.3585e+10 4.4268e+12 61917
## + GarageQual     4 3.0247e+09 4.4002e+12 61917
## + HouseStyle     7 1.1339e+10 4.3919e+12 61918
## + Condition1     8 1.4108e+10 4.3891e+12 61918
## + GarageCond     4 1.8013e+09 4.4014e+12 61918
## - PoolArea       1 1.3673e+10 4.4169e+12 61918
## + SaleType       8 1.3219e+10 4.3900e+12 61918
## - LandSlope      2 1.7074e+10 4.4203e+12 61918
```

```
## + Functional     6 7.0338e+09 4.3962e+12 61919
## + ExterCond      4 9.7096e+08 4.4022e+12 61919
## + HeatingQC      4 7.8226e+08 4.4024e+12 61919
## - MiscVal        1 1.5299e+10 4.4185e+12 61919
## + RoofStyle      5 2.2607e+09 4.4009e+12 61920
## + GarageType     5 1.9980e+09 4.4012e+12 61920
## + Foundation     5 1.6016e+09 4.4016e+12 61920
## - LandContour    3 2.4773e+10 4.4280e+12 61922
## - BsmtUnfSF      1 2.0928e+10 4.4241e+12 61923
## + Exterior1st   14 2.3389e+10 4.3798e+12 61924
## - YearBuilt      1 2.2760e+10 4.4260e+12 61924
## - Condition2     7 4.1288e+10 4.4445e+12 61924
## - KitchenAbvGr   1 2.6216e+10 4.4294e+12 61927
## - TotRmsAbvGrd   1 2.9389e+10 4.4326e+12 61929
## - X1stFlrSF      1 3.0019e+10 4.4332e+12 61929
## + Exterior2nd   15 1.4638e+10 4.3886e+12 61931
## - OverallCond    1 3.4929e+10 4.4381e+12 61932
## - BsmtFinSF1     1 3.8532e+10 4.4417e+12 61935
## - Neighborhood  24 1.1820e+11 4.5214e+12 61941
## - X2ndFlrSF      1 6.9527e+10 4.4727e+12 61955
## - LotArea        1 1.0466e+11 4.5079e+12 61978
## - RoofMatl       7 2.0514e+11 4.6083e+12 62030
##
## Step:  AIC=61909.2
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + BsmtFullBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageCars + GarageArea + PavedDrive +
##     WoodDeckSF + OpenPorchSF + X3SsnPorch + ScreenPorch + PoolArea +
##     MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - WoodDeckSF     1 6.1553e+07 4.4033e+12 61907
## - BedroomAbvGr   1 7.5012e+07 4.4033e+12 61907
## - GarageCars     1 1.5217e+08 4.4034e+12 61907
## - MasVnrArea     1 1.6303e+08 4.4034e+12 61907
## - Fireplaces     1 3.7076e+08 4.4036e+12 61907
## - HalfBath       1 3.9158e+08 4.4036e+12 61907
## - OpenPorchSF    1 3.9580e+08 4.4036e+12 61907
## - BsmtFullBath   1 6.3585e+08 4.4039e+12 61908
## - X3SsnPorch     1 6.8234e+08 4.4039e+12 61908
## - YearRemodAdd   1 1.2903e+09 4.4045e+12 61908
## - Utilities      1 1.5013e+09 4.4047e+12 61908
## - LotFrontage    1 1.8328e+09 4.4051e+12 61908
## - CentralAir     1 2.0083e+09 4.4052e+12 61909
## - Street         1 2.0364e+09 4.4053e+12 61909
## - BsmtQual       3 8.0787e+09 4.4113e+12 61909
## - BsmtFinType2   5 1.4765e+10 4.4180e+12 61909
```

```
## <none>                        4.4032e+12 61909
## - LotShape       3 9.1243e+09 4.4123e+12 61909
## - Heating        5 1.5426e+10 4.4186e+12 61909
## - ScreenPorch    1 3.4950e+09 4.4067e+12 61910
## - GarageYrBlt    1 3.5716e+09 4.4068e+12 61910
## - BsmtExposure   3 9.6787e+09 4.4129e+12 61910
## - BsmtFinSF2     1 3.7405e+09 4.4070e+12 61910
## - GarageArea     1 4.2829e+09 4.4075e+12 61910
## - FullBath       1 4.3303e+09 4.4075e+12 61910
## + EnclosedPorch  1 2.3581e+07 4.4032e+12 61911
## + LowQualFinSF   1 2.1864e+07 4.4032e+12 61911
## + GrLivArea      1 2.1864e+07 4.4032e+12 61911
## + BsmtHalfBath   1 3.2561e+06 4.4032e+12 61911
## + YrSold         1 1.6054e+06 4.4032e+12 61911
## + MSZoning       4 8.0662e+09 4.3952e+12 61912
## + GarageFinish   2 1.3870e+09 4.4018e+12 61912
## + Electrical     4 7.3875e+09 4.3958e+12 61912
## - MoSold         1 8.4689e+09 4.4117e+12 61913
## - PavedDrive     2 1.2034e+10 4.4153e+12 61913
## - MSSubClass     1 9.1038e+09 4.4123e+12 61913
## - KitchenQual    3 1.5198e+10 4.4184e+12 61913
## + BldgType       4 5.6963e+09 4.3975e+12 61913
## - OverallQual    1 9.7679e+09 4.4130e+12 61914
## - ExterQual      3 1.6258e+10 4.4195e+12 61914
## + BsmtCond       3 1.5018e+09 4.4017e+12 61914
## + BsmtFinType1   5 7.1087e+09 4.3961e+12 61914
## + LotConfig      4 4.0736e+09 4.3991e+12 61914
## - SaleCondition  5 2.3563e+10 4.4268e+12 61915
## + GarageQual     4 3.0208e+09 4.4002e+12 61915
## + HouseStyle     7 1.1338e+10 4.3919e+12 61916
## + Condition1     8 1.4129e+10 4.3891e+12 61916
## + GarageCond     4 1.7872e+09 4.4014e+12 61916
## - PoolArea       1 1.3693e+10 4.4169e+12 61916
## + SaleType       8 1.3226e+10 4.3900e+12 61916
## - LandSlope      2 1.7051e+10 4.4203e+12 61916
## + Functional     6 7.0572e+09 4.3962e+12 61917
## + ExterCond      4 9.7970e+08 4.4022e+12 61917
## + HeatingQC      4 7.8799e+08 4.4024e+12 61917
## - MiscVal        1 1.5309e+10 4.4185e+12 61917
## + RoofStyle      5 2.2639e+09 4.4010e+12 61918
## + GarageType     5 2.0073e+09 4.4012e+12 61918
## + Foundation     5 1.5757e+09 4.4016e+12 61918
## - LandContour    3 2.4750e+10 4.4280e+12 61920
## - BsmtUnfSF      1 2.0911e+10 4.4241e+12 61921
## + Exterior1st   14 2.3336e+10 4.3799e+12 61922
## - Condition2     7 4.1270e+10 4.4445e+12 61922
## - YearBuilt      1 2.3688e+10 4.4269e+12 61923
## - KitchenAbvGr   1 2.6194e+10 4.4294e+12 61925
## - TotRmsAbvGrd   1 2.9448e+10 4.4327e+12 61927
## - X1stFlrSF      1 3.0028e+10 4.4332e+12 61927
## + Exterior2nd   15 1.4642e+10 4.3886e+12 61929
## - OverallCond    1 3.5196e+10 4.4384e+12 61930
```

```
## - BsmtFinSF1     1 3.8516e+10 4.4417e+12 61933
## - Neighborhood  24 1.1823e+11 4.5214e+12 61939
## - X2ndFlrSF      1 6.9509e+10 4.4727e+12 61953
## - LotArea        1 1.0464e+11 4.5079e+12 61976
## - RoofMatl       7 2.0512e+11 4.6083e+12 62028
##
## Step:  AIC=61907.24
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + BsmtFullBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageCars + GarageArea + PavedDrive +
##     OpenPorchSF + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BedroomAbvGr   1 7.7632e+07 4.4034e+12 61905
## - GarageCars     1 1.5272e+08 4.4034e+12 61905
## - MasVnrArea     1 1.6449e+08 4.4034e+12 61905
## - Fireplaces     1 3.5087e+08 4.4036e+12 61905
## - OpenPorchSF    1 3.7123e+08 4.4037e+12 61905
## - HalfBath       1 3.9608e+08 4.4037e+12 61906
## - BsmtFullBath   1 6.5139e+08 4.4039e+12 61906
## - X3SsnPorch     1 6.9358e+08 4.4040e+12 61906
## - YearRemodAdd   1 1.2998e+09 4.4046e+12 61906
## - Utilities      1 1.4977e+09 4.4048e+12 61906
## - LotFrontage    1 1.8473e+09 4.4051e+12 61906
## - CentralAir     1 1.9957e+09 4.4053e+12 61907
## - BsmtQual       3 8.0678e+09 4.4113e+12 61907
## - Street         1 2.0499e+09 4.4053e+12 61907
## - BsmtFinType2   5 1.4736e+10 4.4180e+12 61907
## <none>                        4.4033e+12 61907
## - LotShape       3 9.1844e+09 4.4125e+12 61907
## - Heating        5 1.5394e+10 4.4187e+12 61907
## - BsmtExposure   3 9.6172e+09 4.4129e+12 61908
## - GarageYrBlt    1 3.6387e+09 4.4069e+12 61908
## - ScreenPorch    1 3.6457e+09 4.4069e+12 61908
## - BsmtFinSF2     1 3.7149e+09 4.4070e+12 61908
## - GarageArea     1 4.2731e+09 4.4076e+12 61908
## - FullBath       1 4.3061e+09 4.4076e+12 61908
## + WoodDeckSF     1 6.1553e+07 4.4032e+12 61909
## + LowQualFinSF   1 2.2116e+07 4.4033e+12 61909
## + GrLivArea      1 2.2116e+07 4.4033e+12 61909
## + EnclosedPorch  1 1.7111e+07 4.4033e+12 61909
## + BsmtHalfBath   1 4.3710e+06 4.4033e+12 61909
## + YrSold         1 1.6604e+06 4.4033e+12 61909
## + MSZoning       4 8.0251e+09 4.3953e+12 61910
## + GarageFinish   2 1.3738e+09 4.4019e+12 61910
## + Electrical     4 7.3889e+09 4.3959e+12 61910
```

```
## - MoSold         1 8.4350e+09 4.4117e+12 61911
## - PavedDrive     2 1.2028e+10 4.4153e+12 61911
## - MSSubClass     1 9.0670e+09 4.4123e+12 61911
## - KitchenQual    3 1.5199e+10 4.4185e+12 61911
## + BldgType       4 5.7065e+09 4.3976e+12 61911
## - OverallQual    1 9.8150e+09 4.4131e+12 61912
## - ExterQual      3 1.6233e+10 4.4195e+12 61912
## + BsmtCond       3 1.5051e+09 4.4018e+12 61912
## + BsmtFinType1   5 7.1529e+09 4.3961e+12 61912
## + LotConfig      4 4.0789e+09 4.3992e+12 61913
## - SaleCondition  5 2.3796e+10 4.4271e+12 61913
## + GarageQual     4 3.0077e+09 4.4003e+12 61913
## + HouseStyle     7 1.1347e+10 4.3919e+12 61914
## + Condition1     8 1.4148e+10 4.3891e+12 61914
## + GarageCond     4 1.8003e+09 4.4015e+12 61914
## - PoolArea       1 1.3632e+10 4.4169e+12 61914
## + SaleType       8 1.3248e+10 4.3900e+12 61914
## - LandSlope      2 1.7030e+10 4.4203e+12 61915
## + Functional     6 7.0759e+09 4.3962e+12 61915
## + ExterCond      4 9.7382e+08 4.4023e+12 61915
## + HeatingQC      4 7.8777e+08 4.4025e+12 61915
## - MiscVal        1 1.5470e+10 4.4187e+12 61915
## + RoofStyle      5 2.2428e+09 4.4010e+12 61916
## + GarageType     5 2.0280e+09 4.4013e+12 61916
## + Foundation     5 1.5774e+09 4.4017e+12 61916
## - LandContour    3 2.4724e+10 4.4280e+12 61918
## - BsmtUnfSF      1 2.0876e+10 4.4242e+12 61919
## + Exterior1st   14 2.3397e+10 4.3799e+12 61920
## - Condition2     7 4.1256e+10 4.4445e+12 61920
## - YearBuilt      1 2.3735e+10 4.4270e+12 61921
## - KitchenAbvGr   1 2.6134e+10 4.4294e+12 61923
## - TotRmsAbvGrd   1 2.9411e+10 4.4327e+12 61925
## - X1stFlrSF      1 2.9966e+10 4.4332e+12 61925
## + Exterior2nd   15 1.4666e+10 4.3886e+12 61927
## - OverallCond    1 3.5203e+10 4.4385e+12 61928
## - BsmtFinSF1     1 3.8457e+10 4.4417e+12 61931
## - Neighborhood  24 1.1820e+11 4.5215e+12 61937
## - X2ndFlrSF      1 6.9501e+10 4.4728e+12 61951
## - LotArea        1 1.0458e+11 4.5079e+12 61974
## - RoofMatl       7 2.0546e+11 4.6087e+12 62026
##
## Step:  AIC=61905.29
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + BsmtFullBath + FullBath + HalfBath +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageCars + GarageArea + PavedDrive + OpenPorchSF +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     SaleCondition
```

```
##
##                 Df  Sum of Sq        RSS   AIC
## - GarageCars     1 1.5552e+08 4.4035e+12 61903
## - MasVnrArea     1 1.5980e+08 4.4035e+12 61903
## - Fireplaces     1 3.1951e+08 4.4037e+12 61904
## - OpenPorchSF    1 3.8942e+08 4.4037e+12 61904
## - HalfBath       1 4.0092e+08 4.4038e+12 61904
## - BsmtFullBath   1 6.6508e+08 4.4040e+12 61904
## - X3SsnPorch     1 6.7665e+08 4.4040e+12 61904
## - YearRemodAdd   1 1.3346e+09 4.4047e+12 61904
## - Utilities      1 1.4941e+09 4.4049e+12 61904
## - LotFrontage    1 1.8945e+09 4.4053e+12 61905
## - CentralAir     1 1.9600e+09 4.4053e+12 61905
## - BsmtQual       3 8.0662e+09 4.4114e+12 61905
## - Street         1 2.0452e+09 4.4054e+12 61905
## - BsmtFinType2   5 1.4751e+10 4.4181e+12 61905
## <none>                        4.4034e+12 61905
## - LotShape       3 9.1702e+09 4.4125e+12 61905
## - Heating        5 1.5413e+10 4.4188e+12 61905
## - BsmtExposure   3 9.6147e+09 4.4130e+12 61906
## - ScreenPorch    1 3.6340e+09 4.4070e+12 61906
## - GarageYrBlt    1 3.6546e+09 4.4070e+12 61906
## - BsmtFinSF2     1 3.7590e+09 4.4071e+12 61906
## - GarageArea     1 4.2367e+09 4.4076e+12 61906
## - FullBath       1 4.5548e+09 4.4079e+12 61906
## + BedroomAbvGr   1 7.7632e+07 4.4033e+12 61907
## + WoodDeckSF     1 6.4173e+07 4.4033e+12 61907
## + LowQualFinSF   1 2.2745e+07 4.4033e+12 61907
## + GrLivArea      1 2.2745e+07 4.4033e+12 61907
## + EnclosedPorch  1 1.6488e+07 4.4033e+12 61907
## + BsmtHalfBath   1 2.3362e+06 4.4034e+12 61907
## + YrSold         1 1.8471e+06 4.4034e+12 61907
## + MSZoning       4 8.0649e+09 4.3953e+12 61908
## + Electrical     4 7.4317e+09 4.3959e+12 61908
## + GarageFinish   2 1.3716e+09 4.4020e+12 61908
## - MoSold         1 8.5800e+09 4.4119e+12 61909
## - PavedDrive     2 1.1997e+10 4.4154e+12 61909
## - KitchenQual    3 1.5133e+10 4.4185e+12 61909
## - MSSubClass     1 9.1247e+09 4.4125e+12 61909
## + BldgType       4 5.7671e+09 4.3976e+12 61909
## - OverallQual    1 9.7734e+09 4.4131e+12 61910
## - ExterQual      3 1.6184e+10 4.4195e+12 61910
## + BsmtCond       3 1.5189e+09 4.4018e+12 61910
## + BsmtFinType1   5 7.1949e+09 4.3962e+12 61911
## + LotConfig      4 4.0905e+09 4.3993e+12 61911
## - SaleCondition  5 2.3725e+10 4.4271e+12 61911
## + GarageQual     4 2.9881e+09 4.4004e+12 61911
## + HouseStyle     7 1.1333e+10 4.3920e+12 61912
## + Condition1     8 1.4011e+10 4.3893e+12 61912
## + GarageCond     4 1.7940e+09 4.4016e+12 61912
## - PoolArea       1 1.3613e+10 4.4170e+12 61912
## + SaleType       8 1.3212e+10 4.3901e+12 61913
```

```
## + Functional     6 7.1349e+09 4.3962e+12 61913
## + ExterCond      4 9.6940e+08 4.4024e+12 61913
## - LandSlope      2 1.7197e+10 4.4206e+12 61913
## + HeatingQC      4 7.9604e+08 4.4026e+12 61913
## - MiscVal        1 1.5691e+10 4.4190e+12 61914
## + RoofStyle      5 2.2157e+09 4.4011e+12 61914
## + GarageType     5 2.0806e+09 4.4013e+12 61914
## + Foundation     5 1.5811e+09 4.4018e+12 61914
## - LandContour    3 2.4713e+10 4.4281e+12 61916
## - BsmtUnfSF      1 2.1627e+10 4.4250e+12 61918
## + Exterior1st   14 2.3390e+10 4.3800e+12 61918
## - Condition2     7 4.1495e+10 4.4449e+12 61919
## - YearBuilt      1 2.3954e+10 4.4273e+12 61919
## - KitchenAbvGr   1 2.6200e+10 4.4296e+12 61921
## - X1stFlrSF      1 3.0000e+10 4.4334e+12 61923
## + Exterior2nd   15 1.4720e+10 4.3886e+12 61926
## - OverallCond    1 3.5420e+10 4.4388e+12 61927
## - TotRmsAbvGrd   1 3.7161e+10 4.4405e+12 61928
## - BsmtFinSF1     1 3.8877e+10 4.4422e+12 61929
## - Neighborhood  24 1.1813e+11 4.5215e+12 61935
## - X2ndFlrSF      1 7.2686e+10 4.4760e+12 61951
## - LotArea        1 1.0484e+11 4.5082e+12 61972
## - RoofMatl       7 2.0589e+11 4.6092e+12 62025
##
## Step:  AIC=61903.39
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + BsmtFullBath + FullBath + HalfBath +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageArea + PavedDrive + OpenPorchSF + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - MasVnrArea     1 1.5014e+08 4.4037e+12 61901
## - Fireplaces     1 2.9674e+08 4.4038e+12 61902
## - OpenPorchSF    1 3.6669e+08 4.4039e+12 61902
## - HalfBath       1 4.3171e+08 4.4039e+12 61902
## - X3SsnPorch     1 6.7775e+08 4.4042e+12 61902
## - BsmtFullBath   1 6.8263e+08 4.4042e+12 61902
## - YearRemodAdd   1 1.3503e+09 4.4049e+12 61902
## - Utilities      1 1.5039e+09 4.4050e+12 61902
## - LotFrontage    1 1.8859e+09 4.4054e+12 61903
## - BsmtQual       3 7.9981e+09 4.4115e+12 61903
## - CentralAir     1 1.9857e+09 4.4055e+12 61903
## - Street         1 2.0426e+09 4.4056e+12 61903
## - BsmtFinType2   5 1.4773e+10 4.4183e+12 61903
## <none>                        4.4035e+12 61903
## - LotShape       3 9.1232e+09 4.4126e+12 61903
## - Heating        5 1.5427e+10 4.4189e+12 61904
```

```
## - GarageYrBlt    1 3.5311e+09 4.4070e+12 61904
## - BsmtExposure   3 9.6166e+09 4.4131e+12 61904
## - ScreenPorch    1 3.6583e+09 4.4072e+12 61904
## - BsmtFinSF2     1 3.7387e+09 4.4073e+12 61904
## - FullBath       1 4.4395e+09 4.4080e+12 61904
## + GarageCars     1 1.5552e+08 4.4034e+12 61905
## + BedroomAbvGr   1 8.0424e+07 4.4034e+12 61905
## + WoodDeckSF     1 6.4788e+07 4.4034e+12 61905
## + LowQualFinSF   1 2.0786e+07 4.4035e+12 61905
## + GrLivArea      1 2.0786e+07 4.4035e+12 61905
## + EnclosedPorch  1 1.5230e+07 4.4035e+12 61905
## + YrSold         1 2.6845e+06 4.4035e+12 61905
## + BsmtHalfBath   1 2.1695e+06 4.4035e+12 61905
## + MSZoning       4 8.1342e+09 4.3954e+12 61906
## + Electrical     4 7.3976e+09 4.3961e+12 61906
## + GarageFinish   2 1.3368e+09 4.4022e+12 61907
## - MoSold         1 8.5232e+09 4.4120e+12 61907
## - PavedDrive     2 1.1953e+10 4.4155e+12 61907
## - GarageArea     1 8.9290e+09 4.4124e+12 61907
## - KitchenQual    3 1.5083e+10 4.4186e+12 61907
## - MSSubClass     1 9.1745e+09 4.4127e+12 61907
## + BldgType       4 5.8543e+09 4.3977e+12 61908
## - OverallQual    1 9.6780e+09 4.4132e+12 61908
## - ExterQual      3 1.6375e+10 4.4199e+12 61908
## + BsmtCond       3 1.5090e+09 4.4020e+12 61908
## + BsmtFinType1   5 7.2316e+09 4.3963e+12 61909
## + LotConfig      4 4.0572e+09 4.3995e+12 61909
## - SaleCondition  5 2.3751e+10 4.4273e+12 61909
## + GarageQual     4 2.9312e+09 4.4006e+12 61909
## + HouseStyle     7 1.1376e+10 4.3921e+12 61910
## + Condition1     8 1.3950e+10 4.3896e+12 61910
## + GarageCond     4 1.8089e+09 4.4017e+12 61910
## - PoolArea       1 1.3593e+10 4.4171e+12 61910
## + SaleType       8 1.3233e+10 4.3903e+12 61911
## + Functional     6 7.1757e+09 4.3963e+12 61911
## + ExterCond      4 9.5127e+08 4.4026e+12 61911
## - LandSlope      2 1.7251e+10 4.4208e+12 61911
## + HeatingQC      4 7.8711e+08 4.4027e+12 61911
## - MiscVal        1 1.5611e+10 4.4191e+12 61912
## + RoofStyle      5 2.2316e+09 4.4013e+12 61912
## + GarageType     5 2.0791e+09 4.4014e+12 61912
## + Foundation     5 1.5515e+09 4.4020e+12 61912
## - LandContour    3 2.4733e+10 4.4282e+12 61914
## - BsmtUnfSF      1 2.1650e+10 4.4252e+12 61916
## + Exterior1st   14 2.3457e+10 4.3801e+12 61916
## - Condition2     7 4.1397e+10 4.4449e+12 61917
## - YearBuilt      1 2.3805e+10 4.4273e+12 61917
## - KitchenAbvGr   1 2.6318e+10 4.4298e+12 61919
## - X1stFlrSF      1 3.0269e+10 4.4338e+12 61921
## + Exterior2nd   15 1.4727e+10 4.3888e+12 61924
## - OverallCond    1 3.5521e+10 4.4390e+12 61925
## - TotRmsAbvGrd   1 3.7010e+10 4.4405e+12 61926
```

```
## - BsmtFinSF1     1 3.9078e+10 4.4426e+12 61927
## - Neighborhood  24 1.1799e+11 4.5215e+12 61933
## - X2ndFlrSF      1 7.3167e+10 4.4767e+12 61949
## - LotArea        1 1.0511e+11 4.5086e+12 61970
## - RoofMatl       7 2.0660e+11 4.6101e+12 62023
##
## Step:  AIC=61901.49
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + BsmtFullBath + FullBath + HalfBath + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageArea +
##     PavedDrive + OpenPorchSF + X3SsnPorch + ScreenPorch + PoolArea +
##     MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Fireplaces     1 3.1489e+08 4.4040e+12 61900
## - OpenPorchSF    1 3.8956e+08 4.4041e+12 61900
## - HalfBath       1 4.0755e+08 4.4041e+12 61900
## - X3SsnPorch     1 6.7931e+08 4.4043e+12 61900
## - BsmtFullBath   1 7.0879e+08 4.4044e+12 61900
## - YearRemodAdd   1 1.3776e+09 4.4050e+12 61900
## - Utilities      1 1.4501e+09 4.4051e+12 61900
## - LotFrontage    1 1.9006e+09 4.4056e+12 61901
## - CentralAir     1 2.0080e+09 4.4057e+12 61901
## - BsmtQual       3 8.0809e+09 4.4117e+12 61901
## - Street         1 2.0507e+09 4.4057e+12 61901
## - BsmtFinType2   5 1.4710e+10 4.4184e+12 61901
## <none>                        4.4037e+12 61901
## - LotShape       3 9.1166e+09 4.4128e+12 61902
## - Heating        5 1.5470e+10 4.4191e+12 61902
## - GarageYrBlt    1 3.5483e+09 4.4072e+12 61902
## - ScreenPorch    1 3.6861e+09 4.4073e+12 61902
## - BsmtExposure   3 9.7694e+09 4.4134e+12 61902
## - BsmtFinSF2     1 3.7596e+09 4.4074e+12 61902
## - FullBath       1 4.4453e+09 4.4081e+12 61902
## + MasVnrArea     1 1.5014e+08 4.4035e+12 61903
## + GarageCars     1 1.4586e+08 4.4035e+12 61903
## + BedroomAbvGr   1 7.5708e+07 4.4036e+12 61903
## + WoodDeckSF     1 6.6118e+07 4.4036e+12 61903
## + LowQualFinSF   1 2.2803e+07 4.4036e+12 61903
## + GrLivArea      1 2.2803e+07 4.4036e+12 61903
## + EnclosedPorch  1 1.8339e+07 4.4036e+12 61903
## + YrSold         1 1.8977e+06 4.4037e+12 61903
## + BsmtHalfBath   1 1.5459e+06 4.4037e+12 61903
## + MSZoning       4 8.0622e+09 4.3956e+12 61904
## + Electrical     4 7.4140e+09 4.3962e+12 61905
## + GarageFinish   2 1.3565e+09 4.4023e+12 61905
## - MoSold         1 8.4767e+09 4.4121e+12 61905
## - KitchenQual    3 1.5013e+10 4.4187e+12 61905
```

```
## - PavedDrive     2 1.2014e+10 4.4157e+12 61905
## - GarageArea     1 9.0164e+09 4.4127e+12 61905
## - MSSubClass     1 9.1317e+09 4.4128e+12 61906
## + BldgType       4 5.8062e+09 4.3979e+12 61906
## - OverallQual    1 9.8071e+09 4.4135e+12 61906
## + BsmtCond       3 1.5321e+09 4.4021e+12 61906
## + BsmtFinType1   5 7.2851e+09 4.3964e+12 61907
## - ExterQual      3 1.7080e+10 4.4207e+12 61907
## + LotConfig      4 4.0665e+09 4.3996e+12 61907
## - SaleCondition  5 2.3714e+10 4.4274e+12 61907
## + GarageQual     4 2.9149e+09 4.4007e+12 61908
## + HouseStyle     7 1.1473e+10 4.3922e+12 61908
## + Condition1     8 1.3949e+10 4.3897e+12 61908
## + GarageCond     4 1.8077e+09 4.4019e+12 61908
## - PoolArea       1 1.3469e+10 4.4171e+12 61908
## + Functional     6 7.2589e+09 4.3964e+12 61909
## + SaleType       8 1.3165e+10 4.3905e+12 61909
## + ExterCond      4 9.3969e+08 4.4027e+12 61909
## - LandSlope      2 1.7206e+10 4.4209e+12 61909
## + HeatingQC      4 7.9164e+08 4.4029e+12 61909
## - MiscVal        1 1.5538e+10 4.4192e+12 61910
## + RoofStyle      5 2.1785e+09 4.4015e+12 61910
## + GarageType     5 2.0824e+09 4.4016e+12 61910
## + Foundation     5 1.5402e+09 4.4021e+12 61910
## - LandContour    3 2.4746e+10 4.4284e+12 61912
## - BsmtUnfSF      1 2.1759e+10 4.4254e+12 61914
## + Exterior1st   14 2.2944e+10 4.3807e+12 61914
## - Condition2     7 4.1543e+10 4.4452e+12 61915
## - YearBuilt      1 2.4154e+10 4.4278e+12 61915
## - KitchenAbvGr   1 2.6492e+10 4.4302e+12 61917
## - X1stFlrSF      1 3.1095e+10 4.4348e+12 61920
## + Exterior2nd   15 1.4710e+10 4.3890e+12 61922
## - OverallCond    1 3.5439e+10 4.4391e+12 61923
## - TotRmsAbvGrd   1 3.6924e+10 4.4406e+12 61924
## - BsmtFinSF1     1 3.9563e+10 4.4432e+12 61926
## - Neighborhood  24 1.2025e+11 4.5239e+12 61932
## - X2ndFlrSF      1 7.4067e+10 4.4777e+12 61948
## - LotArea        1 1.0521e+11 4.5089e+12 61968
## - RoofMatl       7 2.0645e+11 4.6101e+12 62021
##
## Step:  AIC=61899.7
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + BsmtFullBath + FullBath + HalfBath + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + GarageYrBlt + GarageArea + PavedDrive +
##     OpenPorchSF + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
```

```
## - HalfBath       1 3.5930e+08 4.4043e+12 61898
## - OpenPorchSF    1 3.8478e+08 4.4044e+12 61898
## - X3SsnPorch     1 6.7214e+08 4.4046e+12 61898
## - BsmtFullBath   1 6.9985e+08 4.4047e+12 61898
## - YearRemodAdd   1 1.4274e+09 4.4054e+12 61899
## - Utilities      1 1.4479e+09 4.4054e+12 61899
## - LotFrontage    1 1.8936e+09 4.4059e+12 61899
## - CentralAir     1 1.9524e+09 4.4059e+12 61899
## - BsmtQual       3 8.0409e+09 4.4120e+12 61899
## - Street         1 2.0495e+09 4.4060e+12 61899
## - BsmtFinType2   5 1.4716e+10 4.4187e+12 61899
## - LotShape       3 8.9883e+09 4.4130e+12 61900
## <none>                        4.4040e+12 61900
## - Heating        5 1.5541e+10 4.4195e+12 61900
## - BsmtFinSF2     1 3.7047e+09 4.4077e+12 61900
## - GarageYrBlt    1 3.7455e+09 4.4077e+12 61900
## - BsmtExposure   3 9.9331e+09 4.4139e+12 61900
## - ScreenPorch    1 3.9271e+09 4.4079e+12 61900
## - FullBath       1 4.4147e+09 4.4084e+12 61901
## + Fireplaces     1 3.1489e+08 4.4037e+12 61901
## + MasVnrArea     1 1.6829e+08 4.4038e+12 61902
## + GarageCars     1 1.2269e+08 4.4039e+12 61902
## + WoodDeckSF     1 4.5762e+07 4.4039e+12 61902
## + BedroomAbvGr   1 4.4546e+07 4.4039e+12 61902
## + LowQualFinSF   1 1.9815e+07 4.4040e+12 61902
## + GrLivArea      1 1.9815e+07 4.4040e+12 61902
## + EnclosedPorch  1 1.4454e+07 4.4040e+12 61902
## + YrSold         1 2.4248e+06 4.4040e+12 61902
## + BsmtHalfBath   1 8.5886e+05 4.4040e+12 61902
## + MSZoning       4 8.1696e+09 4.3958e+12 61902
## + Electrical     4 7.4675e+09 4.3965e+12 61903
## + GarageFinish   2 1.4084e+09 4.4026e+12 61903
## - MoSold         1 8.5738e+09 4.4126e+12 61903
## - PavedDrive     2 1.1906e+10 4.4159e+12 61904
## - KitchenQual    3 1.4966e+10 4.4189e+12 61904
## - GarageArea     1 9.0335e+09 4.4130e+12 61904
## - MSSubClass     1 9.1666e+09 4.4131e+12 61904
## + BldgType       4 5.5539e+09 4.3984e+12 61904
## - OverallQual    1 1.0486e+10 4.4145e+12 61905
## + BsmtCond       3 1.5317e+09 4.4024e+12 61905
## + LotConfig      4 4.1059e+09 4.3999e+12 61905
## - ExterQual      3 1.7084e+10 4.4211e+12 61905
## + BsmtFinType1   5 7.0781e+09 4.3969e+12 61905
## - SaleCondition  5 2.3545e+10 4.4275e+12 61905
## + GarageQual     4 2.9037e+09 4.4011e+12 61906
## + HouseStyle     7 1.1454e+10 4.3925e+12 61906
## + GarageCond     4 1.8569e+09 4.4021e+12 61906
## + Condition1     8 1.3816e+10 4.3902e+12 61907
## - PoolArea       1 1.3599e+10 4.4176e+12 61907
## + SaleType       8 1.3203e+10 4.3908e+12 61907
## + Functional     6 7.1758e+09 4.3968e+12 61907
## - LandSlope      2 1.7126e+10 4.4211e+12 61907
```

```
## + ExterCond      4 9.2196e+08 4.4031e+12 61907
## + HeatingQC      4 8.3438e+08 4.4031e+12 61907
## - MiscVal        1 1.5802e+10 4.4198e+12 61908
## + RoofStyle      5 2.1518e+09 4.4018e+12 61908
## + GarageType     5 2.0808e+09 4.4019e+12 61908
## + Foundation     5 1.5656e+09 4.4024e+12 61909
## - LandContour    3 2.4873e+10 4.4289e+12 61910
## - BsmtUnfSF      1 2.1509e+10 4.4255e+12 61912
## + Exterior1st   14 2.3205e+10 4.3808e+12 61912
## - Condition2     7 4.1792e+10 4.4458e+12 61913
## - YearBuilt      1 2.4077e+10 4.4281e+12 61914
## - KitchenAbvGr   1 2.7130e+10 4.4311e+12 61916
## + Exterior2nd   15 1.4881e+10 4.3891e+12 61920
## - X1stFlrSF      1 3.3897e+10 4.4379e+12 61920
## - OverallCond    1 3.5285e+10 4.4393e+12 61921
## - TotRmsAbvGrd   1 3.6855e+10 4.4408e+12 61922
## - BsmtFinSF1     1 3.9655e+10 4.4436e+12 61924
## - Neighborhood  24 1.2056e+11 4.5245e+12 61931
## - X2ndFlrSF      1 7.5851e+10 4.4798e+12 61948
## - LotArea        1 1.0615e+11 4.5101e+12 61967
## - RoofMatl       7 2.0678e+11 4.6108e+12 62020
##
## Step:  AIC=61897.94
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + BsmtFullBath + FullBath + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + GarageYrBlt + GarageArea + PavedDrive + OpenPorchSF +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - OpenPorchSF    1 4.3855e+08 4.4048e+12 61896
## - BsmtFullBath   1 6.7039e+08 4.4050e+12 61896
## - X3SsnPorch     1 6.8634e+08 4.4050e+12 61896
## - Utilities      1 1.4146e+09 4.4058e+12 61897
## - YearRemodAdd   1 1.4468e+09 4.4058e+12 61897
## - LotFrontage    1 1.8760e+09 4.4062e+12 61897
## - BsmtQual       3 8.0523e+09 4.4124e+12 61897
## - CentralAir     1 2.0253e+09 4.4064e+12 61897
## - Street         1 2.0760e+09 4.4064e+12 61897
## - BsmtFinType2   5 1.4617e+10 4.4190e+12 61898
## - LotShape       3 9.0261e+09 4.4134e+12 61898
## <none>                        4.4043e+12 61898
## - Heating        5 1.5451e+10 4.4198e+12 61898
## - BsmtFinSF2     1 3.6217e+09 4.4080e+12 61898
## - GarageYrBlt    1 3.7522e+09 4.4081e+12 61898
## - ScreenPorch    1 3.8802e+09 4.4082e+12 61899
## - BsmtExposure   3 1.0382e+10 4.4147e+12 61899
## + HalfBath       1 3.5930e+08 4.4040e+12 61900
```

```
## + Fireplaces     1 2.6664e+08 4.4041e+12 61900
## + GarageCars     1 1.5021e+08 4.4042e+12 61900
## + MasVnrArea     1 1.4267e+08 4.4042e+12 61900
## + WoodDeckSF     1 5.0963e+07 4.4043e+12 61900
## + BedroomAbvGr   1 5.0633e+07 4.4043e+12 61900
## + LowQualFinSF   1 2.3575e+07 4.4043e+12 61900
## + GrLivArea      1 2.3575e+07 4.4043e+12 61900
## + EnclosedPorch  1 1.4222e+07 4.4043e+12 61900
## + YrSold         1 1.1524e+06 4.4043e+12 61900
## + BsmtHalfBath   1 2.2842e+04 4.4043e+12 61900
## - FullBath       1 6.3008e+09 4.4106e+12 61900
## + MSZoning       4 7.7743e+09 4.3966e+12 61901
## + Electrical     4 7.4916e+09 4.3968e+12 61901
## + GarageFinish   2 1.3684e+09 4.4030e+12 61901
## - MoSold         1 8.6286e+09 4.4130e+12 61902
## - PavedDrive     2 1.1784e+10 4.4161e+12 61902
## - KitchenQual    3 1.4888e+10 4.4192e+12 61902
## - GarageArea     1 9.0408e+09 4.4134e+12 61902
## - MSSubClass     1 9.1743e+09 4.4135e+12 61902
## + BldgType       4 5.4152e+09 4.3989e+12 61902
## - OverallQual    1 1.0462e+10 4.4148e+12 61903
## + BsmtCond       3 1.5469e+09 4.4028e+12 61903
## + BsmtFinType1   5 7.2018e+09 4.3971e+12 61903
## - ExterQual      3 1.7040e+10 4.4214e+12 61903
## + LotConfig      4 4.0559e+09 4.4003e+12 61903
## - SaleCondition  5 2.3452e+10 4.4278e+12 61903
## + GarageQual     4 2.8680e+09 4.4015e+12 61904
## + HouseStyle     7 1.1812e+10 4.3925e+12 61904
## + GarageCond     4 1.8755e+09 4.4025e+12 61905
## + Condition1     8 1.3782e+10 4.3906e+12 61905
## + SaleType       8 1.3258e+10 4.3911e+12 61905
## - PoolArea       1 1.3926e+10 4.4183e+12 61905
## + Functional     6 7.1346e+09 4.3972e+12 61905
## + ExterCond      4 9.3419e+08 4.4034e+12 61905
## - LandSlope      2 1.7248e+10 4.4216e+12 61905
## + HeatingQC      4 8.4428e+08 4.4035e+12 61905
## - MiscVal        1 1.5852e+10 4.4202e+12 61906
## + RoofStyle      5 2.1699e+09 4.4022e+12 61907
## + GarageType     5 2.1457e+09 4.4022e+12 61907
## + Foundation     5 1.6350e+09 4.4027e+12 61907
## - LandContour    3 2.4720e+10 4.4291e+12 61908
## - BsmtUnfSF      1 2.1736e+10 4.4261e+12 61910
## + Exterior1st   14 2.3027e+10 4.3813e+12 61911
## - Condition2     7 4.1685e+10 4.4460e+12 61911
## - YearBuilt      1 2.3739e+10 4.4281e+12 61912
## - KitchenAbvGr   1 2.7494e+10 4.4318e+12 61914
## + Exterior2nd   15 1.4880e+10 4.3895e+12 61918
## - X1stFlrSF      1 3.3542e+10 4.4379e+12 61918
## - OverallCond    1 3.5566e+10 4.4399e+12 61919
## - TotRmsAbvGrd   1 3.7064e+10 4.4414e+12 61920
## - BsmtFinSF1     1 3.9755e+10 4.4441e+12 61922
## - Neighborhood  24 1.2182e+11 4.5262e+12 61930
```

```
## - X2ndFlrSF      1 9.2825e+10 4.4972e+12 61957
## - LotArea        1 1.0616e+11 4.5105e+12 61965
## - RoofMatl       7 2.0649e+11 4.6108e+12 62018
##
## Step:  AIC=61896.23
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + BsmtFullBath + FullBath + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + GarageYrBlt + GarageArea + PavedDrive + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - X3SsnPorch     1 7.0851e+08 4.4055e+12 61895
## - BsmtFullBath   1 7.1612e+08 4.4055e+12 61895
## - Utilities      1 1.4658e+09 4.4062e+12 61895
## - YearRemodAdd   1 1.5307e+09 4.4063e+12 61895
## - LotFrontage    1 1.8387e+09 4.4066e+12 61895
## - CentralAir     1 1.9769e+09 4.4068e+12 61896
## - BsmtQual       3 8.0365e+09 4.4128e+12 61896
## - Street         1 2.1191e+09 4.4069e+12 61896
## - BsmtFinType2   5 1.4578e+10 4.4194e+12 61896
## <none>                        4.4048e+12 61896
## - LotShape       3 9.0703e+09 4.4138e+12 61896
## - BsmtFinSF2     1 3.5578e+09 4.4083e+12 61897
## - Heating        5 1.5718e+10 4.4205e+12 61897
## - ScreenPorch    1 3.8103e+09 4.4086e+12 61897
## - GarageYrBlt    1 3.8427e+09 4.4086e+12 61897
## - BsmtExposure   3 1.0440e+10 4.4152e+12 61897
## + OpenPorchSF    1 4.3855e+08 4.4043e+12 61898
## + HalfBath       1 4.1306e+08 4.4044e+12 61898
## + Fireplaces     1 2.5863e+08 4.4045e+12 61898
## + MasVnrArea     1 1.6440e+08 4.4046e+12 61898
## + GarageCars     1 1.2749e+08 4.4046e+12 61898
## + BedroomAbvGr   1 6.6727e+07 4.4047e+12 61898
## + WoodDeckSF     1 2.7936e+07 4.4047e+12 61898
## + LowQualFinSF   1 2.3161e+07 4.4048e+12 61898
## + GrLivArea      1 2.3161e+07 4.4048e+12 61898
## + EnclosedPorch  1 9.5448e+06 4.4048e+12 61898
## + YrSold         1 3.3171e+06 4.4048e+12 61898
## + BsmtHalfBath   1 4.5360e+03 4.4048e+12 61898
## - FullBath       1 6.2937e+09 4.4111e+12 61898
## + MSZoning       4 7.8082e+09 4.3970e+12 61899
## + Electrical     4 7.4839e+09 4.3973e+12 61899
## + GarageFinish   2 1.3686e+09 4.4034e+12 61899
## - MoSold         1 8.5243e+09 4.4133e+12 61900
## - PavedDrive     2 1.1812e+10 4.4166e+12 61900
## - KitchenQual    3 1.4843e+10 4.4196e+12 61900
## - MSSubClass     1 9.1146e+09 4.4139e+12 61900
## - GarageArea     1 9.1334e+09 4.4139e+12 61900
```

```
## + BldgType       4 5.5677e+09 4.3992e+12 61901
## - OverallQual    1 1.0472e+10 4.4152e+12 61901
## + BsmtCond       3 1.5589e+09 4.4032e+12 61901
## + BsmtFinType1   5 7.1748e+09 4.3976e+12 61901
## - ExterQual      3 1.7014e+10 4.4218e+12 61901
## + LotConfig      4 4.1386e+09 4.4006e+12 61901
## - SaleCondition  5 2.3396e+10 4.4282e+12 61902
## + GarageQual     4 2.8596e+09 4.4019e+12 61902
## + HouseStyle     7 1.1855e+10 4.3929e+12 61902
## + GarageCond     4 1.9012e+09 4.4029e+12 61903
## + Condition1     8 1.3934e+10 4.3908e+12 61903
## - PoolArea       1 1.3813e+10 4.4186e+12 61903
## + SaleType       8 1.3318e+10 4.3915e+12 61903
## - LandSlope      2 1.7089e+10 4.4219e+12 61904
## + ExterCond      4 1.0035e+09 4.4038e+12 61904
## + Functional     6 7.0210e+09 4.3978e+12 61904
## + HeatingQC      4 8.2531e+08 4.4039e+12 61904
## + GarageType     5 2.2161e+09 4.4026e+12 61905
## + RoofStyle      5 2.1658e+09 4.4026e+12 61905
## - MiscVal        1 1.6278e+10 4.4211e+12 61905
## + Foundation     5 1.5697e+09 4.4032e+12 61905
## - LandContour    3 2.4674e+10 4.4294e+12 61907
## - BsmtUnfSF      1 2.1568e+10 4.4263e+12 61908
## + Exterior1st   14 2.3036e+10 4.3817e+12 61909
## - YearBuilt      1 2.3637e+10 4.4284e+12 61910
## - Condition2     7 4.2157e+10 4.4469e+12 61910
## - KitchenAbvGr   1 2.7146e+10 4.4319e+12 61912
## - X1stFlrSF      1 3.3117e+10 4.4379e+12 61916
## + Exterior2nd   15 1.4731e+10 4.3900e+12 61916
## - OverallCond    1 3.5587e+10 4.4404e+12 61918
## - TotRmsAbvGrd   1 3.7092e+10 4.4419e+12 61919
## - BsmtFinSF1     1 3.9645e+10 4.4444e+12 61920
## - Neighborhood  24 1.2352e+11 4.5283e+12 61929
## - X2ndFlrSF      1 9.2703e+10 4.4975e+12 61955
## - LotArea        1 1.0587e+11 4.5106e+12 61964
## - RoofMatl       7 2.0658e+11 4.6114e+12 62016
##
## Step:  AIC=61894.7
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + BsmtFullBath + FullBath + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + GarageYrBlt + GarageArea + PavedDrive + ScreenPorch +
##     PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BsmtFullBath   1 7.0975e+08 4.4062e+12 61893
## - Utilities      1 1.4513e+09 4.4069e+12 61894
## - YearRemodAdd   1 1.4876e+09 4.4070e+12 61894
## - LotFrontage    1 1.8966e+09 4.4074e+12 61894
```

```
## - BsmtQual       3 8.0119e+09 4.4135e+12 61894
## - CentralAir     1 1.9739e+09 4.4075e+12 61894
## - Street         1 2.1212e+09 4.4076e+12 61894
## - BsmtFinType2   5 1.4622e+10 4.4201e+12 61894
## <none>                        4.4055e+12 61895
## - LotShape       3 9.0655e+09 4.4145e+12 61895
## - BsmtFinSF2     1 3.4238e+09 4.4089e+12 61895
## - Heating        5 1.5686e+10 4.4212e+12 61895
## - ScreenPorch    1 3.6671e+09 4.4092e+12 61895
## - GarageYrBlt    1 3.7960e+09 4.4093e+12 61895
## - BsmtExposure   3 1.0310e+10 4.4158e+12 61896
## + X3SsnPorch     1 7.0851e+08 4.4048e+12 61896
## + OpenPorchSF    1 4.6072e+08 4.4050e+12 61896
## + HalfBath       1 4.2993e+08 4.4051e+12 61896
## + Fireplaces     1 2.5092e+08 4.4052e+12 61897
## + MasVnrArea     1 1.6589e+08 4.4053e+12 61897
## + GarageCars     1 1.2872e+08 4.4054e+12 61897
## + BedroomAbvGr   1 5.1733e+07 4.4054e+12 61897
## + WoodDeckSF     1 3.5211e+07 4.4054e+12 61897
## + LowQualFinSF   1 2.0455e+07 4.4055e+12 61897
## + GrLivArea      1 2.0455e+07 4.4055e+12 61897
## + EnclosedPorch  1 1.4886e+07 4.4055e+12 61897
## + YrSold         1 6.2603e+06 4.4055e+12 61897
## + BsmtHalfBath   1 3.0509e+05 4.4055e+12 61897
## - FullBath       1 6.3765e+09 4.4119e+12 61897
## + MSZoning       4 7.7809e+09 4.3977e+12 61898
## + Electrical     4 7.4628e+09 4.3980e+12 61898
## + GarageFinish   2 1.2875e+09 4.4042e+12 61898
## - MoSold         1 8.6352e+09 4.4141e+12 61898
## - KitchenQual    3 1.4823e+10 4.4203e+12 61899
## - PavedDrive     2 1.1907e+10 4.4174e+12 61899
## - MSSubClass     1 9.1600e+09 4.4146e+12 61899
## - GarageArea     1 9.2205e+09 4.4147e+12 61899
## + BldgType       4 5.5181e+09 4.4000e+12 61899
## - OverallQual    1 1.0387e+10 4.4159e+12 61900
## + BsmtCond       3 1.2727e+09 4.4042e+12 61900
## + BsmtFinType1   5 7.2250e+09 4.3983e+12 61900
## + LotConfig      4 4.1874e+09 4.4013e+12 61900
## - ExterQual      3 1.7307e+10 4.4228e+12 61900
## - SaleCondition  5 2.3522e+10 4.4290e+12 61900
## + GarageQual     4 2.8846e+09 4.4026e+12 61901
## + HouseStyle     7 1.1860e+10 4.3936e+12 61901
## + GarageCond     4 1.8975e+09 4.4036e+12 61901
## + Condition1     8 1.3912e+10 4.3916e+12 61901
## - PoolArea       1 1.3729e+10 4.4192e+12 61902
## + SaleType       8 1.3272e+10 4.3922e+12 61902
## - LandSlope      2 1.7128e+10 4.4226e+12 61902
## + ExterCond      4 9.5617e+08 4.4045e+12 61902
## + Functional     6 6.9183e+09 4.3986e+12 61902
## + HeatingQC      4 7.9454e+08 4.4047e+12 61902
## + GarageType     5 2.2052e+09 4.4033e+12 61903
## + RoofStyle      5 2.1652e+09 4.4033e+12 61903
```

```
## - MiscVal        1 1.6366e+10 4.4218e+12 61904
## + Foundation     5 1.4503e+09 4.4040e+12 61904
## - LandContour    3 2.4541e+10 4.4300e+12 61905
## - BsmtUnfSF      1 2.1609e+10 4.4271e+12 61907
## + Exterior1st   14 2.3133e+10 4.3824e+12 61907
## - YearBuilt      1 2.3648e+10 4.4291e+12 61908
## - Condition2     7 4.2521e+10 4.4480e+12 61909
## - KitchenAbvGr   1 2.7187e+10 4.4327e+12 61911
## - X1stFlrSF      1 3.3422e+10 4.4389e+12 61915
## + Exterior2nd   15 1.4747e+10 4.3907e+12 61915
## - OverallCond    1 3.5918e+10 4.4414e+12 61916
## - TotRmsAbvGrd   1 3.6711e+10 4.4422e+12 61917
## - BsmtFinSF1     1 3.9811e+10 4.4453e+12 61919
## - Neighborhood  24 1.2327e+11 4.5288e+12 61927
## - X2ndFlrSF      1 9.2970e+10 4.4985e+12 61954
## - LotArea        1 1.0580e+11 4.5113e+12 61962
## - RoofMatl       7 2.0694e+11 4.6124e+12 62015
##
## Step:  AIC=61893.17
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + Utilities + LandSlope + Neighborhood + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     ExterQual + BsmtQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + FullBath + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     GarageYrBlt + GarageArea + PavedDrive + ScreenPorch + PoolArea +
##     MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Utilities      1 1.4248e+09 4.4076e+12 61892
## - YearRemodAdd   1 1.6285e+09 4.4078e+12 61892
## - LotFrontage    1 1.8784e+09 4.4081e+12 61892
## - BsmtQual       3 7.9398e+09 4.4141e+12 61892
## - CentralAir     1 1.9982e+09 4.4082e+12 61892
## - Street         1 2.1476e+09 4.4083e+12 61893
## - BsmtFinType2   5 1.4656e+10 4.4208e+12 61893
## - LotShape       3 9.0298e+09 4.4152e+12 61893
## <none>                        4.4062e+12 61893
## - BsmtFinSF2     1 3.1065e+09 4.4093e+12 61893
## - ScreenPorch    1 3.6757e+09 4.4099e+12 61894
## - Heating        5 1.5786e+10 4.4220e+12 61894
## - GarageYrBlt    1 3.7470e+09 4.4099e+12 61894
## - BsmtExposure   3 1.0195e+10 4.4164e+12 61894
## + BsmtFullBath   1 7.0975e+08 4.4055e+12 61895
## + X3SsnPorch     1 7.0213e+08 4.4055e+12 61895
## + OpenPorchSF    1 5.0721e+08 4.4057e+12 61895
## + HalfBath       1 3.9965e+08 4.4058e+12 61895
## + Fireplaces     1 2.4438e+08 4.4059e+12 61895
## + MasVnrArea     1 1.9541e+08 4.4060e+12 61895
## + GarageCars     1 1.4219e+08 4.4061e+12 61895
## + BsmtHalfBath   1 7.8522e+07 4.4061e+12 61895
## + BedroomAbvGr   1 6.4214e+07 4.4061e+12 61895
```

```
## + WoodDeckSF     1 4.6749e+07 4.4061e+12 61895
## + LowQualFinSF   1 2.2741e+07 4.4062e+12 61895
## + GrLivArea      1 2.2741e+07 4.4062e+12 61895
## + EnclosedPorch  1 1.6191e+07 4.4062e+12 61895
## + YrSold         1 2.3516e+06 4.4062e+12 61895
## - FullBath       1 6.8139e+09 4.4130e+12 61896
## + MSZoning       4 7.7101e+09 4.3985e+12 61896
## + Electrical     4 7.4627e+09 4.3987e+12 61896
## + GarageFinish   2 1.3013e+09 4.4049e+12 61896
## - MoSold         1 8.6076e+09 4.4148e+12 61897
## - KitchenQual    3 1.4780e+10 4.4210e+12 61897
## - PavedDrive     2 1.1768e+10 4.4180e+12 61897
## - GarageArea     1 9.1431e+09 4.4153e+12 61897
## - MSSubClass     1 9.6660e+09 4.4159e+12 61898
## + BldgType       4 5.3583e+09 4.4008e+12 61898
## - OverallQual    1 1.0324e+10 4.4165e+12 61898
## + BsmtCond       3 1.2726e+09 4.4049e+12 61898
## + LotConfig      4 4.2696e+09 4.4019e+12 61898
## - SaleCondition  5 2.3372e+10 4.4296e+12 61899
## - ExterQual      3 1.7515e+10 4.4237e+12 61899
## + BsmtFinType1   5 6.3270e+09 4.3999e+12 61899
## + GarageQual     4 2.9342e+09 4.4033e+12 61899
## + HouseStyle     7 1.1503e+10 4.3947e+12 61900
## + Condition1     8 1.4155e+10 4.3920e+12 61900
## + GarageCond     4 1.8566e+09 4.4043e+12 61900
## - PoolArea       1 1.3644e+10 4.4198e+12 61900
## + SaleType       8 1.3481e+10 4.3927e+12 61900
## + Functional     6 7.0069e+09 4.3992e+12 61901
## + ExterCond      4 8.9142e+08 4.4053e+12 61901
## - LandSlope      2 1.7292e+10 4.4235e+12 61901
## + HeatingQC      4 7.8108e+08 4.4054e+12 61901
## - MiscVal        1 1.5940e+10 4.4221e+12 61902
## + GarageType     5 2.2007e+09 4.4040e+12 61902
## + RoofStyle      5 2.1098e+09 4.4041e+12 61902
## + Foundation     5 1.4285e+09 4.4048e+12 61902
## - LandContour    3 2.4447e+10 4.4306e+12 61903
## - BsmtUnfSF      1 2.1548e+10 4.4277e+12 61905
## + Exterior1st   14 2.3365e+10 4.3828e+12 61906
## - YearBuilt      1 2.3530e+10 4.4297e+12 61907
## - Condition2     7 4.2340e+10 4.4485e+12 61907
## - KitchenAbvGr   1 2.7976e+10 4.4342e+12 61910
## + Exterior2nd   15 1.4930e+10 4.3913e+12 61913
## - X1stFlrSF      1 3.4050e+10 4.4402e+12 61914
## - OverallCond    1 3.6170e+10 4.4424e+12 61915
## - TotRmsAbvGrd   1 3.6725e+10 4.4429e+12 61915
## - BsmtFinSF1     1 4.2816e+10 4.4490e+12 61919
## - Neighborhood  24 1.2288e+11 4.5291e+12 61925
## - X2ndFlrSF      1 9.3864e+10 4.5001e+12 61953
## - LotArea        1 1.0548e+11 4.5117e+12 61960
## - RoofMatl       7 2.0640e+11 4.6126e+12 62013
##
## Step:  AIC=61892.11
```

```
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + LandSlope + Neighborhood + Condition2 + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofMatl + ExterQual +
##     BsmtQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + CentralAir + X1stFlrSF + X2ndFlrSF +
##     FullBath + KitchenAbvGr + KitchenQual + TotRmsAbvGrd + GarageYrBlt +
##     GarageArea + PavedDrive + ScreenPorch + PoolArea + MiscVal +
##     MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - YearRemodAdd   1 1.5286e+09 4.4091e+12 61891
## - BsmtQual       3 7.9633e+09 4.4156e+12 61891
## - LotFrontage    1 1.9389e+09 4.4096e+12 61891
## - CentralAir     1 2.0336e+09 4.4097e+12 61891
## - Street         1 2.1407e+09 4.4098e+12 61892
## - BsmtFinType2   5 1.4620e+10 4.4222e+12 61892
## <none>                        4.4076e+12 61892
## - LotShape       3 9.1338e+09 4.4168e+12 61892
## - BsmtFinSF2     1 3.1710e+09 4.4108e+12 61892
## - ScreenPorch    1 3.3660e+09 4.4110e+12 61892
## - Heating        5 1.5786e+10 4.4234e+12 61893
## - GarageYrBlt    1 3.6816e+09 4.4113e+12 61893
## - BsmtExposure   3 1.0372e+10 4.4180e+12 61893
## + Utilities      1 1.4248e+09 4.4062e+12 61893
## + X3SsnPorch     1 6.8795e+08 4.4069e+12 61894
## + BsmtFullBath   1 6.8325e+08 4.4069e+12 61894
## + OpenPorchSF    1 5.6002e+08 4.4071e+12 61894
## + HalfBath       1 3.6797e+08 4.4072e+12 61894
## + Fireplaces     1 2.4423e+08 4.4074e+12 61894
## + GarageCars     1 1.5023e+08 4.4075e+12 61894
## + MasVnrArea     1 1.3548e+08 4.4075e+12 61894
## + BedroomAbvGr   1 6.2544e+07 4.4076e+12 61894
## + WoodDeckSF     1 4.1671e+07 4.4076e+12 61894
## + BsmtHalfBath   1 3.5771e+07 4.4076e+12 61894
## + LowQualFinSF   1 2.0815e+07 4.4076e+12 61894
## + GrLivArea      1 2.0815e+07 4.4076e+12 61894
## + EnclosedPorch  1 1.3219e+07 4.4076e+12 61894
## + YrSold         1 1.1570e+06 4.4076e+12 61894
## - FullBath       1 6.9582e+09 4.4146e+12 61895
## + MSZoning       4 7.7612e+09 4.3999e+12 61895
## + Electrical     4 7.5782e+09 4.4000e+12 61895
## + GarageFinish   2 1.3413e+09 4.4063e+12 61895
## - KitchenQual    3 1.4682e+10 4.4223e+12 61896
## - PavedDrive     2 1.1751e+10 4.4194e+12 61896
## - MoSold         1 8.8979e+09 4.4165e+12 61896
## - GarageArea     1 9.0551e+09 4.4167e+12 61896
## - MSSubClass     1 9.6744e+09 4.4173e+12 61897
## + BldgType       4 5.2761e+09 4.4023e+12 61897
## - OverallQual    1 1.0266e+10 4.4179e+12 61897
## + BsmtCond       3 1.2680e+09 4.4063e+12 61897
## + LotConfig      4 3.8813e+09 4.4037e+12 61898
## - SaleCondition  5 2.3489e+10 4.4311e+12 61898
```

```
## - ExterQual      3 1.7516e+10 4.4251e+12 61898
## + BsmtFinType1   5 5.9995e+09 4.4016e+12 61898
## + GarageQual     4 2.9738e+09 4.4046e+12 61898
## + HouseStyle     7 1.1620e+10 4.3960e+12 61898
## + Condition1     8 1.4171e+10 4.3934e+12 61899
## + GarageCond     4 1.8411e+09 4.4058e+12 61899
## - PoolArea       1 1.3684e+10 4.4213e+12 61899
## + SaleType       8 1.3412e+10 4.3942e+12 61899
## - LandSlope      2 1.7201e+10 4.4248e+12 61899
## + Functional     6 6.9700e+09 4.4006e+12 61899
## + ExterCond      4 8.9096e+08 4.4067e+12 61900
## + HeatingQC      4 8.5661e+08 4.4068e+12 61900
## - MiscVal        1 1.6009e+10 4.4236e+12 61901
## + RoofStyle      5 2.0658e+09 4.4056e+12 61901
## + GarageType     5 1.9101e+09 4.4057e+12 61901
## + Foundation     5 1.4666e+09 4.4062e+12 61901
## - LandContour    3 2.4177e+10 4.4318e+12 61902
## + Exterior1st   14 2.3508e+10 4.3841e+12 61905
## - BsmtUnfSF      1 2.1910e+10 4.4295e+12 61905
## - YearBuilt      1 2.3683e+10 4.4313e+12 61906
## - Condition2     7 4.2448e+10 4.4501e+12 61906
## - KitchenAbvGr   1 2.8061e+10 4.4357e+12 61909
## + Exterior2nd   15 1.4830e+10 4.3928e+12 61912
## - X1stFlrSF      1 3.3890e+10 4.4415e+12 61912
## - OverallCond    1 3.5998e+10 4.4436e+12 61914
## - TotRmsAbvGrd   1 3.6432e+10 4.4440e+12 61914
## - BsmtFinSF1     1 4.3461e+10 4.4511e+12 61919
## - Neighborhood  24 1.2252e+11 4.5301e+12 61924
## - X2ndFlrSF      1 9.4329e+10 4.5019e+12 61952
## - LotArea        1 1.0580e+11 4.5134e+12 61959
## - RoofMatl       7 2.0690e+11 4.6145e+12 62012
##
## Step:  AIC=61891.13
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + LandSlope + Neighborhood + Condition2 + OverallQual +
##     OverallCond + YearBuilt + RoofMatl + ExterQual + BsmtQual +
##     BsmtExposure + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + CentralAir + X1stFlrSF + X2ndFlrSF + FullBath +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + GarageYrBlt +
##     GarageArea + PavedDrive + ScreenPorch + PoolArea + MiscVal +
##     MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BsmtQual       3 7.9460e+09 4.4171e+12 61890
## - LotFrontage    1 1.9669e+09 4.4111e+12 61890
## - Street         1 2.0689e+09 4.4112e+12 61890
## - CentralAir     1 2.1583e+09 4.4113e+12 61891
## - BsmtFinType2   5 1.4610e+10 4.4238e+12 61891
## - LotShape       3 8.8440e+09 4.4180e+12 61891
## <none>                        4.4091e+12 61891
## - BsmtFinSF2     1 3.2112e+09 4.4124e+12 61891
## - ScreenPorch    1 3.4743e+09 4.4126e+12 61891
```

```
## - Heating        5 1.5604e+10 4.4248e+12 61891
## - BsmtExposure   3 1.0491e+10 4.4196e+12 61892
## + YearRemodAdd   1 1.5286e+09 4.4076e+12 61892
## - GarageYrBlt    1 4.7036e+09 4.4139e+12 61892
## + Utilities      1 1.3249e+09 4.4078e+12 61892
## + BsmtFullBath   1 8.1810e+08 4.4083e+12 61893
## + OpenPorchSF    1 6.5644e+08 4.4085e+12 61893
## + X3SsnPorch     1 6.4471e+08 4.4085e+12 61893
## + HalfBath       1 3.9122e+08 4.4088e+12 61893
## + Fireplaces     1 2.8756e+08 4.4089e+12 61893
## + MasVnrArea     1 1.7043e+08 4.4090e+12 61893
## + GarageCars     1 1.6349e+08 4.4090e+12 61893
## + BedroomAbvGr   1 9.7348e+07 4.4090e+12 61893
## + WoodDeckSF     1 4.8294e+07 4.4091e+12 61893
## + BsmtHalfBath   1 3.7138e+07 4.4091e+12 61893
## + LowQualFinSF   1 2.3705e+07 4.4091e+12 61893
## + GrLivArea      1 2.3705e+07 4.4091e+12 61893
## + EnclosedPorch  1 6.6299e+06 4.4091e+12 61893
## + YrSold         1 2.0445e+06 4.4091e+12 61893
## - FullBath       1 6.4875e+09 4.4156e+12 61893
## + MSZoning       4 7.9499e+09 4.4012e+12 61894
## + Electrical     4 7.4396e+09 4.4017e+12 61894
## + GarageFinish   2 1.3241e+09 4.4078e+12 61894
## - KitchenQual    3 1.3931e+10 4.4231e+12 61894
## - PavedDrive     2 1.1102e+10 4.4202e+12 61894
## - MoSold         1 8.9984e+09 4.4181e+12 61895
## - GarageArea     1 9.5273e+09 4.4187e+12 61895
## + BldgType       4 5.5008e+09 4.4036e+12 61895
## - MSSubClass     1 9.7970e+09 4.4189e+12 61896
## - OverallQual    1 1.0288e+10 4.4194e+12 61896
## + BsmtCond       3 1.4283e+09 4.4077e+12 61896
## - SaleCondition  5 2.2886e+10 4.4320e+12 61896
## + LotConfig      4 3.9408e+09 4.4052e+12 61897
## - ExterQual      3 1.7715e+10 4.4269e+12 61897
## + GarageQual     4 3.0262e+09 4.4061e+12 61897
## + BsmtFinType1   5 6.0376e+09 4.4031e+12 61897
## + HouseStyle     7 1.1652e+10 4.3975e+12 61897
## + Condition1     8 1.4021e+10 4.3951e+12 61898
## + GarageCond     4 1.8967e+09 4.4072e+12 61898
## + Functional     6 7.4834e+09 4.4017e+12 61898
## + SaleType       8 1.3441e+10 4.3957e+12 61898
## - PoolArea       1 1.3935e+10 4.4231e+12 61898
## - LandSlope      2 1.7152e+10 4.4263e+12 61898
## + HeatingQC      4 9.7359e+08 4.4082e+12 61898
## + ExterCond      4 8.7641e+08 4.4083e+12 61899
## + RoofStyle      5 2.1046e+09 4.4070e+12 61900
## - MiscVal        1 1.6083e+10 4.4252e+12 61900
## + GarageType     5 2.0492e+09 4.4071e+12 61900
## + Foundation     5 1.5325e+09 4.4076e+12 61900
## - LandContour    3 2.4159e+10 4.4333e+12 61901
## + Exterior1st   14 2.4507e+10 4.3846e+12 61903
## - BsmtUnfSF      1 2.2292e+10 4.4314e+12 61904
```

```
## - YearBuilt      1 2.2588e+10 4.4317e+12 61904
## - Condition2     7 4.2527e+10 4.4517e+12 61905
## - KitchenAbvGr   1 2.7483e+10 4.4366e+12 61907
## + Exterior2nd   15 1.5333e+10 4.3938e+12 61911
## - X1stFlrSF      1 3.3292e+10 4.4424e+12 61911
## - OverallCond    1 3.5411e+10 4.4446e+12 61912
## - TotRmsAbvGrd   1 3.6070e+10 4.4452e+12 61913
## - BsmtFinSF1     1 4.4009e+10 4.4532e+12 61918
## - Neighborhood  24 1.2353e+11 4.5327e+12 61924
## - X2ndFlrSF      1 9.4060e+10 4.5032e+12 61951
## - LotArea        1 1.0633e+11 4.5155e+12 61959
## - RoofMatl       7 2.0746e+11 4.6166e+12 62011
##
## Step:  AIC=61890.38
## SalePrice ~ MSSubClass + LotFrontage + LotArea + Street + LotShape +
##     LandContour + LandSlope + Neighborhood + Condition2 + OverallQual +
##     OverallCond + YearBuilt + RoofMatl + ExterQual + BsmtExposure +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     CentralAir + X1stFlrSF + X2ndFlrSF + FullBath + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + GarageYrBlt + GarageArea + PavedDrive +
##     ScreenPorch + PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Street         1 2.0291e+09 4.4191e+12 61890
## - LotFrontage    1 2.0780e+09 4.4192e+12 61890
## - CentralAir     1 2.3420e+09 4.4194e+12 61890
## - BsmtFinType2   5 1.4691e+10 4.4318e+12 61890
## - LotShape       3 8.7033e+09 4.4258e+12 61890
## <none>                        4.4171e+12 61890
## - BsmtFinSF2     1 3.2452e+09 4.4203e+12 61891
## - ScreenPorch    1 3.3074e+09 4.4204e+12 61891
## + BsmtQual       3 7.9460e+09 4.4091e+12 61891
## + YearRemodAdd   1 1.5114e+09 4.4156e+12 61891
## - Heating        5 1.6718e+10 4.4338e+12 61891
## - GarageYrBlt    1 4.6207e+09 4.4217e+12 61891
## + Utilities      1 1.3430e+09 4.4157e+12 61891
## - BsmtExposure   3 1.1030e+10 4.4281e+12 61892
## + BsmtFullBath   1 7.4712e+08 4.4163e+12 61892
## + OpenPorchSF    1 6.4169e+08 4.4165e+12 61892
## + X3SsnPorch     1 6.2167e+08 4.4165e+12 61892
## + HalfBath       1 4.0258e+08 4.4167e+12 61892
## + MasVnrArea     1 2.5317e+08 4.4168e+12 61892
## + Fireplaces     1 2.5116e+08 4.4168e+12 61892
## + BedroomAbvGr   1 1.0860e+08 4.4170e+12 61892
## + GarageCars     1 1.0158e+08 4.4170e+12 61892
## + BsmtHalfBath   1 6.6299e+07 4.4170e+12 61892
## + WoodDeckSF     1 4.4070e+07 4.4170e+12 61892
## + LowQualFinSF   1 4.2251e+07 4.4171e+12 61892
## + GrLivArea      1 4.2251e+07 4.4171e+12 61892
## + EnclosedPorch  1 1.4145e+07 4.4171e+12 61892
## + YrSold         1 3.8967e+06 4.4171e+12 61892
## - FullBath       1 6.3752e+09 4.4235e+12 61893
```

```
## + GarageFinish   2 1.8506e+09 4.4152e+12 61893
## + MSZoning       4 7.3899e+09 4.4097e+12 61893
## + Electrical     4 6.9138e+09 4.4102e+12 61894
## - PavedDrive     2 1.1726e+10 4.4288e+12 61894
## - MoSold         1 8.7191e+09 4.4258e+12 61894
## + BldgType       4 5.6384e+09 4.4115e+12 61895
## - GarageArea     1 1.0354e+10 4.4274e+12 61895
## + LotConfig      4 4.3110e+09 4.4128e+12 61896
## + BsmtCond       3 8.0614e+08 4.4163e+12 61896
## - OverallQual    1 1.1518e+10 4.4286e+12 61896
## + HouseStyle     7 1.2207e+10 4.4049e+12 61896
## - MSSubClass     1 1.2101e+10 4.4292e+12 61896
## - KitchenQual    3 1.8346e+10 4.4354e+12 61896
## + GarageQual     4 2.8220e+09 4.4143e+12 61897
## - SaleCondition  5 2.4822e+10 4.4419e+12 61897
## + BsmtFinType1   5 5.3466e+09 4.4117e+12 61897
## + GarageCond     4 1.9019e+09 4.4152e+12 61897
## + Condition1     8 1.3838e+10 4.4033e+12 61897
## + SaleType       8 1.3681e+10 4.4034e+12 61897
## - PoolArea       1 1.3988e+10 4.4311e+12 61898
## + ExterCond      4 1.0506e+09 4.4160e+12 61898
## + Functional     6 7.0677e+09 4.4100e+12 61898
## + HeatingQC      4 9.9616e+08 4.4161e+12 61898
## - LandSlope      2 1.7275e+10 4.4344e+12 61898
## - ExterQual      3 2.1937e+10 4.4390e+12 61899
## + GarageType     5 2.0265e+09 4.4151e+12 61899
## + RoofStyle      5 2.0135e+09 4.4151e+12 61899
## + Foundation     5 1.9393e+09 4.4152e+12 61899
## - MiscVal        1 1.6360e+10 4.4335e+12 61899
## - LandContour    3 2.3542e+10 4.4406e+12 61900
## - BsmtUnfSF      1 2.1027e+10 4.4381e+12 61902
## + Exterior1st   14 2.2673e+10 4.3944e+12 61903
## - YearBuilt      1 2.3176e+10 4.4403e+12 61904
## - Condition2     7 4.2799e+10 4.4599e+12 61905
## - KitchenAbvGr   1 2.6647e+10 4.4437e+12 61906
## - OverallCond    1 3.3697e+10 4.4508e+12 61911
## - X1stFlrSF      1 3.4562e+10 4.4517e+12 61911
## + Exterior2nd   15 1.3833e+10 4.4033e+12 61911
## - TotRmsAbvGrd   1 3.6714e+10 4.4538e+12 61913
## - BsmtFinSF1     1 4.5225e+10 4.4623e+12 61918
## - Neighborhood  24 1.2775e+11 4.5448e+12 61926
## - X2ndFlrSF      1 9.5664e+10 4.5128e+12 61951
## - LotArea        1 1.0465e+11 4.5217e+12 61957
## - RoofMatl       7 2.0908e+11 4.6262e+12 62011
##
## Step:  AIC=61889.72
## SalePrice ~ MSSubClass + LotFrontage + LotArea + LotShape + LandContour +
##     LandSlope + Neighborhood + Condition2 + OverallQual + OverallCond +
##     YearBuilt + RoofMatl + ExterQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     X1stFlrSF + X2ndFlrSF + FullBath + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + GarageYrBlt + GarageArea + PavedDrive + ScreenPorch +
```

```
##     PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - LotFrontage    1 1.9086e+09 4.4210e+12 61889
## - CentralAir     1 2.3062e+09 4.4214e+12 61889
## - LotShape       3 8.6178e+09 4.4277e+12 61889
## - BsmtFinType2   5 1.5153e+10 4.4343e+12 61890
## <none>                        4.4191e+12 61890
## - BsmtFinSF2     1 3.1954e+09 4.4223e+12 61890
## - ScreenPorch    1 3.2253e+09 4.4223e+12 61890
## + Street         1 2.0291e+09 4.4171e+12 61890
## + BsmtQual       3 7.9061e+09 4.4112e+12 61890
## - GarageYrBlt    1 4.4049e+09 4.4235e+12 61891
## - Heating        5 1.6576e+10 4.4357e+12 61891
## + YearRemodAdd   1 1.4325e+09 4.4177e+12 61891
## + Utilities      1 1.3400e+09 4.4178e+12 61891
## - BsmtExposure   3 1.0873e+10 4.4300e+12 61891
## + BsmtFullBath   1 7.6722e+08 4.4184e+12 61891
## + OpenPorchSF    1 6.8818e+08 4.4184e+12 61891
## + X3SsnPorch     1 6.2454e+08 4.4185e+12 61891
## + HalfBath       1 4.3200e+08 4.4187e+12 61891
## + MasVnrArea     1 2.6298e+08 4.4189e+12 61892
## + Fireplaces     1 2.4699e+08 4.4189e+12 61892
## + BedroomAbvGr   1 1.0141e+08 4.4190e+12 61892
## + GarageCars     1 9.7892e+07 4.4190e+12 61892
## + BsmtHalfBath   1 6.9666e+07 4.4191e+12 61892
## + WoodDeckSF     1 5.3596e+07 4.4191e+12 61892
## + LowQualFinSF   1 2.2420e+07 4.4191e+12 61892
## + GrLivArea      1 2.2420e+07 4.4191e+12 61892
## + EnclosedPorch  1 1.2494e+07 4.4191e+12 61892
## + YrSold         1 6.4833e+06 4.4191e+12 61892
## - FullBath       1 6.4163e+09 4.4255e+12 61892
## + GarageFinish   2 1.9208e+09 4.4172e+12 61892
## + MSZoning       4 7.7642e+09 4.4114e+12 61893
## + Electrical     4 6.7728e+09 4.4123e+12 61893
## - PavedDrive     2 1.1477e+10 4.4306e+12 61893
## - MoSold         1 8.7281e+09 4.4278e+12 61893
## + BldgType       4 5.3688e+09 4.4138e+12 61894
## - GarageArea     1 9.8537e+09 4.4290e+12 61894
## + LotConfig      4 4.4192e+09 4.4147e+12 61895
## + BsmtCond       3 6.8558e+08 4.4184e+12 61895
## - OverallQual    1 1.1679e+10 4.4308e+12 61895
## - SaleCondition  5 2.4208e+10 4.4433e+12 61896
## + GarageQual     4 2.9065e+09 4.4162e+12 61896
## + HouseStyle     7 1.1954e+10 4.4072e+12 61896
## - KitchenQual    3 1.8456e+10 4.4376e+12 61896
## - MSSubClass     1 1.2449e+10 4.4316e+12 61896
## + BsmtFinType1   5 5.4676e+09 4.4137e+12 61896
## + GarageCond     4 1.8484e+09 4.4173e+12 61897
## + SaleType       8 1.3471e+10 4.4057e+12 61897
## + HeatingQC      4 1.0449e+09 4.4181e+12 61897
## + Functional     6 7.0862e+09 4.4120e+12 61897
```

```
## - PoolArea       1 1.4131e+10 4.4333e+12 61897
## + ExterCond      4 1.0256e+09 4.4181e+12 61897
## - LandSlope      2 1.7292e+10 4.4364e+12 61897
## + Condition1     8 1.2876e+10 4.4062e+12 61897
## - ExterQual      3 2.1191e+10 4.4403e+12 61898
## + GarageType     5 2.0465e+09 4.4171e+12 61898
## + RoofStyle      5 1.9552e+09 4.4172e+12 61898
## + Foundation     5 1.8841e+09 4.4172e+12 61898
## - MiscVal        1 1.6375e+10 4.4355e+12 61899
## - LandContour    3 2.3964e+10 4.4431e+12 61900
## - BsmtUnfSF      1 2.0721e+10 4.4398e+12 61901
## + Exterior1st   14 2.2888e+10 4.3962e+12 61903
## - YearBuilt      1 2.2800e+10 4.4419e+12 61903
## - Condition2     7 4.2756e+10 4.4619e+12 61904
## - KitchenAbvGr   1 2.6244e+10 4.4454e+12 61905
## - OverallCond    1 3.3431e+10 4.4526e+12 61910
## + Exterior2nd   15 1.3899e+10 4.4052e+12 61911
## - X1stFlrSF      1 3.5526e+10 4.4546e+12 61911
## - TotRmsAbvGrd   1 3.6984e+10 4.4561e+12 61912
## - BsmtFinSF1     1 4.4962e+10 4.4641e+12 61917
## - Neighborhood  24 1.2776e+11 4.5469e+12 61925
## - X2ndFlrSF      1 9.6518e+10 4.5156e+12 61951
## - LotArea        1 1.0262e+11 4.5217e+12 61955
## - RoofMatl       7 2.0892e+11 4.6280e+12 62011
##
## Step:  AIC=61888.98
## SalePrice ~ MSSubClass + LotArea + LotShape + LandContour + LandSlope +
##     Neighborhood + Condition2 + OverallQual + OverallCond + YearBuilt +
##     RoofMatl + ExterQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + FullBath + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     GarageYrBlt + GarageArea + PavedDrive + ScreenPorch + PoolArea +
##     MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - LotShape       3 8.3931e+09 4.4294e+12 61889
## - CentralAir     1 2.3502e+09 4.4234e+12 61889
## <none>                        4.4210e+12 61889
## - BsmtFinSF2     1 3.0871e+09 4.4241e+12 61889
## - BsmtFinType2   5 1.5359e+10 4.4364e+12 61889
## - ScreenPorch    1 3.3050e+09 4.4243e+12 61889
## - Heating        5 1.6235e+10 4.4373e+12 61890
## + BsmtQual       3 8.0166e+09 4.4130e+12 61890
## + LotFrontage    1 1.9086e+09 4.4191e+12 61890
## + Street         1 1.8597e+09 4.4192e+12 61890
## + YearRemodAdd   1 1.4577e+09 4.4196e+12 61890
## + Utilities      1 1.3967e+09 4.4196e+12 61890
## - GarageYrBlt    1 4.7910e+09 4.4258e+12 61890
## + BsmtFullBath   1 7.4775e+08 4.4203e+12 61890
## + X3SsnPorch     1 6.7809e+08 4.4204e+12 61891
## + OpenPorchSF    1 6.3922e+08 4.4204e+12 61891
## + HalfBath       1 4.0978e+08 4.4206e+12 61891
```

```
## + MasVnrArea     1 2.8116e+08 4.4207e+12 61891
## + Fireplaces     1 2.4346e+08 4.4208e+12 61891
## + BedroomAbvGr   1 1.5256e+08 4.4209e+12 61891
## + GarageCars     1 9.1258e+07 4.4209e+12 61891
## + WoodDeckSF     1 6.8689e+07 4.4210e+12 61891
## + BsmtHalfBath   1 4.6338e+07 4.4210e+12 61891
## + LowQualFinSF   1 1.8404e+07 4.4210e+12 61891
## + GrLivArea      1 1.8404e+07 4.4210e+12 61891
## + EnclosedPorch  1 5.7596e+06 4.4210e+12 61891
## + YrSold         1 4.8142e+06 4.4210e+12 61891
## - BsmtExposure   3 1.2176e+10 4.4332e+12 61891
## - FullBath       1 6.3346e+09 4.4274e+12 61891
## + MSZoning       4 8.3135e+09 4.4127e+12 61891
## + GarageFinish   2 1.8900e+09 4.4191e+12 61892
## + Electrical     4 6.9052e+09 4.4141e+12 61892
## - PavedDrive     2 1.1643e+10 4.4327e+12 61893
## - MoSold         1 8.7011e+09 4.4297e+12 61893
## + BldgType       4 6.4017e+09 4.4146e+12 61893
## - GarageArea     1 1.1263e+10 4.4323e+12 61894
## + BsmtCond       3 7.1435e+08 4.4203e+12 61895
## - OverallQual    1 1.1804e+10 4.4328e+12 61895
## + LotConfig      4 3.2504e+09 4.4178e+12 61895
## - SaleCondition  5 2.4228e+10 4.4453e+12 61895
## + HouseStyle     7 1.2009e+10 4.4090e+12 61895
## + GarageQual     4 2.8157e+09 4.4182e+12 61895
## - KitchenQual    3 1.8761e+10 4.4398e+12 61895
## + BsmtFinType1   5 5.2293e+09 4.4158e+12 61896
## + GarageCond     4 2.0375e+09 4.4190e+12 61896
## + SaleType       8 1.3422e+10 4.4076e+12 61896
## + Functional     6 7.3176e+09 4.4137e+12 61896
## + HeatingQC      4 1.1175e+09 4.4199e+12 61896
## + ExterCond      4 1.0635e+09 4.4200e+12 61896
## + Condition1     8 1.2590e+10 4.4084e+12 61897
## - ExterQual      3 2.0964e+10 4.4420e+12 61897
## - LandSlope      2 1.8088e+10 4.4391e+12 61897
## - PoolArea       1 1.5086e+10 4.4361e+12 61897
## + RoofStyle      5 2.3033e+09 4.4187e+12 61897
## - MSSubClass     1 1.6222e+10 4.4373e+12 61898
## + GarageType     5 1.8877e+09 4.4191e+12 61898
## - MiscVal        1 1.6366e+10 4.4374e+12 61898
## + Foundation     5 1.6303e+09 4.4194e+12 61898
## - LandContour    3 2.3730e+10 4.4448e+12 61899
## - BsmtUnfSF      1 2.0126e+10 4.4412e+12 61900
## - YearBuilt      1 2.2391e+10 4.4434e+12 61902
## + Exterior1st   14 2.2898e+10 4.3981e+12 61902
## - Condition2     7 4.2429e+10 4.4635e+12 61903
## - KitchenAbvGr   1 2.5784e+10 4.4468e+12 61904
## - OverallCond    1 3.3288e+10 4.4543e+12 61909
## + Exterior2nd   15 1.4251e+10 4.4068e+12 61910
## - X1stFlrSF      1 3.7415e+10 4.4584e+12 61912
## - TotRmsAbvGrd   1 3.8839e+10 4.4599e+12 61913
## - BsmtFinSF1     1 4.3985e+10 4.4650e+12 61916
```

```
## - Neighborhood  24 1.2838e+11 4.5494e+12 61925
## - X2ndFlrSF      1 9.7517e+10 4.5185e+12 61951
## - LotArea        1 1.0957e+11 4.5306e+12 61958
## - RoofMatl       7 2.0711e+11 4.6281e+12 62009
##
## Step:  AIC=61888.52
## SalePrice ~ MSSubClass + LotArea + LandContour + LandSlope +
##     Neighborhood + Condition2 + OverallQual + OverallCond + YearBuilt +
##     RoofMatl + ExterQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + X1stFlrSF +
##     X2ndFlrSF + FullBath + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     GarageYrBlt + GarageArea + PavedDrive + ScreenPorch + PoolArea +
##     MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - CentralAir     1 2.3164e+09 4.4317e+12 61888
## - BsmtFinSF2     1 2.8704e+09 4.4323e+12 61888
## <none>                        4.4294e+12 61889
## - BsmtFinType2   5 1.5547e+10 4.4450e+12 61889
## - ScreenPorch    1 3.6158e+09 4.4330e+12 61889
## + LotShape       3 8.3931e+09 4.4210e+12 61889
## + BsmtQual       3 7.8833e+09 4.4215e+12 61889
## + Street         1 1.7892e+09 4.4276e+12 61889
## + LotFrontage    1 1.6838e+09 4.4277e+12 61889
## - Heating        5 1.6565e+10 4.4460e+12 61889
## + Utilities      1 1.5092e+09 4.4279e+12 61890
## - BsmtExposure   3 1.0784e+10 4.4402e+12 61890
## - GarageYrBlt    1 4.7522e+09 4.4342e+12 61890
## + YearRemodAdd   1 1.1665e+09 4.4283e+12 61890
## + BsmtFullBath   1 6.8729e+08 4.4287e+12 61890
## + OpenPorchSF    1 6.8316e+08 4.4287e+12 61890
## + X3SsnPorch     1 6.7294e+08 4.4288e+12 61890
## + HalfBath       1 4.5511e+08 4.4290e+12 61890
## + MasVnrArea     1 2.5343e+08 4.4292e+12 61890
## + BedroomAbvGr   1 1.3388e+08 4.4293e+12 61890
## + WoodDeckSF     1 1.3263e+08 4.4293e+12 61890
## + Fireplaces     1 1.3257e+08 4.4293e+12 61890
## + GarageCars     1 5.9380e+07 4.4294e+12 61890
## + BsmtHalfBath   1 5.1179e+07 4.4294e+12 61890
## + EnclosedPorch  1 1.1784e+07 4.4294e+12 61891
## + LowQualFinSF   1 1.1514e+07 4.4294e+12 61891
## + GrLivArea      1 1.1514e+07 4.4294e+12 61891
## + YrSold         1 5.3294e+06 4.4294e+12 61891
## - FullBath       1 6.1032e+09 4.4355e+12 61891
## + GarageFinish   2 2.0723e+09 4.4274e+12 61891
## + MSZoning       4 7.4406e+09 4.4220e+12 61892
## - PavedDrive     2 1.1274e+10 4.4407e+12 61892
## + Electrical     4 6.8856e+09 4.4225e+12 61892
## + BldgType       4 6.3132e+09 4.4231e+12 61892
## - MoSold         1 9.1821e+09 4.4386e+12 61893
## - GarageArea     1 1.1122e+10 4.4405e+12 61894
## + BsmtCond       3 8.2333e+08 4.4286e+12 61894
```

```
## - OverallQual    1 1.1513e+10 4.4409e+12 61894
## + GarageQual     4 3.1627e+09 4.4263e+12 61894
## - KitchenQual    3 1.8342e+10 4.4478e+12 61895
## - SaleCondition  5 2.4500e+10 4.4539e+12 61895
## + HouseStyle     7 1.1633e+10 4.4178e+12 61895
## + LotConfig      4 2.4696e+09 4.4270e+12 61895
## + BsmtFinType1   5 5.3160e+09 4.4241e+12 61895
## + GarageCond     4 2.1860e+09 4.4272e+12 61895
## + Functional     6 7.5030e+09 4.4219e+12 61896
## + SaleType       8 1.3535e+10 4.4159e+12 61896
## + ExterCond      4 1.1172e+09 4.4283e+12 61896
## + HeatingQC      4 9.7551e+08 4.4284e+12 61896
## - PoolArea       1 1.4976e+10 4.4444e+12 61896
## - LandSlope      2 1.8296e+10 4.4477e+12 61897
## - MSSubClass     1 1.5576e+10 4.4450e+12 61897
## + RoofStyle      5 2.4233e+09 4.4270e+12 61897
## + Condition1     8 1.1379e+10 4.4180e+12 61897
## + GarageType     5 1.9931e+09 4.4274e+12 61897
## + Foundation     5 1.9004e+09 4.4275e+12 61897
## - ExterQual      3 2.2461e+10 4.4519e+12 61897
## - MiscVal        1 1.7136e+10 4.4466e+12 61898
## - LandContour    3 2.5112e+10 4.4545e+12 61899
## - BsmtUnfSF      1 2.0230e+10 4.4497e+12 61900
## - YearBuilt      1 2.2402e+10 4.4518e+12 61901
## + Exterior1st   14 2.2674e+10 4.4067e+12 61902
## - Condition2     7 4.2667e+10 4.4721e+12 61903
## - KitchenAbvGr   1 2.5274e+10 4.4547e+12 61903
## - OverallCond    1 3.3257e+10 4.4627e+12 61908
## + Exterior2nd   15 1.3685e+10 4.4157e+12 61909
## - X1stFlrSF      1 3.7024e+10 4.4664e+12 61911
## - TotRmsAbvGrd   1 3.8388e+10 4.4678e+12 61912
## - BsmtFinSF1     1 4.3543e+10 4.4730e+12 61915
## - Neighborhood  24 1.2476e+11 4.5542e+12 61922
## - X2ndFlrSF      1 9.7783e+10 4.5272e+12 61950
## - LotArea        1 1.2130e+11 4.5507e+12 61965
## - RoofMatl       7 2.1340e+11 4.6428e+12 62012
##
## Step:  AIC=61888.04
## SalePrice ~ MSSubClass + LotArea + LandContour + LandSlope +
##     Neighborhood + Condition2 + OverallQual + OverallCond + YearBuilt +
##     RoofMatl + ExterQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + X1stFlrSF + X2ndFlrSF +
##     FullBath + KitchenAbvGr + KitchenQual + TotRmsAbvGrd + GarageYrBlt +
##     GarageArea + PavedDrive + ScreenPorch + PoolArea + MiscVal +
##     MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - Heating        5 1.4551e+10 4.4463e+12 61888
## - BsmtFinSF2     1 2.7535e+09 4.4345e+12 61888
## <none>                        4.4317e+12 61888
## - BsmtFinType2   5 1.5505e+10 4.4472e+12 61888
## - ScreenPorch    1 3.5248e+09 4.4353e+12 61888
```

```
## + CentralAir     1 2.3164e+09 4.4294e+12 61889
## + LotShape       3 8.3593e+09 4.4234e+12 61889
## + BsmtQual       3 8.0733e+09 4.4237e+12 61889
## + Street         1 1.7588e+09 4.4300e+12 61889
## + LotFrontage    1 1.7246e+09 4.4300e+12 61889
## + Utilities      1 1.5435e+09 4.4302e+12 61889
## + YearRemodAdd   1 1.2728e+09 4.4305e+12 61889
## - GarageYrBlt    1 4.8246e+09 4.4366e+12 61889
## - BsmtExposure   3 1.1035e+10 4.4428e+12 61889
## + BsmtFullBath   1 7.1940e+08 4.4310e+12 61890
## + X3SsnPorch     1 6.6749e+08 4.4311e+12 61890
## + OpenPorchSF    1 6.2164e+08 4.4311e+12 61890
## + HalfBath       1 5.3849e+08 4.4312e+12 61890
## + MasVnrArea     1 2.7625e+08 4.4315e+12 61890
## + WoodDeckSF     1 1.1788e+08 4.4316e+12 61890
## + Fireplaces     1 9.3257e+07 4.4316e+12 61890
## + BedroomAbvGr   1 8.7546e+07 4.4317e+12 61890
## + GarageCars     1 8.0406e+07 4.4317e+12 61890
## + BsmtHalfBath   1 3.9673e+07 4.4317e+12 61890
## + LowQualFinSF   1 3.0501e+07 4.4317e+12 61890
## + GrLivArea      1 3.0501e+07 4.4317e+12 61890
## + EnclosedPorch  1 1.6299e+07 4.4317e+12 61890
## + YrSold         1 6.0771e+06 4.4317e+12 61890
## - FullBath       1 6.1044e+09 4.4378e+12 61890
## + GarageFinish   2 2.1224e+09 4.4296e+12 61891
## + Electrical     4 7.1794e+09 4.4246e+12 61891
## + BldgType       4 6.9804e+09 4.4248e+12 61891
## + MSZoning       4 6.9561e+09 4.4248e+12 61891
## - MoSold         1 9.2736e+09 4.4410e+12 61892
## - PavedDrive     2 1.2641e+10 4.4444e+12 61892
## - GarageArea     1 1.0630e+10 4.4424e+12 61893
## - OverallQual    1 1.1294e+10 4.4430e+12 61893
## + BsmtCond       3 8.3351e+08 4.4309e+12 61893
## + GarageQual     4 3.2016e+09 4.4285e+12 61894
## - KitchenQual    3 1.8488e+10 4.4502e+12 61894
## - SaleCondition  5 2.4738e+10 4.4565e+12 61894
## + LotConfig      4 2.5459e+09 4.4292e+12 61894
## + HouseStyle     7 1.1561e+10 4.4202e+12 61894
## + BsmtFinType1   5 5.2048e+09 4.4265e+12 61895
## + GarageCond     4 1.9536e+09 4.4298e+12 61895
## + Functional     6 7.6680e+09 4.4241e+12 61895
## + SaleType       8 1.3592e+10 4.4181e+12 61895
## + HeatingQC      4 1.2533e+09 4.4305e+12 61895
## + ExterCond      4 1.0681e+09 4.4307e+12 61895
## - PoolArea       1 1.4958e+10 4.4467e+12 61896
## - LandSlope      2 1.8160e+10 4.4499e+12 61896
## - MSSubClass     1 1.5433e+10 4.4472e+12 61896
## + RoofStyle      5 2.4594e+09 4.4293e+12 61896
## + Foundation     5 2.1877e+09 4.4296e+12 61897
## + GarageType     5 2.0396e+09 4.4297e+12 61897
## + Condition1     8 1.0993e+10 4.4207e+12 61897
## - MiscVal        1 1.7000e+10 4.4487e+12 61897
```

```
## - ExterQual      3 2.3800e+10 4.4555e+12 61898
## - LandContour    3 2.4868e+10 4.4566e+12 61898
## - BsmtUnfSF      1 1.9561e+10 4.4513e+12 61899
## - YearBuilt      1 2.0888e+10 4.4526e+12 61900
## + Exterior1st   14 2.2780e+10 4.4090e+12 61901
## - KitchenAbvGr   1 2.3599e+10 4.4553e+12 61902
## - Condition2     7 4.2430e+10 4.4742e+12 61902
## - OverallCond    1 3.1108e+10 4.4628e+12 61906
## + Exterior2nd   15 1.3429e+10 4.4183e+12 61909
## - X1stFlrSF      1 3.6741e+10 4.4685e+12 61910
## - TotRmsAbvGrd   1 3.8250e+10 4.4700e+12 61911
## - BsmtFinSF1     1 4.2702e+10 4.4744e+12 61914
## - Neighborhood  24 1.2550e+11 4.5572e+12 61922
## - X2ndFlrSF      1 9.6755e+10 4.5285e+12 61949
## - LotArea        1 1.2027e+11 4.5520e+12 61964
## - RoofMatl       7 2.1284e+11 4.6446e+12 62011
##
## Step:  AIC=61887.61
## SalePrice ~ MSSubClass + LotArea + LandContour + LandSlope +
##     Neighborhood + Condition2 + OverallQual + OverallCond + YearBuilt +
##     RoofMatl + ExterQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + X1stFlrSF + X2ndFlrSF + FullBath +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + GarageYrBlt +
##     GarageArea + PavedDrive + ScreenPorch + PoolArea + MiscVal +
##     MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## - BsmtFinSF2     1 2.7663e+09 4.4491e+12 61887
## <none>                        4.4463e+12 61888
## + BsmtQual       3 8.9341e+09 4.4374e+12 61888
## - BsmtFinType2   5 1.5457e+10 4.4617e+12 61888
## + LotShape       3 8.6950e+09 4.4376e+12 61888
## - ScreenPorch    1 3.6090e+09 4.4499e+12 61888
## + Heating        5 1.4551e+10 4.4317e+12 61888
## - BsmtExposure   3 1.0458e+10 4.4567e+12 61888
## + Street         1 1.6717e+09 4.4446e+12 61889
## + Utilities      1 1.5212e+09 4.4448e+12 61889
## + LotFrontage    1 1.3791e+09 4.4449e+12 61889
## + YearRemodAdd   1 1.0929e+09 4.4452e+12 61889
## + OpenPorchSF    1 9.8042e+08 4.4453e+12 61889
## - GarageYrBlt    1 5.2697e+09 4.4516e+12 61889
## + BsmtFullBath   1 7.9865e+08 4.4455e+12 61889
## + X3SsnPorch     1 6.3615e+08 4.4457e+12 61889
## + HalfBath       1 3.9531e+08 4.4459e+12 61889
## + MasVnrArea     1 3.4229e+08 4.4459e+12 61889
## + CentralAir     1 3.0208e+08 4.4460e+12 61889
## + Fireplaces     1 1.6277e+08 4.4461e+12 61890
## + BedroomAbvGr   1 1.3865e+08 4.4462e+12 61890
## + GarageCars     1 7.2554e+07 4.4462e+12 61890
## + WoodDeckSF     1 7.0862e+07 4.4462e+12 61890
## + BsmtHalfBath   1 6.7603e+07 4.4462e+12 61890
## + LowQualFinSF   1 2.9777e+07 4.4463e+12 61890
```

```
## + GrLivArea      1 2.9777e+07 4.4463e+12 61890
## + YrSold         1 2.7826e+07 4.4463e+12 61890
## + EnclosedPorch  1 1.2188e+07 4.4463e+12 61890
## - FullBath       1 6.3002e+09 4.4526e+12 61890
## + GarageFinish   2 2.4310e+09 4.4439e+12 61890
## + Electrical     4 7.5816e+09 4.4387e+12 61891
## + MSZoning       4 6.6460e+09 4.4396e+12 61891
## + BldgType       4 6.3785e+09 4.4399e+12 61891
## - MoSold         1 9.4839e+09 4.4558e+12 61892
## - OverallQual    1 1.0782e+10 4.4571e+12 61893
## - PavedDrive     2 1.3857e+10 4.4601e+12 61893
## + GarageQual     4 3.8534e+09 4.4424e+12 61893
## + BsmtCond       3 6.6367e+08 4.4456e+12 61893
## - SaleCondition  5 2.4088e+10 4.4704e+12 61893
## - GarageArea     1 1.2410e+10 4.4587e+12 61894
## + LotConfig      4 2.6131e+09 4.4437e+12 61894
## + GarageCond     4 2.4315e+09 4.4439e+12 61894
## + BsmtFinType1   5 5.2907e+09 4.4410e+12 61894
## + HouseStyle     7 1.1168e+10 4.4351e+12 61894
## + Functional     6 8.0793e+09 4.4382e+12 61894
## - KitchenQual    3 1.9745e+10 4.4660e+12 61895
## + HeatingQC      4 1.2939e+09 4.4450e+12 61895
## + SaleType       8 1.3322e+10 4.4330e+12 61895
## + ExterCond      4 1.1499e+09 4.4451e+12 61895
## - LandSlope      2 1.8123e+10 4.4644e+12 61895
## - PoolArea       1 1.5144e+10 4.4614e+12 61896
## - MSSubClass     1 1.5829e+10 4.4621e+12 61896
## + RoofStyle      5 2.4408e+09 4.4438e+12 61896
## + Foundation     5 2.3785e+09 4.4439e+12 61896
## + GarageType     5 2.1371e+09 4.4442e+12 61896
## + Condition1     8 1.0935e+10 4.4354e+12 61896
## - MiscVal        1 1.7239e+10 4.4635e+12 61897
## - LandContour    3 2.3552e+10 4.4698e+12 61897
## - ExterQual      3 2.3761e+10 4.4701e+12 61897
## - BsmtUnfSF      1 2.0356e+10 4.4666e+12 61899
## + Exterior1st   14 2.2975e+10 4.4233e+12 61900
## - YearBuilt      1 2.3717e+10 4.4700e+12 61901
## - Condition2     7 4.2191e+10 4.4885e+12 61901
## - KitchenAbvGr   1 2.3936e+10 4.4702e+12 61901
## + Exterior2nd   15 1.3831e+10 4.4325e+12 61909
## - OverallCond    1 3.5156e+10 4.4814e+12 61909
## - X1stFlrSF      1 3.6511e+10 4.4828e+12 61909
## - TotRmsAbvGrd   1 3.8328e+10 4.4846e+12 61911
## - BsmtFinSF1     1 4.3840e+10 4.4901e+12 61914
## - Neighborhood  24 1.2678e+11 4.5731e+12 61922
## - X2ndFlrSF      1 9.5702e+10 4.5420e+12 61948
## - LotArea        1 1.1851e+11 4.5648e+12 61962
## - RoofMatl       7 2.1376e+11 4.6600e+12 62011
##
## Step:  AIC=61887.43
## SalePrice ~ MSSubClass + LotArea + LandContour + LandSlope +
##     Neighborhood + Condition2 + OverallQual + OverallCond + YearBuilt +
```

```
##     RoofMatl + ExterQual + BsmtExposure + BsmtFinSF1 + BsmtFinType2 +
##     BsmtUnfSF + X1stFlrSF + X2ndFlrSF + FullBath + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + GarageYrBlt + GarageArea + PavedDrive +
##     ScreenPorch + PoolArea + MiscVal + MoSold + SaleCondition
##
##                 Df  Sum of Sq        RSS   AIC
## <none>                        4.4491e+12 61887
## + BsmtQual       3 8.9358e+09 4.4401e+12 61888
## + BsmtFinSF2     1 2.7663e+09 4.4463e+12 61888
## + TotalBsmtSF    1 2.7663e+09 4.4463e+12 61888
## - BsmtFinType2   5 1.5910e+10 4.4650e+12 61888
## + Heating        5 1.4564e+10 4.4345e+12 61888
## + LotShape       3 8.4769e+09 4.4406e+12 61888
## - ScreenPorch    1 3.8750e+09 4.4529e+12 61888
## + Street         1 1.6394e+09 4.4474e+12 61888
## + Utilities      1 1.5752e+09 4.4475e+12 61888
## - BsmtExposure   3 1.0827e+10 4.4599e+12 61889
## + LotFrontage    1 1.3041e+09 4.4478e+12 61889
## + YearRemodAdd   1 1.1108e+09 4.4479e+12 61889
## + OpenPorchSF    1 8.7552e+08 4.4482e+12 61889
## - GarageYrBlt    1 5.2625e+09 4.4543e+12 61889
## + X3SsnPorch     1 5.1940e+08 4.4485e+12 61889
## + BsmtFullBath   1 4.7745e+08 4.4486e+12 61889
## + MasVnrArea     1 3.5316e+08 4.4487e+12 61889
## + HalfBath       1 3.1958e+08 4.4487e+12 61889
## + CentralAir     1 2.6267e+08 4.4488e+12 61889
## + BedroomAbvGr   1 1.9172e+08 4.4489e+12 61889
## + Fireplaces     1 1.3707e+08 4.4489e+12 61889
## + BsmtHalfBath   1 7.6609e+07 4.4490e+12 61889
## + GarageCars     1 6.0506e+07 4.4490e+12 61889
## + WoodDeckSF     1 4.5619e+07 4.4490e+12 61889
## + YrSold         1 4.2212e+07 4.4490e+12 61889
## + LowQualFinSF   1 2.4861e+07 4.4490e+12 61889
## + GrLivArea      1 2.4861e+07 4.4490e+12 61889
## - FullBath       1 6.0761e+09 4.4551e+12 61889
## + EnclosedPorch  1 9.2291e+06 4.4490e+12 61889
## + GarageFinish   2 2.5544e+09 4.4465e+12 61890
## + Electrical     4 7.5949e+09 4.4415e+12 61890
## + MSZoning       4 6.6830e+09 4.4424e+12 61891
## + BldgType       4 6.3391e+09 4.4427e+12 61891
## - MoSold         1 9.3430e+09 4.4584e+12 61892
## - PavedDrive     2 1.3537e+10 4.4626e+12 61892
## - OverallQual    1 1.1281e+10 4.4603e+12 61893
## + BsmtCond       3 6.8575e+08 4.4484e+12 61893
## + GarageQual     4 3.7153e+09 4.4453e+12 61893
## - SaleCondition  5 2.3836e+10 4.4729e+12 61893
## - GarageArea     1 1.2218e+10 4.4613e+12 61893
## + GarageCond     4 2.5525e+09 4.4465e+12 61894
## + Functional     6 8.6092e+09 4.4404e+12 61894
## + LotConfig      4 2.5059e+09 4.4466e+12 61894
## + BsmtFinType1   5 5.4430e+09 4.4436e+12 61894
## + HouseStyle     7 1.1043e+10 4.4380e+12 61894
```

```
## - KitchenQual    3 1.9517e+10 4.4686e+12 61894
## + SaleType       8 1.3516e+10 4.4355e+12 61895
## + HeatingQC      4 1.3338e+09 4.4477e+12 61895
## + ExterCond      4 1.1560e+09 4.4479e+12 61895
## - PoolArea       1 1.4991e+10 4.4640e+12 61895
## - LandSlope      2 1.8184e+10 4.4672e+12 61895
## + RoofStyle      5 2.5513e+09 4.4465e+12 61896
## + Condition1     8 1.1497e+10 4.4376e+12 61896
## + GarageType     5 2.0524e+09 4.4470e+12 61896
## + Foundation     5 1.8479e+09 4.4472e+12 61896
## - MSSubClass     1 1.6750e+10 4.4658e+12 61896
## - MiscVal        1 1.7183e+10 4.4662e+12 61897
## - ExterQual      3 2.3338e+10 4.4724e+12 61897
## - LandContour    3 2.3807e+10 4.4729e+12 61897
## - BsmtUnfSF      1 1.7732e+10 4.4668e+12 61897
## + Exterior1st   14 2.3275e+10 4.4258e+12 61900
## - Condition2     7 4.2032e+10 4.4911e+12 61901
## - KitchenAbvGr   1 2.3834e+10 4.4729e+12 61901
## - YearBuilt      1 2.3970e+10 4.4730e+12 61901
## - OverallCond    1 3.3951e+10 4.4830e+12 61908
## + Exterior2nd   15 1.3959e+10 4.4351e+12 61908
## - TotRmsAbvGrd   1 3.8240e+10 4.4873e+12 61910
## - BsmtFinSF1     1 4.1229e+10 4.4903e+12 61912
## - X1stFlrSF      1 4.2193e+10 4.4912e+12 61913
## - Neighborhood  24 1.2895e+11 4.5780e+12 61923
## - X2ndFlrSF      1 9.5485e+10 4.5445e+12 61947
## - LotArea        1 1.1892e+11 4.5680e+12 61962
## - RoofMatl       7 2.1222e+11 4.6613e+12 62009
```

```
# Summary of the final model
summary(step_model)
```

```
##
## Call:
## lm(formula = SalePrice ~ MSSubClass + LotArea + LandContour +
##     LandSlope + Neighborhood + Condition2 + OverallQual + OverallCond +
##     YearBuilt + RoofMatl + ExterQual + BsmtExposure + BsmtFinSF1 +
##     BsmtFinType2 + BsmtUnfSF + X1stFlrSF + X2ndFlrSF + FullBath +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + GarageYrBlt +
##     GarageArea + PavedDrive + ScreenPorch + PoolArea + MiscVal +
##     MoSold + SaleCondition, data = imputed_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -150093  -22047   -1844   21384  329534
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -6.039e+05  1.463e+05  -4.128 3.77e-05 ***
## MSSubClass           -8.097e+01  2.478e+01  -3.268 0.001098 **
## LotArea               1.112e+00  1.277e-01   8.707  < 2e-16 ***
## LandContourHLS       -5.086e+03  5.710e+03  -0.891 0.373168
## LandContourLow        7.884e+03  7.319e+03   1.077 0.281469
## LandContourLvl        9.940e+03  4.156e+03   2.392 0.016821 *
## LandSlopeMod          6.167e+02  4.495e+03   0.137 0.890887
## LandSlopeSev         -4.360e+04  1.310e+04  -3.328 0.000885 ***
## NeighborhoodBlueste   5.944e+03  1.503e+04   0.396 0.692473
## NeighborhoodBrDale   -4.920e+03  1.104e+04  -0.446 0.655808
## NeighborhoodBrkSide  -9.044e+02  9.669e+03  -0.094 0.925482
## NeighborhoodClearCr  -3.362e+03  1.062e+04  -0.317 0.751503
## NeighborhoodCollgCr  -6.399e+02  8.262e+03  -0.077 0.938264
## NeighborhoodCrawfor   1.662e+04  9.506e+03   1.748 0.080589 .
## NeighborhoodEdwards  -5.624e+03  8.838e+03  -0.636 0.524579
## NeighborhoodGilbert  -1.594e+03  8.600e+03  -0.185 0.852972
## NeighborhoodIDOTRR   -2.544e+02  9.852e+03  -0.026 0.979401
## NeighborhoodMeadowV   4.270e+03  1.063e+04   0.402 0.687973
## NeighborhoodMitchel  -1.906e+03  8.990e+03  -0.212 0.832093
## NeighborhoodNAmes    -8.738e+02  8.635e+03  -0.101 0.919404
## NeighborhoodNoRidge   3.451e+04  9.529e+03   3.622 0.000298 ***
## NeighborhoodNPkVill   2.590e+03  1.164e+04   0.223 0.823906
## NeighborhoodNridgHt   6.455e+03  8.519e+03   0.758 0.448682
## NeighborhoodNWAmes   -7.500e+03  8.949e+03  -0.838 0.402087
## NeighborhoodOldTown  -2.497e+03  9.335e+03  -0.267 0.789141
## NeighborhoodSawyer    9.775e+02  8.995e+03   0.109 0.913469
## NeighborhoodSawyerW   1.069e+03  8.750e+03   0.122 0.902746
## NeighborhoodSomerst   8.952e+02  8.290e+03   0.108 0.914017
## NeighborhoodStoneBr   2.629e+04  9.739e+03   2.699 0.006987 **
## NeighborhoodSWISU     2.393e+03  1.059e+04   0.226 0.821242
## NeighborhoodTimber    2.707e+03  9.305e+03   0.291 0.771158
## NeighborhoodVeenker  -7.883e+03  1.157e+04  -0.682 0.495564
## Condition2Feedr      -1.034e+04  2.115e+04  -0.489 0.624867
## Condition2Norm       -6.672e+03  1.820e+04  -0.367 0.713975
## Condition2PosA       -6.850e+04  2.739e+04  -2.501 0.012457 *
## Condition2PosN       -9.241e+04  2.735e+04  -3.379 0.000736 ***
```

```
## Condition2RRAe       -1.599e+04  4.571e+04  -0.350 0.726452
## Condition2RRAn       -3.491e+04  4.383e+04  -0.796 0.425865
## Condition2RRNn       -2.466e+04  3.359e+04  -0.734 0.462990
## OverallQual           2.829e+03  1.055e+03   2.682 0.007371 **
## OverallCond           3.700e+03  7.954e+02   4.652 3.44e-06 ***
## YearBuilt             2.656e+02  6.794e+01   3.909 9.49e-05 ***
## RoofMatlCompShg       3.777e+05  4.400e+04   8.583  < 2e-16 ***
## RoofMatlMembran       4.795e+05  6.125e+04   7.829 6.90e-15 ***
## RoofMatlMetal         4.394e+05  6.116e+04   7.184 8.61e-13 ***
## RoofMatlRoll          3.394e+05  5.940e+04   5.713 1.23e-08 ***
## RoofMatlTar&Grv       3.794e+05  4.478e+04   8.472  < 2e-16 ***
## RoofMatlWdShake       3.863e+05  4.610e+04   8.379  < 2e-16 ***
## RoofMatlWdShngl       4.910e+05  4.635e+04  10.593  < 2e-16 ***
## ExterQualFa          -8.593e+03  9.525e+03  -0.902 0.367059
## ExterQualGd          -1.914e+04  5.398e+03  -3.546 0.000398 ***
## ExterQualTA          -1.928e+04  6.034e+03  -3.196 0.001409 **
## BsmtExposureGd       -8.389e+02  3.372e+03  -0.249 0.803548
## BsmtExposureMn       -3.314e+03  3.319e+03  -0.998 0.318138
## BsmtExposureNo       -5.813e+03  2.375e+03  -2.447 0.014465 *
## BsmtFinSF1            1.661e+01  3.240e+00   5.126 3.15e-07 ***
## BsmtFinType2BLQ      -1.968e+04  7.452e+03  -2.641 0.008318 **
## BsmtFinType2GLQ      -2.363e+04  9.042e+03  -2.614 0.009003 **
## BsmtFinType2LwQ      -1.612e+04  7.170e+03  -2.248 0.024671 *
## BsmtFinType2Rec      -1.337e+04  6.881e+03  -1.943 0.052089 .
## BsmtFinType2Unf      -1.598e+04  5.866e+03  -2.723 0.006500 **
## BsmtUnfSF             1.069e+01  3.180e+00   3.362 0.000784 ***
## X1stFlrSF             2.298e+01  4.431e+00   5.186 2.30e-07 ***
## X2ndFlrSF             2.861e+01  3.667e+00   7.802 8.52e-15 ***
## FullBath              4.244e+03  2.156e+03   1.968 0.049161 *
## KitchenAbvGr         -1.751e+04  4.492e+03  -3.898 9.93e-05 ***
## KitchenQualFa        -1.953e+04  6.753e+03  -2.892 0.003852 **
## KitchenQualGd        -1.306e+04  4.011e+03  -3.256 0.001145 **
## KitchenQualTA        -1.463e+04  4.464e+03  -3.278 0.001057 **
## TotRmsAbvGrd          4.479e+03  9.071e+02   4.937 8.39e-07 ***
## GarageYrBlt          -9.336e+01  5.097e+01  -1.832 0.067126 .
## GarageArea            1.366e+01  4.894e+00   2.791 0.005294 **
## PavedDriveP          -1.205e+04  5.934e+03  -2.030 0.042418 *
## PavedDriveY          -1.001e+04  3.556e+03  -2.815 0.004909 **
## ScreenPorch           2.124e+01  1.352e+01   1.572 0.116143
## PoolArea              6.794e+01  2.198e+01   3.091 0.002013 **
## MiscVal              -4.556e+00  1.377e+00  -3.310 0.000946 ***
## MoSold                6.710e+02  2.750e+02   2.440 0.014732 *
## SaleConditionAdjLand  1.277e+04  1.214e+04   1.051 0.293146
## SaleConditionAlloca   1.044e+04  9.032e+03   1.156 0.247739
## SaleConditionFamily   3.506e+02  6.576e+03   0.053 0.957484
## SaleConditionNormal   2.104e+03  3.080e+03   0.683 0.494660
## SaleConditionPartial  1.317e+04  4.306e+03   3.057 0.002253 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 39610 on 2836 degrees of freedom
```

```
## Multiple R-squared:  0.5369, Adjusted R-squared:  0.5236
## F-statistic:  40.1 on 82 and 2836 DF,  p-value: < 2.2e-16
```

```
# Check for normality of residuals
shapiro.test(resid(step_model))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid(step_model)
## W = 0.92902, p-value < 2.2e-16
```

```
# Check for independence of residuals
durbinWatsonTest(step_model)
```

```
##  lag Autocorrelation D-W Statistic p-value
##    1       0.2486434      1.502446       0
##  Alternative hypothesis: rho != 0
```

```
# Check for homoscedasticity

bptest(step_model)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  step_model
## BP = 1119.1, df = 82, p-value < 2.2e-16
```

```
# Check for multicollinearity
vif(step_model)
```

```
##                     GVIF Df GVIF^(1/(2*Df))
## MSSubClass      2.064744  1        1.436922
## LotArea         1.887326  1        1.373800
## LandContour     2.822526  3        1.188793
## LandSlope       2.602580  2        1.270138
## Neighborhood  123.590477 24        1.105562
## Condition2      1.490756  7        1.028931
## OverallQual     4.115337  1        2.028629
## OverallCond     1.457929  1        1.207447
## YearBuilt       7.877778  1        2.806738
## RoofMatl        2.011339  7        1.051181
## ExterQual       6.960366  3        1.381779
## BsmtExposure    1.933101  3        1.116115
## BsmtFinSF1      4.054419  1        2.013559
## BsmtFinType2    1.730427  5        1.056368
## BsmtUnfSF       3.634288  1        1.906381
## X1stFlrSF       5.622070  1        2.371090
## X2ndFlrSF       4.596878  1        2.144033
## FullBath        2.644486  1        1.626188
## KitchenAbvGr    1.726491  1        1.313960
## KitchenQual     4.940356  3        1.305048
## TotRmsAbvGrd    3.769767  1        1.941589
## GarageYrBlt     3.091693  1        1.758321
## GarageArea      2.069857  1        1.438700
## PavedDrive      1.736300  2        1.147906
## ScreenPorch     1.072595  1        1.035662
## PoolArea        1.142784  1        1.069011
## MiscVal         1.134625  1        1.065188
## MoSold          1.036467  1        1.018070
## SaleCondition   1.819925  5        1.061709
```

## Step-wise Regression with log y

```
# Fit the full linear regression model
full_model_log_model <- lm(log(SalePrice) ~ ., data = imputed_data)

# Perform stepwise regression
step_model_log_model <- stepAIC(full_model_log_model, direction = "both")
```

```
## Start:  AIC=-9128.45
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
##     HouseStyle + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofStyle + RoofMatl + Exterior1st + Exterior2nd + MasVnrArea +
##     ExterQual + ExterCond + Foundation + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + TotalBsmtSF + Heating + HeatingQC +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
##     GrLivArea + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Functional + Fireplaces + GarageType + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleType + SaleCondition
##
##
## Step:  AIC=-9128.45
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
##     HouseStyle + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofStyle + RoofMatl + Exterior1st + Exterior2nd + MasVnrArea +
##     ExterQual + ExterCond + Foundation + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + TotalBsmtSF + Heating + HeatingQC +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
##     BsmtFullBath + BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Functional +
##     Fireplaces + GarageType + GarageYrBlt + GarageFinish + GarageCars +
##     GarageArea + GarageQual + GarageCond + PavedDrive + WoodDeckSF +
##     OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleType + SaleCondition
##
##
## Step:  AIC=-9128.45
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
##     HouseStyle + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofStyle + RoofMatl + Exterior1st + Exterior2nd + MasVnrArea +
##     ExterQual + ExterCond + Foundation + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageType +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleType + SaleCondition
```

```
##
##                 Df Sum of Sq    RSS     AIC
## - Exterior2nd   15   0.43134 109.89 -9147.0
## - Neighborhood  24   1.44404 110.90 -9138.2
## - Exterior1st   14   0.69069 110.15 -9138.1
## - HouseStyle     7   0.16573 109.62 -9138.0
## - Functional     6   0.16692 109.62 -9136.0
## - GarageType     5   0.09336 109.55 -9136.0
## - ExterCond      4   0.01995 109.48 -9135.9
## - Foundation     5   0.12424 109.58 -9135.1
## - Condition1     8   0.35015 109.81 -9135.1
## - SaleType       8   0.37239 109.83 -9134.5
## - HeatingQC      4   0.07950 109.54 -9134.3
## - RoofStyle      5   0.15941 109.62 -9134.2
## - BsmtCond       3   0.00998 109.47 -9134.2
## - GarageQual     4   0.10217 109.56 -9133.7
## - BldgType       4   0.10312 109.56 -9133.7
## - BsmtExposure   3   0.02933 109.49 -9133.7
## - LotConfig      4   0.11852 109.58 -9133.3
## - ExterQual      3   0.05440 109.51 -9133.0
## - SaleCondition  5   0.25029 109.71 -9131.8
## - BsmtQual       3   0.10117 109.56 -9131.8
## - BsmtFinType2   5   0.25822 109.71 -9131.6
## - GarageFinish   2   0.03426 109.49 -9131.5
## - CentralAir     1   0.00014 109.46 -9130.5
## - LowQualFinSF   1   0.00065 109.46 -9130.4
## - MSSubClass     1   0.00092 109.46 -9130.4
## - OpenPorchSF    1   0.00166 109.46 -9130.4
## - GarageCars     1   0.00224 109.46 -9130.4
## - MasVnrArea     1   0.00264 109.46 -9130.4
## - HalfBath       1   0.00822 109.46 -9130.2
## - BsmtFullBath   1   0.00958 109.47 -9130.2
## - YearRemodAdd   1   0.01039 109.47 -9130.2
## - X3SsnPorch     1   0.01895 109.47 -9129.9
## - EnclosedPorch  1   0.02036 109.48 -9129.9
## - GarageCond     4   0.24688 109.70 -9129.9
## - BsmtHalfBath   1   0.02641 109.48 -9129.8
## - YrSold         1   0.03664 109.49 -9129.5
## - WoodDeckSF     1   0.03837 109.50 -9129.4
## - Street         1   0.03951 109.50 -9129.4
## - Fireplaces     1   0.04950 109.51 -9129.1
## - LotFrontage    1   0.05247 109.51 -9129.1
## - BsmtFinSF2     1   0.05559 109.51 -9129.0
## - PoolArea       1   0.06106 109.52 -9128.8
## - Utilities      1   0.07005 109.53 -9128.6
## <none>                       109.46 -9128.5
## - LotShape       3   0.22749 109.68 -9128.4
## - Condition2     7   0.54548 110.00 -9127.9
## - BsmtFinType1   5   0.39690 109.85 -9127.9
## - FullBath       1   0.12040 109.58 -9127.2
## - BsmtFinSF1     1   0.13563 109.59 -9126.8
## - MSZoning       4   0.37511 109.83 -9126.5
```

```
## - LandContour    3   0.32996 109.79 -9125.7
## - BedroomAbvGr   1   0.20305 109.66 -9125.0
## - MiscVal        1   0.20413 109.66 -9125.0
## - KitchenQual    3   0.36475 109.82 -9124.7
## - GarageArea     1   0.21765 109.67 -9124.7
## - ScreenPorch    1   0.21871 109.67 -9124.6
## - Electrical     4   0.44609 109.90 -9124.6
## - BsmtUnfSF      1   0.26549 109.72 -9123.4
## - OverallQual    1   0.31272 109.77 -9122.1
## - GarageYrBlt    1   0.31500 109.77 -9122.1
## - MoSold         1   0.35412 109.81 -9121.0
## - LandSlope      2   0.48033 109.94 -9119.7
## - PavedDrive     2   0.55607 110.01 -9117.7
## - X1stFlrSF      1   0.51705 109.97 -9116.7
## - KitchenAbvGr   1   0.54731 110.00 -9115.9
## - TotRmsAbvGrd   1   0.64840 110.11 -9113.2
## - X2ndFlrSF      1   0.66366 110.12 -9112.8
## - YearBuilt      1   0.75206 110.21 -9110.5
## - Heating        5   1.06110 110.52 -9110.3
## - OverallCond    1   0.89873 110.36 -9106.6
## - RoofMatl       7   2.02897 111.48 -9088.8
## - LotArea        1   1.87560 111.33 -9080.9
##
## Step:  AIC=-9146.97
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
##     HouseStyle + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofStyle + RoofMatl + Exterior1st + MasVnrArea + ExterQual +
##     ExterCond + Foundation + BsmtQual + BsmtCond + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + HeatingQC + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Functional + Fireplaces + GarageType + GarageYrBlt +
##     GarageFinish + GarageCars + GarageArea + GarageQual + GarageCond +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleType +
##     SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - Exterior1st   14   0.54856 110.44 -9160.4
## - HouseStyle     7   0.17355 110.06 -9156.4
## - Neighborhood  24   1.52032 111.41 -9154.9
## - ExterCond      4   0.01073 109.90 -9154.7
## - Functional     6   0.16643 110.05 -9154.6
## - Condition1     8   0.33260 110.22 -9154.2
## - GarageType     5   0.11071 110.00 -9154.0
## - SaleType       8   0.36452 110.25 -9153.3
## - Foundation     5   0.14661 110.03 -9153.1
## - BsmtCond       3   0.00710 109.89 -9152.8
## - HeatingQC      4   0.08867 109.98 -9152.6
```

```
## - RoofStyle      5   0.17139 110.06 -9152.4
## - BsmtExposure   3   0.03390 109.92 -9152.1
## - LotConfig      4   0.11187 110.00 -9152.0
## - GarageQual     4   0.11207 110.00 -9152.0
## - BldgType       4   0.11489 110.00 -9151.9
## - ExterQual      3   0.05725 109.94 -9151.5
## - BsmtQual       3   0.09967 109.99 -9150.3
## - SaleCondition  5   0.25491 110.14 -9150.2
## - GarageFinish   2   0.03940 109.93 -9149.9
## - BsmtFinType2   5   0.26818 110.16 -9149.9
## - CentralAir     1   0.00051 109.89 -9149.0
## - LowQualFinSF   1   0.00062 109.89 -9149.0
## - GarageCars     1   0.00181 109.89 -9148.9
## - MSSubClass     1   0.00190 109.89 -9148.9
## - MasVnrArea     1   0.00190 109.89 -9148.9
## - HalfBath       1   0.00355 109.89 -9148.9
## - YearRemodAdd   1   0.00436 109.89 -9148.9
## - OpenPorchSF    1   0.00437 109.89 -9148.9
## - BsmtFullBath   1   0.00517 109.89 -9148.8
## - GarageCond     4   0.24063 110.13 -9148.6
## - BsmtHalfBath   1   0.01597 109.90 -9148.6
## - EnclosedPorch  1   0.02140 109.91 -9148.4
## - X3SsnPorch     1   0.02996 109.92 -9148.2
## - Street         1   0.03198 109.92 -9148.1
## - WoodDeckSF     1   0.03251 109.92 -9148.1
## - YrSold         1   0.03607 109.92 -9148.0
## - BsmtFinSF2     1   0.04887 109.94 -9147.7
## - PoolArea       1   0.05264 109.94 -9147.6
## - Fireplaces     1   0.05347 109.94 -9147.6
## - LotFrontage    1   0.06218 109.95 -9147.3
## - Utilities      1   0.06876 109.96 -9147.1
## <none>                       109.89 -9147.0
## - LotShape       3   0.22649 110.11 -9147.0
## - BsmtFinType1   5   0.39108 110.28 -9146.6
## - Condition2     7   0.56339 110.45 -9146.0
## - FullBath       1   0.11174 110.00 -9146.0
## - BsmtFinSF1     1   0.13817 110.03 -9145.3
## - KitchenQual    3   0.32609 110.21 -9144.3
## - MSZoning       4   0.41513 110.30 -9144.0
## - MiscVal        1   0.20088 110.09 -9143.6
## - Electrical     4   0.42814 110.32 -9143.6
## - LandContour    3   0.35614 110.24 -9143.5
## - GarageArea     1   0.21228 110.10 -9143.3
## - ScreenPorch    1   0.21565 110.10 -9143.3
## - BedroomAbvGr   1   0.22322 110.11 -9143.1
## - BsmtUnfSF      1   0.24005 110.13 -9142.6
## - GarageYrBlt    1   0.27338 110.16 -9141.7
## - MoSold         1   0.31563 110.20 -9140.6
## - OverallQual    1   0.34472 110.23 -9139.8
## - LandSlope      2   0.47625 110.36 -9138.4
## - PavedDrive     2   0.53069 110.42 -9136.9
## - X1stFlrSF      1   0.52662 110.41 -9135.0
```

```
## - KitchenAbvGr   1   0.56285 110.45 -9134.1
## - TotRmsAbvGrd   1   0.64288 110.53 -9131.9
## - X2ndFlrSF      1   0.67249 110.56 -9131.2
## - YearBuilt      1   0.73768 110.62 -9129.4
## + Exterior2nd   15   0.43134 109.46 -9128.5
## - Heating        5   1.09272 110.98 -9128.1
## - OverallCond    1   0.93003 110.82 -9124.4
## - RoofMatl       7   2.05616 111.94 -9106.9
## - LotArea        1   1.90337 111.79 -9098.8
##
## Step:  AIC=-9160.44
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
##     HouseStyle + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofStyle + RoofMatl + MasVnrArea + ExterQual + ExterCond +
##     Foundation + BsmtQual + BsmtCond + BsmtExposure + BsmtFinType1 +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     HeatingQC + CentralAir + Electrical + X1stFlrSF + X2ndFlrSF +
##     LowQualFinSF + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Functional + Fireplaces + GarageType + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleType + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - HouseStyle     7   0.17804 110.61 -9169.7
## - Neighborhood  24   1.50593 111.94 -9168.9
## - ExterCond      4   0.01831 110.45 -9168.0
## - GarageType     5   0.11374 110.55 -9167.4
## - Condition1     8   0.34848 110.78 -9167.2
## - Foundation     5   0.12764 110.56 -9167.1
## - Functional     6   0.20432 110.64 -9167.0
## - HeatingQC      4   0.06490 110.50 -9166.7
## - SaleType       8   0.37126 110.81 -9166.6
## - BsmtCond       3   0.00629 110.44 -9166.3
## - RoofStyle      5   0.15911 110.59 -9166.2
## - BsmtExposure   3   0.02719 110.46 -9165.7
## - GarageQual     4   0.11666 110.55 -9165.4
## - LotConfig      4   0.11992 110.56 -9165.3
## - BldgType       4   0.12110 110.56 -9165.2
## - ExterQual      3   0.04763 110.48 -9165.2
## - BsmtQual       3   0.09051 110.53 -9164.0
## - BsmtFinType2   5   0.26383 110.70 -9163.5
## - SaleCondition  5   0.26418 110.70 -9163.5
## - GarageFinish   2   0.05005 110.49 -9163.1
## - GarageCond     4   0.22247 110.66 -9162.6
## - BsmtFullBath   1   0.00123 110.44 -9162.4
## - LowQualFinSF   1   0.00131 110.44 -9162.4
## - CentralAir     1   0.00170 110.44 -9162.4
## - MSSubClass     1   0.00285 110.44 -9162.4
```

```
## - GarageCars     1   0.00291 110.44 -9162.4
## - OpenPorchSF    1   0.00402 110.44 -9162.3
## - HalfBath       1   0.00579 110.44 -9162.3
## - BsmtHalfBath   1   0.01041 110.45 -9162.2
## - MasVnrArea     1   0.01052 110.45 -9162.2
## - YearRemodAdd   1   0.01697 110.45 -9162.0
## - WoodDeckSF     1   0.02138 110.46 -9161.9
## - EnclosedPorch  1   0.02947 110.47 -9161.7
## - X3SsnPorch     1   0.03258 110.47 -9161.6
## - Street         1   0.04023 110.48 -9161.4
## - YrSold         1   0.04062 110.48 -9161.4
## - PoolArea       1   0.04161 110.48 -9161.3
## - BsmtFinSF2     1   0.05212 110.49 -9161.1
## - LotFrontage    1   0.05435 110.49 -9161.0
## - Utilities      1   0.07043 110.51 -9160.6
## - LotShape       3   0.22301 110.66 -9160.6
## - Fireplaces     1   0.07507 110.51 -9160.5
## <none>                       110.44 -9160.4
## - BsmtFinType1   5   0.38923 110.83 -9160.2
## - Condition2     7   0.55421 110.99 -9159.8
## - FullBath       1   0.12379 110.56 -9159.2
## - BsmtFinSF1     1   0.13753 110.57 -9158.8
## - MSZoning       4   0.41186 110.85 -9157.6
## - Electrical     4   0.42398 110.86 -9157.3
## - GarageArea     1   0.20847 110.64 -9156.9
## - MiscVal        1   0.21235 110.65 -9156.8
## - ScreenPorch    1   0.21957 110.66 -9156.6
## - KitchenQual    3   0.37274 110.81 -9156.6
## - LandContour    3   0.37637 110.81 -9156.5
## - BsmtUnfSF      1   0.22920 110.67 -9156.4
## - BedroomAbvGr   1   0.23279 110.67 -9156.3
## - GarageYrBlt    1   0.30806 110.74 -9154.3
## - MoSold         1   0.31406 110.75 -9154.1
## - OverallQual    1   0.35686 110.79 -9153.0
## - LandSlope      2   0.45746 110.89 -9152.4
## - PavedDrive     2   0.51557 110.95 -9150.8
## + Exterior1st   14   0.54856 109.89 -9147.0
## - KitchenAbvGr   1   0.60358 111.04 -9146.5
## - TotRmsAbvGrd   1   0.61619 111.05 -9146.2
## - X1stFlrSF      1   0.65686 111.09 -9145.1
## - X2ndFlrSF      1   0.68614 111.12 -9144.4
## - YearBuilt      1   0.72308 111.16 -9143.4
## - Heating        5   1.09927 111.53 -9141.5
## + Exterior2nd   15   0.28921 110.15 -9138.1
## - OverallCond    1   0.94591 111.38 -9137.5
## - RoofMatl       7   2.15717 112.59 -9118.0
## - LotArea        1   1.97028 112.41 -9110.8
##
## Step:  AIC=-9169.74
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
```

```
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + MasVnrArea + ExterQual + ExterCond + Foundation +
##     BsmtQual + BsmtCond + BsmtExposure + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
##     BsmtFullBath + BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Functional +
##     Fireplaces + GarageType + GarageYrBlt + GarageFinish + GarageCars +
##     GarageArea + GarageQual + GarageCond + PavedDrive + WoodDeckSF +
##     OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleType + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - ExterCond      4   0.01217 110.63 -9177.4
## - GarageType     5   0.09339 110.71 -9177.3
## - Neighborhood  24   1.55976 112.17 -9176.9
## - Condition1     8   0.34865 110.96 -9176.6
## - Foundation     5   0.12145 110.74 -9176.5
## - HeatingQC      4   0.04780 110.66 -9176.5
## - Functional     6   0.20114 110.81 -9176.4
## - SaleType       8   0.37319 110.99 -9175.9
## - RoofStyle      5   0.15348 110.77 -9175.7
## - BsmtCond       3   0.00653 110.62 -9175.6
## - BsmtExposure   3   0.03135 110.64 -9174.9
## - LotConfig      4   0.11226 110.73 -9174.8
## - GarageQual     4   0.11588 110.73 -9174.7
## - BldgType       4   0.11667 110.73 -9174.7
## - ExterQual      3   0.04194 110.66 -9174.6
## - BsmtQual       3   0.09837 110.71 -9173.1
## - SaleCondition  5   0.25354 110.87 -9173.1
## - BsmtFinType2   5   0.25717 110.87 -9173.0
## - GarageFinish   2   0.04425 110.66 -9172.6
## - GarageCond     4   0.20321 110.82 -9172.4
## - MSSubClass     1   0.00000 110.61 -9171.7
## - LowQualFinSF   1   0.00024 110.61 -9171.7
## - HalfBath       1   0.00164 110.62 -9171.7
## - BsmtFullBath   1   0.00170 110.62 -9171.7
## - CentralAir     1   0.00233 110.62 -9171.7
## - OpenPorchSF    1   0.00272 110.62 -9171.7
## - GarageCars     1   0.00451 110.62 -9171.6
## - MasVnrArea     1   0.00893 110.62 -9171.5
## - BsmtHalfBath   1   0.00995 110.62 -9171.5
## - YearRemodAdd   1   0.01712 110.63 -9171.3
## - WoodDeckSF     1   0.02131 110.64 -9171.2
## - EnclosedPorch  1   0.02676 110.64 -9171.0
## - X3SsnPorch     1   0.03391 110.65 -9170.8
## - YrSold         1   0.03661 110.65 -9170.8
## - Street         1   0.03690 110.65 -9170.8
## - BsmtFinSF2     1   0.04699 110.66 -9170.5
## - PoolArea       1   0.04945 110.66 -9170.4
## - LotFrontage    1   0.05772 110.67 -9170.2
## - LotShape       3   0.21550 110.83 -9170.1
```

```
## - Utilities      1   0.07375 110.69 -9169.8
## - Fireplaces     1   0.07544 110.69 -9169.7
## <none>                       110.61 -9169.7
## - Condition2     7   0.53309 111.15 -9169.7
## - BsmtFinType1   5   0.41461 111.03 -9168.8
## - FullBath       1   0.12736 110.74 -9168.4
## - BsmtFinSF1     1   0.13122 110.75 -9168.3
## - Electrical     4   0.41479 111.03 -9166.8
## - MiscVal        1   0.21415 110.83 -9166.1
## - KitchenQual    3   0.36756 110.98 -9166.1
## - GarageArea     1   0.22265 110.84 -9165.9
## - MSZoning       4   0.45089 111.06 -9165.9
## - ScreenPorch    1   0.22327 110.84 -9165.9
## - BsmtUnfSF      1   0.22715 110.84 -9165.7
## - BedroomAbvGr   1   0.23797 110.85 -9165.5
## - LandContour    3   0.40115 111.02 -9165.2
## - MoSold         1   0.31213 110.93 -9163.5
## - GarageYrBlt    1   0.31578 110.93 -9163.4
## - OverallQual    1   0.33226 110.95 -9163.0
## - LandSlope      2   0.45729 111.07 -9161.7
## + HouseStyle     7   0.17804 110.44 -9160.4
## - PavedDrive     2   0.52465 111.14 -9159.9
## + Exterior1st   14   0.55305 110.06 -9156.4
## - KitchenAbvGr   1   0.60812 111.22 -9155.7
## - TotRmsAbvGrd   1   0.60990 111.22 -9155.7
## - X1stFlrSF      1   0.74159 111.36 -9152.2
## - YearBuilt      1   0.77463 111.39 -9151.4
## - Heating        5   1.16044 111.78 -9149.3
## - X2ndFlrSF      1   0.88116 111.50 -9148.6
## + Exterior2nd   15   0.28626 110.33 -9147.3
## - OverallCond    1   0.93066 111.55 -9147.3
## - RoofMatl       7   2.20515 112.82 -9126.1
## - LotArea        1   1.94950 112.56 -9120.7
##
## Step:  AIC=-9177.42
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + MasVnrArea + ExterQual + Foundation + BsmtQual +
##     BsmtCond + BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageType +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleType + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - GarageType     5   0.09277 110.72 -9185.0
```

```
## - Neighborhood  24   1.56038 112.19 -9184.5
## - HeatingQC      4   0.04728 110.67 -9184.2
## - Condition1     8   0.35108 110.98 -9184.2
## - Foundation     5   0.12475 110.75 -9184.1
## - Functional     6   0.20210 110.83 -9184.1
## - SaleType       8   0.37168 111.00 -9183.6
## - RoofStyle      5   0.15405 110.78 -9183.4
## - BsmtCond       3   0.00610 110.63 -9183.3
## - BsmtExposure   3   0.03219 110.66 -9182.6
## - LotConfig      4   0.11088 110.74 -9182.5
## - ExterQual      3   0.03844 110.67 -9182.4
## - BldgType       4   0.11585 110.74 -9182.4
## - GarageQual     4   0.11780 110.74 -9182.3
## - SaleCondition  5   0.25235 110.88 -9180.8
## - BsmtQual       3   0.10062 110.73 -9180.8
## - BsmtFinType2   5   0.26061 110.89 -9180.5
## - GarageFinish   2   0.04398 110.67 -9180.3
## - GarageCond     4   0.20110 110.83 -9180.1
## - MSSubClass     1   0.00000 110.63 -9179.4
## - LowQualFinSF   1   0.00016 110.63 -9179.4
## - HalfBath       1   0.00144 110.63 -9179.4
## - BsmtFullBath   1   0.00165 110.63 -9179.4
## - OpenPorchSF    1   0.00247 110.63 -9179.4
## - CentralAir     1   0.00252 110.63 -9179.3
## - GarageCars     1   0.00400 110.63 -9179.3
## - BsmtHalfBath   1   0.00984 110.64 -9179.2
## - MasVnrArea     1   0.00999 110.64 -9179.2
## - YearRemodAdd   1   0.01873 110.64 -9178.9
## - WoodDeckSF     1   0.02039 110.65 -9178.9
## - EnclosedPorch  1   0.02565 110.65 -9178.7
## - X3SsnPorch     1   0.03372 110.66 -9178.5
## - Street         1   0.03675 110.66 -9178.4
## - YrSold         1   0.03737 110.66 -9178.4
## - BsmtFinSF2     1   0.04703 110.67 -9178.2
## - PoolArea       1   0.04836 110.67 -9178.1
## - LotFrontage    1   0.05902 110.69 -9177.9
## - LotShape       3   0.21601 110.84 -9177.7
## - Utilities      1   0.07285 110.70 -9177.5
## - Fireplaces     1   0.07428 110.70 -9177.5
## <none>                       110.63 -9177.4
## - Condition2     7   0.53974 111.17 -9177.2
## - BsmtFinType1   5   0.41142 111.04 -9176.6
## - FullBath       1   0.12172 110.75 -9176.2
## - BsmtFinSF1     1   0.13340 110.76 -9175.9
## - Electrical     4   0.42498 111.05 -9174.2
## - KitchenQual    3   0.36557 110.99 -9173.8
## - MiscVal        1   0.21614 110.84 -9173.7
## - GarageArea     1   0.21822 110.84 -9173.7
## - MSZoning       4   0.45031 111.08 -9173.6
## - ScreenPorch    1   0.22264 110.85 -9173.5
## - BsmtUnfSF      1   0.23111 110.86 -9173.3
## - BedroomAbvGr   1   0.23573 110.86 -9173.2
```

```
## - LandContour    3   0.39737 111.02 -9172.9
## - GarageYrBlt    1   0.31290 110.94 -9171.2
## - MoSold         1   0.31653 110.94 -9171.1
## - OverallQual    1   0.34241 110.97 -9170.4
## + ExterCond      4   0.01217 110.61 -9169.7
## - LandSlope      2   0.45767 111.08 -9169.4
## + HouseStyle     7   0.17189 110.45 -9168.0
## - PavedDrive     2   0.52417 111.15 -9167.6
## + Exterior1st   14   0.55905 110.07 -9164.2
## - KitchenAbvGr   1   0.60631 111.23 -9163.5
## - TotRmsAbvGrd   1   0.61079 111.24 -9163.3
## - X1stFlrSF      1   0.74505 111.37 -9159.8
## - YearBuilt      1   0.78879 111.42 -9158.7
## - Heating        5   1.16512 111.79 -9156.8
## - X2ndFlrSF      1   0.88750 111.51 -9156.1
## + Exterior2nd   15   0.29033 110.34 -9155.1
## - OverallCond    1   0.96418 111.59 -9154.1
## - RoofMatl       7   2.20703 112.83 -9133.8
## - LotArea        1   1.95379 112.58 -9128.3
##
## Step:  AIC=-9184.97
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Neighborhood + Condition1 + Condition2 + BldgType +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofStyle +
##     RoofMatl + MasVnrArea + ExterQual + Foundation + BsmtQual +
##     BsmtCond + BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageYrBlt +
##     GarageFinish + GarageCars + GarageArea + GarageQual + GarageCond +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleType +
##     SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - Neighborhood  24   1.55863 112.28 -9192.2
## - Functional     6   0.19538 110.91 -9191.8
## - Condition1     8   0.34794 111.07 -9191.8
## - HeatingQC      4   0.04788 110.77 -9191.7
## - Foundation     5   0.12784 110.85 -9191.6
## - SaleType       8   0.37655 111.10 -9191.1
## - RoofStyle      5   0.15132 110.87 -9191.0
## - BsmtCond       3   0.00633 110.72 -9190.8
## - BldgType       4   0.10335 110.82 -9190.2
## - GarageQual     4   0.10936 110.83 -9190.1
## - BsmtExposure   3   0.03356 110.75 -9190.1
## - ExterQual      3   0.03468 110.75 -9190.1
## - LotConfig      4   0.11565 110.83 -9189.9
## - BsmtQual       3   0.10086 110.82 -9188.3
## - SaleCondition  5   0.25447 110.97 -9188.3
```

```
## - BsmtFinType2   5   0.26168 110.98 -9188.1
## - GarageFinish   2   0.04328 110.76 -9187.8
## - GarageCond     4   0.19614 110.92 -9187.8
## - MSSubClass     1   0.00002 110.72 -9187.0
## - LowQualFinSF   1   0.00015 110.72 -9187.0
## - BsmtFullBath   1   0.00116 110.72 -9186.9
## - HalfBath       1   0.00152 110.72 -9186.9
## - CentralAir     1   0.00209 110.72 -9186.9
## - OpenPorchSF    1   0.00237 110.72 -9186.9
## - GarageCars     1   0.00656 110.73 -9186.8
## - BsmtHalfBath   1   0.00807 110.73 -9186.8
## - MasVnrArea     1   0.00912 110.73 -9186.7
## - WoodDeckSF     1   0.01937 110.74 -9186.5
## - YearRemodAdd   1   0.02246 110.74 -9186.4
## - EnclosedPorch  1   0.02431 110.74 -9186.3
## - X3SsnPorch     1   0.03418 110.75 -9186.1
## - YrSold         1   0.03756 110.76 -9186.0
## - Street         1   0.03884 110.76 -9185.9
## - BsmtFinSF2     1   0.05187 110.77 -9185.6
## - PoolArea       1   0.05266 110.77 -9185.6
## - LotFrontage    1   0.05783 110.78 -9185.4
## - Fireplaces     1   0.07276 110.79 -9185.1
## <none>                       110.72 -9185.0
## - LotShape       3   0.22782 110.95 -9185.0
## - Utilities      1   0.07650 110.80 -9185.0
## - Condition2     7   0.53792 111.26 -9184.8
## - BsmtFinType1   5   0.42389 111.14 -9183.8
## - FullBath       1   0.11994 110.84 -9183.8
## - BsmtFinSF1     1   0.15254 110.87 -9183.0
## - Electrical     4   0.42707 111.15 -9181.7
## - KitchenQual    3   0.36622 111.08 -9181.3
## - MiscVal        1   0.22285 110.94 -9181.1
## - ScreenPorch    1   0.22419 110.94 -9181.1
## - LandContour    3   0.38117 111.10 -9180.9
## - MSZoning       4   0.45838 111.18 -9180.9
## - GarageArea     1   0.23565 110.95 -9180.8
## - BsmtUnfSF      1   0.25519 110.97 -9180.2
## - BedroomAbvGr   1   0.25633 110.97 -9180.2
## - GarageYrBlt    1   0.30535 111.02 -9178.9
## - MoSold         1   0.31475 111.03 -9178.7
## - OverallQual    1   0.35423 111.07 -9177.6
## + GarageType     5   0.09277 110.63 -9177.4
## + ExterCond      4   0.01155 110.71 -9177.3
## - LandSlope      2   0.44897 111.17 -9177.2
## + HouseStyle     7   0.15215 110.57 -9175.0
## - PavedDrive     2   0.54481 111.26 -9174.6
## + Exterior1st   14   0.55876 110.16 -9171.7
## - KitchenAbvGr   1   0.60215 111.32 -9171.1
## - TotRmsAbvGrd   1   0.61642 111.33 -9170.8
## - X1stFlrSF      1   0.73413 111.45 -9167.7
## - YearBuilt      1   0.78974 111.51 -9166.2
## - Heating        5   1.16522 111.88 -9164.4
```

```
## - X2ndFlrSF      1   0.86885 111.59 -9164.2
## + Exterior2nd   15   0.30452 110.42 -9163.0
## - OverallCond    1   0.98872 111.71 -9161.0
## - RoofMatl       7   2.21398 112.93 -9141.2
## - LotArea        1   1.95958 112.68 -9135.8
##
## Step:  AIC=-9192.16
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition1 + Condition2 + BldgType + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofStyle + RoofMatl +
##     MasVnrArea + ExterQual + Foundation + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageYrBlt +
##     GarageFinish + GarageCars + GarageArea + GarageQual + GarageCond +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleType +
##     SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - SaleType       8   0.33454 112.61 -9199.5
## - Functional     6   0.20830 112.49 -9198.8
## - HeatingQC      4   0.06942 112.35 -9198.4
## - RoofStyle      5   0.15063 112.43 -9198.3
## - Condition1     8   0.38226 112.66 -9198.2
## - BsmtCond       3   0.00349 112.28 -9198.1
## - LotConfig      4   0.11461 112.39 -9197.2
## - Foundation     5   0.19368 112.47 -9197.1
## - GarageQual     4   0.12340 112.40 -9197.0
## - BsmtExposure   3   0.04992 112.33 -9196.9
## - ExterQual      3   0.05205 112.33 -9196.8
## - BldgType       4   0.14011 112.42 -9196.5
## - SaleCondition  5   0.22561 112.50 -9196.3
## - BsmtFinType2   5   0.25638 112.53 -9195.5
## - GarageCond     4   0.18296 112.46 -9195.4
## - BsmtQual       3   0.10755 112.39 -9195.4
## - LotShape       3   0.13095 112.41 -9194.8
## - GarageFinish   2   0.05605 112.33 -9194.7
## - GarageCars     1   0.00000 112.28 -9194.2
## - OpenPorchSF    1   0.00001 112.28 -9194.2
## - LowQualFinSF   1   0.00009 112.28 -9194.2
## - MSSubClass     1   0.00046 112.28 -9194.2
## - HalfBath       1   0.00086 112.28 -9194.1
## - BsmtFullBath   1   0.00141 112.28 -9194.1
## - CentralAir     1   0.00379 112.28 -9194.1
## - MasVnrArea     1   0.00501 112.28 -9194.0
## - BsmtHalfBath   1   0.00606 112.28 -9194.0
## - WoodDeckSF     1   0.00865 112.29 -9193.9
## - PoolArea       1   0.03191 112.31 -9193.3
```

```
## - Street         1   0.03714 112.31 -9193.2
## - YearRemodAdd   1   0.03838 112.32 -9193.2
## - X3SsnPorch     1   0.04031 112.32 -9193.1
## - LandContour    3   0.19678 112.47 -9193.1
## - YrSold         1   0.04577 112.32 -9193.0
## - EnclosedPorch  1   0.04611 112.32 -9193.0
## - Utilities      1   0.06602 112.34 -9192.4
## - LotFrontage    1   0.06772 112.34 -9192.4
## <none>                       112.28 -9192.2
## - FullBath       1   0.08147 112.36 -9192.0
## - BsmtFinSF2     1   0.08373 112.36 -9192.0
## - Fireplaces     1   0.11897 112.40 -9191.1
## - Condition2     7   0.59182 112.87 -9190.8
## - BsmtFinType1   5   0.44365 112.72 -9190.7
## - KitchenQual    3   0.32708 112.61 -9189.7
## - BsmtFinSF1     1   0.18447 112.46 -9189.4
## - GarageArea     1   0.18992 112.47 -9189.2
## - Electrical     4   0.44120 112.72 -9188.7
## - BedroomAbvGr   1   0.21743 112.50 -9188.5
## - ScreenPorch    1   0.23554 112.51 -9188.0
## - MiscVal        1   0.26729 112.55 -9187.2
## - BsmtUnfSF      1   0.29023 112.57 -9186.6
## + Neighborhood  24   1.55863 110.72 -9185.0
## - GarageYrBlt    1   0.36002 112.64 -9184.8
## + GarageType     5   0.09102 112.19 -9184.5
## + ExterCond      4   0.01221 112.27 -9184.5
## - MoSold         1   0.38030 112.66 -9184.3
## - MSZoning       4   0.61664 112.89 -9184.2
## - PavedDrive     2   0.47219 112.75 -9183.9
## - LandSlope      2   0.49699 112.78 -9183.3
## + HouseStyle     7   0.19257 112.08 -9183.2
## - OverallQual    1   0.55016 112.83 -9179.9
## - TotRmsAbvGrd   1   0.56150 112.84 -9179.6
## - KitchenAbvGr   1   0.58295 112.86 -9179.0
## + Exterior1st   14   0.55542 111.72 -9178.6
## - YearBuilt      1   0.80463 113.08 -9173.3
## - Heating        5   1.19844 113.48 -9171.2
## - X1stFlrSF      1   0.91802 113.20 -9170.4
## + Exterior2nd   15   0.29472 111.98 -9169.8
## - OverallCond    1   1.10891 113.39 -9165.5
## - X2ndFlrSF      1   1.24387 113.52 -9162.0
## - RoofMatl       7   2.42353 114.70 -9143.8
## - LotArea        1   2.12383 114.40 -9139.5
##
## Step:  AIC=-9199.48
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition1 + Condition2 + BldgType + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofStyle + RoofMatl +
##     MasVnrArea + ExterQual + Foundation + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
```

```
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Functional + Fireplaces + GarageYrBlt +
##     GarageFinish + GarageCars + GarageArea + GarageQual + GarageCond +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - Functional     6   0.20775 112.82 -9206.1
## - HeatingQC      4   0.06178 112.67 -9205.9
## - BsmtCond       3   0.00257 112.61 -9205.4
## - Condition1     8   0.39057 113.00 -9205.4
## - RoofStyle      5   0.16355 112.78 -9205.2
## - BsmtExposure   3   0.04138 112.65 -9204.4
## - GarageQual     4   0.11965 112.73 -9204.4
## - Foundation     5   0.19735 112.81 -9204.4
## - LotConfig      4   0.12635 112.74 -9204.2
## - BldgType       4   0.14229 112.75 -9203.8
## - ExterQual      3   0.07215 112.68 -9203.6
## - GarageCond     4   0.16901 112.78 -9203.1
## - BsmtQual       3   0.11450 112.73 -9202.5
## - BsmtFinType2   5   0.28265 112.89 -9202.2
## - GarageFinish   2   0.05529 112.67 -9202.0
## - LotShape       3   0.14218 112.75 -9201.8
## - GarageCars     1   0.00011 112.61 -9201.5
## - LowQualFinSF   1   0.00019 112.61 -9201.5
## - OpenPorchSF    1   0.00028 112.61 -9201.5
## - BsmtFullBath   1   0.00042 112.61 -9201.5
## - MSSubClass     1   0.00056 112.61 -9201.5
## - HalfBath       1   0.00130 112.61 -9201.4
## - CentralAir     1   0.00241 112.61 -9201.4
## - WoodDeckSF     1   0.00628 112.62 -9201.3
## - BsmtHalfBath   1   0.00661 112.62 -9201.3
## - MasVnrArea     1   0.00670 112.62 -9201.3
## - Street         1   0.03114 112.64 -9200.7
## - PoolArea       1   0.03154 112.64 -9200.7
## - YrSold         1   0.03788 112.65 -9200.5
## - X3SsnPorch     1   0.03810 112.65 -9200.5
## - YearRemodAdd   1   0.03979 112.65 -9200.4
## - EnclosedPorch  1   0.04422 112.66 -9200.3
## - LandContour    3   0.20061 112.81 -9200.3
## - Utilities      1   0.06289 112.67 -9199.8
## - SaleCondition  5   0.37535 112.99 -9199.8
## - LotFrontage    1   0.06834 112.68 -9199.7
## <none>                       112.61 -9199.5
## - FullBath       1   0.07865 112.69 -9199.4
## - BsmtFinSF2     1   0.08675 112.70 -9199.2
## - BsmtFinType1   5   0.42449 113.04 -9198.5
## - Fireplaces     1   0.12364 112.74 -9198.3
## - Condition2     7   0.59723 113.21 -9198.0
## - KitchenQual    3   0.32437 112.94 -9197.1
## - BsmtFinSF1     1   0.18475 112.80 -9196.7
```

```
## - BedroomAbvGr   1   0.20177 112.81 -9196.3
## - GarageArea     1   0.20770 112.82 -9196.1
## - Electrical     4   0.44760 113.06 -9195.9
## - ScreenPorch    1   0.22592 112.84 -9195.6
## - MiscVal        1   0.27370 112.89 -9194.4
## - BsmtUnfSF      1   0.27595 112.89 -9194.3
## + SaleType       8   0.33454 112.28 -9192.2
## + GarageType     5   0.09771 112.52 -9192.0
## + ExterCond      4   0.01043 112.60 -9191.7
## - GarageYrBlt    1   0.38037 112.99 -9191.6
## - MoSold         1   0.38442 113.00 -9191.5
## - MSZoning       4   0.62750 113.24 -9191.3
## + Neighborhood  24   1.51661 111.10 -9191.1
## - PavedDrive     2   0.49200 113.10 -9190.8
## + HouseStyle     7   0.19763 112.42 -9190.6
## - LandSlope      2   0.55312 113.17 -9189.2
## - OverallQual    1   0.55492 113.17 -9187.1
## - KitchenAbvGr   1   0.57959 113.19 -9186.5
## - TotRmsAbvGrd   1   0.58086 113.19 -9186.5
## + Exterior1st   14   0.56167 112.05 -9186.1
## - YearBuilt      1   0.81183 113.42 -9180.5
## - Heating        5   1.19356 113.81 -9178.7
## + Exterior2nd   15   0.31502 112.30 -9177.7
## - X1stFlrSF      1   0.92924 113.54 -9177.5
## - OverallCond    1   1.07453 113.69 -9173.8
## - X2ndFlrSF      1   1.24679 113.86 -9169.3
## - RoofMatl       7   2.44854 115.06 -9150.7
## - LotArea        1   2.16376 114.78 -9145.9
##
## Step:  AIC=-9206.1
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition1 + Condition2 + BldgType + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofStyle + RoofMatl +
##     MasVnrArea + ExterQual + Foundation + BsmtQual + BsmtCond +
##     BsmtExposure + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + HeatingQC + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - Condition1     8   0.37118 113.19 -9212.5
## - HeatingQC      4   0.06232 112.88 -9212.5
## - BsmtCond       3   0.00096 112.82 -9212.1
## - RoofStyle      5   0.16923 112.99 -9211.7
## - BsmtExposure   3   0.04188 112.86 -9211.0
## - Foundation     5   0.19725 113.02 -9211.0
## - LotConfig      4   0.12724 112.95 -9210.8
```

```
## - ExterQual      3   0.05525 112.88 -9210.7
## - GarageQual     4   0.13635 112.96 -9210.6
## - BldgType       4   0.16117 112.98 -9209.9
## - GarageCond     4   0.16740 112.99 -9209.8
## - BsmtQual       3   0.10525 112.92 -9209.4
## - BsmtFinType2   5   0.29455 113.11 -9208.5
## - GarageFinish   2   0.06398 112.88 -9208.4
## - LotShape       3   0.14471 112.97 -9208.4
## - OpenPorchSF    1   0.00001 112.82 -9208.1
## - GarageCars     1   0.00008 112.82 -9208.1
## - BsmtFullBath   1   0.00012 112.82 -9208.1
## - LowQualFinSF   1   0.00033 112.82 -9208.1
## - HalfBath       1   0.00053 112.82 -9208.1
## - MSSubClass     1   0.00133 112.82 -9208.1
## - CentralAir     1   0.00302 112.82 -9208.0
## - MasVnrArea     1   0.00322 112.82 -9208.0
## - WoodDeckSF     1   0.00569 112.83 -9208.0
## - BsmtHalfBath   1   0.00682 112.83 -9207.9
## - Street         1   0.02894 112.85 -9207.4
## - YrSold         1   0.03279 112.85 -9207.3
## - X3SsnPorch     1   0.03451 112.85 -9207.2
## - PoolArea       1   0.03642 112.86 -9207.2
## - EnclosedPorch  1   0.03955 112.86 -9207.1
## - LandContour    3   0.20521 113.03 -9206.8
## - YearRemodAdd   1   0.05528 112.88 -9206.7
## - Utilities      1   0.06329 112.88 -9206.5
## - SaleCondition  5   0.37336 113.19 -9206.5
## - LotFrontage    1   0.06709 112.89 -9206.4
## - FullBath       1   0.07419 112.89 -9206.2
## <none>                       112.82 -9206.1
## - BsmtFinSF2     1   0.09727 112.92 -9205.6
## - Fireplaces     1   0.12038 112.94 -9205.0
## - BsmtFinType1   5   0.43502 113.25 -9204.9
## - Condition2     7   0.59741 113.42 -9204.7
## - KitchenQual    3   0.33046 113.15 -9203.6
## - GarageArea     1   0.20935 113.03 -9202.7
## - BedroomAbvGr   1   0.20979 113.03 -9202.7
## - BsmtFinSF1     1   0.21860 113.04 -9202.4
## - ScreenPorch    1   0.22693 113.05 -9202.2
## - MiscVal        1   0.27390 113.09 -9201.0
## - Electrical     4   0.53698 113.36 -9200.2
## - BsmtUnfSF      1   0.32293 113.14 -9199.8
## + Functional     6   0.20775 112.61 -9199.5
## - GarageYrBlt    1   0.35284 113.17 -9199.0
## + SaleType       8   0.33399 112.49 -9198.8
## + GarageType     5   0.09056 112.73 -9198.4
## + ExterCond      4   0.01151 112.81 -9198.4
## - MSZoning       4   0.62523 113.44 -9198.0
## - PavedDrive     2   0.47552 113.30 -9197.8
## + Neighborhood  24   1.52361 111.30 -9197.8
## - MoSold         1   0.39966 113.22 -9197.8
## + HouseStyle     7   0.19629 112.62 -9197.2
```

```
## - LandSlope      2   0.57486 113.39 -9195.3
## + Exterior1st   14   0.59445 112.23 -9193.5
## - KitchenAbvGr   1   0.56760 113.39 -9193.5
## - TotRmsAbvGrd   1   0.58417 113.40 -9193.0
## - OverallQual    1   0.63891 113.46 -9191.6
## - YearBuilt      1   0.81205 113.63 -9187.2
## - X1stFlrSF      1   0.84290 113.66 -9186.4
## - Heating        5   1.24025 114.06 -9184.2
## + Exterior2nd   15   0.30925 112.51 -9184.1
## - X2ndFlrSF      1   1.19879 114.02 -9177.2
## - OverallCond    1   1.25591 114.08 -9175.8
## - RoofMatl       7   2.47281 115.29 -9156.8
## - LotArea        1   2.19365 115.01 -9151.9
##
## Step:  AIC=-9212.51
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofStyle + RoofMatl + MasVnrArea +
##     ExterQual + Foundation + BsmtQual + BsmtCond + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + HeatingQC + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - HeatingQC      4   0.06429 113.25 -9218.9
## - BsmtCond       3   0.00052 113.19 -9218.5
## - RoofStyle      5   0.16973 113.36 -9218.1
## - BsmtExposure   3   0.03612 113.23 -9217.6
## - LotConfig      4   0.12297 113.31 -9217.3
## - ExterQual      3   0.05031 113.24 -9217.2
## - Foundation     5   0.20671 113.40 -9217.2
## - GarageQual     4   0.15123 113.34 -9216.6
## - GarageCond     4   0.15498 113.35 -9216.5
## - BldgType       4   0.15621 113.35 -9216.5
## - BsmtQual       3   0.10452 113.30 -9215.8
## - LotShape       3   0.12209 113.31 -9215.4
## - BsmtFinType2   5   0.28315 113.47 -9215.2
## - GarageFinish   2   0.06791 113.26 -9214.8
## - GarageCars     1   0.00007 113.19 -9214.5
## - HalfBath       1   0.00010 113.19 -9214.5
## - BsmtFullBath   1   0.00022 113.19 -9214.5
## - LowQualFinSF   1   0.00032 113.19 -9214.5
## - OpenPorchSF    1   0.00079 113.19 -9214.5
## - MSSubClass     1   0.00132 113.19 -9214.5
## - MasVnrArea     1   0.00333 113.19 -9214.4
## - BsmtHalfBath   1   0.00427 113.19 -9214.4
```

```
## - CentralAir     1   0.00438 113.20 -9214.4
## - WoodDeckSF     1   0.00522 113.20 -9214.4
## - Street         1   0.01482 113.21 -9214.1
## - PoolArea       1   0.03054 113.22 -9213.7
## - EnclosedPorch  1   0.03464 113.23 -9213.6
## - SaleCondition  5   0.34597 113.54 -9213.6
## - X3SsnPorch     1   0.03621 113.23 -9213.6
## - YrSold         1   0.04150 113.23 -9213.4
## - YearRemodAdd   1   0.04798 113.24 -9213.3
## - Utilities      1   0.06038 113.25 -9213.0
## - LotFrontage    1   0.06247 113.25 -9212.9
## - LandContour    3   0.21998 113.41 -9212.8
## - FullBath       1   0.07195 113.26 -9212.7
## <none>                       113.19 -9212.5
## - Fireplaces     1   0.11474 113.31 -9211.6
## - BsmtFinSF2     1   0.12017 113.31 -9211.4
## - BsmtFinType1   5   0.45092 113.64 -9210.9
## - KitchenQual    3   0.32557 113.52 -9210.1
## - BedroomAbvGr   1   0.18637 113.38 -9209.7
## - GarageArea     1   0.19732 113.39 -9209.4
## - ScreenPorch    1   0.22857 113.42 -9208.6
## - BsmtFinSF1     1   0.23167 113.42 -9208.5
## - MiscVal        1   0.27061 113.46 -9207.5
## - Electrical     4   0.52049 113.71 -9207.1
## + Condition1     8   0.37118 112.82 -9206.1
## - BsmtUnfSF      1   0.33577 113.53 -9205.9
## - Condition2     7   0.80473 114.00 -9205.8
## - GarageYrBlt    1   0.34653 113.54 -9205.6
## + Functional     6   0.18836 113.00 -9205.4
## + SaleType       8   0.34137 112.85 -9205.3
## + Neighborhood  24   1.55570 111.64 -9204.9
## + GarageType     5   0.08894 113.10 -9204.8
## + ExterCond      4   0.01083 113.18 -9204.8
## - PavedDrive     2   0.45581 113.65 -9204.8
## - MoSold         1   0.38459 113.58 -9204.6
## - MSZoning       4   0.63077 113.82 -9204.3
## + HouseStyle     7   0.19807 112.99 -9203.6
## - LandSlope      2   0.55364 113.75 -9202.3
## - KitchenAbvGr   1   0.55519 113.75 -9200.2
## + Exterior1st   14   0.60776 112.58 -9200.2
## - TotRmsAbvGrd   1   0.61182 113.80 -9198.8
## - OverallQual    1   0.61240 113.80 -9198.8
## - X1stFlrSF      1   0.83335 114.03 -9193.1
## - YearBuilt      1   0.85485 114.05 -9192.5
## - Heating        5   1.23866 114.43 -9190.7
## + Exterior2nd   15   0.31532 112.88 -9190.7
## - X2ndFlrSF      1   1.18571 114.38 -9184.1
## - OverallCond    1   1.26388 114.45 -9182.1
## - RoofMatl       7   2.42123 115.61 -9164.7
## - LotArea        1   2.18875 115.38 -9158.6
##
## Step:  AIC=-9218.85
```

```
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofStyle + RoofMatl + MasVnrArea +
##     ExterQual + Foundation + BsmtQual + BsmtCond + BsmtExposure +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + CentralAir + Electrical + X1stFlrSF + X2ndFlrSF +
##     LowQualFinSF + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageFinish + GarageCars + GarageArea +
##     GarageQual + GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - BsmtCond       3   0.00063 113.26 -9224.8
## - RoofStyle      5   0.17072 113.43 -9224.5
## - BsmtExposure   3   0.03894 113.29 -9223.9
## - LotConfig      4   0.12154 113.38 -9223.7
## - ExterQual      3   0.06290 113.32 -9223.2
## - GarageQual     4   0.15424 113.41 -9222.9
## - BldgType       4   0.15484 113.41 -9222.9
## - GarageCond     4   0.15504 113.41 -9222.9
## - Foundation     5   0.23967 113.50 -9222.7
## - BsmtQual       3   0.09723 113.35 -9222.3
## - LotShape       3   0.12411 113.38 -9221.7
## - BsmtFinType2   5   0.28239 113.54 -9221.6
## - GarageFinish   2   0.07091 113.33 -9221.0
## - GarageCars     1   0.00000 113.25 -9220.9
## - BsmtFullBath   1   0.00012 113.26 -9220.9
## - HalfBath       1   0.00019 113.26 -9220.8
## - LowQualFinSF   1   0.00038 113.26 -9220.8
## - OpenPorchSF    1   0.00043 113.26 -9220.8
## - MSSubClass     1   0.00192 113.26 -9220.8
## - MasVnrArea     1   0.00311 113.26 -9220.8
## - BsmtHalfBath   1   0.00417 113.26 -9220.7
## - WoodDeckSF     1   0.00468 113.26 -9220.7
## - CentralAir     1   0.00544 113.26 -9220.7
## - Street         1   0.01485 113.27 -9220.5
## - SaleCondition  5   0.34079 113.60 -9220.1
## - PoolArea       1   0.02992 113.28 -9220.1
## - EnclosedPorch  1   0.03503 113.29 -9220.0
## - X3SsnPorch     1   0.03626 113.29 -9219.9
## - YrSold         1   0.04251 113.30 -9219.8
## - YearRemodAdd   1   0.04297 113.30 -9219.7
## - Utilities      1   0.06190 113.32 -9219.3
## - LotFrontage    1   0.06333 113.32 -9219.2
## - LandContour    3   0.22722 113.48 -9219.0
## - FullBath       1   0.07359 113.33 -9219.0
## <none>                       113.25 -9218.9
## - Fireplaces     1   0.11780 113.37 -9217.8
## - BsmtFinSF2     1   0.12123 113.38 -9217.7
```

```
## - BsmtFinType1   5   0.45169 113.71 -9217.2
## - KitchenQual    3   0.32339 113.58 -9216.5
## - BedroomAbvGr   1   0.18429 113.44 -9216.1
## - GarageArea     1   0.19139 113.45 -9215.9
## - BsmtFinSF1     1   0.22789 113.48 -9215.0
## - ScreenPorch    1   0.23321 113.49 -9214.8
## - MiscVal        1   0.26972 113.53 -9213.9
## - Electrical     4   0.51217 113.77 -9213.7
## + HeatingQC      4   0.06429 113.19 -9212.5
## + Condition1     8   0.37315 112.88 -9212.5
## - Condition2     7   0.79851 114.05 -9212.3
## - BsmtUnfSF      1   0.33765 113.59 -9212.2
## - GarageYrBlt    1   0.34731 113.60 -9211.9
## + Neighborhood  24   1.58182 111.67 -9211.9
## + Functional     6   0.18951 113.07 -9211.7
## + SaleType       8   0.33339 112.92 -9211.5
## - PavedDrive     2   0.44481 113.70 -9211.4
## + GarageType     5   0.09042 113.17 -9211.2
## + ExterCond      4   0.01256 113.24 -9211.2
## - MoSold         1   0.38331 113.64 -9211.0
## - MSZoning       4   0.62571 113.88 -9210.8
## + HouseStyle     7   0.17850 113.08 -9209.5
## - LandSlope      2   0.56151 113.82 -9208.4
## - KitchenAbvGr   1   0.56781 113.82 -9206.3
## + Exterior1st   14   0.57724 112.68 -9205.8
## - OverallQual    1   0.60881 113.86 -9205.2
## - TotRmsAbvGrd   1   0.62114 113.88 -9204.9
## - X1stFlrSF      1   0.83091 114.09 -9199.5
## - YearBuilt      1   0.87573 114.13 -9198.4
## + Exterior2nd   15   0.28627 112.97 -9196.2
## - Heating        5   1.31254 114.57 -9195.2
## - X2ndFlrSF      1   1.19715 114.45 -9190.2
## - OverallCond    1   1.30390 114.56 -9187.4
## - RoofMatl       7   2.41124 115.67 -9171.4
## - LotArea        1   2.19287 115.45 -9164.9
##
## Step:  AIC=-9224.84
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofStyle + RoofMatl + MasVnrArea +
##     ExterQual + Foundation + BsmtQual + BsmtExposure + BsmtFinType1 +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
##     BsmtFullBath + BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
```

```
## - RoofStyle      5   0.17305 113.43 -9230.4
## - BsmtExposure   3   0.03905 113.30 -9229.8
## - LotConfig      4   0.12193 113.38 -9229.7
## - ExterQual      3   0.06279 113.32 -9229.2
## - GarageQual     4   0.15503 113.41 -9228.8
## - BldgType       4   0.15533 113.41 -9228.8
## - GarageCond     4   0.15562 113.41 -9228.8
## - Foundation     5   0.24020 113.50 -9228.7
## - BsmtQual       3   0.09849 113.36 -9228.3
## - LotShape       3   0.12452 113.38 -9227.6
## - BsmtFinType2   5   0.28306 113.54 -9227.6
## - GarageFinish   2   0.07099 113.33 -9227.0
## - GarageCars     1   0.00000 113.26 -9226.8
## - BsmtFullBath   1   0.00011 113.26 -9226.8
## - HalfBath       1   0.00021 113.26 -9226.8
## - LowQualFinSF   1   0.00036 113.26 -9226.8
## - OpenPorchSF    1   0.00044 113.26 -9226.8
## - MSSubClass     1   0.00198 113.26 -9226.8
## - MasVnrArea     1   0.00312 113.26 -9226.8
## - BsmtHalfBath   1   0.00424 113.26 -9226.7
## - WoodDeckSF     1   0.00472 113.26 -9226.7
## - CentralAir     1   0.00552 113.26 -9226.7
## - Street         1   0.01475 113.27 -9226.5
## - PoolArea       1   0.02980 113.29 -9226.1
## - SaleCondition  5   0.34230 113.60 -9226.0
## - EnclosedPorch  1   0.03505 113.29 -9225.9
## - X3SsnPorch     1   0.03579 113.29 -9225.9
## - YrSold         1   0.04308 113.30 -9225.7
## - YearRemodAdd   1   0.04350 113.30 -9225.7
## - Utilities      1   0.06203 113.32 -9225.2
## - LotFrontage    1   0.06332 113.32 -9225.2
## - LandContour    3   0.22709 113.48 -9225.0
## - FullBath       1   0.07403 113.33 -9224.9
## <none>                       113.26 -9224.8
## - Fireplaces     1   0.11804 113.37 -9223.8
## - BsmtFinSF2     1   0.12111 113.38 -9223.7
## - BsmtFinType1   5   0.45221 113.71 -9223.2
## - KitchenQual    3   0.32809 113.58 -9222.4
## - BedroomAbvGr   1   0.18450 113.44 -9222.1
## - GarageArea     1   0.19183 113.45 -9221.9
## - BsmtFinSF1     1   0.22816 113.48 -9221.0
## - ScreenPorch    1   0.23341 113.49 -9220.8
## - MiscVal        1   0.26963 113.53 -9219.9
## + BsmtCond       3   0.00063 113.25 -9218.9
## + HeatingQC      4   0.06441 113.19 -9218.5
## + Condition1     8   0.37309 112.88 -9218.5
## - Condition2     7   0.80023 114.06 -9218.3
## - BsmtUnfSF      1   0.33784 113.59 -9218.1
## + Neighborhood  24   1.58121 111.67 -9217.9
## - GarageYrBlt    1   0.34887 113.61 -9217.9
## + Functional     6   0.18976 113.07 -9217.7
## + SaleType       8   0.33375 112.92 -9217.5
```

```
## - PavedDrive     2   0.44506 113.70 -9217.4
## + GarageType     5   0.09045 113.17 -9217.2
## + ExterCond      4   0.01248 113.24 -9217.2
## - Electrical     4   0.61538 113.87 -9217.0
## - MoSold         1   0.38321 113.64 -9217.0
## - MSZoning       4   0.62621 113.88 -9216.7
## + HouseStyle     7   0.17820 113.08 -9215.4
## - LandSlope      2   0.57790 113.83 -9214.0
## - KitchenAbvGr   1   0.57154 113.83 -9212.1
## + Exterior1st   14   0.57698 112.68 -9211.7
## - OverallQual    1   0.61128 113.87 -9211.1
## - TotRmsAbvGrd   1   0.62125 113.88 -9210.9
## - X1stFlrSF      1   0.83116 114.09 -9205.5
## - YearBuilt      1   0.87869 114.14 -9204.3
## + Exterior2nd   15   0.28649 112.97 -9202.2
## - Heating        5   1.31267 114.57 -9201.2
## - X2ndFlrSF      1   1.19724 114.45 -9196.1
## - OverallCond    1   1.34304 114.60 -9192.4
## - RoofMatl       7   2.41415 115.67 -9177.3
## - LotArea        1   2.20496 115.46 -9170.6
##
## Step:  AIC=-9230.38
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea + ExterQual +
##     Foundation + BsmtQual + BsmtExposure + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - BsmtExposure   3   0.03722 113.47 -9235.4
## - LotConfig      4   0.12247 113.55 -9235.2
## - ExterQual      3   0.05638 113.49 -9234.9
## - GarageCond     4   0.14415 113.57 -9234.7
## - GarageQual     4   0.15092 113.58 -9234.5
## - BldgType       4   0.16185 113.59 -9234.2
## - Foundation     5   0.24212 113.67 -9234.2
## - BsmtQual       3   0.09544 113.53 -9233.9
## - BsmtFinType2   5   0.27324 113.70 -9233.4
## - LotShape       3   0.12489 113.55 -9233.2
## - GarageFinish   2   0.06840 113.50 -9232.6
## - BsmtFullBath   1   0.00004 113.43 -9232.4
## - OpenPorchSF    1   0.00009 113.43 -9232.4
## - GarageCars     1   0.00012 113.43 -9232.4
## - LowQualFinSF   1   0.00021 113.43 -9232.4
## - HalfBath       1   0.00036 113.43 -9232.4
```

```
## - MSSubClass     1   0.00201 113.43 -9232.3
## - BsmtHalfBath   1   0.00301 113.43 -9232.3
## - CentralAir     1   0.00489 113.43 -9232.3
## - WoodDeckSF     1   0.00611 113.44 -9232.2
## - MasVnrArea     1   0.00648 113.44 -9232.2
## - Street         1   0.01417 113.44 -9232.0
## - SaleCondition  5   0.33757 113.77 -9231.7
## - PoolArea       1   0.03053 113.46 -9231.6
## - X3SsnPorch     1   0.03549 113.47 -9231.5
## - EnclosedPorch  1   0.03592 113.47 -9231.5
## - YearRemodAdd   1   0.04411 113.47 -9231.2
## - YrSold         1   0.04578 113.47 -9231.2
## - Utilities      1   0.05897 113.49 -9230.9
## - FullBath       1   0.07031 113.50 -9230.6
## - LandContour    3   0.23130 113.66 -9230.4
## <none>                       113.43 -9230.4
## - LotFrontage    1   0.08196 113.51 -9230.3
## - Fireplaces     1   0.11564 113.55 -9229.4
## - BsmtFinSF2     1   0.12389 113.55 -9229.2
## - BsmtFinType1   5   0.45901 113.89 -9228.6
## - KitchenQual    3   0.32057 113.75 -9228.1
## - BedroomAbvGr   1   0.17234 113.60 -9227.9
## - GarageArea     1   0.19508 113.62 -9227.4
## - BsmtFinSF1     1   0.21770 113.65 -9226.8
## - ScreenPorch    1   0.22103 113.65 -9226.7
## - Condition2     7   0.74638 114.18 -9225.2
## - MiscVal        1   0.27804 113.71 -9225.2
## + RoofStyle      5   0.17305 113.26 -9224.8
## + BsmtCond       3   0.00296 113.43 -9224.5
## + Condition1     8   0.37607 113.05 -9224.1
## + HeatingQC      4   0.06562 113.36 -9224.1
## - BsmtUnfSF      1   0.32932 113.76 -9223.9
## - GarageYrBlt    1   0.33922 113.77 -9223.7
## + Neighborhood  24   1.58782 111.84 -9223.5
## + Functional     6   0.19482 113.23 -9223.4
## + SaleType       8   0.34608 113.08 -9223.3
## - PavedDrive     2   0.43541 113.86 -9223.2
## + ExterCond      4   0.01322 113.42 -9222.7
## - MoSold         1   0.37616 113.81 -9222.7
## + GarageType     5   0.08900 113.34 -9222.7
## - Electrical     4   0.61309 114.04 -9222.6
## - LandSlope      2   0.46136 113.89 -9222.5
## - MSZoning       4   0.63594 114.06 -9222.1
## + HouseStyle     7   0.17416 113.25 -9220.9
## - KitchenAbvGr   1   0.57671 114.01 -9217.6
## - TotRmsAbvGrd   1   0.59700 114.03 -9217.1
## + Exterior1st   14   0.56798 112.86 -9217.0
## - OverallQual    1   0.61800 114.05 -9216.5
## - X1stFlrSF      1   0.86163 114.29 -9210.3
## - YearBuilt      1   0.90390 114.33 -9209.2
## + Exterior2nd   15   0.28718 113.14 -9207.8
## - Heating        5   1.33209 114.76 -9206.3
```

```
## - X2ndFlrSF      1   1.24328 114.67 -9200.6
## - OverallCond    1   1.34486 114.77 -9198.0
## - RoofMatl       7   2.50545 115.94 -9180.6
## - LotArea        1   2.09535 115.52 -9179.0
##
## Step:  AIC=-9235.42
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LotConfig +
##     LandSlope + Condition2 + BldgType + OverallQual + OverallCond +
##     YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea + ExterQual +
##     Foundation + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - LotConfig      4   0.12463 113.59 -9240.2
## - ExterQual      3   0.05505 113.52 -9240.0
## - GarageCond     4   0.14511 113.61 -9239.7
## - GarageQual     4   0.14961 113.62 -9239.6
## - BldgType       4   0.15970 113.63 -9239.3
## - Foundation     5   0.24574 113.71 -9239.1
## - BsmtQual       3   0.09360 113.56 -9239.0
## - BsmtFinType2   5   0.26809 113.73 -9238.5
## - LotShape       3   0.12228 113.59 -9238.3
## - GarageFinish   2   0.06330 113.53 -9237.8
## - BsmtFullBath   1   0.00004 113.47 -9237.4
## - OpenPorchSF    1   0.00014 113.47 -9237.4
## - LowQualFinSF   1   0.00027 113.47 -9237.4
## - GarageCars     1   0.00032 113.47 -9237.4
## - HalfBath       1   0.00044 113.47 -9237.4
## - MSSubClass     1   0.00225 113.47 -9237.4
## - BsmtHalfBath   1   0.00246 113.47 -9237.4
## - WoodDeckSF     1   0.00487 113.47 -9237.3
## - CentralAir     1   0.00490 113.47 -9237.3
## - MasVnrArea     1   0.00677 113.47 -9237.2
## - Street         1   0.01417 113.48 -9237.1
## - SaleCondition  5   0.34271 113.81 -9236.6
## - PoolArea       1   0.03164 113.50 -9236.6
## - X3SsnPorch     1   0.03452 113.50 -9236.5
## - EnclosedPorch  1   0.03515 113.50 -9236.5
## - YearRemodAdd   1   0.04484 113.51 -9236.3
## - YrSold         1   0.04525 113.51 -9236.3
## - Utilities      1   0.05876 113.53 -9235.9
## - FullBath       1   0.07069 113.54 -9235.6
## - LandContour    3   0.23157 113.70 -9235.5
## <none>                       113.47 -9235.4
## - LotFrontage    1   0.08040 113.55 -9235.4
```

```
## - Fireplaces     1   0.11716 113.58 -9234.4
## - BsmtFinSF2     1   0.11777 113.58 -9234.4
## - BsmtFinType1   5   0.47269 113.94 -9233.3
## - KitchenQual    3   0.32915 113.80 -9233.0
## - BedroomAbvGr   1   0.17742 113.64 -9232.9
## - GarageArea     1   0.20317 113.67 -9232.2
## - BsmtFinSF1     1   0.20721 113.67 -9232.1
## - ScreenPorch    1   0.21781 113.68 -9231.8
## + BsmtExposure   3   0.03722 113.43 -9230.4
## - MiscVal        1   0.27457 113.74 -9230.4
## - Condition2     7   0.75379 114.22 -9230.1
## + RoofStyle      5   0.17121 113.30 -9229.8
## + BsmtCond       3   0.00289 113.46 -9229.5
## + HeatingQC      4   0.06838 113.40 -9229.2
## + Neighborhood  24   1.60349 111.86 -9229.0
## + Condition1     8   0.36887 113.10 -9228.9
## - BsmtUnfSF      1   0.33124 113.80 -9228.9
## - GarageYrBlt    1   0.33530 113.80 -9228.8
## + Functional     6   0.19592 113.27 -9228.5
## - PavedDrive     2   0.43429 113.90 -9228.3
## + SaleType       8   0.33766 113.13 -9228.1
## + GarageType     5   0.09222 113.37 -9227.8
## + ExterCond      4   0.01444 113.45 -9227.8
## - Electrical     4   0.61059 114.08 -9227.8
## - MoSold         1   0.38008 113.85 -9227.7
## - LandSlope      2   0.47101 113.94 -9227.3
## - MSZoning       4   0.64629 114.11 -9226.8
## + HouseStyle     7   0.17432 113.29 -9225.9
## - KitchenAbvGr   1   0.57097 114.04 -9222.8
## - TotRmsAbvGrd   1   0.60329 114.07 -9221.9
## + Exterior1st   14   0.56171 112.91 -9221.9
## - OverallQual    1   0.60665 114.07 -9221.9
## - X1stFlrSF      1   0.85077 114.32 -9215.6
## - YearBuilt      1   0.90771 114.37 -9214.2
## + Exterior2nd   15   0.27911 113.19 -9212.6
## - Heating        5   1.34272 114.81 -9211.1
## - X2ndFlrSF      1   1.25575 114.72 -9205.3
## - OverallCond    1   1.34851 114.81 -9202.9
## - RoofMatl       7   2.49238 115.96 -9186.0
## - LotArea        1   2.07536 115.54 -9184.5
##
## Step:  AIC=-9240.22
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LandSlope +
##     Condition2 + BldgType + OverallQual + OverallCond + YearBuilt +
##     YearRemodAdd + RoofMatl + MasVnrArea + ExterQual + Foundation +
##     BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + GarageCond + PavedDrive +
```

```
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - ExterQual      3   0.05388 113.64 -9244.8
## - GarageCond     4   0.13738 113.73 -9244.7
## - GarageQual     4   0.15097 113.74 -9244.3
## - Foundation     5   0.23138 113.82 -9244.3
## - BldgType       4   0.16578 113.76 -9244.0
## - BsmtQual       3   0.10277 113.69 -9243.6
## - LotShape       3   0.10729 113.70 -9243.5
## - BsmtFinType2   5   0.26400 113.86 -9243.4
## - GarageFinish   2   0.06380 113.66 -9242.6
## - GarageCars     1   0.00003 113.59 -9242.2
## - BsmtFullBath   1   0.00008 113.59 -9242.2
## - OpenPorchSF    1   0.00022 113.59 -9242.2
## - LowQualFinSF   1   0.00026 113.59 -9242.2
## - HalfBath       1   0.00045 113.59 -9242.2
## - MSSubClass     1   0.00161 113.59 -9242.2
## - BsmtHalfBath   1   0.00239 113.59 -9242.2
## - CentralAir     1   0.00407 113.59 -9242.1
## - WoodDeckSF     1   0.00467 113.60 -9242.1
## - MasVnrArea     1   0.00600 113.60 -9242.1
## - Street         1   0.01622 113.61 -9241.8
## - SaleCondition  5   0.33989 113.93 -9241.5
## - PoolArea       1   0.03211 113.62 -9241.4
## - X3SsnPorch     1   0.03619 113.63 -9241.3
## - EnclosedPorch  1   0.04056 113.63 -9241.2
## - YrSold         1   0.04423 113.64 -9241.1
## - YearRemodAdd   1   0.04462 113.64 -9241.1
## - Utilities      1   0.04838 113.64 -9241.0
## - LotFrontage    1   0.05827 113.65 -9240.7
## - FullBath       1   0.06270 113.65 -9240.6
## <none>                       113.59 -9240.2
## - LandContour    3   0.24003 113.83 -9240.1
## - Fireplaces     1   0.11510 113.71 -9239.3
## - BsmtFinSF2     1   0.11644 113.71 -9239.2
## - BsmtFinType1   5   0.46165 114.05 -9238.4
## - KitchenQual    3   0.31720 113.91 -9238.1
## - BedroomAbvGr   1   0.17915 113.77 -9237.6
## - GarageArea     1   0.20280 113.79 -9237.0
## - BsmtFinSF1     1   0.20627 113.80 -9236.9
## - ScreenPorch    1   0.21524 113.81 -9236.7
## + LotConfig      4   0.12463 113.47 -9235.4
## - MiscVal        1   0.27155 113.86 -9235.2
## + BsmtExposure   3   0.03937 113.55 -9235.2
## + RoofStyle      5   0.17144 113.42 -9234.6
## - Condition2     7   0.77008 114.36 -9234.5
## + BsmtCond       3   0.00385 113.59 -9234.3
## - BsmtUnfSF      1   0.32124 113.91 -9234.0
## + HeatingQC      4   0.06749 113.52 -9234.0
## + Neighborhood  24   1.60372 111.99 -9233.7
```

```
## + Condition1     8   0.36633 113.22 -9233.6
## - GarageYrBlt    1   0.33695 113.93 -9233.6
## + Functional     6   0.19743 113.39 -9233.3
## + SaleType       8   0.34987 113.24 -9233.2
## - PavedDrive     2   0.44355 114.03 -9232.8
## + GarageType     5   0.09558 113.50 -9232.7
## - MoSold         1   0.37214 113.96 -9232.7
## - Electrical     4   0.60715 114.20 -9232.7
## + ExterCond      4   0.01348 113.58 -9232.6
## - LandSlope      2   0.47352 114.06 -9232.1
## - MSZoning       4   0.65073 114.24 -9231.5
## + HouseStyle     7   0.17021 113.42 -9230.6
## - KitchenAbvGr   1   0.56710 114.16 -9227.7
## - TotRmsAbvGrd   1   0.59464 114.19 -9227.0
## + Exterior1st   14   0.56623 113.03 -9226.8
## - OverallQual    1   0.61597 114.21 -9226.4
## - X1stFlrSF      1   0.89293 114.48 -9219.4
## - YearBuilt      1   0.93033 114.52 -9218.4
## + Exterior2nd   15   0.28567 113.31 -9217.6
## - Heating        5   1.34055 114.93 -9216.0
## - X2ndFlrSF      1   1.27775 114.87 -9209.6
## - OverallCond    1   1.37850 114.97 -9207.0
## - RoofMatl       7   2.49551 116.09 -9190.8
## - LotArea        1   2.19369 115.78 -9186.4
##
## Step:  AIC=-9244.83
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LandSlope +
##     Condition2 + BldgType + OverallQual + OverallCond + YearBuilt +
##     YearRemodAdd + RoofMatl + MasVnrArea + Foundation + BsmtQual +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + CentralAir + Electrical + X1stFlrSF + X2ndFlrSF +
##     LowQualFinSF + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageFinish + GarageCars + GarageArea +
##     GarageQual + GarageCond + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - GarageCond     4   0.14953 113.79 -9249.0
## - Foundation     5   0.23232 113.88 -9248.9
## - GarageQual     4   0.15820 113.80 -9248.8
## - BldgType       4   0.16599 113.81 -9248.6
## - BsmtFinType2   5   0.25849 113.90 -9248.2
## - LotShape       3   0.11049 113.75 -9248.0
## - BsmtQual       3   0.11593 113.76 -9247.9
## - GarageFinish   2   0.06759 113.71 -9247.1
## - LowQualFinSF   1   0.00000 113.64 -9246.8
## - GarageCars     1   0.00007 113.64 -9246.8
## - OpenPorchSF    1   0.00017 113.64 -9246.8
## - BsmtFullBath   1   0.00020 113.64 -9246.8
```

```
## - HalfBath       1   0.00064 113.65 -9246.8
## - CentralAir     1   0.00141 113.65 -9246.8
## - BsmtHalfBath   1   0.00228 113.65 -9246.8
## - MSSubClass     1   0.00266 113.65 -9246.8
## - WoodDeckSF     1   0.00404 113.65 -9246.7
## - MasVnrArea     1   0.00453 113.65 -9246.7
## - Street         1   0.00832 113.65 -9246.6
## - PoolArea       1   0.03220 113.68 -9246.0
## - X3SsnPorch     1   0.03737 113.68 -9245.9
## - YrSold         1   0.03882 113.68 -9245.8
## - SaleCondition  5   0.35464 114.00 -9245.7
## - EnclosedPorch  1   0.04271 113.69 -9245.7
## - YearRemodAdd   1   0.04482 113.69 -9245.7
## - Utilities      1   0.05041 113.69 -9245.5
## - LotFrontage    1   0.06039 113.70 -9245.3
## - FullBath       1   0.06329 113.71 -9245.2
## <none>                       113.64 -9244.8
## - LandContour    3   0.24894 113.89 -9244.4
## - Fireplaces     1   0.11365 113.76 -9243.9
## - BsmtFinSF2     1   0.11919 113.76 -9243.8
## - BsmtFinType1   5   0.46621 114.11 -9242.9
## - BedroomAbvGr   1   0.16913 113.81 -9242.5
## - GarageArea     1   0.20516 113.85 -9241.6
## - ScreenPorch    1   0.21146 113.86 -9241.4
## - BsmtFinSF1     1   0.21532 113.86 -9241.3
## - KitchenQual    3   0.37823 114.02 -9241.1
## + ExterQual      3   0.05388 113.59 -9240.2
## + LotConfig      4   0.12345 113.52 -9240.0
## - MiscVal        1   0.26963 113.91 -9239.9
## + BsmtExposure   3   0.03804 113.61 -9239.8
## + RoofStyle      5   0.16437 113.48 -9239.1
## - Condition2     7   0.77670 114.42 -9239.0
## + Neighborhood  24   1.62717 112.02 -9238.9
## + BsmtCond       3   0.00260 113.64 -9238.9
## + HeatingQC      4   0.07918 113.57 -9238.9
## - GarageYrBlt    1   0.32724 113.97 -9238.4
## + SaleType       8   0.36592 113.28 -9238.2
## + Condition1     8   0.36079 113.28 -9238.1
## - BsmtUnfSF      1   0.34117 113.99 -9238.1
## + Functional     6   0.17868 113.47 -9237.4
## - MoSold         1   0.37144 114.02 -9237.3
## - Electrical     4   0.60988 114.25 -9237.2
## + ExterCond      4   0.01109 113.63 -9237.1
## + GarageType     5   0.08859 113.56 -9237.1
## - LandSlope      2   0.47170 114.12 -9236.7
## - PavedDrive     2   0.47476 114.12 -9236.7
## - MSZoning       4   0.63267 114.28 -9236.6
## + HouseStyle     7   0.16469 113.48 -9235.1
## - KitchenAbvGr   1   0.56401 114.21 -9232.4
## - TotRmsAbvGrd   1   0.60063 114.25 -9231.4
## + Exterior1st   14   0.54517 113.10 -9230.9
## - OverallQual    1   0.67351 114.32 -9229.6
```

```
## - X1stFlrSF      1   0.89545 114.54 -9223.9
## - YearBuilt      1   0.95682 114.60 -9222.4
## + Exterior2nd   15   0.27778 113.37 -9222.0
## - Heating        5   1.34931 114.99 -9220.4
## - X2ndFlrSF      1   1.32703 114.97 -9212.9
## - OverallCond    1   1.35595 115.00 -9212.2
## - RoofMatl       7   2.49740 116.14 -9195.4
## - LotArea        1   2.16281 115.81 -9191.8
##
## Step:  AIC=-9249
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LandSlope +
##     Condition2 + BldgType + OverallQual + OverallCond + YearBuilt +
##     YearRemodAdd + RoofMatl + MasVnrArea + Foundation + BsmtQual +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + CentralAir + Electrical + X1stFlrSF + X2ndFlrSF +
##     LowQualFinSF + BsmtFullBath + BsmtHalfBath + FullBath + HalfBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageFinish + GarageCars + GarageArea +
##     GarageQual + PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - Foundation     5   0.22237 114.02 -9253.3
## - BldgType       4   0.15530 113.95 -9253.0
## - GarageQual     4   0.16566 113.96 -9252.8
## - BsmtFinType2   5   0.26465 114.06 -9252.2
## - LotShape       3   0.11518 113.91 -9252.0
## - BsmtQual       3   0.12773 113.92 -9251.7
## - GarageFinish   2   0.06982 113.86 -9251.2
## - LowQualFinSF   1   0.00000 113.79 -9251.0
## - BsmtFullBath   1   0.00001 113.79 -9251.0
## - GarageCars     1   0.00020 113.80 -9251.0
## - OpenPorchSF    1   0.00027 113.80 -9251.0
## - HalfBath       1   0.00116 113.80 -9251.0
## - MSSubClass     1   0.00245 113.80 -9250.9
## - CentralAir     1   0.00252 113.80 -9250.9
## - BsmtHalfBath   1   0.00315 113.80 -9250.9
## - WoodDeckSF     1   0.00405 113.80 -9250.9
## - MasVnrArea     1   0.00495 113.80 -9250.9
## - Street         1   0.00802 113.80 -9250.8
## - PoolArea       1   0.02800 113.82 -9250.3
## - SaleCondition  5   0.34970 114.14 -9250.0
## - YrSold         1   0.03833 113.83 -9250.0
## - X3SsnPorch     1   0.03876 113.83 -9250.0
## - EnclosedPorch  1   0.04454 113.84 -9249.9
## - YearRemodAdd   1   0.04878 113.84 -9249.7
## - Utilities      1   0.05099 113.84 -9249.7
## - FullBath       1   0.06058 113.86 -9249.4
## - LotFrontage    1   0.06563 113.86 -9249.3
## <none>                       113.79 -9249.0
```

```
## - LandContour    3   0.23922 114.03 -9248.9
## - BsmtFinSF2     1   0.12017 113.92 -9247.9
## - Fireplaces     1   0.12193 113.92 -9247.9
## - BsmtFinType1   5   0.47604 114.27 -9246.8
## - BedroomAbvGr   1   0.17063 113.97 -9246.6
## - GarageArea     1   0.20372 114.00 -9245.8
## - BsmtFinSF1     1   0.21224 114.01 -9245.6
## - ScreenPorch    1   0.21878 114.01 -9245.4
## - KitchenQual    3   0.38711 114.18 -9245.1
## + GarageCond     4   0.14953 113.64 -9244.8
## + ExterQual      3   0.06604 113.73 -9244.7
## - MiscVal        1   0.27010 114.06 -9244.1
## + BsmtExposure   3   0.03885 113.76 -9244.0
## + LotConfig      4   0.11597 113.68 -9244.0
## - GarageYrBlt    1   0.28805 114.08 -9243.6
## - Condition2     7   0.76791 114.56 -9243.4
## + HeatingQC      4   0.08130 113.71 -9243.1
## + BsmtCond       3   0.00177 113.79 -9243.0
## + RoofStyle      5   0.15092 113.64 -9242.9
## + Neighborhood  24   1.61879 112.18 -9242.8
## + SaleType       8   0.36230 113.43 -9242.3
## - BsmtUnfSF      1   0.34132 114.14 -9242.3
## - Electrical     4   0.58632 114.38 -9242.0
## + Condition1     8   0.34816 113.45 -9241.9
## - PavedDrive     2   0.44492 114.24 -9241.6
## - MoSold         1   0.36915 114.16 -9241.5
## + Functional     6   0.17374 113.62 -9241.5
## + ExterCond      4   0.01261 113.78 -9241.3
## + GarageType     5   0.08344 113.71 -9241.1
## - LandSlope      2   0.46400 114.26 -9241.1
## - MSZoning       4   0.64442 114.44 -9240.5
## + HouseStyle     7   0.14591 113.65 -9238.7
## - KitchenAbvGr   1   0.54353 114.34 -9237.1
## - TotRmsAbvGrd   1   0.60442 114.40 -9235.5
## + Exterior1st   14   0.53575 113.26 -9234.8
## - OverallQual    1   0.70017 114.50 -9233.1
## - X1stFlrSF      1   0.89943 114.69 -9228.0
## - YearBuilt      1   0.96048 114.75 -9226.5
## + Exterior2nd   15   0.26028 113.53 -9225.7
## - Heating        5   1.32594 115.12 -9225.2
## - X2ndFlrSF      1   1.33347 115.13 -9217.0
## - OverallCond    1   1.41673 115.21 -9214.9
## - RoofMatl       7   2.53994 116.33 -9198.6
## - LotArea        1   2.15523 115.95 -9196.2
##
## Step:  AIC=-9253.3
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LandSlope +
##     Condition2 + BldgType + OverallQual + OverallCond + YearBuilt +
##     YearRemodAdd + RoofMatl + MasVnrArea + BsmtQual + BsmtFinType1 +
##     BsmtFinSF1 + BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     CentralAir + Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF +
```

```
##     BsmtFullBath + BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageFinish + GarageCars + GarageArea + GarageQual +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - BldgType       4   0.15489 114.17 -9257.3
## - BsmtFinType2   5   0.27003 114.29 -9256.4
## - LotShape       3   0.13490 114.15 -9255.8
## - GarageQual     4   0.21742 114.23 -9255.7
## - LowQualFinSF   1   0.00001 114.02 -9255.3
## - OpenPorchSF    1   0.00009 114.02 -9255.3
## - GarageFinish   2   0.07836 114.09 -9255.3
## - GarageCars     1   0.00035 114.02 -9255.3
## - BsmtFullBath   1   0.00047 114.02 -9255.3
## - HalfBath       1   0.00081 114.02 -9255.3
## - MSSubClass     1   0.00105 114.02 -9255.3
## - CentralAir     1   0.00206 114.02 -9255.2
## - BsmtHalfBath   1   0.00251 114.02 -9255.2
## - WoodDeckSF     1   0.00418 114.02 -9255.2
## - Street         1   0.00573 114.02 -9255.2
## - MasVnrArea     1   0.00951 114.03 -9255.1
## - BsmtQual       3   0.17068 114.19 -9254.9
## - YearRemodAdd   1   0.02585 114.04 -9254.6
## - PoolArea       1   0.02640 114.04 -9254.6
## - X3SsnPorch     1   0.03509 114.05 -9254.4
## - YrSold         1   0.03874 114.06 -9254.3
## - SaleCondition  5   0.35592 114.37 -9254.2
## - EnclosedPorch  1   0.05046 114.07 -9254.0
## - Utilities      1   0.05053 114.07 -9254.0
## - LotFrontage    1   0.05217 114.07 -9254.0
## - LandContour    3   0.22685 114.24 -9253.5
## - FullBath       1   0.07231 114.09 -9253.4
## <none>                       114.02 -9253.3
## - Fireplaces     1   0.11618 114.13 -9252.3
## - BsmtFinType1   5   0.46705 114.48 -9251.4
## - BsmtFinSF2     1   0.15567 114.17 -9251.3
## - BedroomAbvGr   1   0.15641 114.17 -9251.3
## - GarageArea     1   0.18841 114.20 -9250.5
## - ScreenPorch    1   0.20710 114.22 -9250.0
## + Neighborhood  24   1.72646 112.29 -9249.8
## + Foundation     5   0.22237 113.79 -9249.0
## + ExterQual      3   0.06427 113.95 -9248.9
## + GarageCond     4   0.13959 113.88 -9248.9
## - GarageYrBlt    1   0.25525 114.27 -9248.8
## - KitchenQual    3   0.42409 114.44 -9248.5
## + BsmtExposure   3   0.04244 113.97 -9248.4
## - MiscVal        1   0.27572 114.29 -9248.2
## + HeatingQC      4   0.11317 113.90 -9248.2
## + LotConfig      4   0.10060 113.92 -9247.9
## - Condition2     7   0.77397 114.79 -9247.5
```

```
## + BsmtCond       3   0.00318 114.01 -9247.4
## + RoofStyle      5   0.15273 113.86 -9247.2
## - BsmtFinSF1     1   0.33798 114.36 -9246.7
## + Condition1     8   0.35828 113.66 -9246.5
## + SaleType       8   0.35818 113.66 -9246.5
## - LandSlope      2   0.42687 114.44 -9246.4
## - MSZoning       4   0.59094 114.61 -9246.2
## - MoSold         1   0.37063 114.39 -9245.8
## - PavedDrive     2   0.45274 114.47 -9245.7
## + ExterCond      4   0.01604 114.00 -9245.7
## + Functional     6   0.17167 113.84 -9245.7
## + GarageType     5   0.09335 113.92 -9245.7
## - Electrical     4   0.62037 114.64 -9245.5
## + HouseStyle     7   0.13874 113.88 -9242.9
## - KitchenAbvGr   1   0.52651 114.54 -9241.8
## - TotRmsAbvGrd   1   0.62844 114.64 -9239.3
## - BsmtUnfSF      1   0.63497 114.65 -9239.1
## + Exterior1st   14   0.50304 113.51 -9238.2
## - OverallQual    1   0.77790 114.80 -9235.5
## - X1stFlrSF      1   0.85623 114.87 -9233.5
## - Heating        5   1.32052 115.34 -9229.7
## - YearBuilt      1   1.00696 115.02 -9229.6
## + Exterior2nd   15   0.23880 113.78 -9229.4
## - OverallCond    1   1.32536 115.34 -9221.6
## - X2ndFlrSF      1   1.37083 115.39 -9220.4
## - RoofMatl       7   2.52034 116.54 -9203.5
## - LotArea        1   2.06321 116.08 -9202.9
##
## Step:  AIC=-9257.34
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LandSlope +
##     Condition2 + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofMatl + MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinType2 + BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir +
##     Electrical + X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath +
##     BsmtHalfBath + FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + PavedDrive + WoodDeckSF +
##     OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - BsmtFinType2   5   0.26640 114.44 -9260.5
## - GarageQual     4   0.20242 114.37 -9260.2
## - LotShape       3   0.13363 114.31 -9259.9
## - CentralAir     1   0.00002 114.17 -9259.3
## - GarageCars     1   0.00004 114.17 -9259.3
## - LowQualFinSF   1   0.00005 114.17 -9259.3
## - OpenPorchSF    1   0.00018 114.17 -9259.3
## - GarageFinish   2   0.07849 114.25 -9259.3
## - HalfBath       1   0.00111 114.17 -9259.3
## - BsmtFullBath   1   0.00298 114.17 -9259.3
```

```
## - Street         1   0.00359 114.17 -9259.2
## - WoodDeckSF     1   0.00412 114.18 -9259.2
## - BsmtHalfBath   1   0.00481 114.18 -9259.2
## - MSSubClass     1   0.01405 114.19 -9259.0
## - MasVnrArea     1   0.01511 114.19 -9258.9
## - BsmtQual       3   0.17578 114.35 -9258.8
## - PoolArea       1   0.02166 114.19 -9258.8
## - YearRemodAdd   1   0.02705 114.20 -9258.6
## - X3SsnPorch     1   0.03431 114.21 -9258.5
## - YrSold         1   0.04377 114.22 -9258.2
## - EnclosedPorch  1   0.04950 114.22 -9258.1
## - Utilities      1   0.04979 114.22 -9258.1
## - LandContour    3   0.21307 114.39 -9257.9
## - FullBath       1   0.07111 114.24 -9257.5
## <none>                       114.17 -9257.3
## - SaleCondition  5   0.39577 114.57 -9257.2
## - Fireplaces     1   0.10473 114.28 -9256.7
## - LotFrontage    1   0.11470 114.29 -9256.4
## - BsmtFinType1   5   0.44806 114.62 -9255.9
## - BsmtFinSF2     1   0.14584 114.32 -9255.6
## + Neighborhood  24   1.78581 112.39 -9255.4
## - ScreenPorch    1   0.20043 114.37 -9254.2
## - GarageArea     1   0.20476 114.38 -9254.1
## - BedroomAbvGr   1   0.22475 114.40 -9253.6
## + BldgType       4   0.15489 114.02 -9253.3
## + Foundation     5   0.22196 113.95 -9253.0
## + ExterQual      3   0.06516 114.11 -9253.0
## - GarageYrBlt    1   0.24813 114.42 -9253.0
## - KitchenQual    3   0.40675 114.58 -9253.0
## + GarageCond     4   0.13076 114.04 -9252.7
## + BsmtExposure   3   0.04227 114.13 -9252.4
## + HeatingQC      4   0.11383 114.06 -9252.2
## - MiscVal        1   0.27904 114.45 -9252.2
## + LotConfig      4   0.10817 114.06 -9252.1
## - Condition2     7   0.76738 114.94 -9251.8
## + BsmtCond       3   0.00425 114.17 -9251.4
## + RoofStyle      5   0.15912 114.01 -9251.4
## + SaleType       8   0.36864 113.80 -9250.8
## + Condition1     8   0.35736 113.81 -9250.5
## - MoSold         1   0.35552 114.53 -9250.3
## + Functional     6   0.18903 113.98 -9250.2
## - LandSlope      2   0.44567 114.62 -9250.0
## - BsmtFinSF1     1   0.37406 114.55 -9249.8
## + ExterCond      4   0.01707 114.16 -9249.8
## - Electrical     4   0.61219 114.78 -9249.7
## - PavedDrive     2   0.45567 114.63 -9249.7
## + GarageType     5   0.08112 114.09 -9249.4
## - MSZoning       4   0.66548 114.84 -9248.4
## + HouseStyle     7   0.11996 114.05 -9246.4
## - KitchenAbvGr   1   0.57776 114.75 -9244.6
## - TotRmsAbvGrd   1   0.65147 114.82 -9242.7
## - BsmtUnfSF      1   0.65314 114.83 -9242.7
```

```
## + Exterior1st   14   0.50239 113.67 -9242.2
## - OverallQual    1   0.76668 114.94 -9239.8
## - X1stFlrSF      1   0.82628 115.00 -9238.3
## - YearBuilt      1   0.96768 115.14 -9234.7
## + Exterior2nd   15   0.25709 113.92 -9233.9
## - Heating        5   1.31662 115.49 -9233.9
## - OverallCond    1   1.37259 115.54 -9224.5
## - X2ndFlrSF      1   1.42920 115.60 -9223.0
## - RoofMatl       7   2.61212 116.78 -9205.3
## - LotArea        1   2.20788 116.38 -9203.4
##
## Step:  AIC=-9260.53
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LandSlope +
##     Condition2 + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofMatl + MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + GarageQual + PavedDrive + WoodDeckSF +
##     OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - GarageQual     4   0.19389 114.63 -9263.6
## - LotShape       3   0.13472 114.57 -9263.1
## - LowQualFinSF   1   0.00001 114.44 -9262.5
## - CentralAir     1   0.00002 114.44 -9262.5
## - GarageCars     1   0.00010 114.44 -9262.5
## - OpenPorchSF    1   0.00011 114.44 -9262.5
## - HalfBath       1   0.00029 114.44 -9262.5
## - GarageFinish   2   0.07906 114.52 -9262.5
## - BsmtFullBath   1   0.00128 114.44 -9262.5
## - BsmtHalfBath   1   0.00310 114.44 -9262.5
## - WoodDeckSF     1   0.00598 114.44 -9262.4
## - Street         1   0.00746 114.45 -9262.3
## - MSSubClass     1   0.01632 114.45 -9262.1
## - PoolArea       1   0.01663 114.45 -9262.1
## - BsmtQual       3   0.17397 114.61 -9262.1
## - MasVnrArea     1   0.01751 114.46 -9262.1
## - YearRemodAdd   1   0.02695 114.47 -9261.8
## - X3SsnPorch     1   0.03469 114.47 -9261.6
## - YrSold         1   0.04725 114.48 -9261.3
## - LandContour    3   0.20556 114.64 -9261.3
## - EnclosedPorch  1   0.04996 114.49 -9261.3
## - Utilities      1   0.05087 114.49 -9261.2
## - FullBath       1   0.07370 114.51 -9260.7
## <none>                       114.44 -9260.5
## - SaleCondition  5   0.39697 114.83 -9260.4
## - Fireplaces     1   0.10681 114.55 -9259.8
## - LotFrontage    1   0.12062 114.56 -9259.5
```

```
## - BsmtFinSF2     1   0.14802 114.59 -9258.8
## - BsmtFinType1   5   0.47239 114.91 -9258.5
## + Neighborhood  24   1.77352 112.67 -9258.1
## - GarageArea     1   0.20048 114.64 -9257.4
## + BsmtFinType2   5   0.26640 114.17 -9257.3
## - ScreenPorch    1   0.21013 114.65 -9257.2
## - BedroomAbvGr   1   0.22908 114.67 -9256.7
## - KitchenQual    3   0.38882 114.83 -9256.6
## + Foundation     5   0.22999 114.21 -9256.4
## + BldgType       4   0.15126 114.29 -9256.4
## + ExterQual      3   0.05999 114.38 -9256.1
## + GarageCond     4   0.13690 114.30 -9256.0
## - GarageYrBlt    1   0.26059 114.70 -9255.9
## + BsmtExposure   3   0.03726 114.40 -9255.5
## + HeatingQC      4   0.11311 114.33 -9255.4
## - MiscVal        1   0.28692 114.72 -9255.2
## + LotConfig      4   0.10440 114.33 -9255.2
## - Condition2     7   0.76580 115.20 -9255.1
## + BsmtCond       3   0.00613 114.43 -9254.7
## + SaleType       8   0.39085 114.05 -9254.5
## + RoofStyle      5   0.15036 114.29 -9254.4
## - MoSold         1   0.34460 114.78 -9253.8
## + Functional     6   0.20220 114.24 -9253.7
## + Condition1     8   0.34735 114.09 -9253.4
## - Electrical     4   0.60395 115.04 -9253.2
## + ExterCond      4   0.01850 114.42 -9253.0
## - LandSlope      2   0.46252 114.90 -9252.8
## + GarageType     5   0.08209 114.36 -9252.6
## - PavedDrive     2   0.47275 114.91 -9252.5
## - BsmtFinSF1     1   0.40969 114.85 -9252.1
## - MSZoning       4   0.67153 115.11 -9251.5
## + HouseStyle     7   0.11540 114.32 -9249.5
## - KitchenAbvGr   1   0.56868 115.01 -9248.1
## - TotRmsAbvGrd   1   0.64591 115.08 -9246.1
## + Exterior1st   14   0.49627 113.94 -9245.2
## - BsmtUnfSF      1   0.68480 115.12 -9245.1
## - OverallQual    1   0.76105 115.20 -9243.2
## - X1stFlrSF      1   0.81144 115.25 -9241.9
## - YearBuilt      1   0.99451 115.43 -9237.3
## - Heating        5   1.31186 115.75 -9237.3
## + Exterior2nd   15   0.24657 114.19 -9236.8
## - OverallCond    1   1.42080 115.86 -9226.5
## - X2ndFlrSF      1   1.42761 115.87 -9226.3
## - RoofMatl       7   2.58884 117.03 -9209.2
## - LotArea        1   2.22270 116.66 -9206.4
##
## Step:  AIC=-9263.59
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LotShape + LandContour + Utilities + LandSlope +
##     Condition2 + OverallQual + OverallCond + YearBuilt + YearRemodAdd +
##     RoofMatl + MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + CentralAir + Electrical +
```

```
##     X1stFlrSF + X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - LotShape       3   0.14083 114.77 -9266.0
## - GarageFinish   2   0.07804 114.71 -9265.6
## - HalfBath       1   0.00006 114.63 -9265.6
## - LowQualFinSF   1   0.00021 114.63 -9265.6
## - OpenPorchSF    1   0.00036 114.63 -9265.6
## - BsmtFullBath   1   0.00109 114.63 -9265.6
## - GarageCars     1   0.00136 114.63 -9265.6
## - CentralAir     1   0.00147 114.63 -9265.6
## - BsmtHalfBath   1   0.00287 114.64 -9265.5
## - WoodDeckSF     1   0.00902 114.64 -9265.4
## - Street         1   0.01031 114.64 -9265.3
## - PoolArea       1   0.01477 114.65 -9265.2
## - MasVnrArea     1   0.01940 114.65 -9265.1
## - YearRemodAdd   1   0.02207 114.65 -9265.0
## - MSSubClass     1   0.02274 114.66 -9265.0
## - X3SsnPorch     1   0.03717 114.67 -9264.6
## - BsmtQual       3   0.19595 114.83 -9264.6
## - LandContour    3   0.20767 114.84 -9264.3
## - Utilities      1   0.05209 114.68 -9264.3
## - EnclosedPorch  1   0.05339 114.69 -9264.2
## - YrSold         1   0.05694 114.69 -9264.1
## - FullBath       1   0.07748 114.71 -9263.6
## <none>                       114.63 -9263.6
## - SaleCondition  5   0.39952 115.03 -9263.4
## - Fireplaces     1   0.09503 114.73 -9263.2
## - LotFrontage    1   0.10828 114.74 -9262.8
## + Neighborhood  24   1.80640 112.83 -9262.0
## - BsmtFinSF2     1   0.15080 114.78 -9261.8
## - BsmtFinType1   5   0.47220 115.10 -9261.6
## + Foundation     5   0.28200 114.35 -9260.8
## + GarageQual     4   0.19389 114.44 -9260.5
## - GarageArea     1   0.20204 114.83 -9260.5
## - ScreenPorch    1   0.21246 114.84 -9260.2
## + BsmtFinType2   5   0.25788 114.37 -9260.2
## - BedroomAbvGr   1   0.22052 114.85 -9260.0
## - KitchenQual    3   0.39690 115.03 -9259.5
## + GarageCond     4   0.14374 114.49 -9259.3
## + ExterQual      3   0.06496 114.57 -9259.2
## + BldgType       4   0.13696 114.50 -9259.1
## + HeatingQC      4   0.13102 114.50 -9258.9
## + BsmtExposure   3   0.03714 114.59 -9258.5
## - Condition2     7   0.75458 115.39 -9258.4
## - MiscVal        1   0.28523 114.92 -9258.3
## + LotConfig      4   0.10344 114.53 -9258.2
```

```
## + BsmtCond       3   0.00806 114.62 -9257.8
## + SaleType       8   0.39092 114.24 -9257.6
## + RoofStyle      5   0.15369 114.48 -9257.5
## + Functional     6   0.21710 114.42 -9257.1
## - MoSold         1   0.33518 114.97 -9257.1
## + Condition1     8   0.36897 114.26 -9257.0
## - Electrical     4   0.59186 115.22 -9256.6
## + ExterCond      4   0.02513 114.61 -9256.2
## - LandSlope      2   0.46934 115.10 -9255.7
## + GarageType     5   0.07961 114.55 -9255.6
## - BsmtFinSF1     1   0.41899 115.05 -9254.9
## - MSZoning       4   0.66527 115.30 -9254.7
## - PavedDrive     2   0.51589 115.15 -9254.5
## - GarageYrBlt    1   0.44171 115.07 -9254.4
## + HouseStyle     7   0.10706 114.53 -9252.3
## - KitchenAbvGr   1   0.58307 115.22 -9250.8
## - TotRmsAbvGrd   1   0.66098 115.29 -9248.8
## + Exterior1st   14   0.49373 114.14 -9248.2
## - BsmtUnfSF      1   0.73171 115.36 -9247.0
## - OverallQual    1   0.73947 115.37 -9246.8
## - X1stFlrSF      1   0.77576 115.41 -9245.9
## - Heating        5   1.28784 115.92 -9241.0
## - YearBuilt      1   0.98682 115.62 -9240.6
## + Exterior2nd   15   0.24917 114.38 -9239.9
## - OverallCond    1   1.39189 116.02 -9230.4
## - X2ndFlrSF      1   1.42258 116.06 -9229.6
## - RoofMatl       7   2.57213 117.20 -9212.8
## - LotArea        1   2.22639 116.86 -9209.4
##
## Step:  AIC=-9266.01
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageFinish +
##     GarageCars + GarageArea + PavedDrive + WoodDeckSF + OpenPorchSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - GarageFinish   2   0.07368 114.85 -9268.1
## - HalfBath       1   0.00002 114.77 -9268.0
## - LowQualFinSF   1   0.00004 114.77 -9268.0
## - OpenPorchSF    1   0.00029 114.77 -9268.0
## - GarageCars     1   0.00123 114.77 -9268.0
## - CentralAir     1   0.00125 114.77 -9268.0
## - BsmtFullBath   1   0.00134 114.77 -9268.0
## - BsmtHalfBath   1   0.00306 114.78 -9267.9
## - WoodDeckSF     1   0.00835 114.78 -9267.8
```

```
## - Street         1   0.00879 114.78 -9267.8
## - PoolArea       1   0.01491 114.79 -9267.6
## - MasVnrArea     1   0.01827 114.79 -9267.5
## - YearRemodAdd   1   0.01980 114.79 -9267.5
## - MSSubClass     1   0.02465 114.80 -9267.4
## - BsmtQual       3   0.19472 114.97 -9267.1
## - X3SsnPorch     1   0.03811 114.81 -9267.0
## - EnclosedPorch  1   0.05336 114.83 -9266.6
## - Utilities      1   0.05502 114.83 -9266.6
## - YrSold         1   0.05722 114.83 -9266.6
## - LandContour    3   0.22707 115.00 -9266.2
## - FullBath       1   0.07556 114.85 -9266.1
## <none>                       114.77 -9266.0
## - Fireplaces     1   0.08153 114.85 -9265.9
## - SaleCondition  5   0.40437 115.18 -9265.7
## - LotFrontage    1   0.09464 114.87 -9265.6
## - BsmtFinSF2     1   0.14797 114.92 -9264.2
## - BsmtFinType1   5   0.47563 115.25 -9263.9
## + Foundation     5   0.30353 114.47 -9263.7
## + LotShape       3   0.14083 114.63 -9263.6
## + GarageQual     4   0.19999 114.57 -9263.1
## - GarageArea     1   0.20699 114.98 -9262.7
## + BsmtFinType2   5   0.25908 114.51 -9262.6
## - BedroomAbvGr   1   0.21306 114.99 -9262.6
## + Neighborhood  24   1.73701 113.04 -9262.5
## - ScreenPorch    1   0.22433 115.00 -9262.3
## - KitchenQual    3   0.39571 115.17 -9262.0
## + ExterQual      3   0.07045 114.70 -9261.8
## + GarageCond     4   0.14730 114.62 -9261.8
## + HeatingQC      4   0.13716 114.64 -9261.5
## + BldgType       4   0.13602 114.64 -9261.5
## + BsmtExposure   3   0.03141 114.74 -9260.8
## - Condition2     7   0.76208 115.53 -9260.7
## - MiscVal        1   0.28953 115.06 -9260.7
## + LotConfig      4   0.09261 114.68 -9260.4
## + SaleType       8   0.40080 114.37 -9260.2
## + BsmtCond       3   0.00737 114.77 -9260.2
## + RoofStyle      5   0.15361 114.62 -9259.9
## + Functional     6   0.22140 114.55 -9259.6
## - MoSold         1   0.34030 115.11 -9259.4
## - Electrical     4   0.58908 115.36 -9259.1
## + Condition1     8   0.34704 114.43 -9258.8
## + ExterCond      4   0.02464 114.75 -9258.6
## - LandSlope      2   0.46622 115.24 -9258.2
## + GarageType     5   0.08484 114.69 -9258.2
## - BsmtFinSF1     1   0.40390 115.18 -9257.8
## - MSZoning       4   0.64180 115.42 -9257.7
## - PavedDrive     2   0.50703 115.28 -9257.1
## - GarageYrBlt    1   0.45007 115.22 -9256.6
## + HouseStyle     7   0.10080 114.67 -9254.6
## - KitchenAbvGr   1   0.56941 115.34 -9253.6
## - TotRmsAbvGrd   1   0.65827 115.43 -9251.3
```

```
## + Exterior1st   14   0.49233 114.28 -9250.6
## - BsmtUnfSF      1   0.72634 115.50 -9249.6
## - OverallQual    1   0.73469 115.51 -9249.4
## - X1stFlrSF      1   0.78816 115.56 -9248.0
## - YearBuilt      1   0.97256 115.75 -9243.4
## - Heating        5   1.29582 116.07 -9243.2
## + Exterior2nd   15   0.23723 114.54 -9242.0
## - OverallCond    1   1.38790 116.16 -9232.9
## - X2ndFlrSF      1   1.43312 116.21 -9231.8
## - RoofMatl       7   2.73547 117.51 -9211.3
## - LotArea        1   2.40115 117.17 -9207.6
##
## Step:  AIC=-9268.13
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageCars + GarageArea +
##     PavedDrive + WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - HalfBath       1   0.00005 114.85 -9270.1
## - OpenPorchSF    1   0.00012 114.85 -9270.1
## - LowQualFinSF   1   0.00017 114.85 -9270.1
## - CentralAir     1   0.00117 114.85 -9270.1
## - BsmtFullBath   1   0.00188 114.85 -9270.1
## - GarageCars     1   0.00191 114.85 -9270.1
## - BsmtHalfBath   1   0.00356 114.85 -9270.0
## - Street         1   0.00847 114.86 -9269.9
## - WoodDeckSF     1   0.01050 114.86 -9269.9
## - PoolArea       1   0.01430 114.86 -9269.8
## - MasVnrArea     1   0.01860 114.86 -9269.7
## - YearRemodAdd   1   0.01880 114.86 -9269.7
## - MSSubClass     1   0.02497 114.87 -9269.5
## - X3SsnPorch     1   0.03864 114.89 -9269.2
## - BsmtQual       3   0.20147 115.05 -9269.0
## - Utilities      1   0.05333 114.90 -9268.8
## - EnclosedPorch  1   0.05545 114.90 -9268.7
## - YrSold         1   0.05802 114.90 -9268.7
## - LandContour    3   0.23552 115.08 -9268.2
## <none>                       114.85 -9268.1
## - FullBath       1   0.08368 114.93 -9268.0
## - Fireplaces     1   0.09471 114.94 -9267.7
## - SaleCondition  5   0.41577 115.26 -9267.6
## - LotFrontage    1   0.10812 114.95 -9267.4
## - BsmtFinSF2     1   0.14693 114.99 -9266.4
## + Foundation     5   0.31660 114.53 -9266.2
## - BsmtFinType1   5   0.47134 115.32 -9266.2
```

```
## + GarageFinish   2   0.07368 114.77 -9266.0
## + LotShape       3   0.13648 114.71 -9265.6
## - GarageArea     1   0.18363 115.03 -9265.5
## + GarageQual     4   0.19923 114.65 -9265.2
## + Neighborhood  24   1.75842 113.09 -9265.2
## - BedroomAbvGr   1   0.20106 115.05 -9265.0
## + BsmtFinType2   5   0.25989 114.59 -9264.7
## - ScreenPorch    1   0.23544 115.08 -9264.2
## - KitchenQual    3   0.39590 115.24 -9264.1
## + ExterQual      3   0.07558 114.77 -9264.1
## + GarageCond     4   0.14815 114.70 -9263.9
## + HeatingQC      4   0.14364 114.70 -9263.8
## + BldgType       4   0.13819 114.71 -9263.6
## + BsmtExposure   3   0.02792 114.82 -9262.8
## - MiscVal        1   0.28878 115.14 -9262.8
## - Condition2     7   0.76448 115.61 -9262.8
## + LotConfig      4   0.09535 114.75 -9262.6
## + SaleType       8   0.40072 114.45 -9262.3
## + BsmtCond       3   0.00581 114.84 -9262.3
## + Functional     6   0.23073 114.62 -9262.0
## + RoofStyle      5   0.15048 114.70 -9262.0
## - MoSold         1   0.32451 115.17 -9261.9
## - Electrical     4   0.58772 115.43 -9261.2
## + Condition1     8   0.35249 114.49 -9261.1
## + ExterCond      4   0.02428 114.82 -9260.8
## - MSZoning       4   0.62066 115.47 -9260.4
## - LandSlope      2   0.46866 115.31 -9260.2
## + GarageType     5   0.08042 114.77 -9260.2
## - BsmtFinSF1     1   0.39912 115.25 -9260.0
## - GarageYrBlt    1   0.39961 115.25 -9260.0
## - PavedDrive     2   0.52164 115.37 -9258.9
## + HouseStyle     7   0.10039 114.75 -9256.7
## - KitchenAbvGr   1   0.59145 115.44 -9255.1
## + Exterior1st   14   0.50573 114.34 -9253.0
## - TotRmsAbvGrd   1   0.67675 115.52 -9253.0
## - BsmtUnfSF      1   0.70793 115.55 -9252.2
## - OverallQual    1   0.76641 115.61 -9250.7
## - X1stFlrSF      1   0.81064 115.66 -9249.6
## - Heating        5   1.31386 116.16 -9244.9
## + Exterior2nd   15   0.24592 114.60 -9244.4
## - YearBuilt      1   1.03983 115.89 -9243.8
## - OverallCond    1   1.37663 116.22 -9235.4
## - X2ndFlrSF      1   1.44444 116.29 -9233.6
## - RoofMatl       7   2.74201 117.59 -9213.3
## - LotArea        1   2.40678 117.25 -9209.6
##
## Step:  AIC=-9270.13
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + CentralAir + Electrical + X1stFlrSF +
```

```
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageCars + GarageArea + PavedDrive +
##     WoodDeckSF + OpenPorchSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - OpenPorchSF    1   0.00011 114.85 -9272.1
## - LowQualFinSF   1   0.00017 114.85 -9272.1
## - CentralAir     1   0.00115 114.85 -9272.1
## - BsmtFullBath   1   0.00185 114.85 -9272.1
## - GarageCars     1   0.00196 114.85 -9272.1
## - BsmtHalfBath   1   0.00353 114.85 -9272.0
## - Street         1   0.00846 114.86 -9271.9
## - WoodDeckSF     1   0.01050 114.86 -9271.9
## - PoolArea       1   0.01425 114.86 -9271.8
## - MasVnrArea     1   0.01856 114.86 -9271.7
## - YearRemodAdd   1   0.01879 114.86 -9271.7
## - MSSubClass     1   0.02494 114.87 -9271.5
## - X3SsnPorch     1   0.03865 114.89 -9271.1
## - BsmtQual       3   0.20142 115.05 -9271.0
## - Utilities      1   0.05340 114.90 -9270.8
## - EnclosedPorch  1   0.05543 114.90 -9270.7
## - YrSold         1   0.05798 114.90 -9270.7
## - LandContour    3   0.23576 115.08 -9270.1
## <none>                       114.85 -9270.1
## - FullBath       1   0.09423 114.94 -9269.7
## - Fireplaces     1   0.09539 114.94 -9269.7
## - SaleCondition  5   0.41596 115.26 -9269.6
## - LotFrontage    1   0.10808 114.95 -9269.4
## - BsmtFinSF2     1   0.14701 114.99 -9268.4
## + Foundation     5   0.31659 114.53 -9268.2
## - BsmtFinType1   5   0.47261 115.32 -9268.1
## + HalfBath       1   0.00005 114.85 -9268.1
## + GarageFinish   2   0.07371 114.77 -9268.0
## + LotShape       3   0.13651 114.71 -9267.6
## - GarageArea     1   0.18417 115.03 -9267.5
## + GarageQual     4   0.19928 114.65 -9267.2
## - BedroomAbvGr   1   0.20102 115.05 -9267.0
## + Neighborhood  24   1.74796 113.10 -9266.9
## + BsmtFinType2   5   0.25987 114.59 -9266.7
## - ScreenPorch    1   0.23568 115.08 -9266.1
## - KitchenQual    3   0.39589 115.24 -9266.1
## + ExterQual      3   0.07539 114.77 -9266.0
## + GarageCond     4   0.14783 114.70 -9265.9
## + HeatingQC      4   0.14326 114.70 -9265.8
## + BldgType       4   0.13795 114.71 -9265.6
## + BsmtExposure   3   0.02770 114.82 -9264.8
## - MiscVal        1   0.28873 115.14 -9264.8
## - Condition2     7   0.76456 115.61 -9264.8
## + LotConfig      4   0.09539 114.75 -9264.6
## + SaleType       8   0.40050 114.45 -9264.3
```

```
## + BsmtCond       3   0.00579 114.84 -9264.3
## + Functional     6   0.23076 114.62 -9264.0
## + RoofStyle      5   0.15042 114.70 -9264.0
## - MoSold         1   0.32446 115.17 -9263.9
## - Electrical     4   0.58770 115.43 -9263.2
## + Condition1     8   0.35253 114.49 -9263.1
## + ExterCond      4   0.02430 114.82 -9262.7
## - LandSlope      2   0.46864 115.31 -9262.2
## - MSZoning       4   0.62693 115.47 -9262.2
## + GarageType     5   0.08038 114.77 -9262.2
## - BsmtFinSF1     1   0.39933 115.25 -9262.0
## - GarageYrBlt    1   0.39957 115.25 -9262.0
## - PavedDrive     2   0.52207 115.37 -9260.9
## + HouseStyle     7   0.09780 114.75 -9258.6
## - KitchenAbvGr   1   0.59202 115.44 -9257.1
## + Exterior1st   14   0.50558 114.34 -9255.0
## - TotRmsAbvGrd   1   0.67672 115.52 -9255.0
## - BsmtUnfSF      1   0.70889 115.56 -9254.2
## - OverallQual    1   0.76639 115.61 -9252.7
## - X1stFlrSF      1   0.81626 115.66 -9251.5
## - Heating        5   1.31541 116.16 -9246.9
## + Exterior2nd   15   0.24532 114.60 -9246.4
## - YearBuilt      1   1.08928 115.94 -9244.6
## - OverallCond    1   1.37687 116.22 -9237.3
## - X2ndFlrSF      1   1.86151 116.71 -9225.2
## - RoofMatl       7   2.74298 117.59 -9215.2
## - LotArea        1   2.40673 117.25 -9211.6
##
## Step:  AIC=-9272.13
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + LowQualFinSF + BsmtFullBath + BsmtHalfBath +
##     FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageCars + GarageArea + PavedDrive +
##     WoodDeckSF + EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea +
##     MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - LowQualFinSF   1   0.00017 114.85 -9274.1
## - CentralAir     1   0.00114 114.85 -9274.1
## - BsmtFullBath   1   0.00182 114.85 -9274.1
## - GarageCars     1   0.00203 114.85 -9274.1
## - BsmtHalfBath   1   0.00353 114.85 -9274.0
## - Street         1   0.00848 114.86 -9273.9
## - WoodDeckSF     1   0.01076 114.86 -9273.9
## - PoolArea       1   0.01419 114.86 -9273.8
## - MasVnrArea     1   0.01847 114.86 -9273.7
## - YearRemodAdd   1   0.01896 114.87 -9273.6
## - MSSubClass     1   0.02484 114.87 -9273.5
```

```
## - X3SsnPorch     1   0.03880 114.89 -9273.1
## - BsmtQual       3   0.20133 115.05 -9273.0
## - Utilities      1   0.05358 114.90 -9272.8
## - EnclosedPorch  1   0.05572 114.90 -9272.7
## - YrSold         1   0.05786 114.90 -9272.7
## - LandContour    3   0.23567 115.08 -9272.1
## <none>                       114.85 -9272.1
## - FullBath       1   0.09414 114.94 -9271.7
## - Fireplaces     1   0.09534 114.94 -9271.7
## - SaleCondition  5   0.41602 115.26 -9271.6
## - LotFrontage    1   0.10797 114.95 -9271.4
## - BsmtFinSF2     1   0.14690 114.99 -9270.4
## + Foundation     5   0.31624 114.53 -9270.2
## - BsmtFinType1   5   0.47258 115.32 -9270.1
## + OpenPorchSF    1   0.00011 114.85 -9270.1
## + HalfBath       1   0.00004 114.85 -9270.1
## + GarageFinish   2   0.07353 114.77 -9270.0
## + LotShape       3   0.13648 114.71 -9269.6
## - GarageArea     1   0.18417 115.03 -9269.5
## + GarageQual     4   0.19939 114.65 -9269.2
## - BedroomAbvGr   1   0.20168 115.05 -9269.0
## + Neighborhood  24   1.74595 113.10 -9268.8
## + BsmtFinType2   5   0.25976 114.59 -9268.7
## - ScreenPorch    1   0.23558 115.08 -9268.1
## - KitchenQual    3   0.39578 115.24 -9268.1
## + ExterQual      3   0.07528 114.77 -9268.0
## + GarageCond     4   0.14792 114.70 -9267.9
## + HeatingQC      4   0.14291 114.70 -9267.8
## + BldgType       4   0.13799 114.71 -9267.6
## + BsmtExposure   3   0.02773 114.82 -9266.8
## - MiscVal        1   0.29096 115.14 -9266.7
## - Condition2     7   0.76661 115.61 -9266.7
## + LotConfig      4   0.09541 114.75 -9266.6
## + SaleType       8   0.40060 114.45 -9266.3
## + BsmtCond       3   0.00577 114.84 -9266.3
## + Functional     6   0.22996 114.62 -9266.0
## + RoofStyle      5   0.15023 114.70 -9266.0
## - MoSold         1   0.32435 115.17 -9265.9
## - Electrical     4   0.58761 115.43 -9265.2
## + Condition1     8   0.35241 114.49 -9265.1
## + ExterCond      4   0.02439 114.82 -9264.7
## - LandSlope      2   0.46862 115.31 -9264.2
## + GarageType     5   0.08041 114.77 -9264.2
## - MSZoning       4   0.62958 115.48 -9264.2
## - BsmtFinSF1     1   0.39925 115.25 -9264.0
## - GarageYrBlt    1   0.40022 115.25 -9264.0
## - PavedDrive     2   0.52199 115.37 -9262.9
## + HouseStyle     7   0.09791 114.75 -9260.6
## - KitchenAbvGr   1   0.59325 115.44 -9259.1
## + Exterior1st   14   0.50569 114.34 -9257.0
## - TotRmsAbvGrd   1   0.67661 115.52 -9257.0
## - BsmtUnfSF      1   0.70974 115.56 -9256.1
```

```
## - OverallQual    1   0.76640 115.61 -9254.7
## - X1stFlrSF      1   0.82169 115.67 -9253.3
## - Heating        5   1.32025 116.17 -9248.8
## + Exterior2nd   15   0.24523 114.60 -9248.4
## - YearBuilt      1   1.08929 115.94 -9246.6
## - OverallCond    1   1.37676 116.22 -9239.3
## - X2ndFlrSF      1   1.89394 116.74 -9226.4
## - RoofMatl       7   2.74327 117.59 -9217.2
## - LotArea        1   2.40742 117.25 -9213.6
##
## Step:  AIC=-9274.12
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + CentralAir + Electrical + X1stFlrSF +
##     X2ndFlrSF + BsmtFullBath + BsmtHalfBath + FullBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageCars + GarageArea + PavedDrive + WoodDeckSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - CentralAir     1   0.00115 114.85 -9276.1
## - BsmtFullBath   1   0.00181 114.85 -9276.1
## - GarageCars     1   0.00206 114.85 -9276.1
## - BsmtHalfBath   1   0.00354 114.85 -9276.0
## - Street         1   0.00844 114.86 -9275.9
## - WoodDeckSF     1   0.01074 114.86 -9275.9
## - PoolArea       1   0.01409 114.86 -9275.8
## - MasVnrArea     1   0.01840 114.86 -9275.7
## - YearRemodAdd   1   0.01895 114.87 -9275.6
## - MSSubClass     1   0.02518 114.87 -9275.5
## - X3SsnPorch     1   0.03874 114.89 -9275.1
## - BsmtQual       3   0.20259 115.05 -9275.0
## - Utilities      1   0.05354 114.90 -9274.8
## - EnclosedPorch  1   0.05558 114.90 -9274.7
## - YrSold         1   0.05789 114.91 -9274.7
## <none>                       114.85 -9274.1
## - LandContour    3   0.23793 115.08 -9274.1
## - FullBath       1   0.09398 114.94 -9273.7
## - Fireplaces     1   0.09527 114.94 -9273.7
## - SaleCondition  5   0.41724 115.26 -9273.5
## - LotFrontage    1   0.10787 114.95 -9273.4
## - BsmtFinSF2     1   0.14689 114.99 -9272.4
## + Foundation     5   0.31617 114.53 -9272.2
## - BsmtFinType1   5   0.47247 115.32 -9272.1
## + LowQualFinSF   1   0.00017 114.85 -9272.1
## + GrLivArea      1   0.00017 114.85 -9272.1
## + OpenPorchSF    1   0.00012 114.85 -9272.1
## + HalfBath       1   0.00004 114.85 -9272.1
## + GarageFinish   2   0.07366 114.77 -9272.0
```

```
## + LotShape       3   0.13616 114.71 -9271.6
## - GarageArea     1   0.18403 115.03 -9271.5
## + GarageQual     4   0.19954 114.65 -9271.2
## - BedroomAbvGr   1   0.20169 115.05 -9271.0
## + Neighborhood  24   1.74302 113.10 -9270.8
## + BsmtFinType2   5   0.25963 114.59 -9270.7
## - ScreenPorch    1   0.23546 115.08 -9270.1
## - KitchenQual    3   0.39565 115.24 -9270.1
## + ExterQual      3   0.07447 114.77 -9270.0
## + GarageCond     4   0.14805 114.70 -9269.9
## + HeatingQC      4   0.14282 114.70 -9269.8
## + BldgType       4   0.13814 114.71 -9269.6
## + BsmtExposure   3   0.02778 114.82 -9268.8
## - MiscVal        1   0.29089 115.14 -9268.7
## - Condition2     7   0.76664 115.61 -9268.7
## + LotConfig      4   0.09532 114.75 -9268.5
## + SaleType       8   0.40022 114.45 -9268.3
## + BsmtCond       3   0.00576 114.84 -9268.3
## + Functional     6   0.22972 114.62 -9268.0
## + RoofStyle      5   0.15023 114.70 -9267.9
## - MoSold         1   0.32427 115.17 -9267.9
## - Electrical     4   0.58744 115.43 -9267.2
## + Condition1     8   0.35245 114.49 -9267.1
## + ExterCond      4   0.02436 114.82 -9266.7
## - LandSlope      2   0.46857 115.31 -9266.2
## + GarageType     5   0.08037 114.77 -9266.2
## - MSZoning       4   0.62998 115.48 -9266.2
## - BsmtFinSF1     1   0.39914 115.25 -9266.0
## - GarageYrBlt    1   0.40332 115.25 -9265.9
## - PavedDrive     2   0.52414 115.37 -9264.8
## + HouseStyle     7   0.09753 114.75 -9262.6
## - KitchenAbvGr   1   0.59695 115.44 -9261.0
## + Exterior1st   14   0.50582 114.34 -9259.0
## - TotRmsAbvGrd   1   0.69066 115.54 -9258.6
## - BsmtUnfSF      1   0.70959 115.56 -9258.1
## - OverallQual    1   0.76711 115.61 -9256.7
## - X1stFlrSF      1   0.83075 115.68 -9255.1
## - Heating        5   1.32008 116.17 -9250.8
## + Exterior2nd   15   0.24506 114.60 -9250.4
## - YearBuilt      1   1.11124 115.96 -9248.0
## - OverallCond    1   1.38167 116.23 -9241.2
## - X2ndFlrSF      1   1.93690 116.78 -9227.3
## - RoofMatl       7   2.74637 117.59 -9219.1
## - LotArea        1   2.40729 117.25 -9215.6
##
## Step:  AIC=-9276.1
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + Electrical + X1stFlrSF + X2ndFlrSF +
##     BsmtFullBath + BsmtHalfBath + FullBath + BedroomAbvGr + KitchenAbvGr +
```

```
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageCars +
##     GarageArea + PavedDrive + WoodDeckSF + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - BsmtFullBath   1   0.00178 114.85 -9278.1
## - GarageCars     1   0.00199 114.85 -9278.0
## - BsmtHalfBath   1   0.00347 114.85 -9278.0
## - Street         1   0.00830 114.86 -9277.9
## - WoodDeckSF     1   0.01085 114.86 -9277.8
## - PoolArea       1   0.01408 114.86 -9277.7
## - MasVnrArea     1   0.01822 114.87 -9277.6
## - YearRemodAdd   1   0.01913 114.87 -9277.6
## - MSSubClass     1   0.02502 114.87 -9277.5
## - X3SsnPorch     1   0.03865 114.89 -9277.1
## - BsmtQual       3   0.20386 115.05 -9276.9
## - Utilities      1   0.05375 114.90 -9276.7
## - EnclosedPorch  1   0.05536 114.90 -9276.7
## - YrSold         1   0.05776 114.91 -9276.6
## <none>                       114.85 -9276.1
## - LandContour    3   0.23698 115.08 -9276.1
## - FullBath       1   0.09441 114.94 -9275.7
## - Fireplaces     1   0.09462 114.94 -9275.7
## - SaleCondition  5   0.41764 115.27 -9275.5
## - LotFrontage    1   0.10848 114.96 -9275.3
## - BsmtFinSF2     1   0.14617 114.99 -9274.4
## + Foundation     5   0.31717 114.53 -9274.2
## - BsmtFinType1   5   0.47179 115.32 -9274.1
## + CentralAir     1   0.00115 114.85 -9274.1
## + LowQualFinSF   1   0.00019 114.85 -9274.1
## + GrLivArea      1   0.00019 114.85 -9274.1
## + OpenPorchSF    1   0.00010 114.85 -9274.1
## + HalfBath       1   0.00002 114.85 -9274.1
## + GarageFinish   2   0.07356 114.77 -9274.0
## + LotShape       3   0.13597 114.71 -9273.6
## - GarageArea     1   0.18354 115.03 -9273.4
## + GarageQual     4   0.20059 114.65 -9273.2
## - BedroomAbvGr   1   0.20072 115.05 -9273.0
## + Neighborhood  24   1.74416 113.10 -9272.8
## + BsmtFinType2   5   0.25953 114.59 -9272.7
## - ScreenPorch    1   0.23491 115.08 -9272.1
## - KitchenQual    3   0.39715 115.25 -9272.0
## + ExterQual      3   0.07562 114.77 -9272.0
## + GarageCond     4   0.14794 114.70 -9271.9
## + HeatingQC      4   0.14204 114.71 -9271.7
## + BldgType       4   0.13927 114.71 -9271.6
## + BsmtExposure   3   0.02787 114.82 -9270.8
## - MiscVal        1   0.29072 115.14 -9270.7
## - Condition2     7   0.76569 115.61 -9270.7
## + LotConfig      4   0.09584 114.75 -9270.5
## + SaleType       8   0.40122 114.45 -9270.3
## + BsmtCond       3   0.00576 114.84 -9270.2
```

```
## + Functional     6   0.22962 114.62 -9269.9
## + RoofStyle      5   0.15041 114.70 -9269.9
## - MoSold         1   0.32465 115.17 -9269.9
## - Electrical     4   0.58748 115.44 -9269.2
## + Condition1     8   0.35177 114.50 -9269.0
## + ExterCond      4   0.02424 114.82 -9268.7
## - LandSlope      2   0.46795 115.32 -9268.2
## - MSZoning       4   0.62911 115.48 -9268.1
## + GarageType     5   0.08030 114.77 -9268.1
## - BsmtFinSF1     1   0.39831 115.25 -9268.0
## - GarageYrBlt    1   0.40276 115.25 -9267.9
## - PavedDrive     2   0.53811 115.39 -9266.5
## + HouseStyle     7   0.09700 114.75 -9264.6
## - KitchenAbvGr   1   0.59912 115.45 -9262.9
## + Exterior1st   14   0.50504 114.34 -9261.0
## - TotRmsAbvGrd   1   0.69198 115.54 -9260.6
## - BsmtUnfSF      1   0.70850 115.56 -9260.1
## - OverallQual    1   0.76690 115.61 -9258.7
## - X1stFlrSF      1   0.83023 115.68 -9257.1
## + Exterior2nd   15   0.24290 114.61 -9252.3
## - Heating        5   1.36647 116.21 -9251.6
## - YearBuilt      1   1.12295 115.97 -9249.7
## - OverallCond    1   1.40813 116.26 -9242.5
## - X2ndFlrSF      1   1.93575 116.78 -9229.3
## - RoofMatl       7   2.74588 117.59 -9221.1
## - LotArea        1   2.40680 117.25 -9217.6
##
## Step:  AIC=-9278.05
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + Electrical + X1stFlrSF + X2ndFlrSF +
##     BsmtHalfBath + FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageCars + GarageArea +
##     PavedDrive + WoodDeckSF + EnclosedPorch + X3SsnPorch + ScreenPorch +
##     PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - GarageCars     1   0.00205 114.85 -9280.0
## - BsmtHalfBath   1   0.00230 114.85 -9280.0
## - Street         1   0.00822 114.86 -9279.8
## - WoodDeckSF     1   0.01127 114.86 -9279.8
## - PoolArea       1   0.01431 114.86 -9279.7
## - YearRemodAdd   1   0.01861 114.87 -9279.6
## - MasVnrArea     1   0.01869 114.87 -9279.6
## - MSSubClass     1   0.02422 114.87 -9279.4
## - X3SsnPorch     1   0.03888 114.89 -9279.1
## - BsmtQual       3   0.20458 115.05 -9278.9
## - Utilities      1   0.05392 114.90 -9278.7
## - EnclosedPorch  1   0.05548 114.91 -9278.6
## - YrSold         1   0.05736 114.91 -9278.6
```

```
## <none>                       114.85 -9278.1
## - LandContour    3   0.23751 115.09 -9278.0
## - FullBath       1   0.09266 114.94 -9277.7
## - Fireplaces     1   0.09498 114.94 -9277.6
## - SaleCondition  5   0.42086 115.27 -9277.4
## - LotFrontage    1   0.10924 114.96 -9277.3
## + Foundation     5   0.31865 114.53 -9276.2
## + BsmtFullBath   1   0.00178 114.85 -9276.1
## + CentralAir     1   0.00112 114.85 -9276.1
## + LowQualFinSF   1   0.00018 114.85 -9276.1
## + GrLivArea      1   0.00018 114.85 -9276.1
## + OpenPorchSF    1   0.00007 114.85 -9276.1
## + HalfBath       1   0.00000 114.85 -9276.1
## + GarageFinish   2   0.07404 114.78 -9275.9
## - BsmtFinSF2     1   0.16839 115.02 -9275.8
## - BsmtFinType1   5   0.49281 115.34 -9275.6
## + LotShape       3   0.13620 114.71 -9275.5
## - GarageArea     1   0.18330 115.03 -9275.4
## + GarageQual     4   0.20015 114.65 -9275.1
## - BedroomAbvGr   1   0.20089 115.05 -9274.9
## + Neighborhood  24   1.74558 113.10 -9274.8
## + BsmtFinType2   5   0.25783 114.59 -9274.6
## - ScreenPorch    1   0.23526 115.08 -9274.1
## - KitchenQual    3   0.39697 115.25 -9274.0
## + ExterQual      3   0.07499 114.78 -9274.0
## + GarageCond     4   0.14849 114.70 -9273.8
## + HeatingQC      4   0.14269 114.71 -9273.7
## + BldgType       4   0.14090 114.71 -9273.6
## + BsmtExposure   3   0.02801 114.82 -9272.8
## - Condition2     7   0.76748 115.62 -9272.6
## - MiscVal        1   0.29474 115.14 -9272.6
## + LotConfig      4   0.09531 114.75 -9272.5
## + SaleType       8   0.39883 114.45 -9272.2
## + BsmtCond       3   0.00598 114.84 -9272.2
## + RoofStyle      5   0.15077 114.70 -9271.9
## + Functional     6   0.22894 114.62 -9271.9
## - MoSold         1   0.32559 115.17 -9271.8
## - Electrical     4   0.58783 115.44 -9271.1
## + Condition1     8   0.34887 114.50 -9270.9
## + ExterCond      4   0.02451 114.83 -9270.7
## - LandSlope      2   0.46670 115.32 -9270.2
## + GarageType     5   0.08024 114.77 -9270.1
## - MSZoning       4   0.63394 115.48 -9270.0
## - GarageYrBlt    1   0.40488 115.25 -9269.8
## - PavedDrive     2   0.53899 115.39 -9268.4
## - BsmtFinSF1     1   0.47864 115.33 -9267.9
## + HouseStyle     7   0.09775 114.75 -9266.5
## - KitchenAbvGr   1   0.59772 115.45 -9264.9
## + Exterior1st   14   0.50092 114.35 -9262.8
## - TotRmsAbvGrd   1   0.69166 115.54 -9262.5
## - BsmtUnfSF      1   0.71128 115.56 -9262.0
## - OverallQual    1   0.76552 115.61 -9260.7
```

```
## - X1stFlrSF      1   0.82845 115.68 -9259.1
## + Exterior2nd   15   0.24259 114.61 -9254.2
## - Heating        5   1.36550 116.22 -9253.5
## - YearBuilt      1   1.12222 115.97 -9251.7
## - OverallCond    1   1.40694 116.26 -9244.5
## - X2ndFlrSF      1   1.93402 116.78 -9231.3
## - RoofMatl       7   2.78142 117.63 -9222.2
## - LotArea        1   2.41005 117.26 -9219.4
##
## Step:  AIC=-9280
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + Electrical + X1stFlrSF + X2ndFlrSF +
##     BsmtHalfBath + FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageArea + PavedDrive +
##     WoodDeckSF + EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea +
##     MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - BsmtHalfBath   1   0.00227 114.85 -9281.9
## - Street         1   0.00824 114.86 -9281.8
## - WoodDeckSF     1   0.01128 114.86 -9281.7
## - PoolArea       1   0.01427 114.87 -9281.6
## - YearRemodAdd   1   0.01827 114.87 -9281.5
## - MasVnrArea     1   0.01863 114.87 -9281.5
## - MSSubClass     1   0.02415 114.88 -9281.4
## - X3SsnPorch     1   0.03888 114.89 -9281.0
## - BsmtQual       3   0.20803 115.06 -9280.7
## - Utilities      1   0.05356 114.91 -9280.6
## - EnclosedPorch  1   0.05529 114.91 -9280.6
## - YrSold         1   0.05776 114.91 -9280.5
## <none>                       114.85 -9280.0
## - LandContour    3   0.23639 115.09 -9280.0
## - FullBath       1   0.09660 114.95 -9279.5
## - Fireplaces     1   0.09830 114.95 -9279.5
## - SaleCondition  5   0.42091 115.27 -9279.3
## - LotFrontage    1   0.11012 114.96 -9279.2
## + Foundation     5   0.32070 114.53 -9278.2
## + GarageCars     1   0.00205 114.85 -9278.1
## + BsmtFullBath   1   0.00184 114.85 -9278.0
## + CentralAir     1   0.00105 114.85 -9278.0
## + LowQualFinSF   1   0.00021 114.85 -9278.0
## + GrLivArea      1   0.00021 114.85 -9278.0
## + OpenPorchSF    1   0.00013 114.85 -9278.0
## + HalfBath       1   0.00003 114.85 -9278.0
## + GarageFinish   2   0.07481 114.78 -9277.9
## - BsmtFinSF2     1   0.16775 115.02 -9277.7
## - BsmtFinType1   5   0.49158 115.34 -9277.5
## + LotShape       3   0.13618 114.72 -9277.5
## + GarageQual     4   0.20188 114.65 -9277.1
```

```
## - BedroomAbvGr   1   0.19967 115.05 -9276.9
## + Neighborhood  24   1.74622 113.11 -9276.7
## + BsmtFinType2   5   0.25811 114.59 -9276.6
## - ScreenPorch    1   0.23465 115.09 -9276.0
## - KitchenQual    3   0.39719 115.25 -9275.9
## + ExterQual      3   0.07442 114.78 -9275.9
## + GarageCond     4   0.14876 114.70 -9275.8
## + HeatingQC      4   0.14399 114.71 -9275.7
## + BldgType       4   0.13982 114.71 -9275.6
## + BsmtExposure   3   0.02743 114.82 -9274.7
## - MiscVal        1   0.29708 115.15 -9274.5
## - Condition2     7   0.77171 115.62 -9274.5
## + LotConfig      4   0.09632 114.75 -9274.4
## + BsmtCond       3   0.00593 114.85 -9274.1
## + SaleType       8   0.39795 114.45 -9274.1
## + Functional     6   0.22896 114.62 -9273.8
## + RoofStyle      5   0.14977 114.70 -9273.8
## - MoSold         1   0.32762 115.18 -9273.7
## - Electrical     4   0.58952 115.44 -9273.1
## + Condition1     8   0.34958 114.50 -9272.9
## + ExterCond      4   0.02490 114.83 -9272.6
## - LandSlope      2   0.46583 115.32 -9272.2
## + GarageType     5   0.07909 114.77 -9272.0
## - MSZoning       4   0.63447 115.49 -9271.9
## - GarageYrBlt    1   0.41617 115.27 -9271.4
## - PavedDrive     2   0.54020 115.39 -9270.3
## - BsmtFinSF1     1   0.47736 115.33 -9269.9
## + HouseStyle     7   0.09681 114.75 -9268.5
## - KitchenAbvGr   1   0.59620 115.45 -9266.9
## - GarageArea     1   0.62966 115.48 -9266.0
## + Exterior1st   14   0.50164 114.35 -9264.8
## - TotRmsAbvGrd   1   0.69993 115.55 -9264.3
## - BsmtUnfSF      1   0.71064 115.56 -9264.0
## - OverallQual    1   0.77555 115.63 -9262.4
## - X1stFlrSF      1   0.82647 115.68 -9261.1
## + Exterior2nd   15   0.24358 114.61 -9256.2
## - Heating        5   1.36592 116.22 -9255.5
## - YearBuilt      1   1.14994 116.00 -9252.9
## - OverallCond    1   1.40501 116.26 -9246.5
## - X2ndFlrSF      1   1.93207 116.78 -9233.3
## - RoofMatl       7   2.83376 117.69 -9222.9
## - LotArea        1   2.40904 117.26 -9221.4
##
## Step:  AIC=-9281.94
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     Street + LandContour + Utilities + LandSlope + Condition2 +
##     OverallQual + OverallCond + YearBuilt + YearRemodAdd + RoofMatl +
##     MasVnrArea + BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + Electrical + X1stFlrSF + X2ndFlrSF +
##     FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + WoodDeckSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
```

```
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - Street         1   0.00839 114.86 -9283.7
## - WoodDeckSF     1   0.01155 114.87 -9283.6
## - PoolArea       1   0.01514 114.87 -9283.6
## - MasVnrArea     1   0.01833 114.87 -9283.5
## - YearRemodAdd   1   0.01834 114.87 -9283.5
## - MSSubClass     1   0.02353 114.88 -9283.3
## - X3SsnPorch     1   0.03942 114.89 -9282.9
## - BsmtQual       3   0.20809 115.06 -9282.7
## - Utilities      1   0.05232 114.91 -9282.6
## - EnclosedPorch  1   0.05510 114.91 -9282.5
## - YrSold         1   0.05798 114.91 -9282.5
## <none>                       114.85 -9281.9
## - LandContour    3   0.23779 115.09 -9281.9
## - FullBath       1   0.09729 114.95 -9281.5
## - Fireplaces     1   0.09913 114.95 -9281.4
## - SaleCondition  5   0.42238 115.28 -9281.2
## - LotFrontage    1   0.10925 114.96 -9281.2
## + Foundation     5   0.31860 114.53 -9280.0
## + BsmtHalfBath   1   0.00227 114.85 -9280.0
## + GarageCars     1   0.00202 114.85 -9280.0
## + CentralAir     1   0.00101 114.85 -9280.0
## + BsmtFullBath   1   0.00064 114.85 -9280.0
## + LowQualFinSF   1   0.00023 114.85 -9279.9
## + GrLivArea      1   0.00023 114.85 -9279.9
## + OpenPorchSF    1   0.00014 114.85 -9279.9
## + HalfBath       1   0.00001 114.85 -9279.9
## + GarageFinish   2   0.07500 114.78 -9279.8
## - BsmtFinSF2     1   0.17071 115.03 -9279.6
## - BsmtFinType1   5   0.49154 115.35 -9279.5
## + LotShape       3   0.13628 114.72 -9279.4
## + GarageQual     4   0.20165 114.65 -9279.1
## - BedroomAbvGr   1   0.20418 115.06 -9278.8
## + Neighborhood  24   1.74323 113.11 -9278.6
## + BsmtFinType2   5   0.25706 114.60 -9278.5
## - ScreenPorch    1   0.23528 115.09 -9278.0
## - KitchenQual    3   0.39531 115.25 -9277.9
## + ExterQual      3   0.07426 114.78 -9277.8
## + GarageCond     4   0.14912 114.70 -9277.7
## + HeatingQC      4   0.14303 114.71 -9277.6
## + BldgType       4   0.14056 114.71 -9277.5
## + BsmtExposure   3   0.02717 114.83 -9276.6
## - MiscVal        1   0.29560 115.15 -9276.4
## + LotConfig      4   0.09674 114.76 -9276.4
## - Condition2     7   0.77294 115.63 -9276.4
## + BsmtCond       3   0.00615 114.85 -9276.1
## + SaleType       8   0.39852 114.46 -9276.1
## + Functional     6   0.22952 114.62 -9275.8
## + RoofStyle      5   0.14851 114.71 -9275.7
## - MoSold         1   0.32912 115.18 -9275.6
```

```
## - Electrical     4   0.58907 115.44 -9275.0
## + Condition1     8   0.34842 114.51 -9274.8
## + ExterCond      4   0.02497 114.83 -9274.6
## - LandSlope      2   0.46438 115.32 -9274.2
## + GarageType     5   0.07907 114.78 -9274.0
## - MSZoning       4   0.63576 115.49 -9273.8
## - GarageYrBlt    1   0.41662 115.27 -9273.4
## - PavedDrive     2   0.53919 115.39 -9272.3
## - BsmtFinSF1     1   0.47937 115.33 -9271.8
## + HouseStyle     7   0.09716 114.76 -9270.4
## - KitchenAbvGr   1   0.60177 115.46 -9268.7
## - GarageArea     1   0.63109 115.48 -9267.9
## + Exterior1st   14   0.50122 114.35 -9266.7
## - TotRmsAbvGrd   1   0.69880 115.55 -9266.2
## - BsmtUnfSF      1   0.71030 115.56 -9265.9
## - OverallQual    1   0.77476 115.63 -9264.3
## - X1stFlrSF      1   0.82487 115.68 -9263.1
## + Exterior2nd   15   0.24347 114.61 -9258.1
## - Heating        5   1.36732 116.22 -9257.4
## - YearBuilt      1   1.14975 116.00 -9254.9
## - OverallCond    1   1.41796 116.27 -9248.1
## - X2ndFlrSF      1   1.93057 116.78 -9235.3
## - RoofMatl       7   2.83565 117.69 -9224.7
## - LotArea        1   2.40843 117.26 -9223.4
##
## Step:  AIC=-9283.73
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     LandContour + Utilities + LandSlope + Condition2 + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea +
##     BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + Electrical + X1stFlrSF + X2ndFlrSF + FullBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + WoodDeckSF +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - WoodDeckSF     1   0.01129 114.87 -9285.4
## - PoolArea       1   0.01555 114.88 -9285.3
## - YearRemodAdd   1   0.01727 114.88 -9285.3
## - MasVnrArea     1   0.01822 114.88 -9285.3
## - MSSubClass     1   0.02516 114.89 -9285.1
## - X3SsnPorch     1   0.03952 114.90 -9284.7
## - BsmtQual       3   0.20755 115.07 -9284.5
## - Utilities      1   0.05235 114.92 -9284.4
## - EnclosedPorch  1   0.05497 114.92 -9284.3
## - YrSold         1   0.05858 114.92 -9284.2
## <none>                       114.86 -9283.7
## - LandContour    3   0.24547 115.11 -9283.5
## - FullBath       1   0.09677 114.96 -9283.3
## - Fireplaces     1   0.09949 114.96 -9283.2
## - SaleCondition  5   0.41547 115.28 -9283.2
```

```
## - LotFrontage    1   0.10629 114.97 -9283.0
## + Street         1   0.00839 114.85 -9281.9
## + Foundation     5   0.31760 114.55 -9281.8
## + BsmtHalfBath   1   0.00243 114.86 -9281.8
## + GarageCars     1   0.00204 114.86 -9281.8
## + CentralAir     1   0.00088 114.86 -9281.7
## + BsmtFullBath   1   0.00057 114.86 -9281.7
## + LowQualFinSF   1   0.00018 114.86 -9281.7
## + GrLivArea      1   0.00018 114.86 -9281.7
## + OpenPorchSF    1   0.00017 114.86 -9281.7
## + HalfBath       1   0.00001 114.86 -9281.7
## + GarageFinish   2   0.07463 114.79 -9281.6
## - BsmtFinSF2     1   0.16861 115.03 -9281.4
## - BsmtFinType1   5   0.49492 115.36 -9281.2
## + LotShape       3   0.13488 114.73 -9281.2
## + GarageQual     4   0.20405 114.66 -9280.9
## - BedroomAbvGr   1   0.20408 115.07 -9280.5
## + BsmtFinType2   5   0.26095 114.60 -9280.4
## + Neighborhood  24   1.73959 113.12 -9280.3
## - ScreenPorch    1   0.23348 115.10 -9279.8
## - KitchenQual    3   0.39814 115.26 -9279.6
## + GarageCond     4   0.14997 114.71 -9279.5
## + ExterQual      3   0.06595 114.80 -9279.4
## + HeatingQC      4   0.14272 114.72 -9279.4
## + BldgType       4   0.13787 114.72 -9279.2
## + BsmtExposure   3   0.02709 114.83 -9278.4
## + LotConfig      4   0.09806 114.76 -9278.2
## - MiscVal        1   0.29596 115.16 -9278.2
## - Condition2     7   0.77149 115.63 -9278.2
## + BsmtCond       3   0.00635 114.86 -9277.9
## + SaleType       8   0.39188 114.47 -9277.7
## + Functional     6   0.22944 114.63 -9277.6
## + RoofStyle      5   0.14825 114.71 -9277.5
## - MoSold         1   0.33069 115.19 -9277.3
## - Electrical     4   0.58957 115.45 -9276.8
## + ExterCond      4   0.02501 114.84 -9276.4
## + Condition1     8   0.33795 114.52 -9276.3
## - LandSlope      2   0.46600 115.33 -9275.9
## + GarageType     5   0.08059 114.78 -9275.8
## - GarageYrBlt    1   0.41469 115.28 -9275.2
## - MSZoning       4   0.68615 115.55 -9274.3
## - PavedDrive     2   0.53396 115.40 -9274.2
## - BsmtFinSF1     1   0.47645 115.34 -9273.6
## + HouseStyle     7   0.09586 114.77 -9272.2
## - KitchenAbvGr   1   0.59638 115.46 -9270.6
## - GarageArea     1   0.62536 115.49 -9269.9
## + Exterior1st   14   0.50335 114.36 -9268.5
## - TotRmsAbvGrd   1   0.70097 115.56 -9268.0
## - BsmtUnfSF      1   0.70641 115.57 -9267.8
## - OverallQual    1   0.77584 115.64 -9266.1
## - X1stFlrSF      1   0.83703 115.70 -9264.5
## + Exterior2nd   15   0.24268 114.62 -9259.9
```

```
## - Heating        5   1.36441 116.23 -9259.3
## - YearBuilt      1   1.14336 116.01 -9256.8
## - OverallCond    1   1.41588 116.28 -9250.0
## - X2ndFlrSF      1   1.94150 116.80 -9236.8
## - RoofMatl       7   2.83184 117.69 -9226.6
## - LotArea        1   2.41522 117.28 -9225.0
##
## Step:  AIC=-9285.44
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     LandContour + Utilities + LandSlope + Condition2 + OverallQual +
##     OverallCond + YearBuilt + YearRemodAdd + RoofMatl + MasVnrArea +
##     BsmtQual + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + Electrical + X1stFlrSF + X2ndFlrSF + FullBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + EnclosedPorch +
##     X3SsnPorch + ScreenPorch + PoolArea + MiscVal + MoSold +
##     YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - YearRemodAdd   1   0.01699 114.89 -9287.0
## - PoolArea       1   0.01773 114.89 -9287.0
## - MasVnrArea     1   0.01795 114.89 -9287.0
## - MSSubClass     1   0.02470 114.90 -9286.8
## - X3SsnPorch     1   0.03836 114.91 -9286.5
## - EnclosedPorch  1   0.05138 114.92 -9286.1
## - BsmtQual       3   0.20954 115.08 -9286.1
## - Utilities      1   0.05257 114.93 -9286.1
## - YrSold         1   0.05840 114.93 -9286.0
## <none>                       114.87 -9285.4
## - LandContour    3   0.24943 115.12 -9285.1
## - SaleCondition  5   0.41008 115.28 -9285.0
## - FullBath       1   0.09678 114.97 -9285.0
## - Fireplaces     1   0.10459 114.98 -9284.8
## - LotFrontage    1   0.10535 114.98 -9284.8
## + WoodDeckSF     1   0.01129 114.86 -9283.7
## + Street         1   0.00813 114.87 -9283.6
## + Foundation     5   0.31926 114.55 -9283.6
## + BsmtHalfBath   1   0.00270 114.87 -9283.5
## + GarageCars     1   0.00205 114.87 -9283.5
## + CentralAir     1   0.00097 114.87 -9283.5
## + BsmtFullBath   1   0.00076 114.87 -9283.5
## + OpenPorchSF    1   0.00048 114.87 -9283.5
## + LowQualFinSF   1   0.00015 114.87 -9283.4
## + GrLivArea      1   0.00015 114.87 -9283.4
## + HalfBath       1   0.00000 114.87 -9283.4
## + GarageFinish   2   0.07680 114.80 -9283.4
## - BsmtFinType1   5   0.49016 115.36 -9283.0
## - BsmtFinSF2     1   0.17992 115.05 -9282.9
## + LotShape       3   0.13417 114.74 -9282.9
## + GarageQual     4   0.20754 114.67 -9282.7
## - BedroomAbvGr   1   0.20232 115.08 -9282.3
## + BsmtFinType2   5   0.26313 114.61 -9282.1
```

```
## - ScreenPorch    1   0.22428 115.10 -9281.7
## + Neighborhood  24   1.73024 113.14 -9281.7
## - KitchenQual    3   0.40004 115.27 -9281.3
## + GarageCond     4   0.15037 114.72 -9281.3
## + ExterQual      3   0.06535 114.81 -9281.1
## + HeatingQC      4   0.14131 114.73 -9281.0
## + BldgType       4   0.13895 114.73 -9281.0
## + BsmtExposure   3   0.02685 114.85 -9280.1
## - MiscVal        1   0.29173 115.17 -9280.0
## + LotConfig      4   0.09845 114.78 -9279.9
## - Condition2     7   0.77235 115.65 -9279.9
## + BsmtCond       3   0.00688 114.87 -9279.6
## + SaleType       8   0.39001 114.48 -9279.4
## + RoofStyle      5   0.15028 114.72 -9279.3
## + Functional     6   0.22824 114.64 -9279.2
## - MoSold         1   0.33353 115.21 -9279.0
## - Electrical     4   0.58937 115.46 -9278.5
## + ExterCond      4   0.02573 114.85 -9278.1
## + Condition1     8   0.33877 114.53 -9278.1
## - LandSlope      2   0.46617 115.34 -9277.6
## + GarageType     5   0.08039 114.79 -9277.5
## - GarageYrBlt    1   0.40853 115.28 -9277.1
## - MSZoning       4   0.69095 115.56 -9275.9
## - PavedDrive     2   0.53585 115.41 -9275.9
## - BsmtFinSF1     1   0.48817 115.36 -9275.1
## + HouseStyle     7   0.09673 114.78 -9273.9
## - KitchenAbvGr   1   0.60746 115.48 -9272.0
## - GarageArea     1   0.63145 115.50 -9271.4
## + Exterior1st   14   0.49200 114.38 -9270.0
## - TotRmsAbvGrd   1   0.70326 115.58 -9269.6
## - BsmtUnfSF      1   0.70853 115.58 -9269.5
## - OverallQual    1   0.77267 115.65 -9267.9
## - X1stFlrSF      1   0.84634 115.72 -9266.0
## + Exterior2nd   15   0.24315 114.63 -9261.6
## - Heating        5   1.36869 116.24 -9260.9
## - YearBuilt      1   1.14228 116.02 -9258.6
## - OverallCond    1   1.44195 116.32 -9251.0
## - X2ndFlrSF      1   1.96795 116.84 -9237.9
## - RoofMatl       7   2.86877 117.74 -9227.4
## - LotArea        1   2.42472 117.30 -9226.5
##
## Step:  AIC=-9287.01
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     LandContour + Utilities + LandSlope + Condition2 + OverallQual +
##     OverallCond + YearBuilt + RoofMatl + MasVnrArea + BsmtQual +
##     BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF + Heating +
##     Electrical + X1stFlrSF + X2ndFlrSF + FullBath + BedroomAbvGr +
##     KitchenAbvGr + KitchenQual + TotRmsAbvGrd + Fireplaces +
##     GarageYrBlt + GarageArea + PavedDrive + EnclosedPorch + X3SsnPorch +
##     ScreenPorch + PoolArea + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
```

```
## - MasVnrArea     1   0.01597 114.91 -9288.6
## - PoolArea       1   0.01887 114.91 -9288.5
## - MSSubClass     1   0.02391 114.92 -9288.4
## - X3SsnPorch     1   0.03770 114.93 -9288.1
## - BsmtQual       3   0.20543 115.10 -9287.8
## - Utilities      1   0.05113 114.94 -9287.7
## - EnclosedPorch  1   0.05246 114.94 -9287.7
## - YrSold         1   0.06200 114.95 -9287.4
## <none>                       114.89 -9287.0
## - SaleCondition  5   0.40006 115.29 -9286.9
## - FullBath       1   0.09066 114.98 -9286.7
## - LandContour    3   0.24855 115.14 -9286.7
## - LotFrontage    1   0.10495 115.00 -9286.3
## - Fireplaces     1   0.10825 115.00 -9286.3
## + YearRemodAdd   1   0.01699 114.87 -9285.4
## + WoodDeckSF     1   0.01100 114.88 -9285.3
## + Street         1   0.00710 114.88 -9285.2
## + BsmtHalfBath   1   0.00276 114.89 -9285.1
## + GarageCars     1   0.00172 114.89 -9285.1
## + CentralAir     1   0.00115 114.89 -9285.0
## + OpenPorchSF    1   0.00076 114.89 -9285.0
## + BsmtFullBath   1   0.00045 114.89 -9285.0
## + LowQualFinSF   1   0.00015 114.89 -9285.0
## + GrLivArea      1   0.00015 114.89 -9285.0
## + HalfBath       1   0.00000 114.89 -9285.0
## + GarageFinish   2   0.07556 114.81 -9284.9
## - BsmtFinType1   5   0.48956 115.38 -9284.6
## - BsmtFinSF2     1   0.17969 115.07 -9284.4
## + Foundation     5   0.29109 114.60 -9284.4
## + LotShape       3   0.13220 114.76 -9284.4
## + GarageQual     4   0.20306 114.69 -9284.2
## + BsmtFinType2   5   0.26259 114.63 -9283.7
## - BedroomAbvGr   1   0.21150 115.10 -9283.6
## - KitchenQual    3   0.38318 115.27 -9283.3
## + Neighborhood  24   1.72770 113.16 -9283.2
## - ScreenPorch    1   0.22778 115.12 -9283.2
## + GarageCond     4   0.14963 114.74 -9282.8
## + ExterQual      3   0.06491 114.83 -9282.7
## + BldgType       4   0.13930 114.75 -9282.6
## + HeatingQC      4   0.13031 114.76 -9282.3
## + BsmtExposure   3   0.02701 114.86 -9281.7
## - MiscVal        1   0.29087 115.18 -9281.6
## + LotConfig      4   0.09889 114.79 -9281.5
## - Condition2     7   0.77194 115.66 -9281.5
## + BsmtCond       3   0.00715 114.88 -9281.2
## + Functional     6   0.23600 114.66 -9281.0
## + SaleType       8   0.39211 114.50 -9281.0
## + RoofStyle      5   0.15043 114.74 -9280.8
## - MoSold         1   0.33431 115.22 -9280.5
## - Electrical     4   0.58456 115.47 -9280.2
## + ExterCond      4   0.02554 114.86 -9279.7
## + Condition1     8   0.33486 114.56 -9279.5
```

```
## - LandSlope      2   0.46304 115.35 -9279.3
## + GarageType     5   0.08222 114.81 -9279.1
## - PavedDrive     2   0.52481 115.42 -9277.7
## - GarageYrBlt    1   0.45302 115.34 -9277.5
## - MSZoning       4   0.69655 115.59 -9277.4
## - BsmtFinSF1     1   0.49218 115.38 -9276.5
## + HouseStyle     7   0.09751 114.79 -9275.5
## - KitchenAbvGr   1   0.59884 115.49 -9273.8
## - GarageArea     1   0.64495 115.54 -9272.7
## + Exterior1st   14   0.50555 114.39 -9271.9
## - TotRmsAbvGrd   1   0.69333 115.58 -9271.4
## - BsmtUnfSF      1   0.71114 115.60 -9271.0
## - OverallQual    1   0.76948 115.66 -9269.5
## - X1stFlrSF      1   0.83845 115.73 -9267.8
## + Exterior2nd   15   0.24670 114.64 -9263.3
## - Heating        5   1.36046 116.25 -9262.6
## - YearBuilt      1   1.12830 116.02 -9260.5
## - OverallCond    1   1.51384 116.41 -9250.8
## - X2ndFlrSF      1   1.95635 116.85 -9239.7
## - RoofMatl       7   2.87280 117.76 -9228.9
## - LotArea        1   2.42026 117.31 -9228.2
##
## Step:  AIC=-9288.6
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     LandContour + Utilities + LandSlope + Condition2 + OverallQual +
##     OverallCond + YearBuilt + RoofMatl + BsmtQual + BsmtFinType1 +
##     BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF + Heating + Electrical +
##     X1stFlrSF + X2ndFlrSF + FullBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageArea +
##     PavedDrive + EnclosedPorch + X3SsnPorch + ScreenPorch + PoolArea +
##     MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - PoolArea       1   0.02105 114.93 -9290.1
## - MSSubClass     1   0.02621 114.93 -9289.9
## - X3SsnPorch     1   0.03786 114.94 -9289.6
## - BsmtQual       3   0.20709 115.11 -9289.3
## - Utilities      1   0.05482 114.96 -9289.2
## - EnclosedPorch  1   0.05516 114.96 -9289.2
## - YrSold         1   0.06088 114.97 -9289.1
## <none>                       114.91 -9288.6
## - SaleCondition  5   0.40177 115.31 -9288.4
## - LandContour    3   0.24488 115.15 -9288.4
## - FullBath       1   0.09444 115.00 -9288.2
## - LotFrontage    1   0.10252 115.01 -9288.0
## - Fireplaces     1   0.10654 115.01 -9287.9
## + MasVnrArea     1   0.01597 114.89 -9287.0
## + YearRemodAdd   1   0.01500 114.89 -9287.0
## + WoodDeckSF     1   0.01076 114.90 -9286.9
## + Street         1   0.00707 114.90 -9286.8
## + BsmtHalfBath   1   0.00243 114.90 -9286.7
## + GarageCars     1   0.00169 114.91 -9286.6
```

```
## + CentralAir     1   0.00096 114.91 -9286.6
## + BsmtFullBath   1   0.00074 114.91 -9286.6
## + OpenPorchSF    1   0.00046 114.91 -9286.6
## + LowQualFinSF   1   0.00009 114.91 -9286.6
## + GrLivArea      1   0.00009 114.91 -9286.6
## + HalfBath       1   0.00004 114.91 -9286.6
## + GarageFinish   2   0.07584 114.83 -9286.5
## - BsmtFinType1   5   0.48629 115.39 -9286.3
## + Foundation     5   0.30035 114.61 -9286.2
## - BsmtFinSF2     1   0.17900 115.09 -9286.1
## + LotShape       3   0.13139 114.78 -9285.9
## + GarageQual     4   0.20451 114.70 -9285.8
## + BsmtFinType2   5   0.26459 114.64 -9285.3
## - BedroomAbvGr   1   0.21468 115.12 -9285.2
## + Neighborhood  24   1.73368 113.17 -9285.0
## - ScreenPorch    1   0.22672 115.13 -9284.8
## - KitchenQual    3   0.38552 115.29 -9284.8
## + GarageCond     4   0.14982 114.76 -9284.4
## + BldgType       4   0.14570 114.76 -9284.3
## + ExterQual      3   0.06151 114.84 -9284.2
## + HeatingQC      4   0.13180 114.78 -9284.0
## + BsmtExposure   3   0.02778 114.88 -9283.3
## - MiscVal        1   0.29346 115.20 -9283.2
## - Condition2     7   0.76980 115.68 -9283.1
## + LotConfig      4   0.09863 114.81 -9283.1
## + BsmtCond       3   0.00784 114.90 -9282.8
## + SaleType       8   0.39470 114.51 -9282.6
## + RoofStyle      5   0.15574 114.75 -9282.6
## + Functional     6   0.22910 114.68 -9282.4
## - MoSold         1   0.33724 115.24 -9282.0
## - Electrical     4   0.58332 115.49 -9281.8
## + ExterCond      4   0.02631 114.88 -9281.3
## + Condition1     8   0.33384 114.57 -9281.1
## - LandSlope      2   0.46678 115.37 -9280.8
## + GarageType     5   0.08212 114.83 -9280.7
## - PavedDrive     2   0.51992 115.43 -9279.4
## - GarageYrBlt    1   0.44862 115.36 -9279.2
## - MSZoning       4   0.70723 115.61 -9278.7
## - BsmtFinSF1     1   0.48272 115.39 -9278.4
## + HouseStyle     7   0.09726 114.81 -9277.1
## - KitchenAbvGr   1   0.59092 115.50 -9275.6
## - GarageArea     1   0.63492 115.54 -9274.5
## + Exterior1st   14   0.51606 114.39 -9273.7
## - TotRmsAbvGrd   1   0.69575 115.60 -9273.0
## - BsmtUnfSF      1   0.70277 115.61 -9272.8
## - OverallQual    1   0.75536 115.66 -9271.5
## - X1stFlrSF      1   0.82252 115.73 -9269.8
## + Exterior2nd   15   0.24072 114.67 -9264.7
## - Heating        5   1.35842 116.27 -9264.3
## - YearBuilt      1   1.11335 116.02 -9262.5
## - OverallCond    1   1.52949 116.44 -9252.0
## - X2ndFlrSF      1   1.94391 116.85 -9241.6
```

```
## - RoofMatl       7   2.87550 117.78 -9230.5
## - LotArea        1   2.42078 117.33 -9229.7
##
## Step:  AIC=-9290.07
## log(SalePrice) ~ MSSubClass + MSZoning + LotFrontage + LotArea +
##     LandContour + Utilities + LandSlope + Condition2 + OverallQual +
##     OverallCond + YearBuilt + RoofMatl + BsmtQual + BsmtFinType1 +
##     BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF + Heating + Electrical +
##     X1stFlrSF + X2ndFlrSF + FullBath + BedroomAbvGr + KitchenAbvGr +
##     KitchenQual + TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageArea +
##     PavedDrive + EnclosedPorch + X3SsnPorch + ScreenPorch + MiscVal +
##     MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - MSSubClass     1   0.02445 114.95 -9291.4
## - X3SsnPorch     1   0.03759 114.97 -9291.1
## - BsmtQual       3   0.20374 115.13 -9290.9
## - Utilities      1   0.05526 114.98 -9290.7
## - EnclosedPorch  1   0.06179 114.99 -9290.5
## - YrSold         1   0.06548 114.99 -9290.4
## <none>                       114.93 -9290.1
## - SaleCondition  5   0.40113 115.33 -9289.9
## - LandContour    3   0.24793 115.18 -9289.8
## - FullBath       1   0.09281 115.02 -9289.7
## - Fireplaces     1   0.10835 115.04 -9289.3
## - LotFrontage    1   0.11036 115.04 -9289.3
## + PoolArea       1   0.02105 114.91 -9288.6
## + MasVnrArea     1   0.01815 114.91 -9288.5
## + YearRemodAdd   1   0.01601 114.91 -9288.5
## + WoodDeckSF     1   0.01308 114.92 -9288.4
## + Street         1   0.00746 114.92 -9288.3
## + BsmtHalfBath   1   0.00353 114.92 -9288.2
## + GarageCars     1   0.00163 114.93 -9288.1
## + CentralAir     1   0.00094 114.93 -9288.1
## + BsmtFullBath   1   0.00075 114.93 -9288.1
## + OpenPorchSF    1   0.00033 114.93 -9288.1
## + HalfBath       1   0.00021 114.93 -9288.1
## + LowQualFinSF   1   0.00001 114.93 -9288.1
## + GrLivArea      1   0.00001 114.93 -9288.1
## + GarageFinish   2   0.07516 114.85 -9288.0
## - BsmtFinType1   5   0.48018 115.41 -9287.9
## + Foundation     5   0.29594 114.63 -9287.6
## - BsmtFinSF2     1   0.17945 115.11 -9287.5
## + LotShape       3   0.13133 114.80 -9287.4
## + GarageQual     4   0.20238 114.72 -9287.2
## + BsmtFinType2   5   0.25871 114.67 -9286.6
## - BedroomAbvGr   1   0.21383 115.14 -9286.6
## - ScreenPorch    1   0.22890 115.16 -9286.3
## - KitchenQual    3   0.39058 115.32 -9286.2
## + Neighborhood  24   1.70966 113.22 -9285.8
## + GarageCond     4   0.14504 114.78 -9285.8
## + BldgType       4   0.14170 114.79 -9285.7
```

```
## + ExterQual      3   0.06092 114.87 -9285.6
## + HeatingQC      4   0.12871 114.80 -9285.3
## + BsmtExposure   3   0.02847 114.90 -9284.8
## - MiscVal        1   0.29295 115.22 -9284.6
## + LotConfig      4   0.09921 114.83 -9284.6
## - Condition2     7   0.77195 115.70 -9284.5
## + BsmtCond       3   0.00713 114.92 -9284.2
## + SaleType       8   0.39325 114.53 -9284.1
## + RoofStyle      5   0.15736 114.77 -9284.1
## + Functional     6   0.23139 114.70 -9284.0
## - MoSold         1   0.33052 115.26 -9283.7
## - Electrical     4   0.58442 115.51 -9283.3
## + ExterCond      4   0.02642 114.90 -9282.7
## + Condition1     8   0.32819 114.60 -9282.4
## + GarageType     5   0.08468 114.84 -9282.2
## - LandSlope      2   0.47266 115.40 -9282.1
## - PavedDrive     2   0.52359 115.45 -9280.8
## - GarageYrBlt    1   0.45237 115.38 -9280.6
## - MSZoning       4   0.70740 115.64 -9280.2
## - BsmtFinSF1     1   0.47593 115.40 -9280.0
## + HouseStyle     7   0.10180 114.83 -9278.7
## - KitchenAbvGr   1   0.59964 115.53 -9276.9
## - GarageArea     1   0.63127 115.56 -9276.1
## + Exterior1st   14   0.50907 114.42 -9275.0
## - TotRmsAbvGrd   1   0.69106 115.62 -9274.6
## - BsmtUnfSF      1   0.69125 115.62 -9274.6
## - OverallQual    1   0.74988 115.68 -9273.1
## - X1stFlrSF      1   0.85419 115.78 -9270.5
## + Exterior2nd   15   0.23915 114.69 -9266.1
## - Heating        5   1.35984 116.29 -9265.7
## - YearBuilt      1   1.12248 116.05 -9263.7
## - OverallCond    1   1.52388 116.45 -9253.6
## - X2ndFlrSF      1   1.97331 116.90 -9242.4
## - RoofMatl       7   2.88511 117.81 -9231.7
## - LotArea        1   2.42118 117.35 -9231.2
##
## Step:  AIC=-9291.45
## log(SalePrice) ~ MSZoning + LotFrontage + LotArea + LandContour +
##     Utilities + LandSlope + Condition2 + OverallQual + OverallCond +
##     YearBuilt + RoofMatl + BsmtQual + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + Electrical + X1stFlrSF +
##     X2ndFlrSF + FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageArea + PavedDrive +
##     EnclosedPorch + X3SsnPorch + ScreenPorch + MiscVal + MoSold +
##     YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - X3SsnPorch     1   0.03876 114.99 -9292.5
## - BsmtQual       3   0.20443 115.16 -9292.3
## - Utilities      1   0.05511 115.01 -9292.0
## - YrSold         1   0.06146 115.01 -9291.9
## - EnclosedPorch  1   0.06180 115.01 -9291.9
```

```
## <none>                      114.95 -9291.4
## - FullBath       1   0.08930 115.04 -9291.2
## - SaleCondition  5   0.40965 115.36 -9291.1
## - LandContour    3   0.25208 115.20 -9291.1
## - Fireplaces     1   0.10557 115.06 -9290.8
## + MSSubClass     1   0.02445 114.93 -9290.1
## + MasVnrArea     1   0.02040 114.93 -9290.0
## + PoolArea       1   0.01930 114.93 -9289.9
## + YearRemodAdd   1   0.01507 114.94 -9289.8
## + WoodDeckSF     1   0.01247 114.94 -9289.8
## + Street         1   0.00899 114.94 -9289.7
## + BsmtHalfBath   1   0.00267 114.95 -9289.5
## - LotFrontage    1   0.15554 115.11 -9289.5
## - BsmtFinType1   5   0.47182 115.42 -9289.5
## + GarageCars     1   0.00159 114.95 -9289.5
## + CentralAir     1   0.00079 114.95 -9289.5
## + BsmtFullBath   1   0.00038 114.95 -9289.5
## + HalfBath       1   0.00027 114.95 -9289.5
## + LowQualFinSF   1   0.00015 114.95 -9289.5
## + GrLivArea      1   0.00015 114.95 -9289.5
## + OpenPorchSF    1   0.00014 114.95 -9289.5
## + GarageFinish   2   0.07549 114.88 -9289.4
## + Foundation     5   0.30851 114.64 -9289.3
## + LotShape       3   0.13337 114.82 -9288.8
## + GarageQual     4   0.20966 114.74 -9288.8
## - BsmtFinSF2     1   0.18476 115.14 -9288.8
## + BsmtFinType2   5   0.26215 114.69 -9288.1
## + Neighborhood  24   1.73386 113.22 -9287.8
## - BedroomAbvGr   1   0.22467 115.18 -9287.7
## - ScreenPorch    1   0.22854 115.18 -9287.6
## + BldgType       4   0.16517 114.79 -9287.6
## - KitchenQual    3   0.39757 115.35 -9287.4
## + GarageCond     4   0.14161 114.81 -9287.0
## + ExterQual      3   0.05907 114.89 -9286.9
## + HeatingQC      4   0.13532 114.82 -9286.9
## + LotConfig      4   0.10471 114.85 -9286.1
## + BsmtExposure   3   0.02560 114.93 -9286.1
## - MiscVal        1   0.29266 115.25 -9286.0
## - Condition2     7   0.77081 115.72 -9285.9
## + BsmtCond       3   0.00800 114.94 -9285.7
## + RoofStyle      5   0.15847 114.79 -9285.5
## + Functional     6   0.23667 114.72 -9285.5
## + SaleType       8   0.38258 114.57 -9285.2
## - MoSold         1   0.33142 115.28 -9285.0
## - Electrical     4   0.57581 115.53 -9284.9
## + ExterCond      4   0.02525 114.93 -9284.1
## + Condition1     8   0.33216 114.62 -9283.9
## + GarageType     5   0.08524 114.87 -9283.6
## - LandSlope      2   0.47877 115.43 -9283.3
## - PavedDrive     2   0.52878 115.48 -9282.1
## - GarageYrBlt    1   0.45571 115.41 -9281.9
## - BsmtFinSF1     1   0.49270 115.44 -9281.0
```

```
## + HouseStyle     7   0.11430 114.84 -9280.4
## - MSZoning       4   0.76765 115.72 -9280.0
## - GarageArea     1   0.64680 115.60 -9277.1
## + Exterior1st   14   0.49779 114.45 -9276.1
## - TotRmsAbvGrd   1   0.70616 115.66 -9275.6
## - BsmtUnfSF      1   0.72376 115.68 -9275.1
## - OverallQual    1   0.73505 115.69 -9274.8
## - KitchenAbvGr   1   0.74619 115.70 -9274.6
## - X1stFlrSF      1   0.84980 115.80 -9271.9
## + Exterior2nd   15   0.24515 114.71 -9267.7
## - Heating        5   1.36539 116.32 -9267.0
## - YearBuilt      1   1.12131 116.07 -9265.1
## - OverallCond    1   1.56063 116.51 -9254.1
## - X2ndFlrSF      1   1.96175 116.91 -9244.1
## - LotArea        1   2.44747 117.40 -9232.0
## - RoofMatl       7   2.95544 117.91 -9231.3
##
## Step:  AIC=-9292.46
## log(SalePrice) ~ MSZoning + LotFrontage + LotArea + LandContour +
##     Utilities + LandSlope + Condition2 + OverallQual + OverallCond +
##     YearBuilt + RoofMatl + BsmtQual + BsmtFinType1 + BsmtFinSF1 +
##     BsmtFinSF2 + BsmtUnfSF + Heating + Electrical + X1stFlrSF +
##     X2ndFlrSF + FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual +
##     TotRmsAbvGrd + Fireplaces + GarageYrBlt + GarageArea + PavedDrive +
##     EnclosedPorch + ScreenPorch + MiscVal + MoSold + YrSold +
##     SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - BsmtQual       3   0.20367 115.19 -9293.3
## - Utilities      1   0.05475 115.05 -9293.1
## - YrSold         1   0.05844 115.05 -9293.0
## - EnclosedPorch  1   0.05889 115.05 -9293.0
## <none>                       114.99 -9292.5
## - FullBath       1   0.09291 115.08 -9292.1
## - LandContour    3   0.25121 115.24 -9292.1
## - SaleCondition  5   0.41351 115.40 -9292.0
## - Fireplaces     1   0.10512 115.10 -9291.8
## + X3SsnPorch     1   0.03876 114.95 -9291.4
## + MSSubClass     1   0.02563 114.97 -9291.1
## + MasVnrArea     1   0.02061 114.97 -9291.0
## + PoolArea       1   0.01900 114.97 -9290.9
## + YearRemodAdd   1   0.01440 114.98 -9290.8
## + WoodDeckSF     1   0.01123 114.98 -9290.7
## + Street         1   0.00916 114.98 -9290.7
## + BsmtHalfBath   1   0.00320 114.99 -9290.5
## + GarageCars     1   0.00159 114.99 -9290.5
## - BsmtFinType1   5   0.47255 115.46 -9290.5
## + CentralAir     1   0.00071 114.99 -9290.5
## + BsmtFullBath   1   0.00040 114.99 -9290.5
## + HalfBath       1   0.00026 114.99 -9290.5
## + OpenPorchSF    1   0.00026 114.99 -9290.5
## + LowQualFinSF   1   0.00010 114.99 -9290.5
```

```
## + GrLivArea      1   0.00010 114.99 -9290.5
## - LotFrontage    1   0.15986 115.15 -9290.4
## + GarageFinish   2   0.07570 114.92 -9290.4
## + Foundation     5   0.30408 114.69 -9290.2
## - BsmtFinSF2     1   0.18070 115.17 -9289.9
## + LotShape       3   0.13436 114.86 -9289.9
## + GarageQual     4   0.21224 114.78 -9289.9
## + BsmtFinType2   5   0.26239 114.73 -9289.1
## + Neighborhood  24   1.73893 113.25 -9288.9
## - BedroomAbvGr   1   0.21776 115.21 -9288.9
## - ScreenPorch    1   0.22087 115.21 -9288.9
## + BldgType       4   0.16514 114.83 -9288.7
## - KitchenQual    3   0.39805 115.39 -9288.4
## + GarageCond     4   0.14335 114.85 -9288.1
## + ExterQual      3   0.06107 114.93 -9288.0
## + HeatingQC      4   0.13592 114.86 -9287.9
## + LotConfig      4   0.10641 114.89 -9287.2
## + BsmtExposure   3   0.02422 114.97 -9287.1
## - MiscVal        1   0.29519 115.29 -9287.0
## - Condition2     7   0.77867 115.77 -9286.8
## + BsmtCond       3   0.00821 114.98 -9286.7
## + RoofStyle      5   0.15736 114.83 -9286.5
## + Functional     6   0.22965 114.76 -9286.3
## + SaleType       8   0.38052 114.61 -9286.1
## - Electrical     4   0.57445 115.56 -9285.9
## - MoSold         1   0.33832 115.33 -9285.9
## + ExterCond      4   0.02360 114.97 -9285.1
## + Condition1     8   0.33111 114.66 -9284.9
## + GarageType     5   0.08632 114.91 -9284.7
## - LandSlope      2   0.48481 115.48 -9284.2
## - PavedDrive     2   0.53171 115.52 -9283.0
## - GarageYrBlt    1   0.45261 115.44 -9283.0
## - BsmtFinSF1     1   0.49423 115.48 -9281.9
## + HouseStyle     7   0.11603 114.88 -9281.4
## - MSZoning       4   0.76539 115.76 -9281.1
## - GarageArea     1   0.64748 115.64 -9278.1
## + Exterior1st   14   0.49962 114.49 -9277.2
## - TotRmsAbvGrd   1   0.69756 115.69 -9276.8
## - BsmtUnfSF      1   0.72659 115.72 -9276.1
## - OverallQual    1   0.73038 115.72 -9276.0
## - KitchenAbvGr   1   0.74912 115.74 -9275.5
## - X1stFlrSF      1   0.86435 115.86 -9272.6
## + Exterior2nd   15   0.24458 114.75 -9268.7
## - Heating        5   1.36544 116.36 -9268.0
## - YearBuilt      1   1.11989 116.11 -9266.2
## - OverallCond    1   1.59108 116.58 -9254.4
## - X2ndFlrSF      1   1.97971 116.97 -9244.6
## - LotArea        1   2.44763 117.44 -9233.0
## - RoofMatl       7   2.97525 117.97 -9231.9
##
## Step:  AIC=-9293.3
## log(SalePrice) ~ MSZoning + LotFrontage + LotArea + LandContour +
```

```
##     Utilities + LandSlope + Condition2 + OverallQual + OverallCond +
##     YearBuilt + RoofMatl + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + Electrical + X1stFlrSF + X2ndFlrSF +
##     FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + EnclosedPorch +
##     ScreenPorch + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - Utilities      1   0.05768 115.25 -9293.8
## - EnclosedPorch  1   0.05804 115.25 -9293.8
## - YrSold         1   0.05972 115.25 -9293.8
## <none>                       115.19 -9293.3
## - LandContour    3   0.24270 115.44 -9293.2
## - Fireplaces     1   0.09632 115.29 -9292.9
## + Foundation     5   0.37070 114.82 -9292.7
## + BsmtQual       3   0.20367 114.99 -9292.5
## + X3SsnPorch     1   0.03799 115.16 -9292.3
## - FullBath       1   0.12466 115.32 -9292.1
## + MSSubClass     1   0.02632 115.17 -9292.0
## - SaleCondition  5   0.45167 115.65 -9291.9
## + MasVnrArea     1   0.02187 115.17 -9291.9
## + PoolArea       1   0.01583 115.18 -9291.7
## - BsmtFinType1   5   0.46136 115.66 -9291.6
## + WoodDeckSF     1   0.01272 115.18 -9291.6
## + YearRemodAdd   1   0.01070 115.18 -9291.6
## + Street         1   0.00863 115.19 -9291.5
## + GarageFinish   2   0.08360 115.11 -9291.4
## + GarageCars     1   0.00467 115.19 -9291.4
## + BsmtHalfBath   1   0.00314 115.19 -9291.4
## + GarageQual     4   0.23956 114.95 -9291.4
## + CentralAir     1   0.00165 115.19 -9291.3
## + LowQualFinSF   1   0.00134 115.19 -9291.3
## + GrLivArea      1   0.00134 115.19 -9291.3
## + BsmtFullBath   1   0.00082 115.19 -9291.3
## + HalfBath       1   0.00059 115.19 -9291.3
## + OpenPorchSF    1   0.00014 115.19 -9291.3
## - LotFrontage    1   0.15985 115.35 -9291.2
## + Neighborhood  24   1.79129 113.40 -9291.0
## - BsmtFinSF2     1   0.17513 115.37 -9290.9
## + LotShape       3   0.13352 115.06 -9290.7
## - BedroomAbvGr   1   0.18572 115.38 -9290.6
## - ScreenPorch    1   0.21373 115.41 -9289.9
## + BsmtFinType2   5   0.25969 114.94 -9289.9
## + BldgType       4   0.17469 115.02 -9289.7
## + ExterQual      3   0.08426 115.11 -9289.4
## + GarageCond     4   0.16292 115.03 -9289.4
## + HeatingQC      4   0.13799 115.06 -9288.8
## - KitchenQual    3   0.42999 115.62 -9288.4
## + LotConfig      4   0.12067 115.07 -9288.4
## + BsmtExposure   3   0.02495 115.17 -9287.9
## + BsmtCond       3   0.01611 115.18 -9287.7
## - MiscVal        1   0.30450 115.50 -9287.6
```

```
## - Electrical     4   0.54271 115.74 -9287.6
## + RoofStyle      5   0.15583 115.04 -9287.2
## + SaleType       8   0.39174 114.80 -9287.2
## - Condition2     7   0.79463 115.99 -9287.2
## - MoSold         1   0.32768 115.52 -9287.0
## + Functional     6   0.21717 114.98 -9286.8
## + Condition1     8   0.34095 114.85 -9286.0
## + ExterCond      4   0.02457 115.17 -9285.9
## + GarageType     5   0.07930 115.11 -9285.3
## - LandSlope      2   0.47970 115.67 -9285.2
## - GarageYrBlt    1   0.43420 115.63 -9284.3
## + HouseStyle     7   0.12054 115.07 -9282.4
## - PavedDrive     2   0.59472 115.79 -9282.3
## - BsmtFinSF1     1   0.51939 115.71 -9282.2
## - MSZoning       4   0.75817 115.95 -9282.1
## - GarageArea     1   0.68565 115.88 -9278.0
## + Exterior1st   14   0.48896 114.71 -9277.7
## - BsmtUnfSF      1   0.69651 115.89 -9277.7
## - TotRmsAbvGrd   1   0.70172 115.90 -9277.6
## - OverallQual    1   0.79853 115.99 -9275.1
## - KitchenAbvGr   1   0.80258 116.00 -9275.0
## - X1stFlrSF      1   0.89571 116.09 -9272.7
## + Exterior2nd   15   0.22874 114.97 -9269.1
## - Heating        5   1.40476 116.60 -9267.9
## - YearBuilt      1   1.38678 116.58 -9260.4
## - OverallCond    1   1.52677 116.72 -9256.9
## - X2ndFlrSF      1   2.07246 117.27 -9243.2
## - LotArea        1   2.44083 117.64 -9234.1
## - RoofMatl       7   3.01568 118.21 -9231.9
##
## Step:  AIC=-9293.84
## log(SalePrice) ~ MSZoning + LotFrontage + LotArea + LandContour +
##     LandSlope + Condition2 + OverallQual + OverallCond + YearBuilt +
##     RoofMatl + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + Electrical + X1stFlrSF + X2ndFlrSF + FullBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + EnclosedPorch +
##     ScreenPorch + MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - EnclosedPorch  1   0.05899 115.31 -9294.3
## - YrSold         1   0.06128 115.31 -9294.3
## <none>                       115.25 -9293.8
## - LandContour    3   0.24068 115.49 -9293.7
## - Fireplaces     1   0.09425 115.35 -9293.5
## + Foundation     5   0.37594 114.88 -9293.4
## + Utilities      1   0.05768 115.19 -9293.3
## + BsmtQual       3   0.20660 115.05 -9293.1
## + X3SsnPorch     1   0.03762 115.22 -9292.8
## - FullBath       1   0.12807 115.38 -9292.6
## - BsmtFinType1   5   0.44813 115.70 -9292.5
## + MasVnrArea     1   0.02637 115.23 -9292.5
```

```
## + MSSubClass     1   0.02603 115.23 -9292.5
## + PoolArea       1   0.01620 115.24 -9292.2
## - SaleCondition  5   0.46097 115.71 -9292.2
## + WoodDeckSF     1   0.01304 115.24 -9292.2
## + YearRemodAdd   1   0.00926 115.24 -9292.1
## + Street         1   0.00871 115.24 -9292.1
## + GarageQual     4   0.24222 115.01 -9292.0
## + GarageCars     1   0.00420 115.25 -9291.9
## + GarageFinish   2   0.08202 115.17 -9291.9
## + CentralAir     1   0.00191 115.25 -9291.9
## + BsmtHalfBath   1   0.00155 115.25 -9291.9
## + BsmtFullBath   1   0.00136 115.25 -9291.9
## + LowQualFinSF   1   0.00120 115.25 -9291.9
## + GrLivArea      1   0.00120 115.25 -9291.9
## + HalfBath       1   0.00046 115.25 -9291.8
## + OpenPorchSF    1   0.00027 115.25 -9291.8
## - LotFrontage    1   0.16319 115.42 -9291.7
## - BsmtFinSF2     1   0.17252 115.42 -9291.5
## + Neighborhood  24   1.78460 113.47 -9291.4
## + LotShape       3   0.13625 115.12 -9291.3
## - BedroomAbvGr   1   0.18219 115.43 -9291.2
## - ScreenPorch    1   0.19951 115.45 -9290.8
## + BsmtFinType2   5   0.26142 114.99 -9290.5
## + BldgType       4   0.17543 115.08 -9290.3
## + ExterQual      3   0.08682 115.17 -9290.0
## + GarageCond     4   0.16348 115.09 -9290.0
## + HeatingQC      4   0.14152 115.11 -9289.4
## - KitchenQual    3   0.42515 115.68 -9289.1
## + LotConfig      4   0.10930 115.14 -9288.6
## + BsmtExposure   3   0.02572 115.23 -9288.5
## + BsmtCond       3   0.01653 115.24 -9288.3
## - MiscVal        1   0.30609 115.56 -9288.1
## - Electrical     4   0.54710 115.80 -9288.0
## + RoofStyle      5   0.15494 115.10 -9287.8
## + SaleType       8   0.38975 114.86 -9287.7
## - Condition2     7   0.79770 116.05 -9287.7
## + Functional     6   0.21686 115.03 -9287.3
## - MoSold         1   0.33923 115.59 -9287.3
## + ExterCond      4   0.02443 115.23 -9286.5
## + Condition1     8   0.33937 114.91 -9286.4
## + GarageType     5   0.08108 115.17 -9285.9
## - LandSlope      2   0.47694 115.73 -9285.8
## - GarageYrBlt    1   0.43103 115.68 -9284.9
## + HouseStyle     7   0.12272 115.13 -9282.9
## - MSZoning       4   0.75267 116.00 -9282.8
## - PavedDrive     2   0.59575 115.85 -9282.8
## - BsmtFinSF1     1   0.53742 115.79 -9282.3
## - GarageArea     1   0.67967 115.93 -9278.7
## + Exterior1st   14   0.49373 114.76 -9278.4
## - TotRmsAbvGrd   1   0.69292 115.94 -9278.3
## - BsmtUnfSF      1   0.70816 115.96 -9278.0
## - OverallQual    1   0.79689 116.05 -9275.7
```

```
## - KitchenAbvGr   1   0.80247 116.06 -9275.6
## - X1stFlrSF      1   0.89570 116.15 -9273.2
## + Exterior2nd   15   0.22762 115.03 -9269.6
## - Heating        5   1.40229 116.66 -9268.5
## - YearBuilt      1   1.38663 116.64 -9260.9
## - OverallCond    1   1.52302 116.78 -9257.5
## - X2ndFlrSF      1   2.09697 117.35 -9243.2
## - LotArea        1   2.43056 117.68 -9234.9
## - RoofMatl       7   3.02881 118.28 -9232.1
##
## Step:  AIC=-9294.34
## log(SalePrice) ~ MSZoning + LotFrontage + LotArea + LandContour +
##     LandSlope + Condition2 + OverallQual + OverallCond + YearBuilt +
##     RoofMatl + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + Electrical + X1stFlrSF + X2ndFlrSF + FullBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + ScreenPorch +
##     MiscVal + MoSold + YrSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
## - YrSold         1   0.06170 115.37 -9294.8
## <none>                       115.31 -9294.3
## + Foundation     5   0.38688 114.92 -9294.2
## - LandContour    3   0.24504 115.56 -9294.1
## + EnclosedPorch  1   0.05899 115.25 -9293.8
## + Utilities      1   0.05863 115.25 -9293.8
## - Fireplaces     1   0.10004 115.41 -9293.8
## + BsmtQual       3   0.20577 115.11 -9293.6
## + X3SsnPorch     1   0.03477 115.28 -9293.2
## + MasVnrArea     1   0.03048 115.28 -9293.1
## - FullBath       1   0.12881 115.44 -9293.1
## + MSSubClass     1   0.02595 115.28 -9293.0
## - BsmtFinType1   5   0.45018 115.76 -9293.0
## + PoolArea       1   0.02196 115.29 -9292.9
## + Neighborhood  24   1.81773 113.49 -9292.7
## + YearRemodAdd   1   0.01012 115.30 -9292.6
## + WoodDeckSF     1   0.00920 115.30 -9292.6
## + Street         1   0.00863 115.30 -9292.6
## - SaleCondition  5   0.46694 115.78 -9292.5
## + GarageQual     4   0.24426 115.07 -9292.5
## + GarageFinish   2   0.08286 115.23 -9292.4
## + GarageCars     1   0.00388 115.31 -9292.4
## + CentralAir     1   0.00157 115.31 -9292.4
## + BsmtFullBath   1   0.00149 115.31 -9292.4
## + BsmtHalfBath   1   0.00141 115.31 -9292.4
## + LowQualFinSF   1   0.00065 115.31 -9292.4
## + GrLivArea      1   0.00065 115.31 -9292.4
## + HalfBath       1   0.00062 115.31 -9292.4
## + OpenPorchSF    1   0.00056 115.31 -9292.4
## - LotFrontage    1   0.17259 115.48 -9292.0
## - ScreenPorch    1   0.17900 115.49 -9291.8
## + LotShape       3   0.13629 115.17 -9291.8
```

```
## - BedroomAbvGr   1   0.18367 115.50 -9291.7
## - BsmtFinSF2     1   0.18687 115.50 -9291.6
## + BsmtFinType2   5   0.26095 115.05 -9291.0
## + BldgType       4   0.17561 115.14 -9290.8
## + ExterQual      3   0.09071 115.22 -9290.6
## + GarageCond     4   0.16276 115.15 -9290.5
## + HeatingQC      4   0.14475 115.17 -9290.0
## + LotConfig      4   0.11519 115.20 -9289.3
## - KitchenQual    3   0.44034 115.75 -9289.2
## + BsmtExposure   3   0.02649 115.28 -9289.0
## + BsmtCond       3   0.01733 115.29 -9288.8
## - MiscVal        1   0.30617 115.62 -9288.6
## - Electrical     4   0.55470 115.87 -9288.3
## + RoofStyle      5   0.15585 115.16 -9288.3
## + SaleType       8   0.38968 114.92 -9288.2
## - Condition2     7   0.80269 116.11 -9288.1
## - MoSold         1   0.33509 115.65 -9287.9
## + Functional     6   0.21226 115.10 -9287.7
## + ExterCond      4   0.02219 115.29 -9286.9
## + Condition1     8   0.33289 114.98 -9286.8
## + GarageType     5   0.08215 115.23 -9286.4
## - LandSlope      2   0.48651 115.80 -9286.1
## - GarageYrBlt    1   0.43854 115.75 -9285.3
## - MSZoning       4   0.73584 116.05 -9283.8
## + HouseStyle     7   0.12120 115.19 -9283.4
## - PavedDrive     2   0.59986 115.91 -9283.2
## - BsmtFinSF1     1   0.55055 115.86 -9282.4
## + Exterior1st   14   0.51083 114.80 -9279.3
## - TotRmsAbvGrd   1   0.68160 115.99 -9279.1
## - GarageArea     1   0.69185 116.00 -9278.9
## - BsmtUnfSF      1   0.73024 116.04 -9277.9
## - KitchenAbvGr   1   0.81724 116.13 -9275.7
## - OverallQual    1   0.81768 116.13 -9275.7
## - X1stFlrSF      1   0.89155 116.20 -9273.9
## + Exterior2nd   15   0.23392 115.08 -9270.3
## - Heating        5   1.40750 116.72 -9268.9
## - YearBuilt      1   1.32915 116.64 -9262.9
## - OverallCond    1   1.48247 116.79 -9259.1
## - X2ndFlrSF      1   2.12430 117.44 -9243.1
## - LotArea        1   2.44088 117.75 -9235.2
## - RoofMatl       7   3.03770 118.35 -9232.4
##
## Step:  AIC=-9294.78
## log(SalePrice) ~ MSZoning + LotFrontage + LotArea + LandContour +
##     LandSlope + Condition2 + OverallQual + OverallCond + YearBuilt +
##     RoofMatl + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 + BsmtUnfSF +
##     Heating + Electrical + X1stFlrSF + X2ndFlrSF + FullBath +
##     BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + ScreenPorch +
##     MiscVal + MoSold + SaleCondition
##
##                 Df Sum of Sq    RSS     AIC
```

```
## <none>                      115.37 -9294.8
## - LandContour    3   0.24467 115.62 -9294.6
## + Foundation     5   0.38602 114.99 -9294.6
## + YrSold         1   0.06170 115.31 -9294.3
## + Utilities      1   0.06021 115.31 -9294.3
## - Fireplaces     1   0.09811 115.47 -9294.3
## + EnclosedPorch  1   0.05941 115.31 -9294.3
## + BsmtQual       3   0.20717 115.17 -9294.0
## - BsmtFinType1   5   0.43936 115.81 -9293.7
## - FullBath       1   0.12543 115.50 -9293.6
## + X3SsnPorch     1   0.03186 115.34 -9293.6
## + MasVnrArea     1   0.02925 115.34 -9293.5
## + Neighborhood  24   1.83049 113.54 -9293.5
## + PoolArea       1   0.02668 115.35 -9293.5
## + MSSubClass     1   0.02176 115.35 -9293.3
## + GarageQual     4   0.25418 115.12 -9293.2
## + YearRemodAdd   1   0.01321 115.36 -9293.1
## + WoodDeckSF     1   0.00927 115.36 -9293.0
## + Street         1   0.00903 115.36 -9293.0
## + GarageCars     1   0.00439 115.37 -9292.9
## + GarageFinish   2   0.08315 115.29 -9292.9
## + BsmtHalfBath   1   0.00170 115.37 -9292.8
## + CentralAir     1   0.00144 115.37 -9292.8
## + HalfBath       1   0.00108 115.37 -9292.8
## + BsmtFullBath   1   0.00106 115.37 -9292.8
## + LowQualFinSF   1   0.00064 115.37 -9292.8
## + GrLivArea      1   0.00064 115.37 -9292.8
## + OpenPorchSF    1   0.00026 115.37 -9292.8
## - LotFrontage    1   0.17027 115.54 -9292.5
## - ScreenPorch    1   0.17924 115.55 -9292.2
## + LotShape       3   0.13663 115.24 -9292.2
## - BedroomAbvGr   1   0.18264 115.56 -9292.2
## - BsmtFinSF2     1   0.18570 115.56 -9292.1
## - SaleCondition  5   0.50978 115.88 -9291.9
## + BsmtFinType2   5   0.26344 115.11 -9291.5
## + BldgType       4   0.17634 115.20 -9291.2
## + GarageCond     4   0.16346 115.21 -9290.9
## + ExterQual      3   0.08271 115.29 -9290.9
## + HeatingQC      4   0.14507 115.23 -9290.5
## - KitchenQual    3   0.42853 115.80 -9290.0
## + LotConfig      4   0.11547 115.26 -9289.7
## + BsmtExposure   3   0.02878 115.34 -9289.5
## + BsmtCond       3   0.02029 115.35 -9289.3
## - MiscVal        1   0.31146 115.68 -9288.9
## - Electrical     4   0.55268 115.93 -9288.8
## + RoofStyle      5   0.15946 115.21 -9288.8
## + SaleType       8   0.38217 114.99 -9288.5
## - Condition2     7   0.80738 116.18 -9288.4
## + Functional     6   0.20650 115.17 -9288.0
## + Condition1     8   0.34273 115.03 -9287.5
## + ExterCond      4   0.02033 115.35 -9287.3
## - MoSold         1   0.38515 115.76 -9287.1
```

```
## + GarageType     5   0.08324 115.29 -9286.9
## - LandSlope      2   0.49783 115.87 -9286.2
## - GarageYrBlt    1   0.44170 115.81 -9285.6
## - MSZoning       4   0.74232 116.11 -9284.1
## + HouseStyle     7   0.11650 115.26 -9283.7
## - PavedDrive     2   0.60143 115.97 -9283.6
## - BsmtFinSF1     1   0.55089 115.92 -9282.9
## + Exterior1st   14   0.52000 114.85 -9280.0
## - TotRmsAbvGrd   1   0.68681 116.06 -9279.5
## - GarageArea     1   0.68844 116.06 -9279.4
## - BsmtUnfSF      1   0.72660 116.10 -9278.5
## - OverallQual    1   0.82266 116.20 -9276.0
## - KitchenAbvGr   1   0.84001 116.21 -9275.6
## - X1stFlrSF      1   0.89799 116.27 -9274.1
## + Exterior2nd   15   0.24264 115.13 -9270.9
## - Heating        5   1.42266 116.80 -9269.0
## - YearBuilt      1   1.33835 116.71 -9263.1
## - OverallCond    1   1.47390 116.85 -9259.7
## - X2ndFlrSF      1   2.13844 117.51 -9243.2
## - LotArea        1   2.46045 117.83 -9235.2
## - RoofMatl       7   3.05984 118.43 -9232.4
```

```
# Summary of the final model
summary(step_model_log_model)
```

```
##
## Call:
## lm(formula = log(SalePrice) ~ MSZoning + LotFrontage + LotArea +
##     LandContour + LandSlope + Condition2 + OverallQual + OverallCond +
##     YearBuilt + RoofMatl + BsmtFinType1 + BsmtFinSF1 + BsmtFinSF2 +
##     BsmtUnfSF + Heating + Electrical + X1stFlrSF + X2ndFlrSF +
##     FullBath + BedroomAbvGr + KitchenAbvGr + KitchenQual + TotRmsAbvGrd +
##     Fireplaces + GarageYrBlt + GarageArea + PavedDrive + ScreenPorch +
##     MiscVal + MoSold + SaleCondition, data = imputed_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.1606 -0.1227  0.0055  0.1346  0.7356
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           7.543e+00  6.217e-01  12.133  < 2e-16 ***
## MSZoningFV            1.161e-01  4.727e-02   2.457 0.014071 *
## MSZoningRH            4.533e-02  5.827e-02   0.778 0.436748
## MSZoningRL            1.290e-01  4.343e-02   2.972 0.002987 **
## MSZoningRM            9.090e-02  4.327e-02   2.101 0.035761 *
## LotFrontage           4.472e-04  2.180e-04   2.051 0.040336 *
## LotArea               5.016e-06  6.432e-07   7.797 8.79e-15 ***
## LandContourHLS        2.732e-02  2.739e-02   0.997 0.318608
## LandContourLow        6.046e-02  3.620e-02   1.670 0.095020 .
## LandContourLvl        4.621e-02  2.075e-02   2.227 0.026007 *
## LandSlopeMod         -4.574e-03  2.216e-02  -0.206 0.836505
## LandSlopeSev         -2.298e-01  6.585e-02  -3.490 0.000491 ***
## Condition2Feedr      -1.864e-02  1.073e-01  -0.174 0.862111
## Condition2Norm        7.813e-03  9.166e-02   0.085 0.932083
## Condition2PosA       -1.954e-01  1.374e-01  -1.422 0.155105
## Condition2PosN       -3.338e-01  1.373e-01  -2.431 0.015118 *
## Condition2RRAe       -7.205e-02  2.303e-01  -0.313 0.754426
## Condition2RRAn       -2.171e-01  2.216e-01  -0.980 0.327324
## Condition2RRNn       -2.709e-01  1.705e-01  -1.588 0.112303
## OverallQual           2.282e-02  5.061e-03   4.509 6.78e-06 ***
## OverallCond           2.461e-02  4.077e-03   6.035 1.79e-09 ***
## YearBuilt             1.600e-03  2.781e-04   5.751 9.82e-09 ***
## RoofMatlCompShg       1.672e+00  2.202e-01   7.591 4.26e-14 ***
## RoofMatlMembran       2.086e+00  3.067e-01   6.800 1.27e-11 ***
## RoofMatlMetal         2.008e+00  3.069e-01   6.542 7.18e-11 ***
## RoofMatlRoll          1.437e+00  2.997e-01   4.794 1.72e-06 ***
## RoofMatlTar&Grv       1.693e+00  2.243e-01   7.550 5.84e-14 ***
## RoofMatlWdShake       1.705e+00  2.303e-01   7.404 1.73e-13 ***
## RoofMatlWdShngl       1.925e+00  2.318e-01   8.304  < 2e-16 ***
## BsmtFinType1BLQ      -5.695e-03  1.588e-02  -0.359 0.719906
## BsmtFinType1GLQ       2.659e-02  1.368e-02   1.943 0.052055 .
## BsmtFinType1LwQ       6.596e-03  1.995e-02   0.331 0.740922
## BsmtFinType1Rec       1.643e-02  1.583e-02   1.038 0.299221
## BsmtFinType1Unf      -1.584e-02  1.559e-02  -1.016 0.309793
## BsmtFinSF1            7.092e-05  1.922e-05   3.690 0.000229 ***
## BsmtFinSF2            6.128e-05  2.860e-05   2.142 0.032265 *
```

```
## BsmtUnfSF             7.180e-05  1.694e-05   4.237 2.33e-05 ***
## HeatingGasA           4.865e-01  2.019e-01   2.409 0.016051 *
## HeatingGasW           4.413e-01  2.060e-01   2.142 0.032269 *
## HeatingGrav           1.321e-01  2.138e-01   0.618 0.536629
## HeatingOthW           2.480e-01  2.489e-01   0.996 0.319098
## HeatingWall           4.400e-01  2.187e-01   2.012 0.044355 *
## ElectricalFuseF      -1.702e-02  3.322e-02  -0.512 0.608387
## ElectricalFuseP       7.615e-02  7.441e-02   1.023 0.306228
## ElectricalMix        -6.701e-01  2.036e-01  -3.291 0.001009 **
## ElectricalSBrkr       1.538e-02  1.657e-02   0.928 0.353673
## X1stFlrSF             1.074e-04  2.279e-05   4.711 2.59e-06 ***
## X2ndFlrSF             1.293e-04  1.778e-05   7.269 4.64e-13 ***
## FullBath              1.861e-02  1.057e-02   1.761 0.078428 .
## BedroomAbvGr          1.513e-02  7.123e-03   2.124 0.033721 *
## KitchenAbvGr         -9.746e-02  2.139e-02  -4.556 5.43e-06 ***
## KitchenQualFa        -8.897e-02  3.341e-02  -2.663 0.007793 **
## KitchenQualGd        -2.514e-02  1.753e-02  -1.435 0.151499
## KitchenQualTA        -5.161e-02  2.027e-02  -2.547 0.010920 *
## TotRmsAbvGrd          2.061e-02  5.003e-03   4.120 3.90e-05 ***
## Fireplaces            1.134e-02  7.282e-03   1.557 0.119568
## GarageYrBlt          -8.563e-04  2.592e-04  -3.304 0.000966 ***
## GarageArea            1.006e-04  2.439e-05   4.125 3.82e-05 ***
## PavedDriveP          -6.521e-02  3.004e-02  -2.171 0.030047 *
## PavedDriveY          -6.835e-02  1.786e-02  -3.826 0.000133 ***
## ScreenPorch           1.443e-04  6.858e-05   2.105 0.035413 *
## MiscVal              -1.935e-05  6.974e-06  -2.774 0.005568 **
## MoSold                4.307e-03  1.396e-03   3.085 0.002055 **
## SaleConditionAdjLand  8.637e-02  6.214e-02   1.390 0.164674
## SaleConditionAlloca   7.270e-02  4.551e-02   1.598 0.110246
## SaleConditionFamily   7.529e-03  3.346e-02   0.225 0.821969
## SaleConditionNormal   2.353e-02  1.579e-02   1.490 0.136362
## SaleConditionPartial  6.471e-02  2.163e-02   2.991 0.002804 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2012 on 2851 degrees of freedom
## Multiple R-squared:  0.5348, Adjusted R-squared:  0.5239
## F-statistic: 48.92 on 67 and 2851 DF,  p-value: < 2.2e-16
```

## check assumptions for Step-wise Regression with log y

```
# Check for normality of residuals
shapiro.test(resid(step_model_log_model))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid(step_model_log_model)
## W = 0.98572, p-value < 2.2e-16
```

```
# Check for independence of residuals
durbinWatsonTest(step_model_log_model)
```

```
##  lag Autocorrelation D-W Statistic p-value
##    1       0.2741922      1.451466       0
##  Alternative hypothesis: rho != 0
```

```
# Check for homoscedasticity

bptest(step_model_log_model)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  step_model_log_model
## BP = 804.69, df = 67, p-value < 2.2e-16
```

```
# Check for multicollinearity
vif(step_model_log_model)
```

```
##                   GVIF Df GVIF^(1/(2*Df))
## MSZoning      2.111983  4        1.097959
## LotFrontage   1.648060  1        1.283768
## LotArea       1.855840  1        1.362292
## LandContour   2.082849  3        1.130081
## LandSlope     2.413371  2        1.246396
## Condition2    1.313826  7        1.019687
## OverallQual   3.671483  1        1.916112
## OverallCond   1.485373  1        1.218759
## YearBuilt     5.118815  1        2.262480
## RoofMatl      1.771826  7        1.041704
## BsmtFinType1  5.059404  5        1.176007
## BsmtFinSF1    5.529822  1        2.351557
## BsmtFinSF2    1.688637  1        1.299476
## BsmtUnfSF     4.000374  1        2.000094
## Heating       1.343721  5        1.029985
## Electrical    1.496067  4        1.051644
## X1stFlrSF     5.765807  1        2.401209
## X2ndFlrSF     4.191155  1        2.047231
## FullBath      2.464568  1        1.569894
## BedroomAbvGr  2.476165  1        1.573584
## KitchenAbvGr  1.517563  1        1.231894
## KitchenQual   3.127337  3        1.209286
## TotRmsAbvGrd  4.444861  1        2.108284
## Fireplaces    1.596118  1        1.263376
## GarageYrBlt   3.098583  1        1.760279
## GarageArea    1.993042  1        1.411752
## PavedDrive    1.684602  2        1.139264
## ScreenPorch   1.070476  1        1.034638
## MiscVal       1.129155  1        1.062617
## MoSold        1.035699  1        1.017693
## SaleCondition 1.760605  5        1.058196
```

## Lasso Regression

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.3.3
```

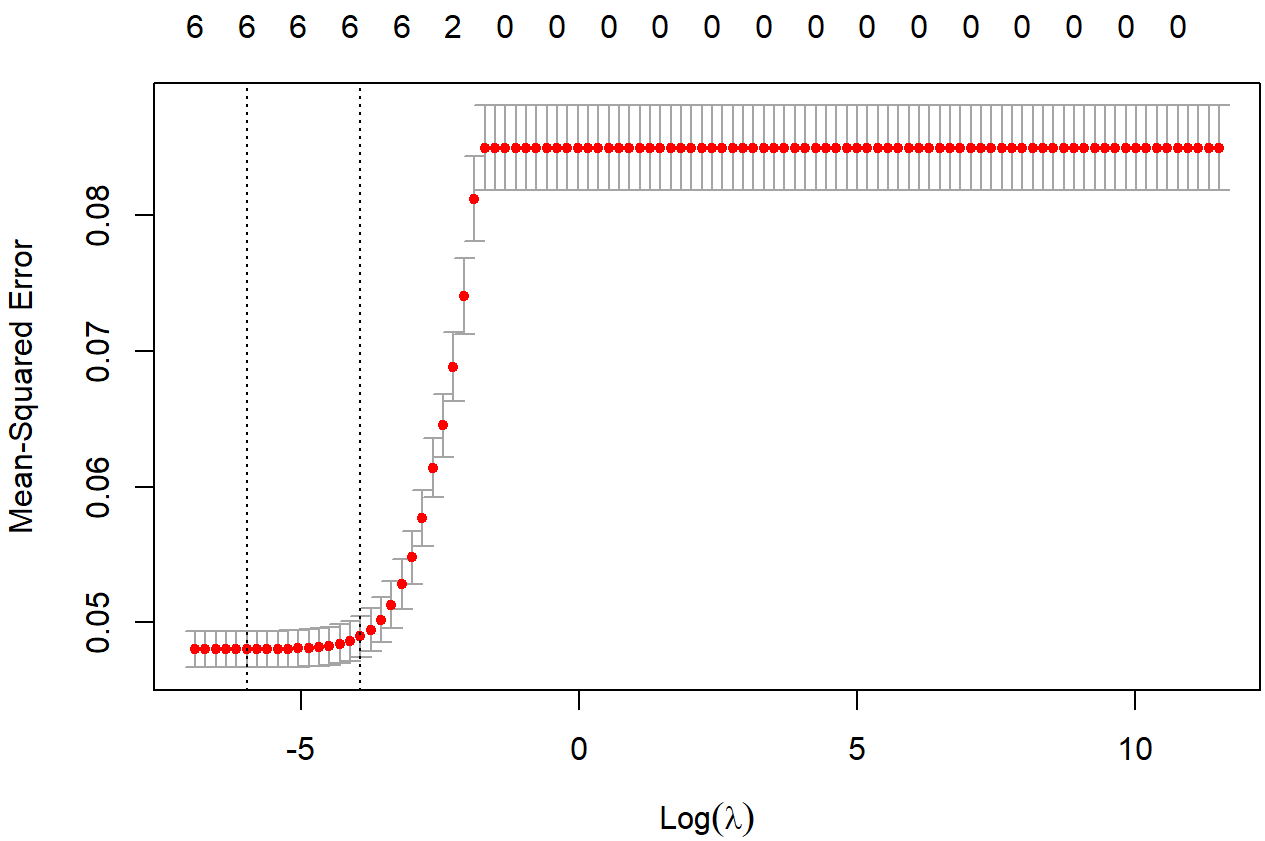
```
## Loading required package: Matrix
```

```
##
## Attaching package: 'Matrix'
```

```
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

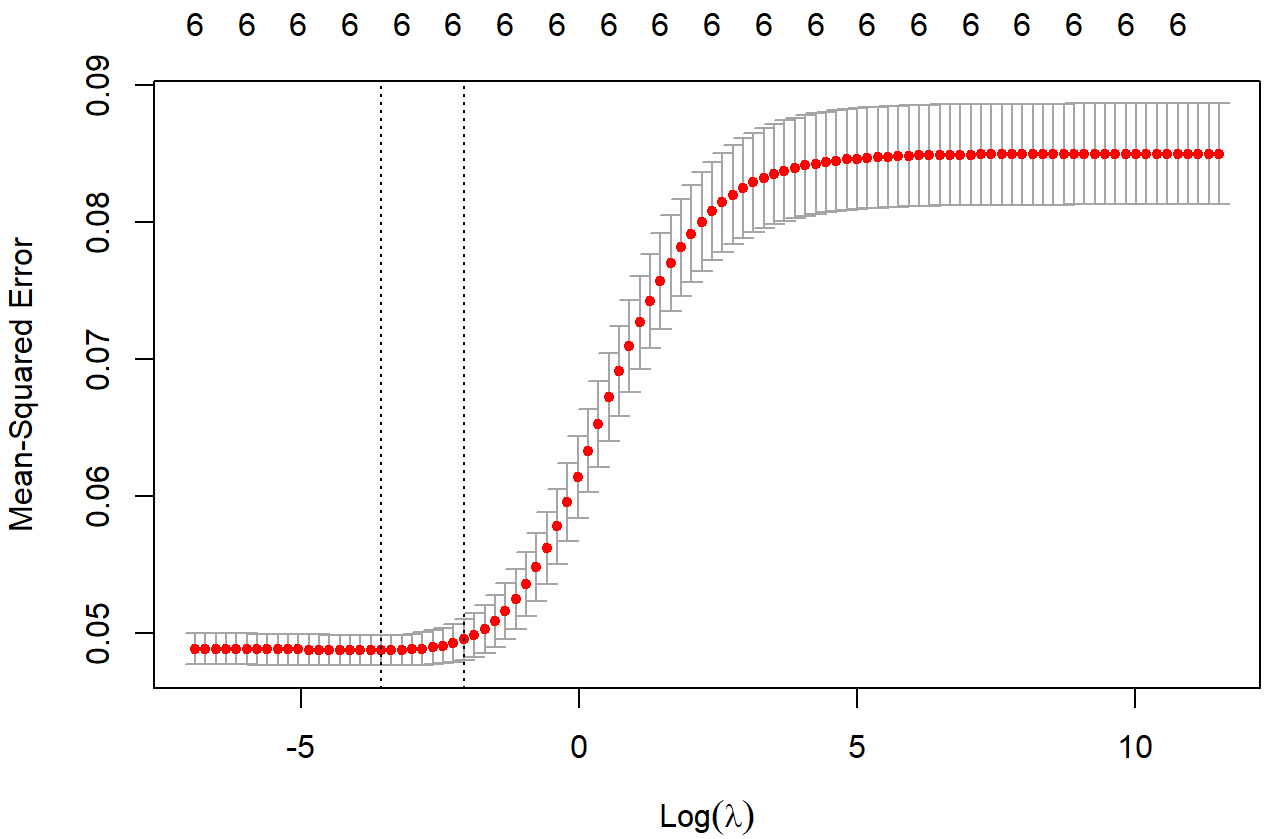
```
## Loaded glmnet 4.1-8
```

```
# Set a seed for reproducibility
set.seed(42)
X <- as.matrix(X)
lambdas_to_try <- 10^seq(-3, 5, length.out = 100)

lasso_cv <- cv.glmnet(X, y, alpha = 1, lambda = lambdas_to_try,
                      standardize = TRUE, nfolds = 10)
# Plot cross-validation results
plot(lasso_cv)
```




## Ridge Regression

```
# Set a seed for reproducibility
set.seed(42)
X <- as.matrix(X)
lambdas_to_try <- 10^seq(-3, 5, length.out = 100)

ridge_cv <- cv.glmnet(X, y, alpha = 0, lambda = lambdas_to_try,
                      standardize = TRUE, nfolds = 5)
# Plot cross-validation results
plot(ridge_cv)
```




####

Random forest Regressor

```
set.seed(123)  # For reproducibility
n <- 100

# Train the Random Forest model
rf_model <- randomForest(x = X, y = y, ntree = n)

# Print the summary of the model
print(rf_model)
```

```
##
## Call:
##  randomForest(x = X, y = y, ntree = n)
##                Type of random forest: regression
##                      Number of trees: 100
## No. of variables tried at each split: 2
##
##           Mean of squared residuals: 0.05074531
##                     % Var explained: 40.27
```

```
# Make predictions
predictions <- predict(rf_model, X)

# Evaluate the model (for example, using mean squared error)
mse <- mean((predictions - y)^2)
print(paste("Mean Squared Error:", mse))
```

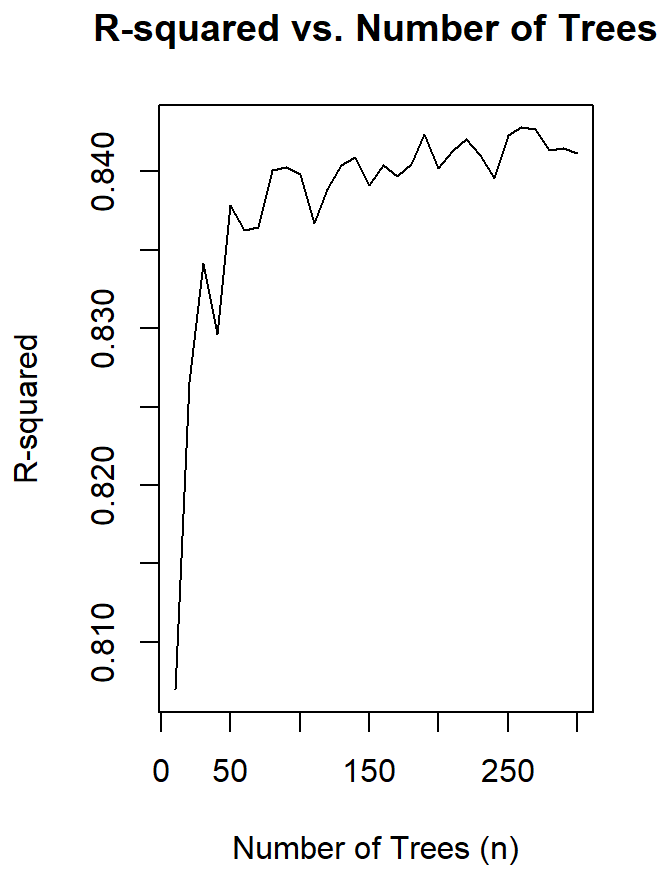
```
## [1] "Mean Squared Error: 0.0134874964639802"
```

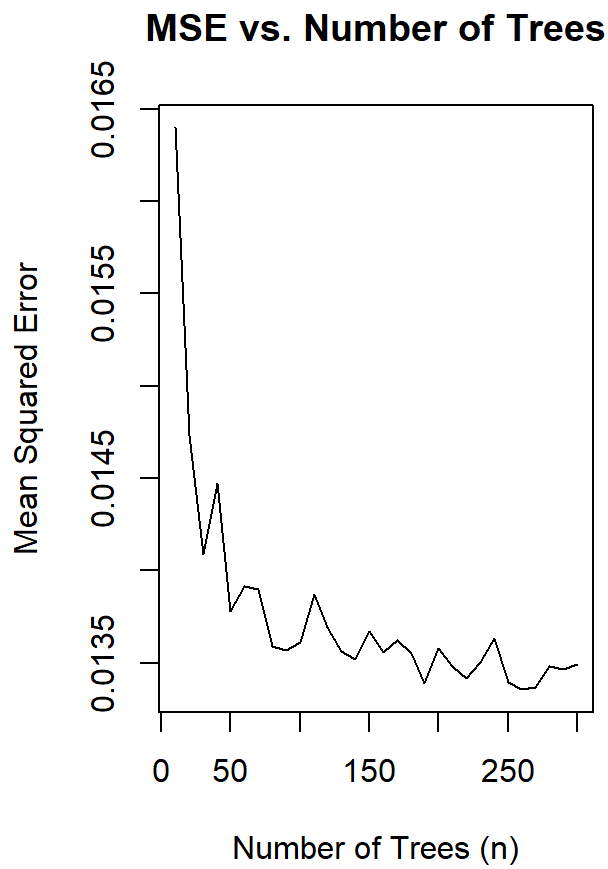
```
# Calculate mean of actual response values
mean_actual <- mean(y)
# Calculate total sum of squares
total_ss <- sum((y - mean_actual)^2)
# Calculate residual sum of squares
residual_ss <- sum((y - predictions)^2)
# Calculate R-squared
r_squared_manual <- 1 - (residual_ss / total_ss)
print(paste("R-squared (Manual):", r_squared_manual))
```

```
## [1] "R-squared (Manual): 0.841250050791717"
```

## HyperTuning the N to achive the best model

```
# Range of n values to try
n_values <- seq(10, 300, by = 10)

# Initialize vectors to store R-squared and MSE values
rsquared_values <- numeric(length(n_values))
mse_values <- numeric(length(n_values))

# Loop through each value of n
for (i in seq_along(n_values)) {
  # Train the Random Forest model
  rf_model <- randomForest(x = X, y = y, ntree = n_values[i])

  # Make predictions
  predictions <- predict(rf_model, X)


   # Calculate R-squared
  mean_y <- mean(y)
  total_ss <- sum((y - mean_y)^2)
  residual_ss <- sum((y - predictions)^2)
  rsquared_values[i] <- 1 - (residual_ss / total_ss)

  # Calculate MSE
  mse_values[i] <- mean((predictions - y)^2)
}

# Plot both R-squared and MSE values against n
par(mfrow=c(1,2))  # Set up a 1x2 plotting grid
plot(n_values, rsquared_values, type = "l",
     xlab = "Number of Trees (n)", ylab = "R-squared",
     main = "R-squared vs. Number of Trees")
plot(n_values, mse_values, type = "l",
     xlab = "Number of Trees (n)", ylab = "Mean Squared Error",
     main = "MSE vs. Number of Trees")
```




# Best Model

```
set.seed(123)  # For reproducibility
n <- 260

# Train the Random Forest model
rf_model <- randomForest(x = X, y = y, ntree = n)

# Print the summary of the model
print(rf_model)
```

```
##
## Call:
##  randomForest(x = X, y = y, ntree = n)
##                Type of random forest: regression
##                      Number of trees: 260
## No. of variables tried at each split: 2
##
##           Mean of squared residuals: 0.04990841
##                     % Var explained: 41.26
```

```
# Make predictions
predictions <- predict(rf_model, X)

# Evaluate the model (for example, using mean squared error)
mse <- mean((predictions - y)^2)
print(paste("Mean Squared Error:", mse))
```

```
## [1] "Mean Squared Error: 0.0134870257671303"
```

```
# Calculate mean of actual response values
mean_actual <- mean(y)
# Calculate total sum of squares
total_ss <- sum((y - mean_actual)^2)
# Calculate residual sum of squares
residual_ss <- sum((y - predictions)^2)
# Calculate R-squared
r_squared_manual <- 1 - (residual_ss / total_ss)
print(paste("R-squared (Manual):", r_squared_manual))
```

```
## [1] "R-squared (Manual): 0.84125559096748"
```